Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Smart Factory & Industry 4.0/5.0

technologies Cyber-Physical System (CPS)

Industry/Factory Actuators

PLC, SCADA and HMI System

Core Automation: PLC Programming



Training

UMES x KMITL x TIRA x ABAC

SCHOOL OF ENGINEERING KMITL

VINCENT MARY SCHOOL OF ENGINEERING SCIENCE AND TECHNOLOGY

ASSUMPTION UNIVERSITY OF THAILAND
SINCE 1969
มหาวิทยาลัยอัสสัมชัญ

TIRA
THAI INNOVATION AND ROBOTICS ASSOCIATION
สมาคมการสร้างนวัตกรรมและหุ่นยนต์

THAILAND BOARD OF INVESTMENT

กระทรวงการอุดมศึกษา วิทยาศาสตร์ วิจัยและนวัตกรรม
Ministry of Higher Education, Science, Research and Innovation

สอวช
สำนักงานสภานโยบายการอุดมศึกษา
วิทยาศาสตร์ วิจัย
และนวัตกรรมแห่งชาติ

Training

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Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Day	Topics
Day 1	Smart Factory & Industry 4.0/5.0 technologies Cyber-Physical System (CPS) Industry/Factory Actuators PLC, SCADA and HMI System Core Automation: PLC Programming Predictive maintenance with AI
	Industry/Factory Sensor Sensor Technology Industry Communication Industry IoT (IIOT)
Day 2	Workshop : PLC Programming and HMI
	Workshop : Hydraulics and Pneumatics System



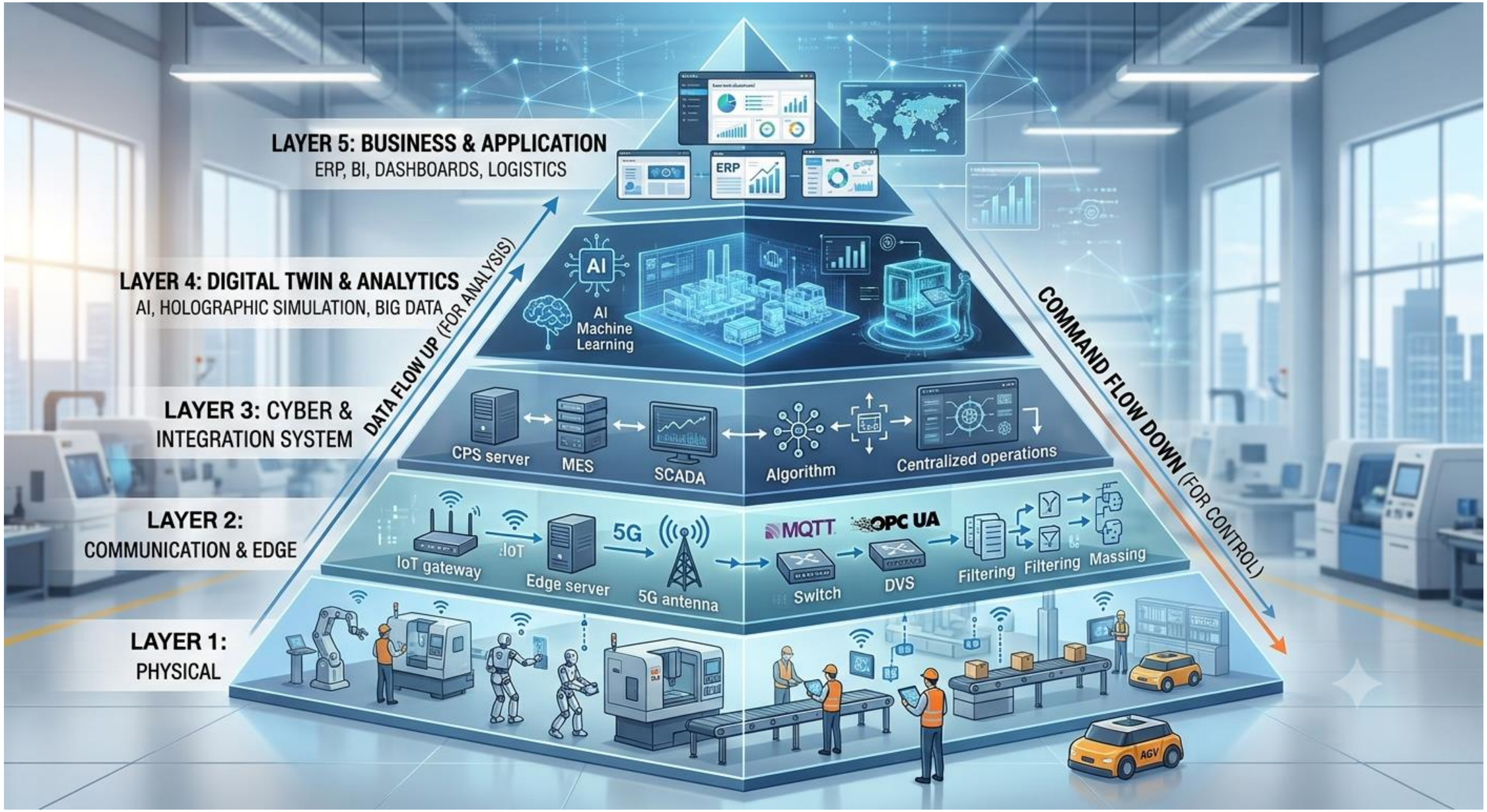
Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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Smart Factory

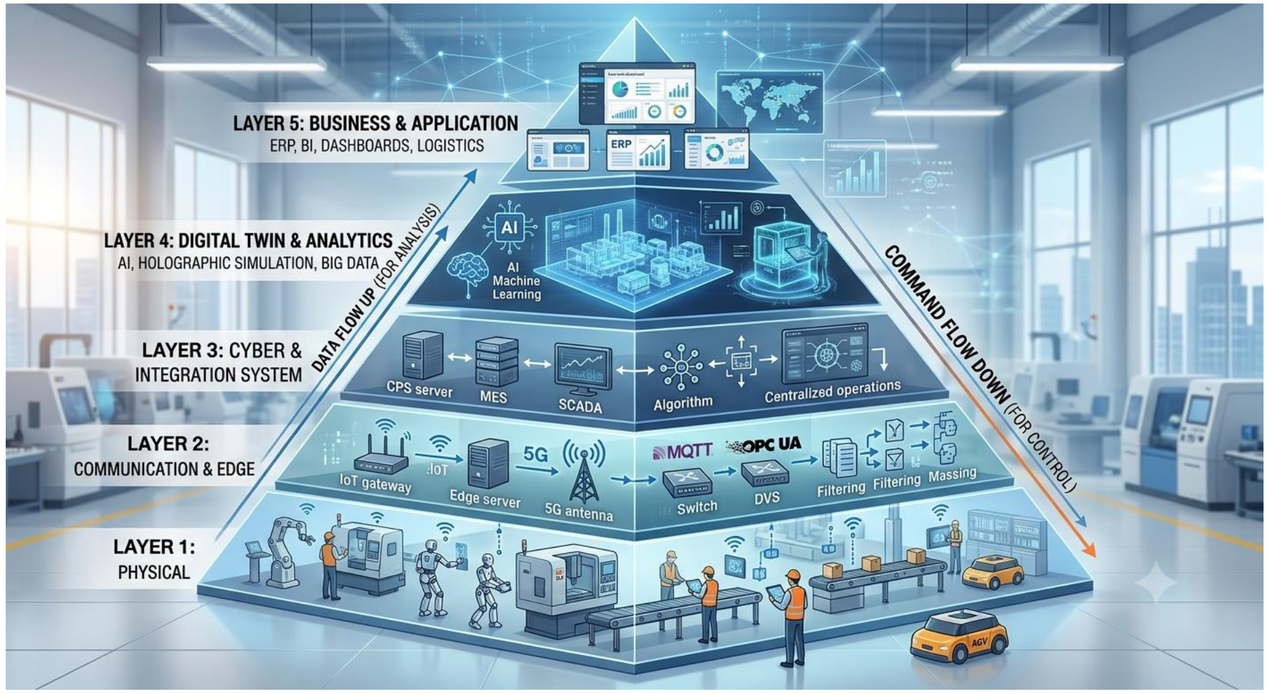


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Training

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#2 Intelligent Industrial Robotics and Flexible Automation

- Industrial Robots, Cobot, Kinematics
- Robot Control and Path Planning
- หุ่นยนต์เคลื่อนที่ (AGV/AMR)
- Robot Operating System (ROS/ROS2)/ SLAM
- Integration with Machine Vision / Sensors
- AI and Deep learning for manipulation
- Flexible Manufacturing Systems (FMS)
- Reconfigurable Robotic Cells
- Virtual Commissioning & Digital Twin

#3 Advanced Robot and Automation Control and Human–Robot Collaboration

- Precision robot control (micron-level)
- Advanced PID for robotics and automation
- Force control/Impedance Control in robot and automation
- Adaptive, robust & sliding-mode control
- Machine learning/ Reinforcement Learning (RL) for control optimization
- Safe Human–Robot Interaction (HRC)
- Collaborative Robot Standards (ISO/TS 15066)

#4 Machine Vision, AI Robotics and Autonomous System

- AI-based inspection
- Industrial Vision System
- 3D/2D Machine Vision
- SLAM, Localization, Mapping
- Sensor Fusion
- Sensor Fusion (Camera, LiDAR, IMU, Force/Depth) Autonomous Navigation & Planning, Optimal planning (CHOMP, STOMP)

Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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1ST INDUSTRIAL REVOLUTION

1784

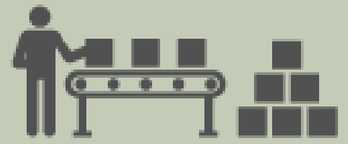


MECHANISATION

MECHANISATION
STEAM POWER

2ND INDUSTRIAL REVOLUTION

1870

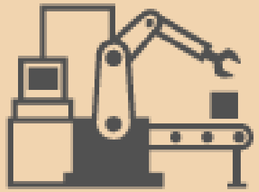


ELECTRIFICATION

ASSEMBLY LINE
MASS PRODUCTION

3RD INDUSTRIAL REVOLUTION

1969



AUTOMATION

AUTOMATION
ELECTRONICS

4TH INDUSTRIAL REVOLUTION

2011



DIGITISATION

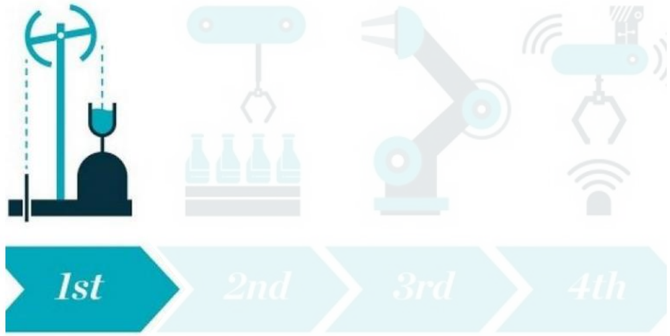
CYBER PHYSICAL SYSTEMS
NETWORKS

Smart Automation

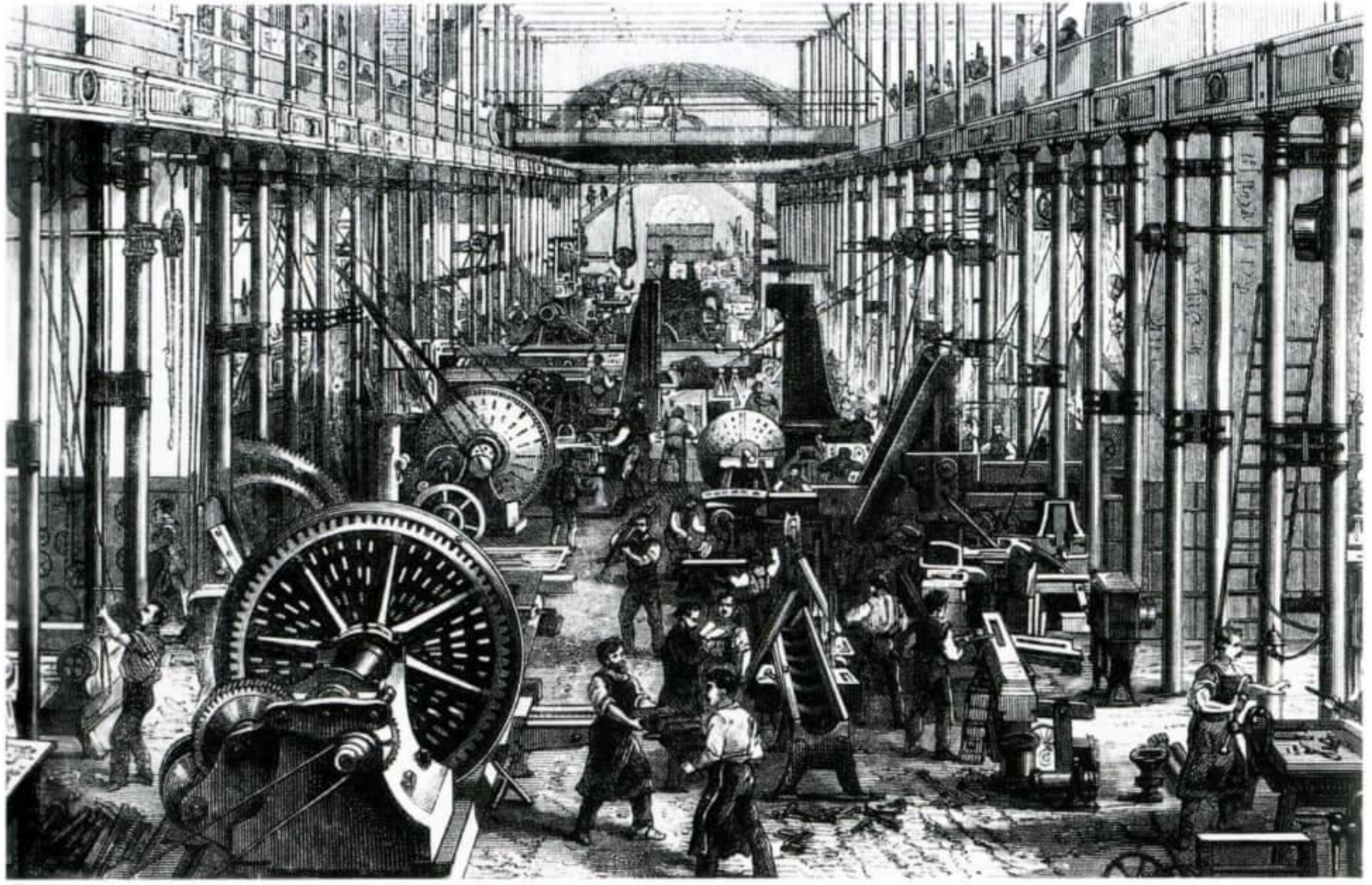
IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC



MECHANISATION
STEAM POWER

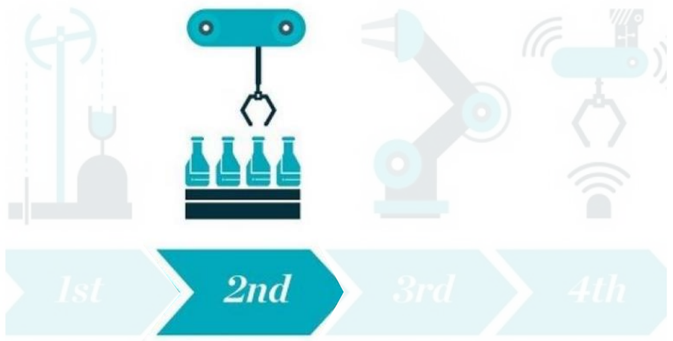


Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC



ASSEMBLY LINE
MASS PRODUCTION

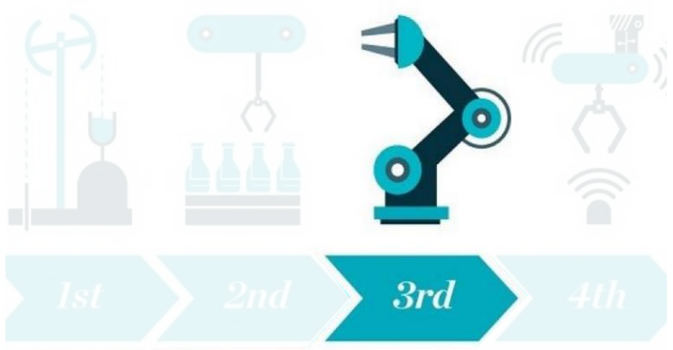


Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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AUTOMATION
ELECTRONICS

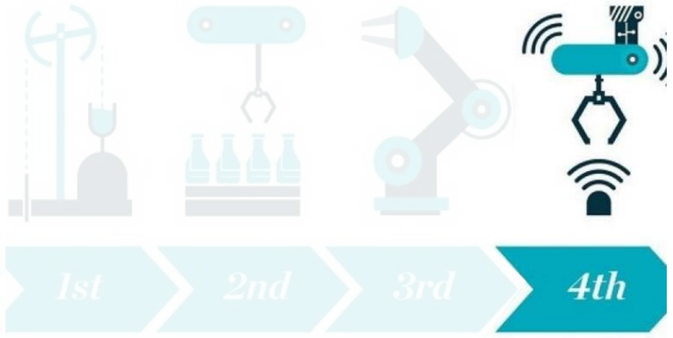


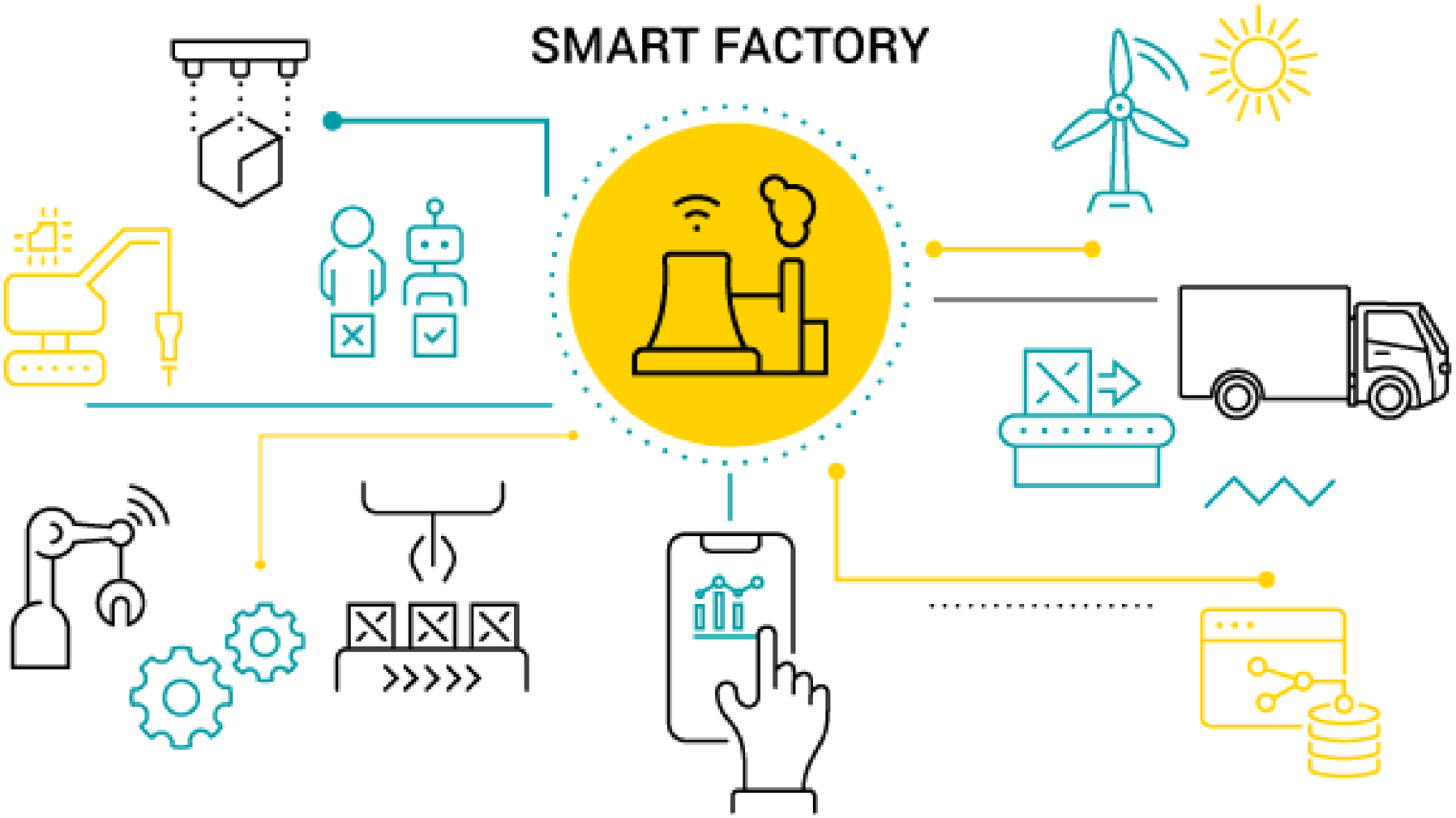
Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC





CYBER PHYSICAL SYSTEMS

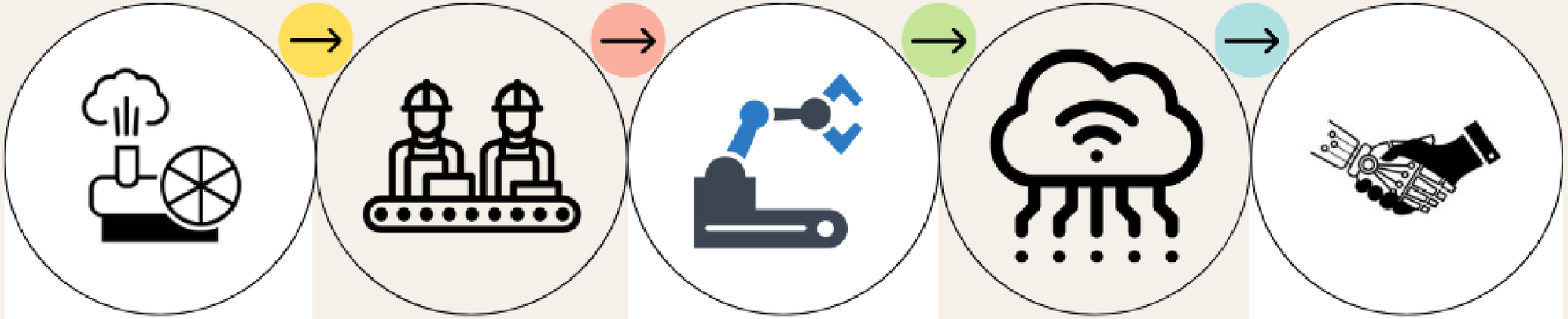
NETWORKS

Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC



Industry 1.0

Marked by mechanization and powered by water and steam

Industry 2.0

Assembly lines and electricity allow for mass production

Industry 3.0

Computers and electronics make automation possible

Industry 4.0

Advances in cyber-physical systems, IoT, and network communications

Industry 5.0

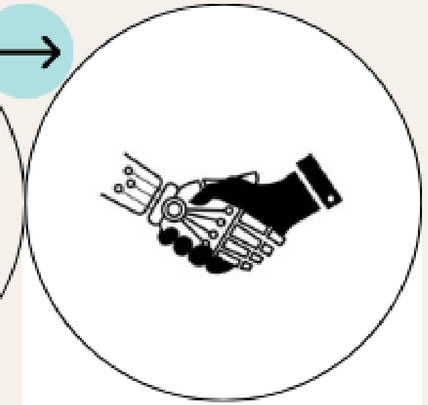
Human-robot collaboration, customizations, cognitive systems

Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

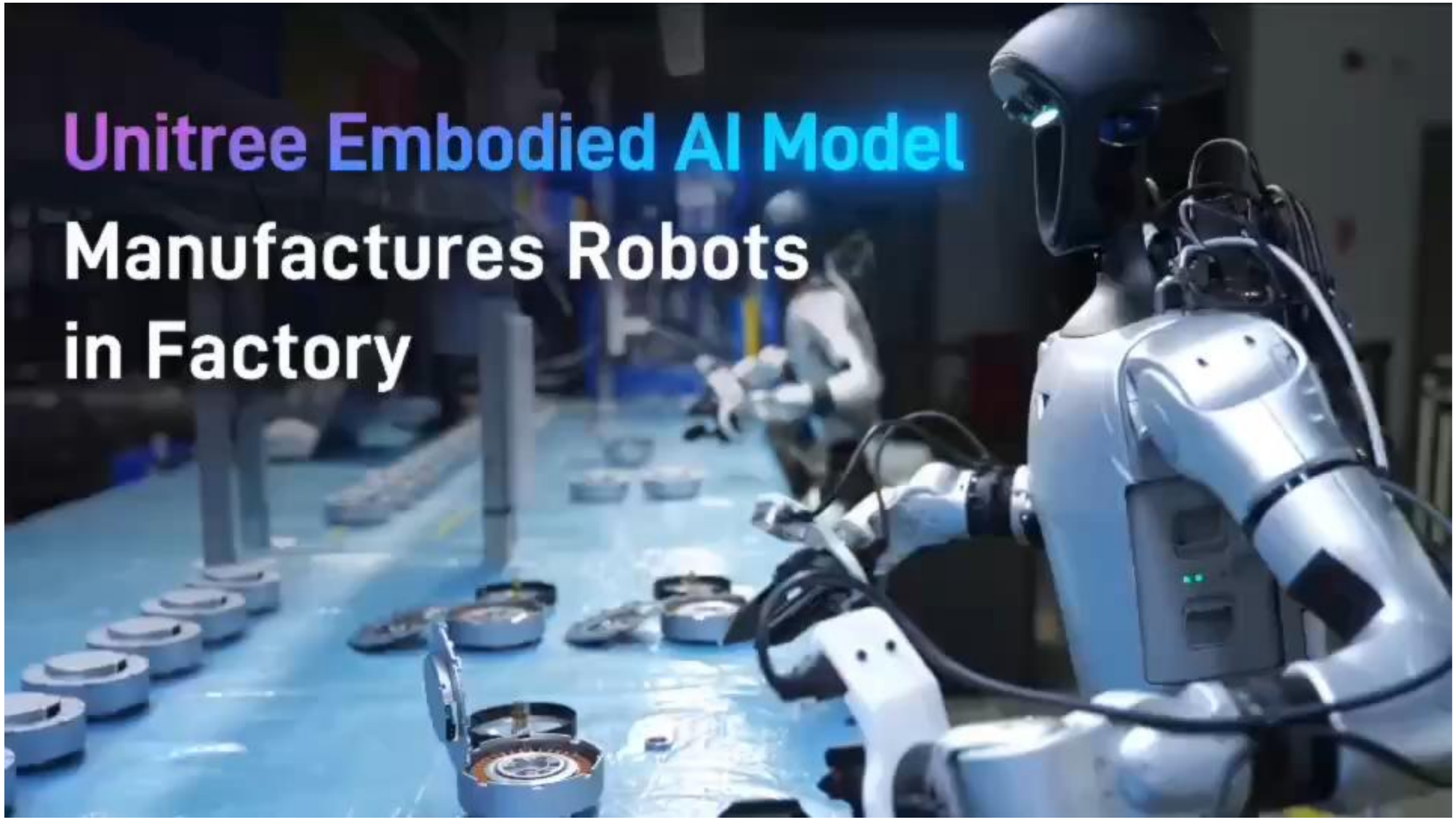
Training

UMES x KMITL x TIRA x ABAC



Industry 5.0

Human-robot collaboration, customizations, cognitive systems



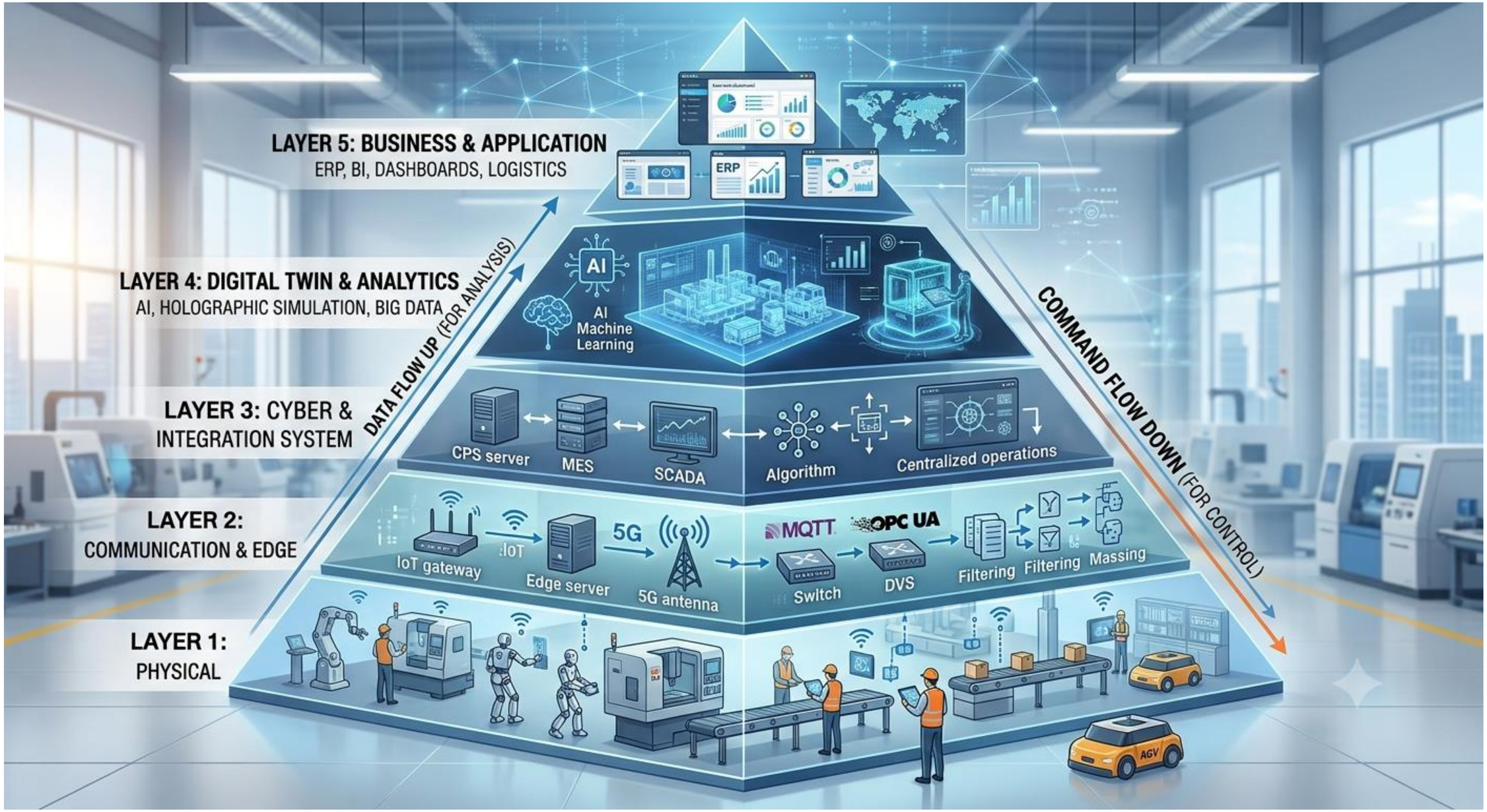
Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC

Smart Factory



Smart Automation

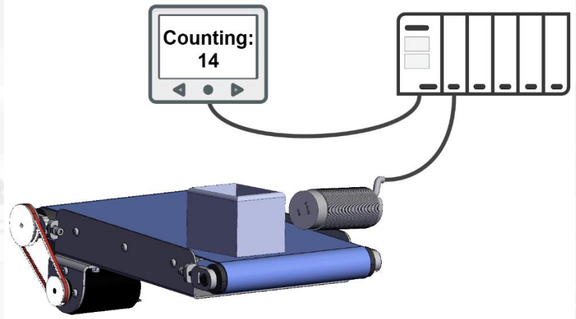
IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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Sensors



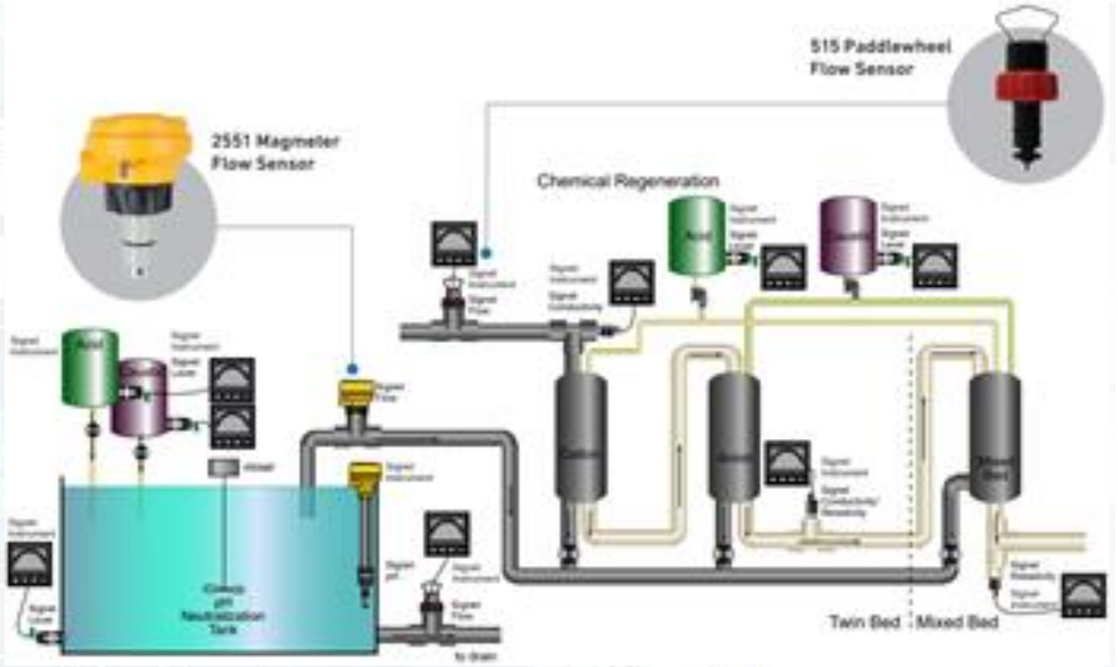


Proximity Sensors





Flow Sensors





Level Sensors



Temperature Sensors



Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC

Actuators

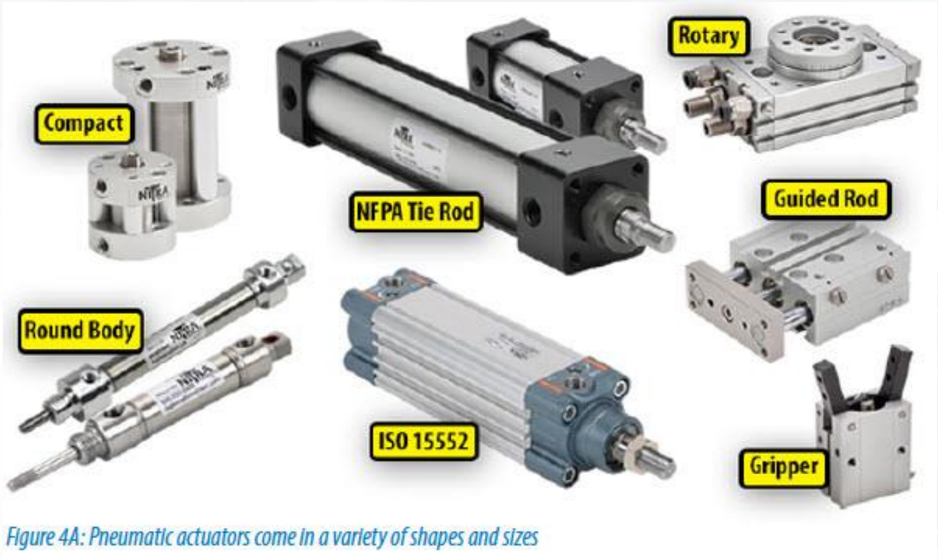
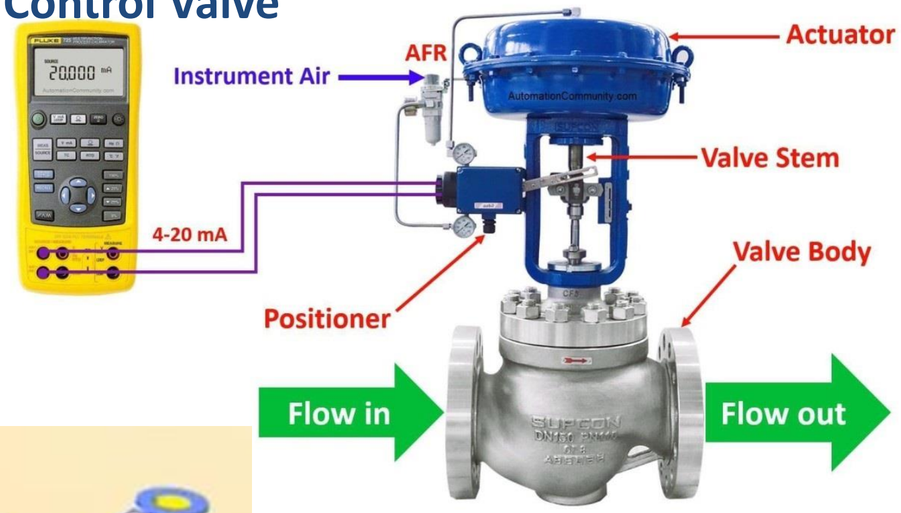
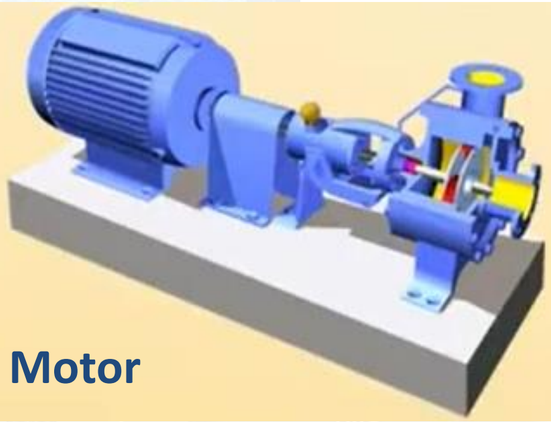


Figure 4A: Pneumatic actuators come in a variety of shapes and sizes

Control Valve





Motor



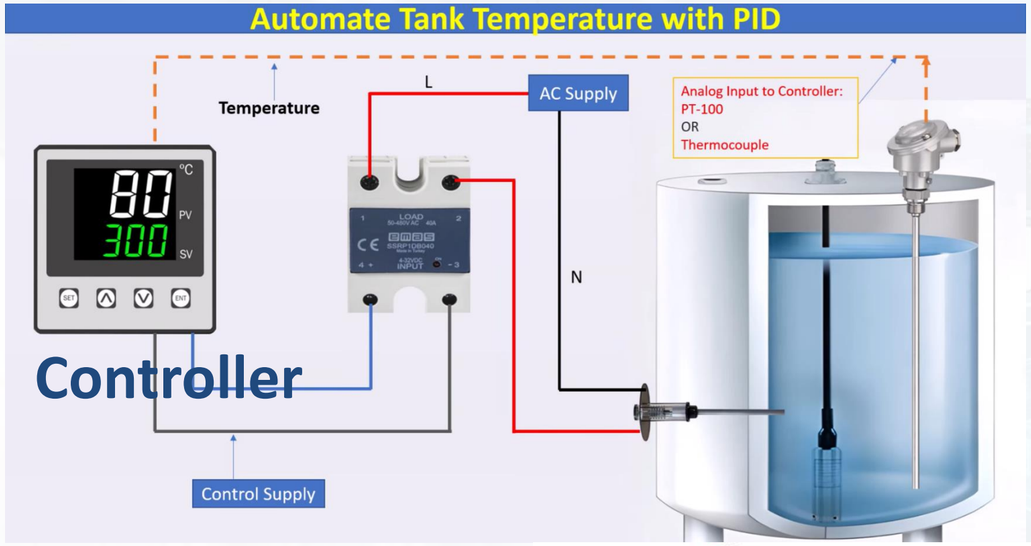
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Training

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Controller&HMI







DCS



distributed control system





PLC

Programmable Logic Controller



HMI





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Training

UMES x KMITL x TIRA x ABAC

Robotics

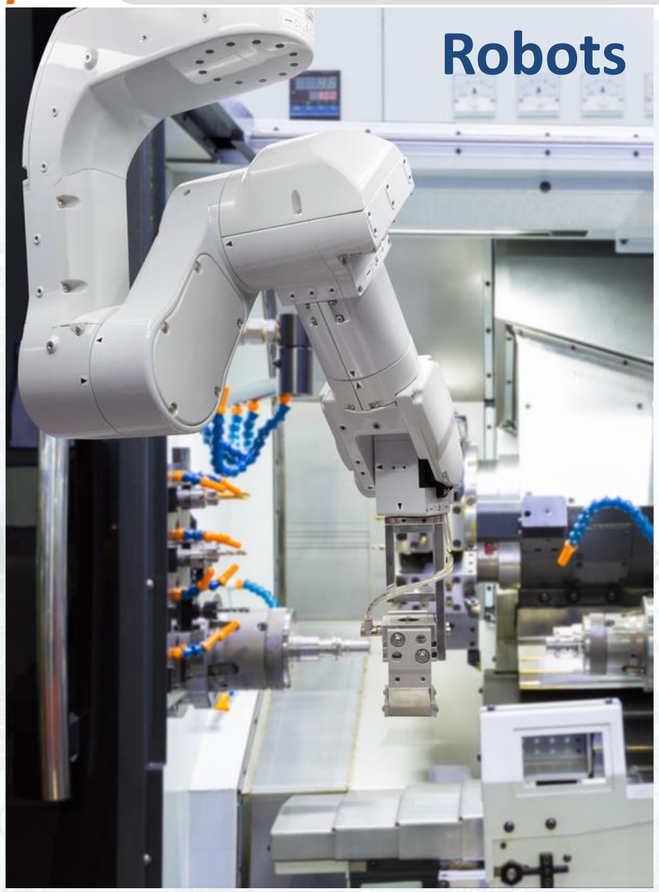
HIKROBOT THAILAND AUTHORIZED DEALER



AMR (Autonomous Mobile Robot)

AGV (Automated Guided Vehicle)

Robots



Humanoids





Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

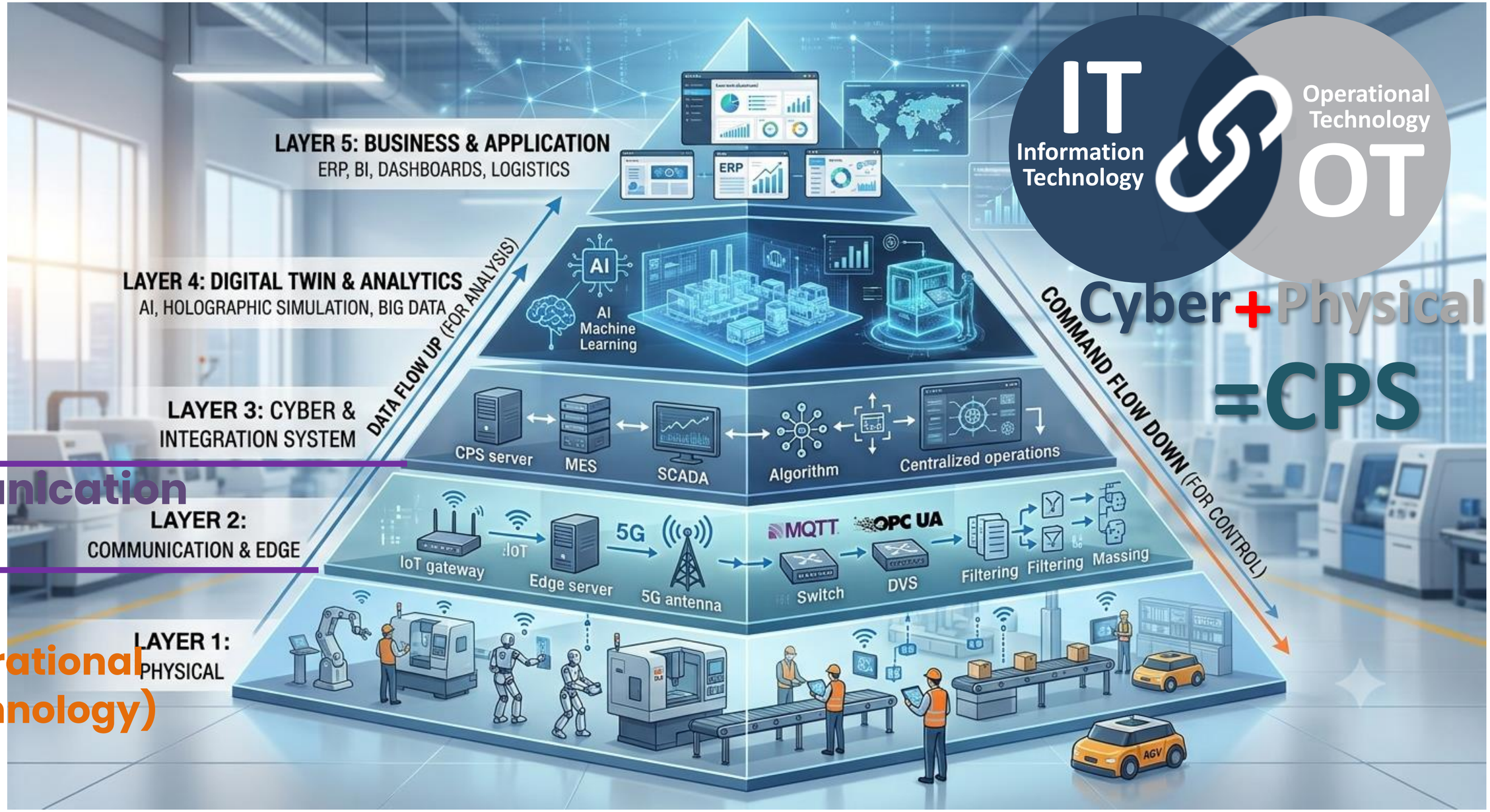
Training

UMES x KMITL x TIRA x ABAC

IT (Information Technology)

Communication

OT (Operational Technology)



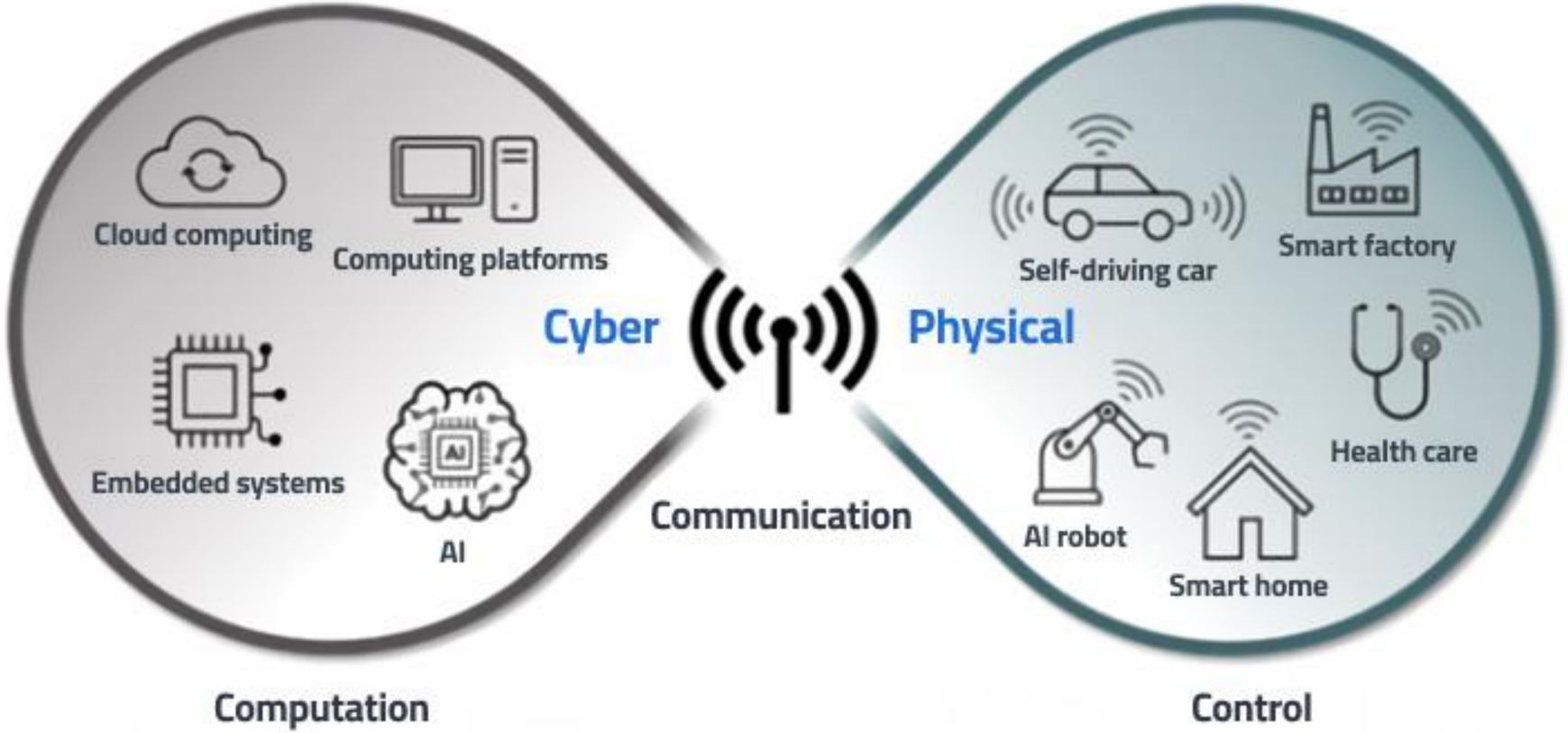
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Training

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Cyber-Physical Systems



Real-time AI service	Cyber-physical security	Intelligent transportation systems
Real-time computing	Learning-enabled CPS	Self-driving car
		Resilient control systems

https://www.dgist.ac.kr/en_eecs/sub03_01_04.do

Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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Process

Sensors → Input → Controller → Output → Actuators



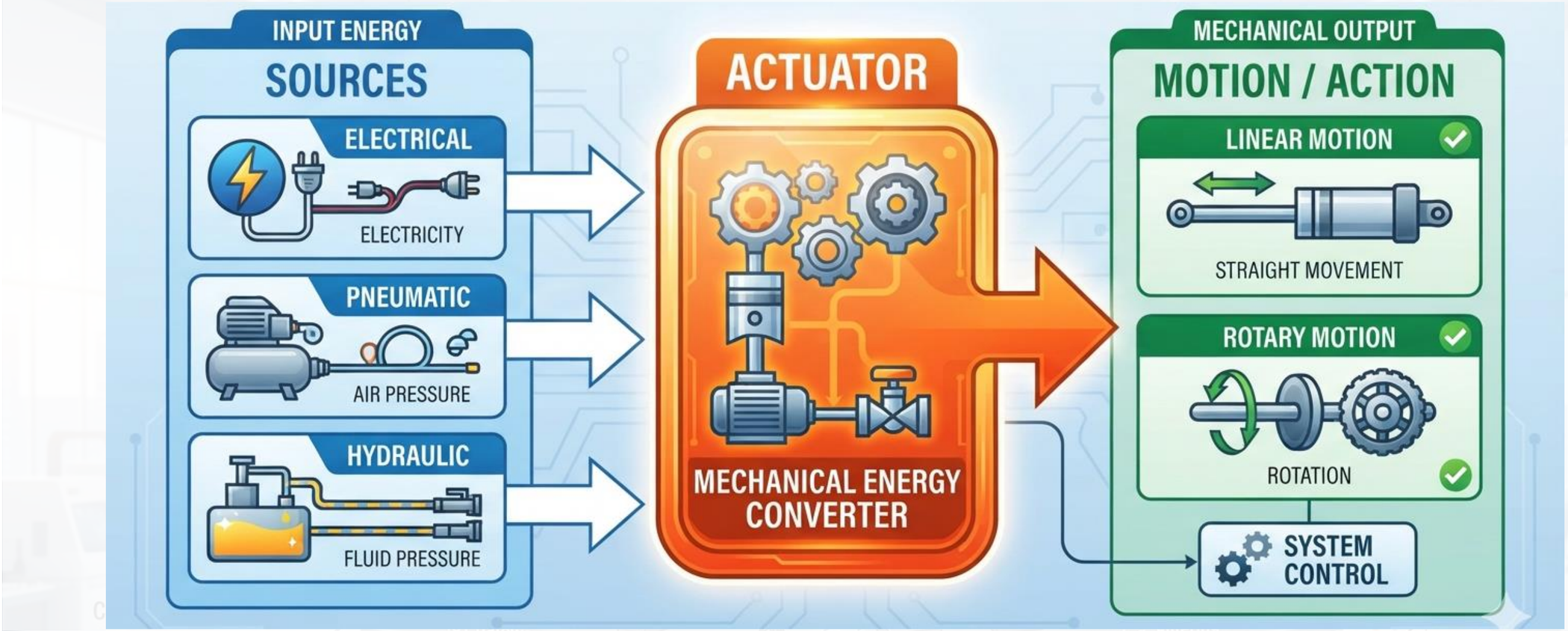
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Actuators





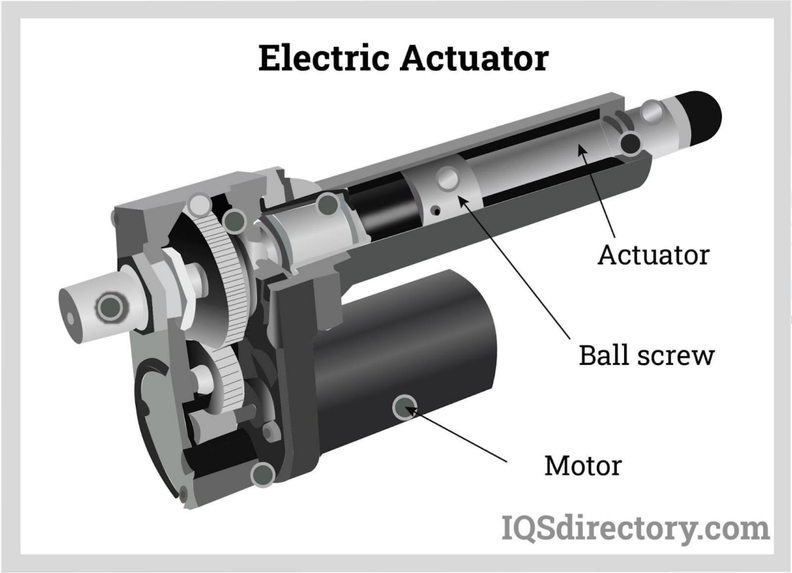
Smart Automation

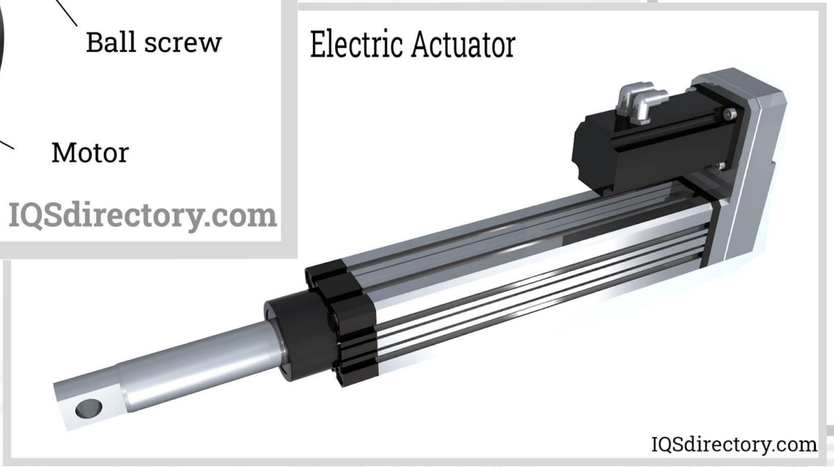
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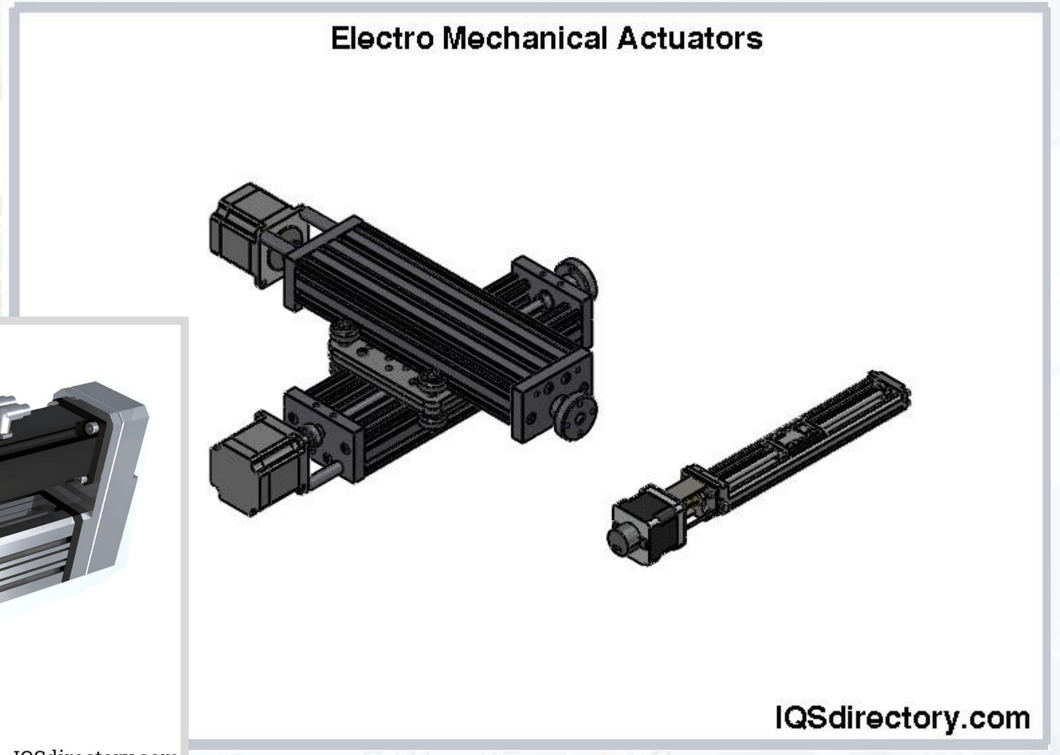
Training

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Actuators







https://www.iqsdirectory.com/articles/linear-actuator/electric-actuators.html



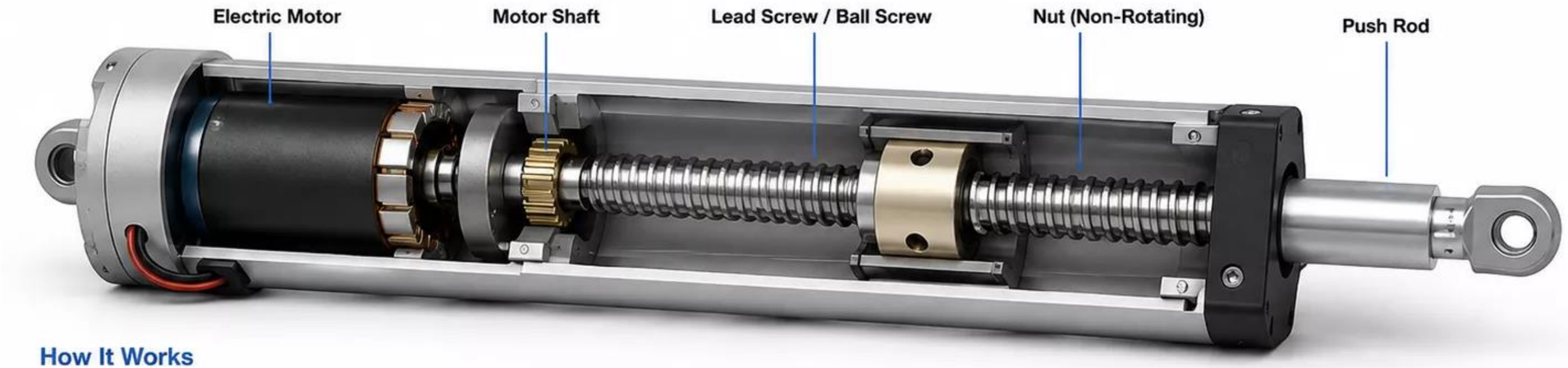
Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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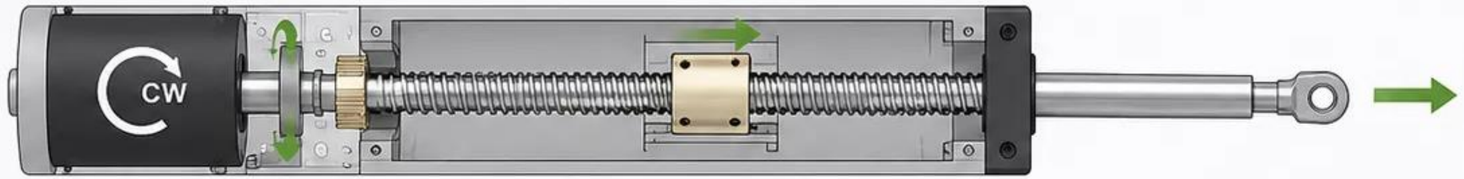
Actuators



How It Works

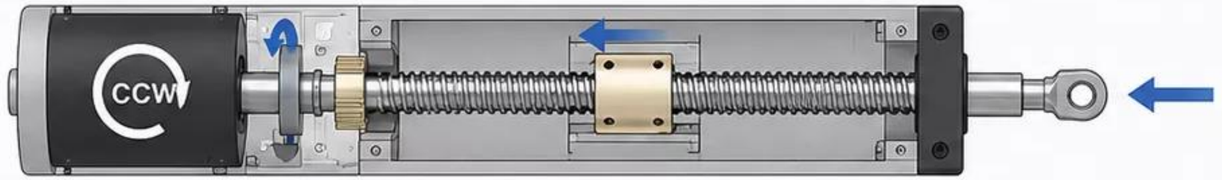
1. Extend (Forward Motion)

Motor rotates clockwise.
The nut moves to the right
along the screw,
pushing the rod outward.



2. Retract (Backward Motion)

Motor rotates counterclockwise.
The nut moves to the left
along the screw,
pulling the rod inward.



Energy Conversion Chain



Electrical Energy (Power In) → Motor Rotation (Rotational Motion) → Screw Rotation (Rotational Motion) → Nut Linear Movement (Linear Motion) → Push Rod Extension / Retraction (Linear Motion Output)

Key Features

- Converts rotational motion to precise linear motion
- Bidirectional movement (extend & retract)
- High efficiency and accurate positioning
- Compact, reliable and easy to control

https://www.jiecang.com/article/what-is-a-linear-actuator-system-working-principle-applications.html

Smart Automation

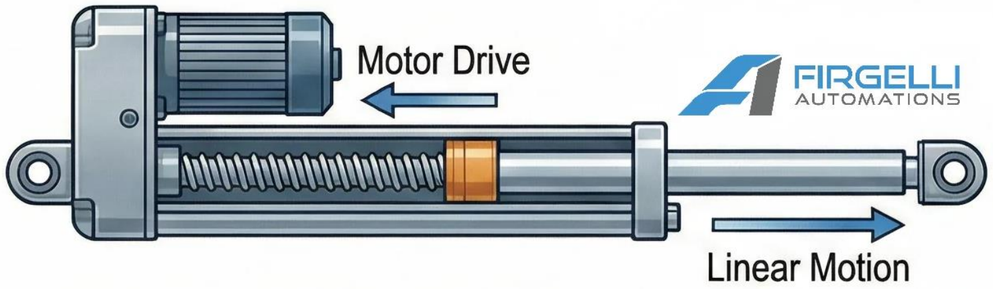
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Training

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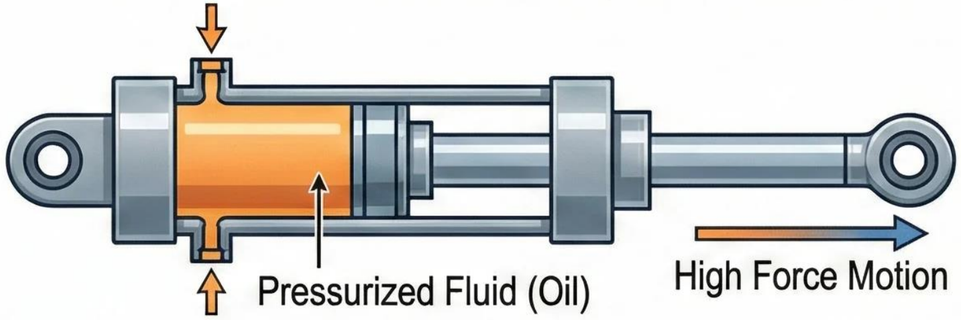
Actuators

LINEAR ACTUATOR (Electric)



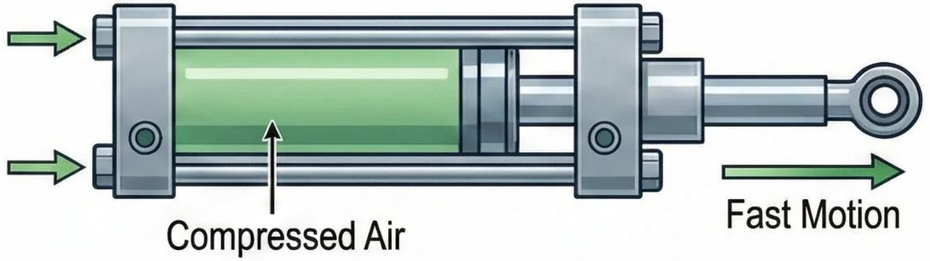
Pros: Precise Control, Clean, Quiet, Programmable
Cons: Lower Force than Hydraulic, Slower Speed, Requires Electricity
Applications: Automation, Medical Devices, Adjustable Furniture

HYDRAULIC CYLINDER (Fluid Power)



Pros: Extremely High Force, Smooth Operation, Durable
Cons: Complex System (Pump, Lines), Potential for Leaks, Maintenance Intensive
Applications: Heavy Machinery, Construction, Industrial Presses.

PNEUMATIC CYLINDER (Compressed Air)



Pros: Fast Speed, Clean, Safe (No Sparks), Cost-Effective
Cons: Compressible Air (Less Precise), Lower Force than Hydraulic, Noisy Exhaust
Applications: Assembly Lines, Packaging, Automation, Robotics.

Actuators

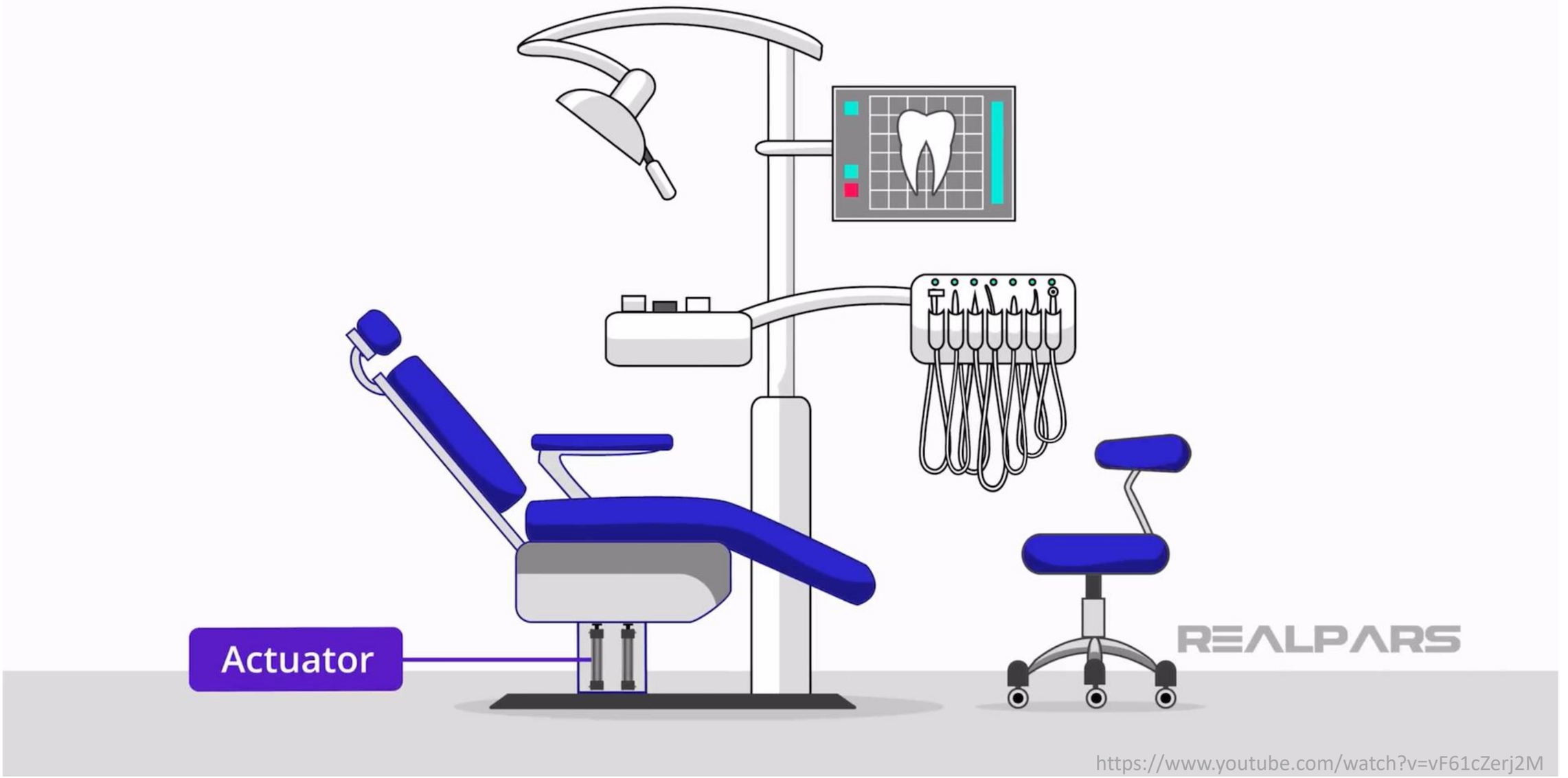


Linear pneumatic actuator

REALPARS

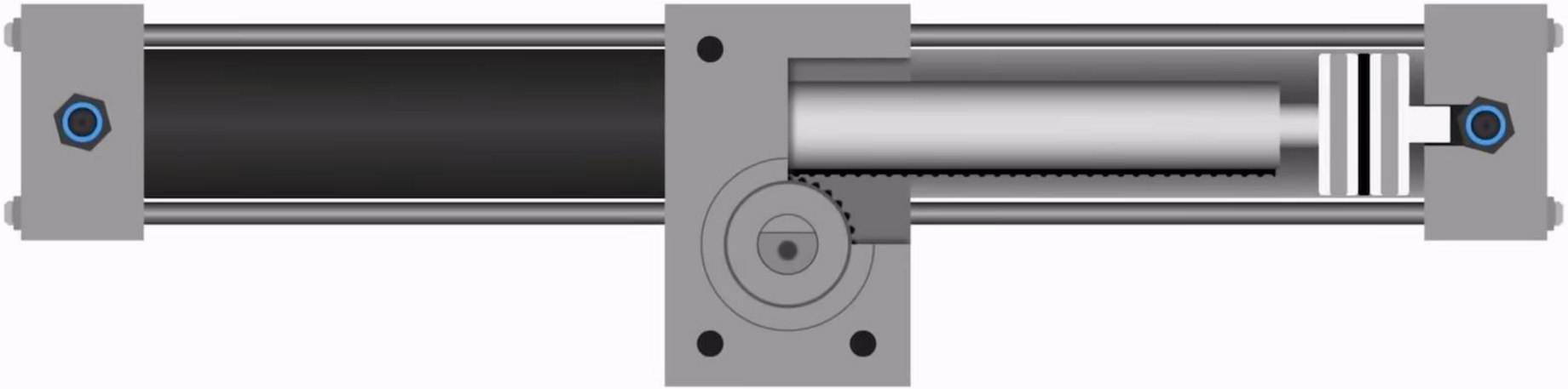
https://www.youtube.com/watch?v=vF61cZerj2M

Actuators



https://www.youtube.com/watch?v=vF61cZerj2M

Actuators



Rotary pneumatic actuator

REALPARS

https://www.youtube.com/watch?v=vF61cZerj2M

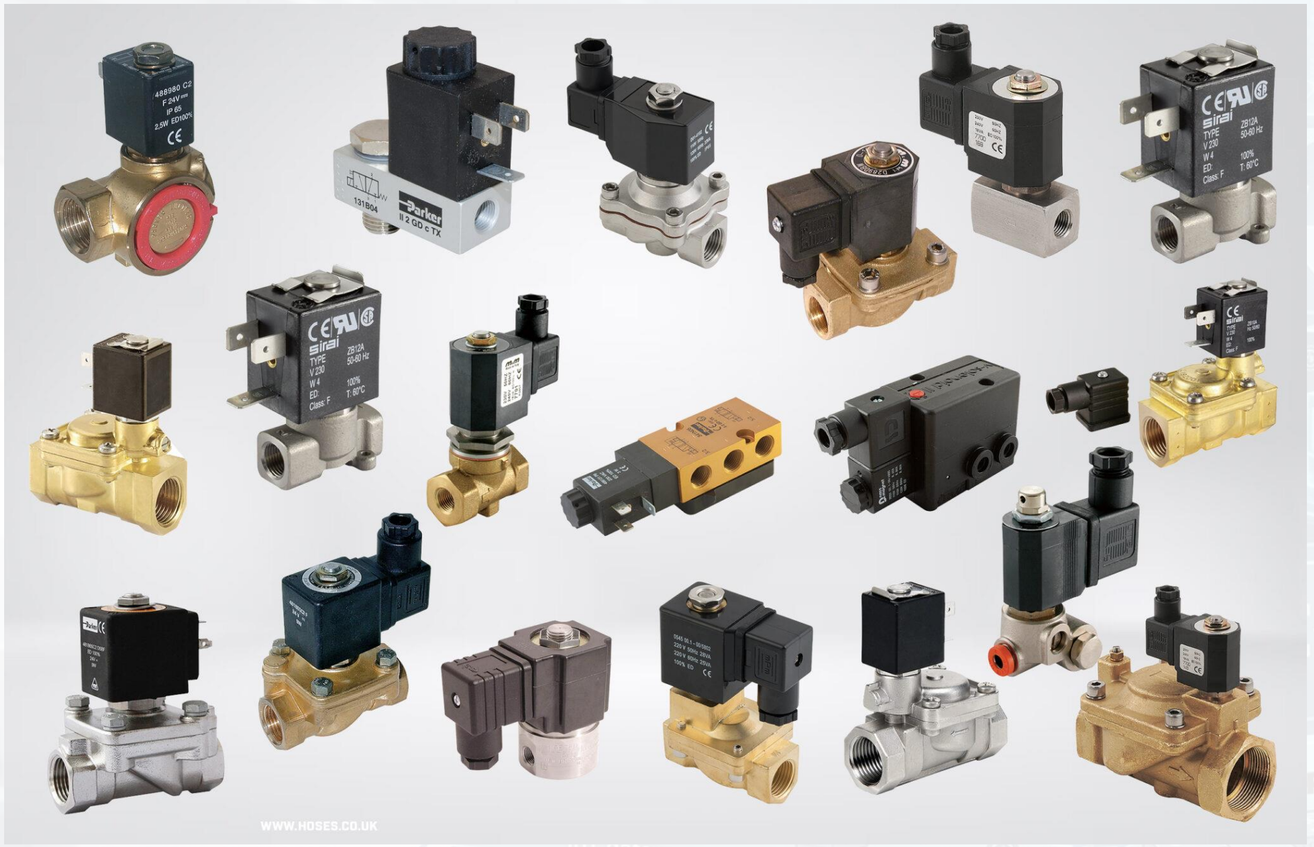
Smart Automation

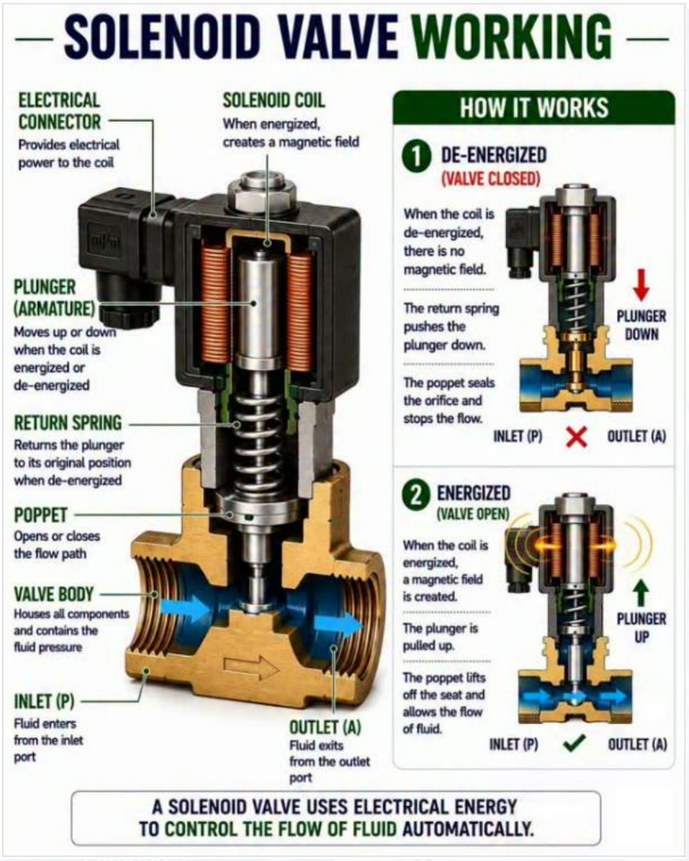
IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC

Actuators







Smart Automation

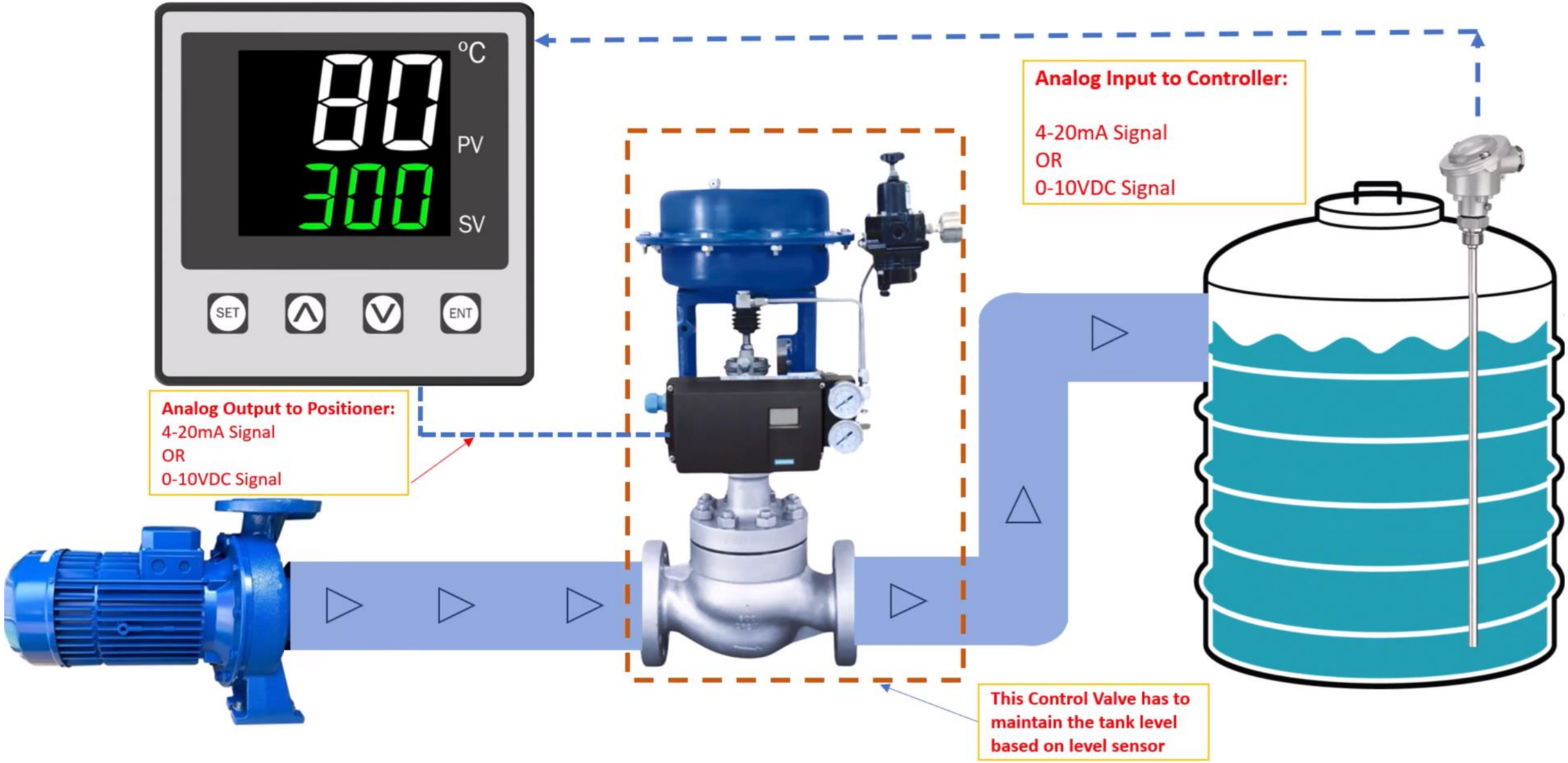
IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC

Actuators

PID in Action: How to Maintain Precise Tank Level Automatically



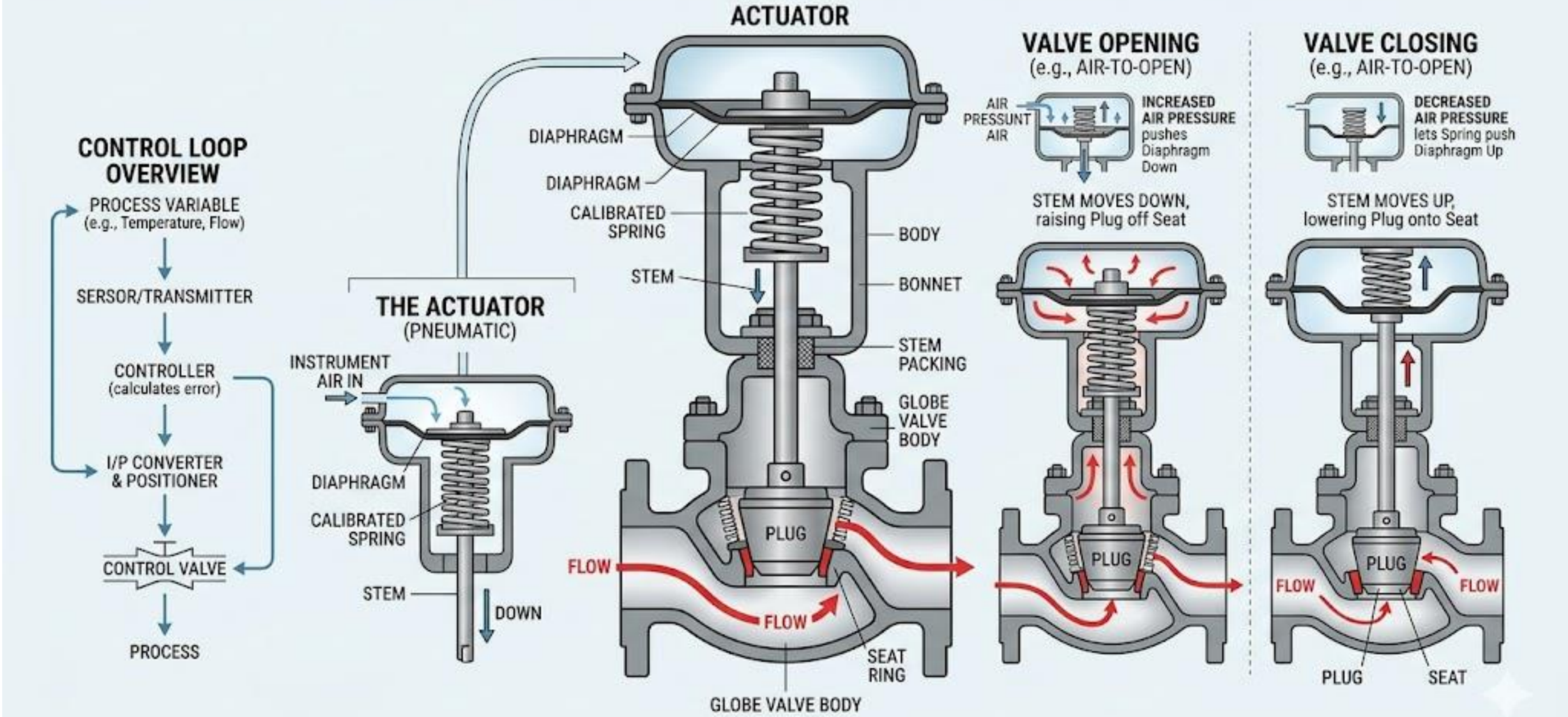
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IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC

Actuators



https://www.xm-valveactuator.com/n/news/process-control-valve-selection-guide

Actuators



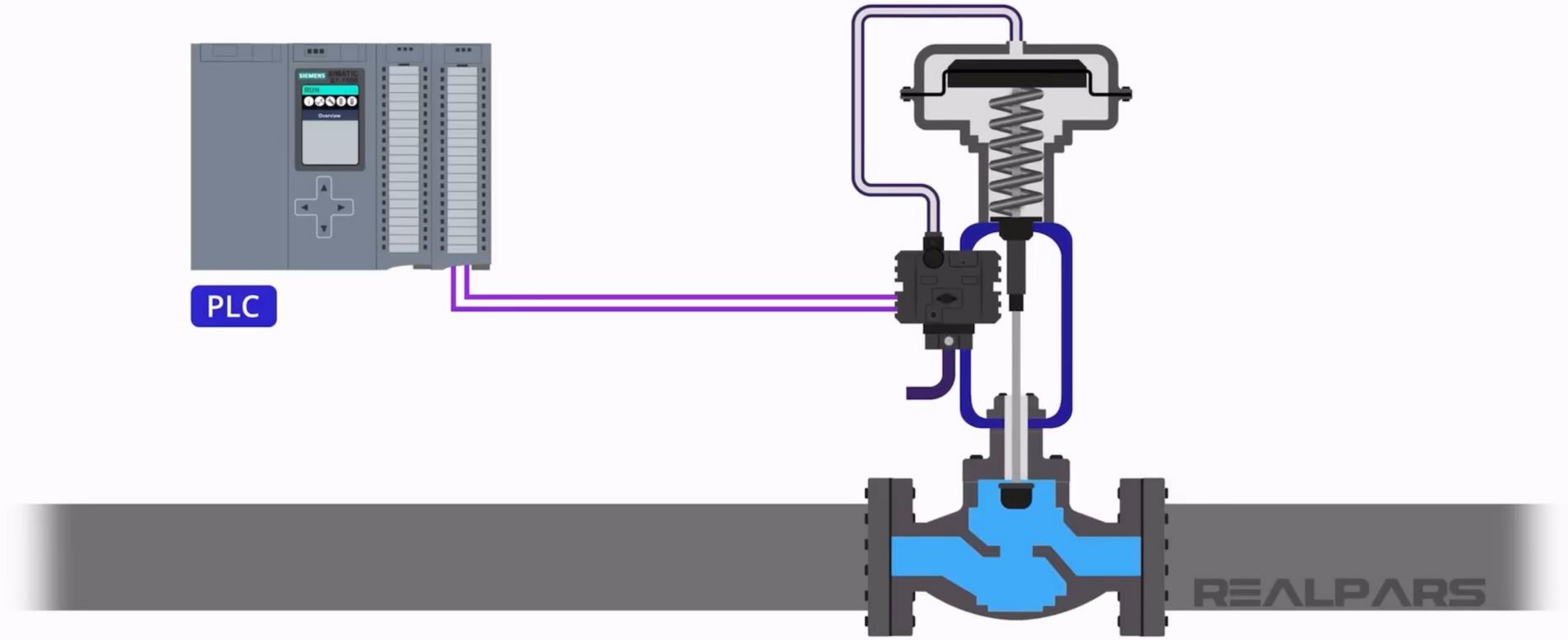
REALPARS

https://www.youtube.com/watch?v=vF61cZerj2M

Actuators

Linear pneumatic actuator

Spring/diaphragm



https://www.youtube.com/watch?v=vF61cZerj2M

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Actuators

TYPES OF PROCESS CONTROL VALVES

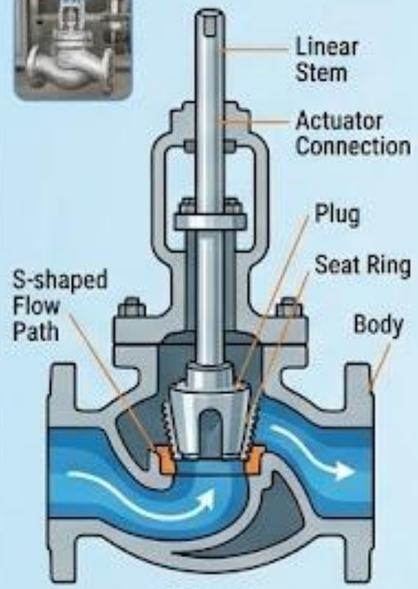
Chemical

Oil & gas

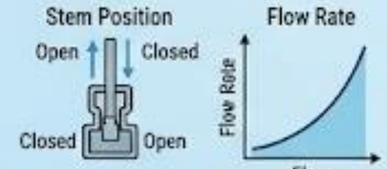
GLOBE VALVES

High pressure, precise throttling, linear plug motion.



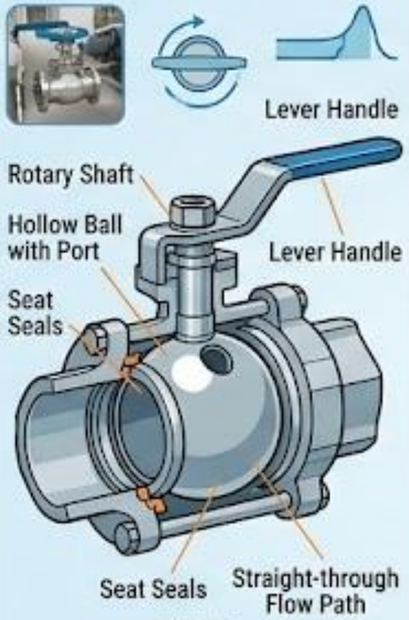


Globe Valve

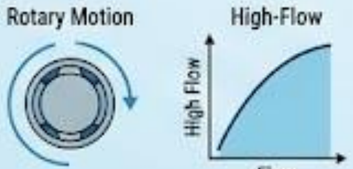


BALL VALVES

Quarter-turn, high flow capacity, fast operation.



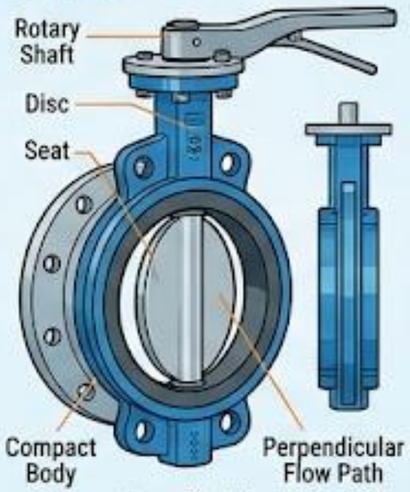
Ball Valve



BUTTERFLY VALVES

Quarter-turn, compact design, design, low pressure drop.



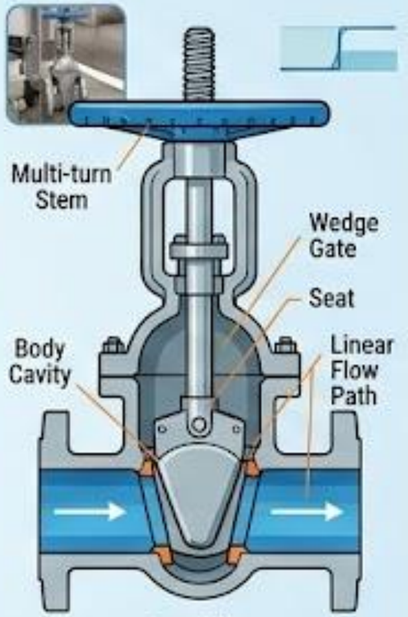


Butterfly Valve

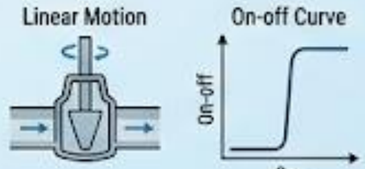


GATE VALVES

On-off service, minimal pressure drop, multi-turn stem.



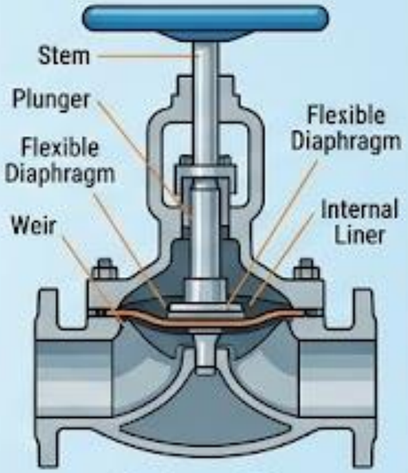
Gate Valve



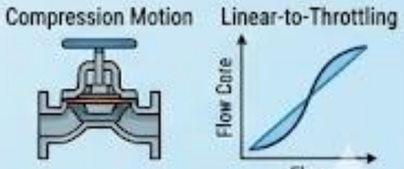
DIAPHRAGM VALVES

Weir-type, sanitary design, flexible diaphragm control.





Diaphragm Valve



ENGINEERED FOR EXCELLENCE

ASME ANSI API

https://www.xm-valveactuator.com/n/news/process-control-valve-selection-guide

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Actuators



OIL & GAS

- Flow control in pipelines
- Pressure regulation
- Safety and shutdown systems



CHEMICAL PROCESSING

- Accurate flow control
- Temperature regulation
- Corrosive fluid handling



WATER & WASTEWATER TREATMENT

- Flow and level control
- Pressure management
- Treatment process optimization



FOOD & BEVERAGE

- Hygienic flow control
- Temperature control
- Clean-in-place (CIP) systems



POWER GENERATION

- Steam and water control
- Pressure and temperature control
- Boiler feedwater regulation



PHARMACEUTICAL

- Sterile process control
- Precise flow regulation
- Compliance with strict standards



PULP & PAPER

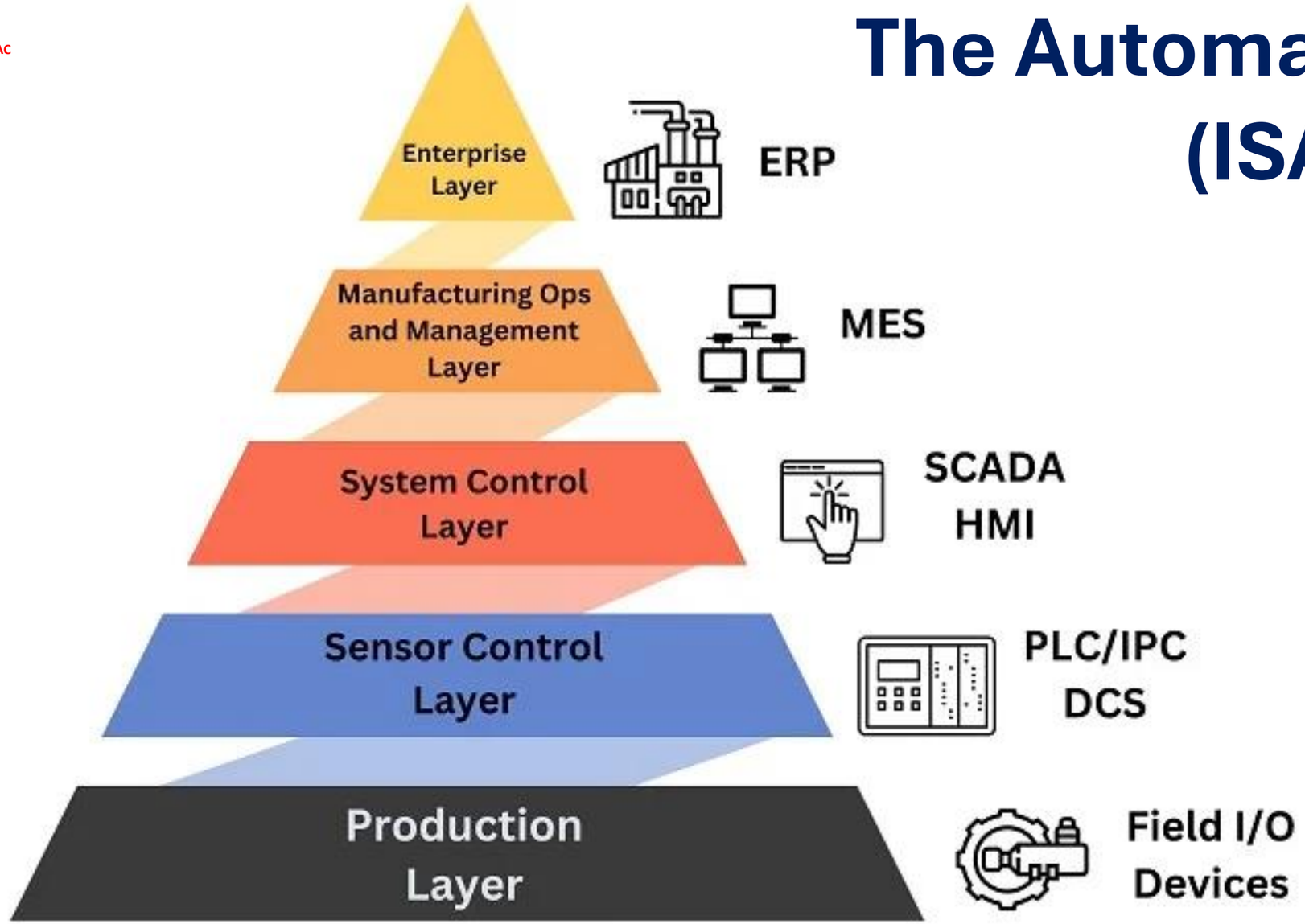
- Flow and consistency control
- Pressure regulation
- Chemical dosing control



GENERAL INDUSTRY

- Multi-process applications
- System automation
- Improved efficiency and safety

The Automation Pyramid (ISA-95)



https://control.com/technical-articles/machine-to-machine-integration-with-opc-ua/

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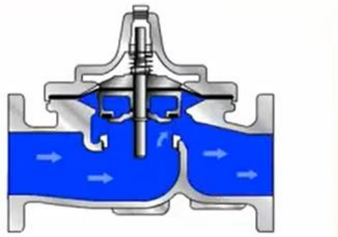
Training

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Level 0 – Field Level

- The field level is the production floor where physical work and monitoring happens
- Sensors and actuator takes major role in the Field Level .
- Sensors are very important role to collect data from the filed such as temperature, Level , pressure, speeds ,flow , feeds and so on, convert it to electrical signals, and relay it up to the next level.
- The main task of these field devices is to transfer data on processes and machines for monitoring and analysis.
- It also includes the actuators, which are controlled by the next level through electrical or pneumatic signals, converting them into actions.
- Actuators turn valves and other devices such as relays, motors, pumps, VFDs and other devices on or off, or adjust their outputs to control the processes.















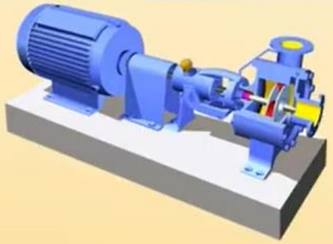












https://www.youtube.com/watch?v=3bwAZPTBPgs

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IIoT & Cyber-Physical Systems for industry 4.0/5.0

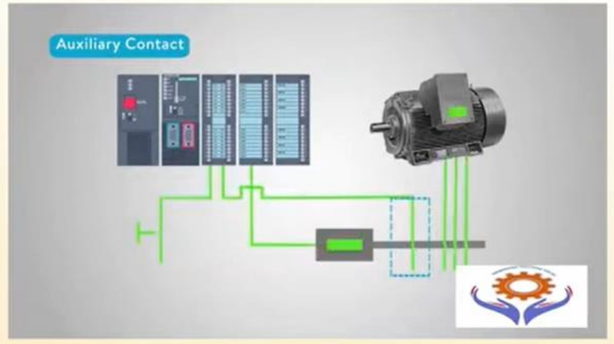
Training

UMES x KMITL x TIRA x ABAC

Level 1 – Controller Level



- At this level, you control and manipulate the devices in the field level that actually do the physical work.
- The controller level consist of PLC , RTU , PAC , DCS , Robots , CNC, PID controllers etc..
- **Programmable Logic controllers (PLC's)** also called as industrial computers are capable of being programmed to perform certain control functions. It consists of a CPU or processor, I/O modules (both analog and digital) to connect the various input/output devices and relay modules.
- Programmable logic controllers (PLCs) are the most widely used and durable industrial controllers that can deliver automatic control functions based on sensor inputs
- These may be modular which is of fixed type or integrated types to extend modules based on the inputs available.
- Along with the PLC's, conventional PC's are used to control the process by online or by changing the programs. PLC's comes with dedicated software to program the control strategy.









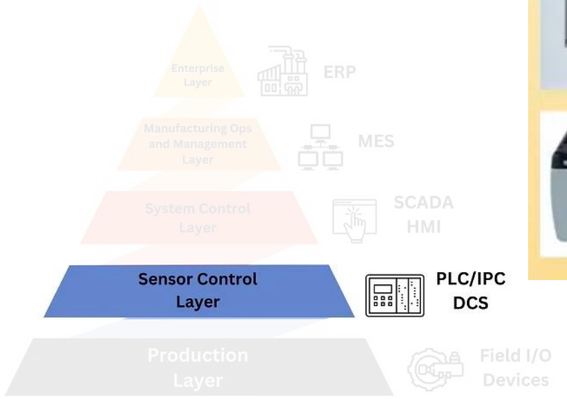












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Level 2 – Supervisory Level



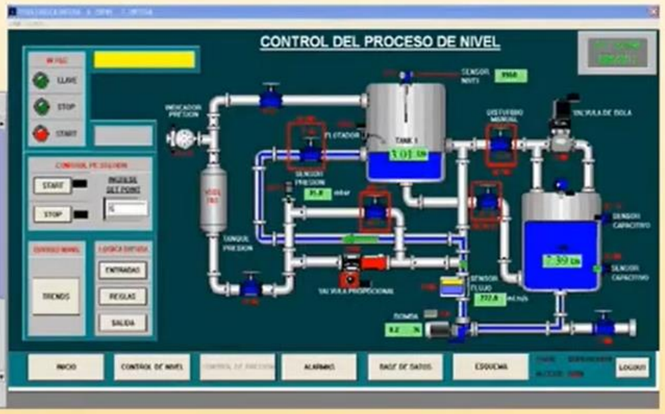
- In this level, automatic devices and monitoring systems facilitate the control and adjustment functions.
- These include Human Machine Interfaces (HMI), Distribution Control Systems (DCS), and supervisory control and data acquisition (SCADA) devices for monitoring various parameters, setting production targets, historical archiving, and setting machine start and shutdown.
- **Supervisory control and data acquisition** (SCADA) is a control system architecture comprising computers, networked data communications and graphical user interfaces (GUI) for high-level process supervisory management

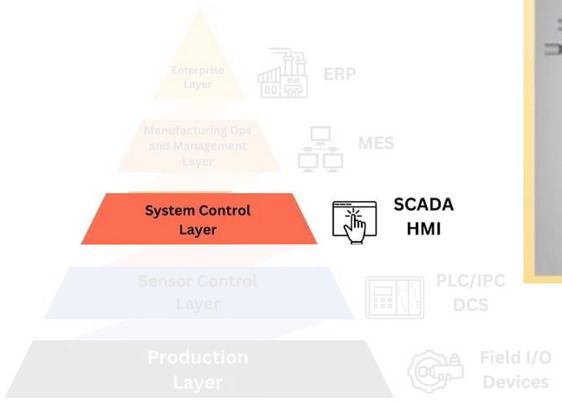
- ❖ Data acquisition
- ❖ Plant monitoring
- ❖ Data checking
- ❖ Open and closed loop control
- ❖ Reporting











https://www.youtube.com/watch?v=3bwAZPTBPgs

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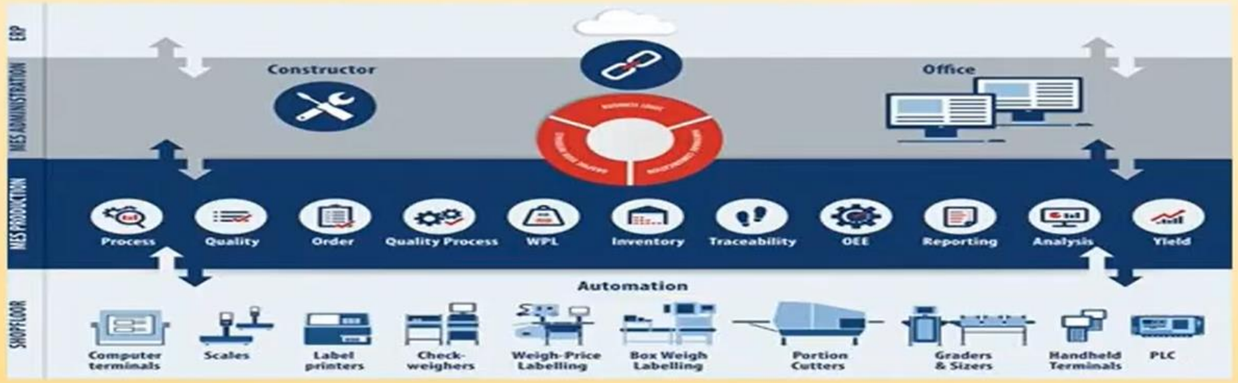
Training

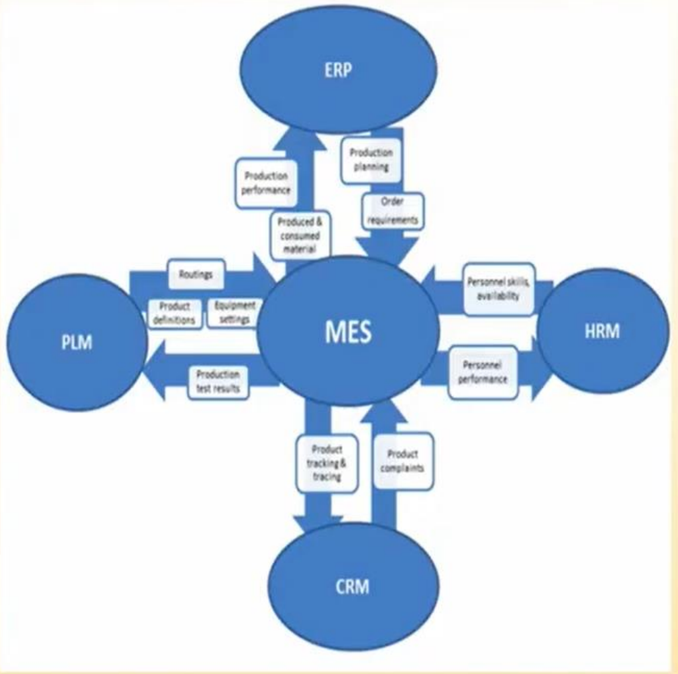
UMES x KMITL x TIRA x ABAC

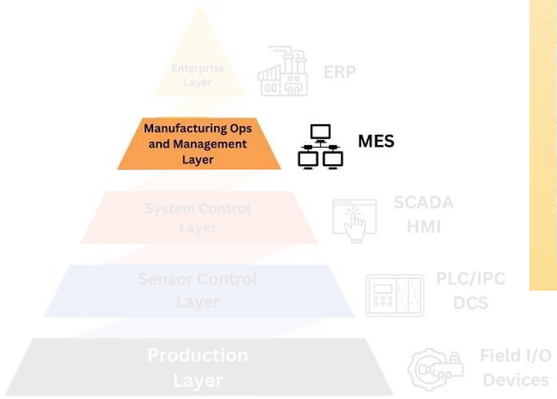
Level 3 – Planning Level - Manufacturing Execution systems



- The fourth level of the automation pyramid is called the planning level.
- This level utilizes a computer management system known as MES or manufacturing execution system.
- MES monitors the entire manufacturing process in a plant or factory from the raw materials to the finished product.
- This allows management to see exactly what is happening and allows them to make decisions based on that information. They can adjust raw material orders or shipment plans based on real data received from the systems.
- Industrial communication networks tie all these levels together, sending data from one level to the other. Production dispatch
- Inventory control
- Production supervision,
- production reporting, etc.
- Product Life cycle ,
- Resource schedule ,
- Production Analysis etc..







https://www.youtube.com/watch?v=3bwAZPTBPgs

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Level 4– Management Level - Enterprise Resource Planning

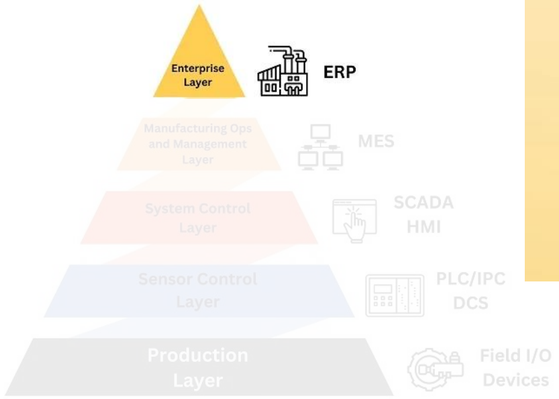
- The top of the pyramid is what is called the management level. which is also called Information-level automation manages the whole automation system
- This level uses the companies integrated management system which is known as the ERP or enterprise resource planning.
- This is where a company's top management can see and control their operations.
- ERP is usually a suite of different computer applications that can see everything going on inside a company.
- It utilizes all of the previous levels technology plus some more software to accomplish this level of integration.
- This allows the business to be able to monitor all levels of the business from manufacturing, to sales, to purchasing, to finance and payroll, plus many others.
- The integration of the ERP promotes efficiency and transparency within a company by keeping everyone in the same page.
- The level handles production planning, customer and market analysis, and orders and sales. So it deals more with commercial activities and less with technical aspects.
- Market and Customer analysis
- Orders and sale statistics
- Business planning, Logistics
- Capacity and order balance, etc.











https://www.youtube.com/watch?v=3bwAZPTBPgs

PLC Applications

- Discrete Control
- Process Control
- Safety Control

PLC Uses in Industry



Automotive Sector



Water Treatment Plants



Manufacturing Automation





Oil & Gas



Packaging Machines



Food & Beverage Industry

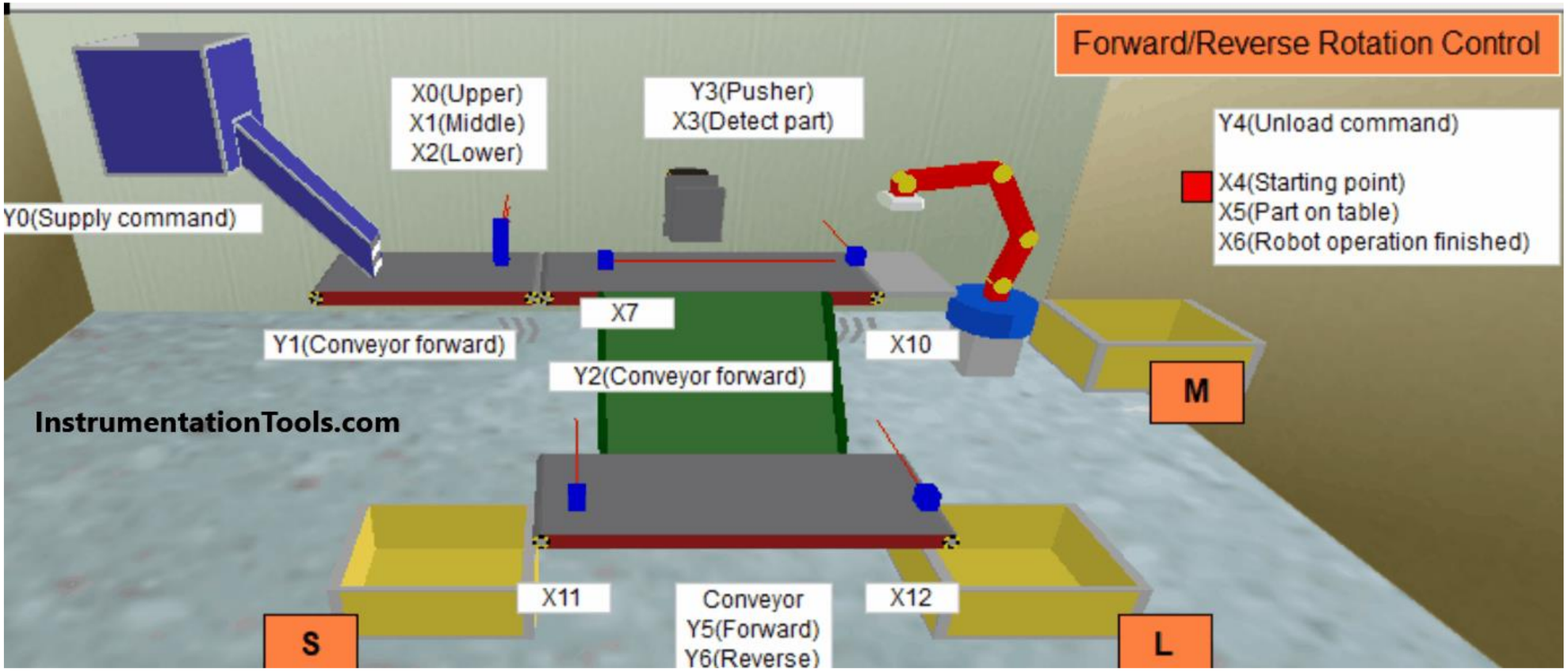


Textile Industry

https://anadiautomation.com/blog-details/plc-applications-how-plc-power-modern-industry

PLC Applications

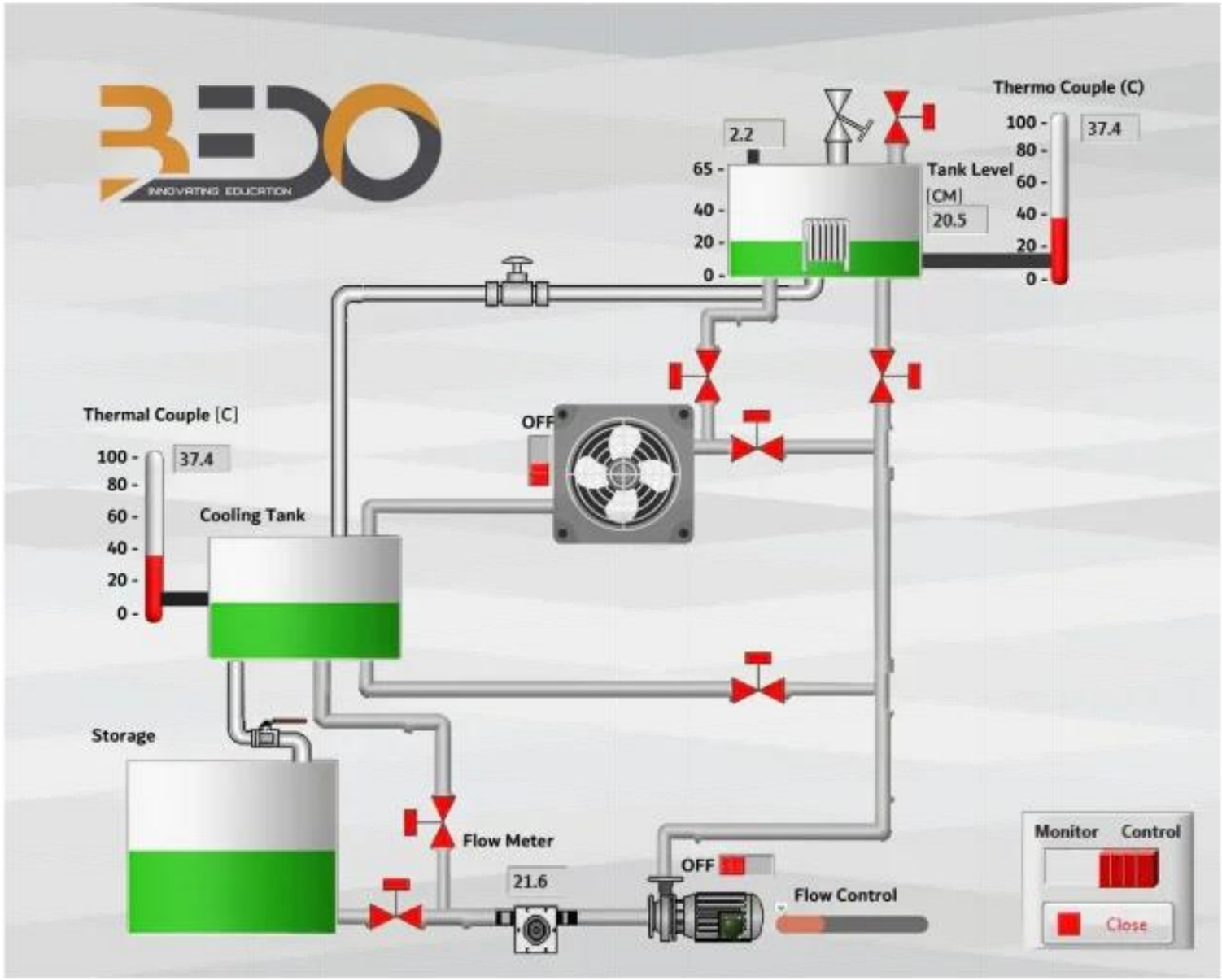
- **Discrete Control**
- Process Control
- Safety Control



https://instrumentationtools.com/advanced-plc-conveyor-control/

PLC Applications

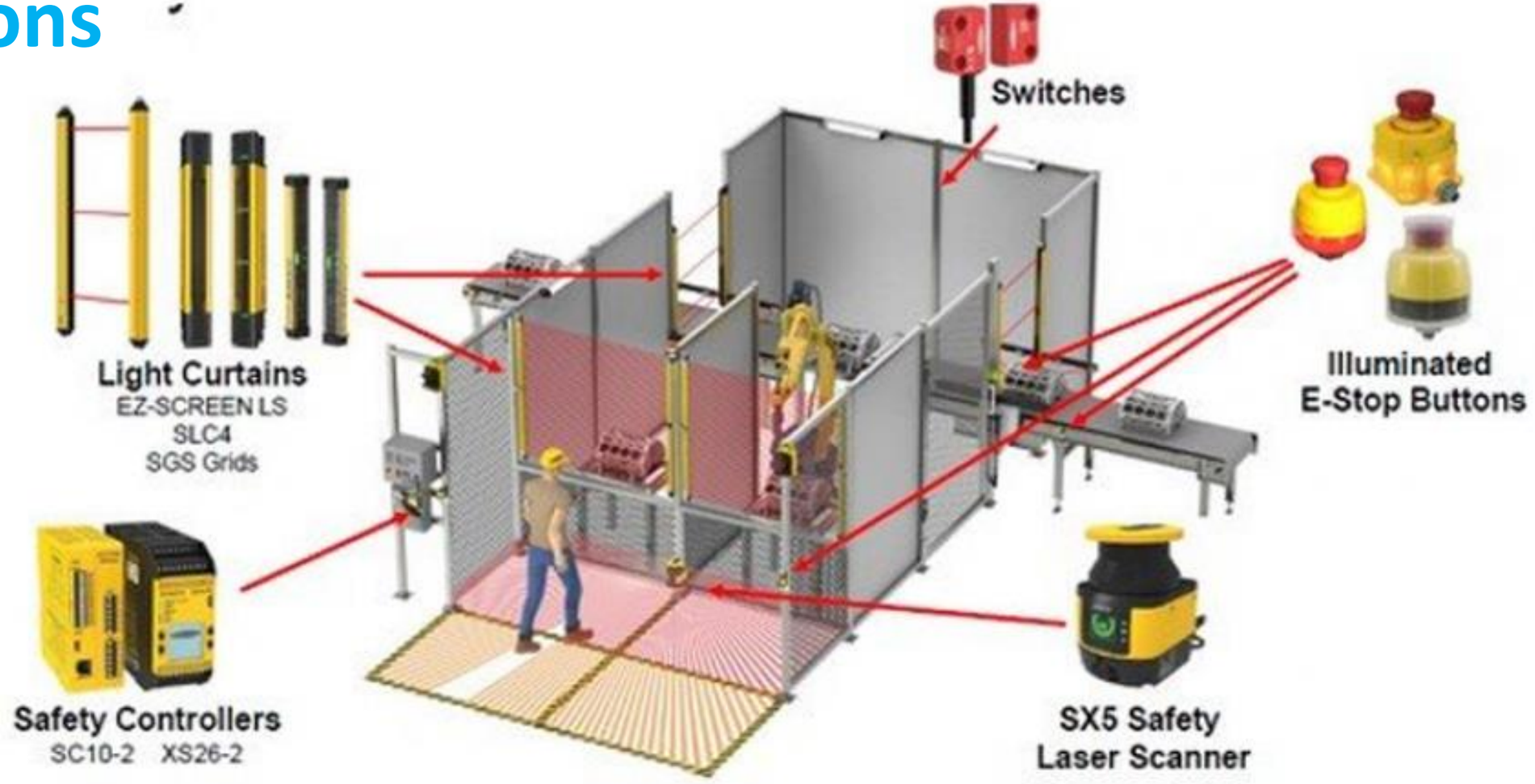
- Discrete Control
- **Process Control**
- Safety Control



https://bedoeg.com/product/mpc106/

PLC Applications

- Discrete Control
- Process Control
- **Safety Control**



https://www.tic.co.th/en/news/detail/601/2

PLC stands for:

Programmable
Logic
Controller



CLICK PLC Module

Definition:

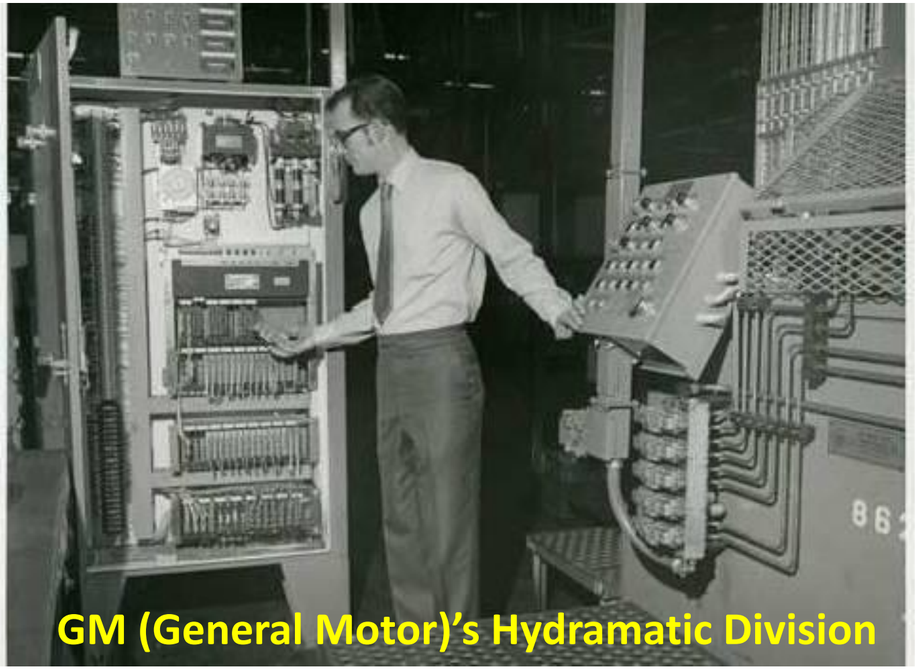
A PLC can be defined as an industrialized computer that includes the hardware and software required to automatically control production equipment, machinery, processes, and other types of mechanical, electrical, and electronic devices typically found in industry. It also includes the flexibility to reprogram its decision making behavior through the software as often as needed. The PLC is the key element behind today's industrial automation.

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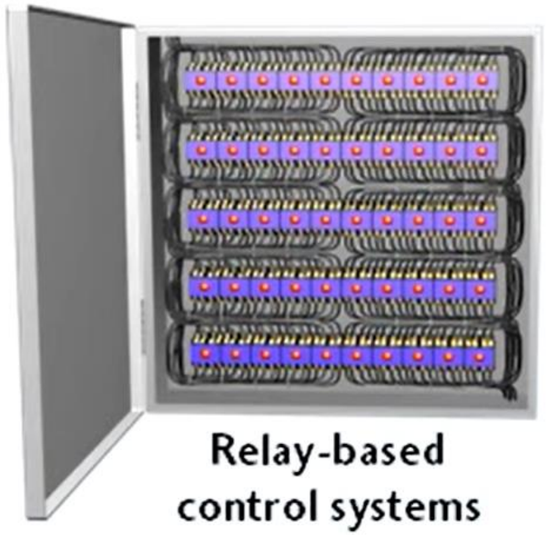
CLICK Koyo

PLC History:

Although PLCs evolved concurrently by different organizations, it is recognized as first being introduced by Bedford Associates in 1968. It was the product of choice to meet GM's Hydra-matic Division's specifications as a replacement for traditional relay-based machine control systems. The Modicon, as it was called, being an electronic device, also reduced wiring and troubleshooting time. Because it is programmable, the PLC also allows quicker changes to the equipment's control behavior.

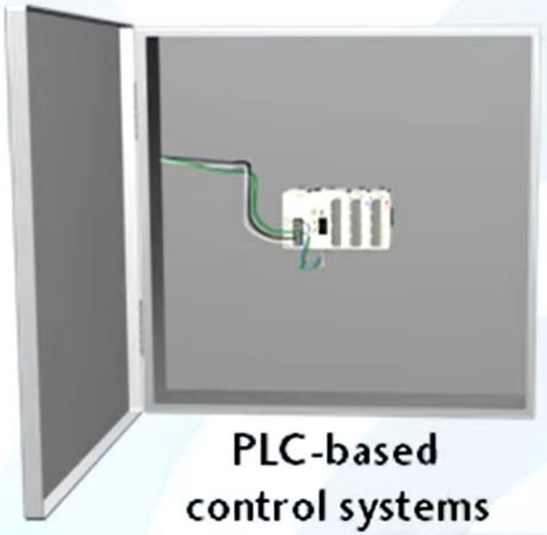


GM (General Motor)'s Hydramatic Division



Relay-based control systems

Replaced by



PLC-based control systems

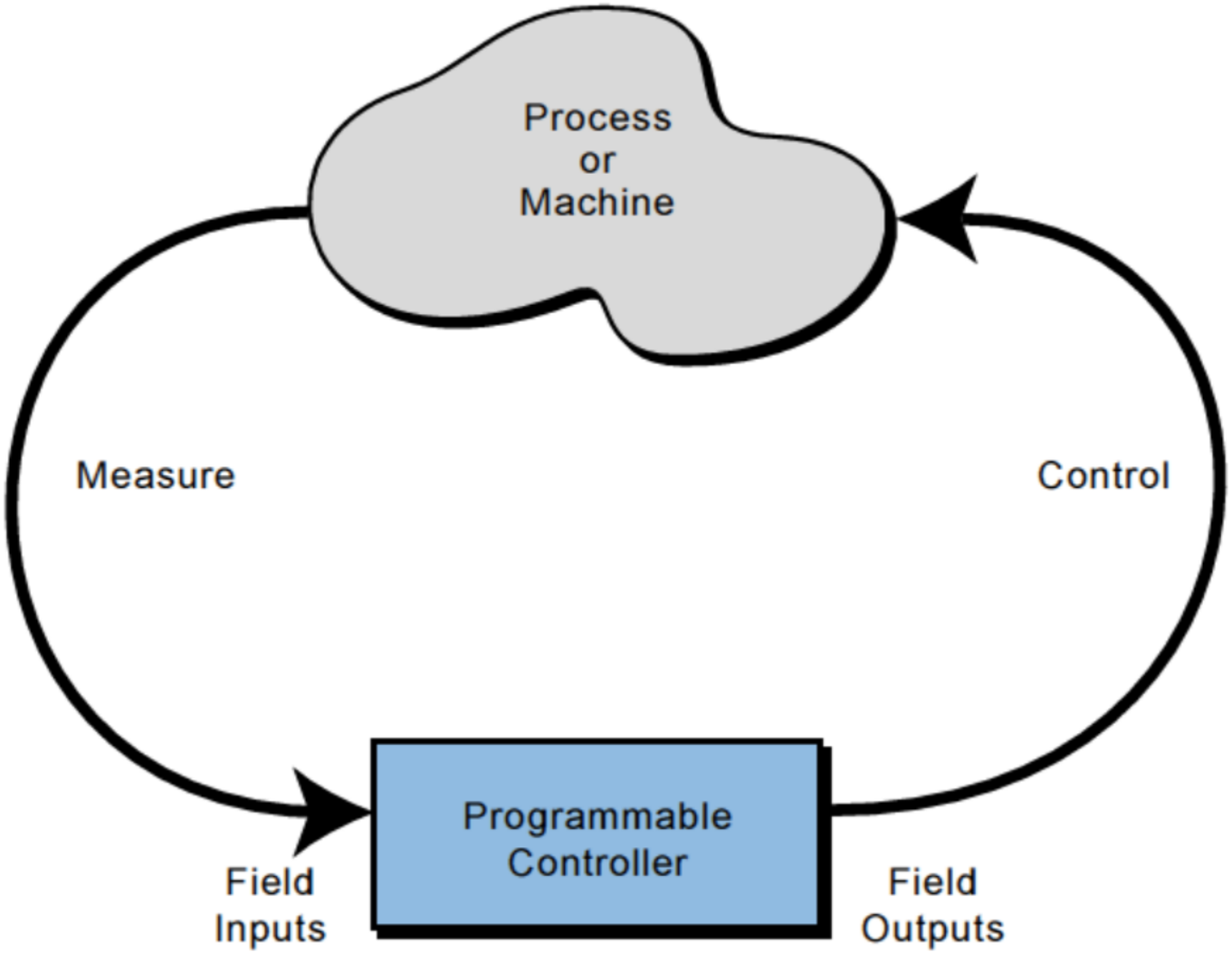


Figure 1-1. PLC conceptual application diagram.

PLC Hardware Components

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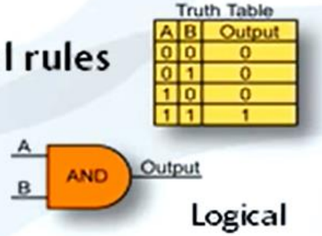
CLICK Koyo

What do the individual words mean?

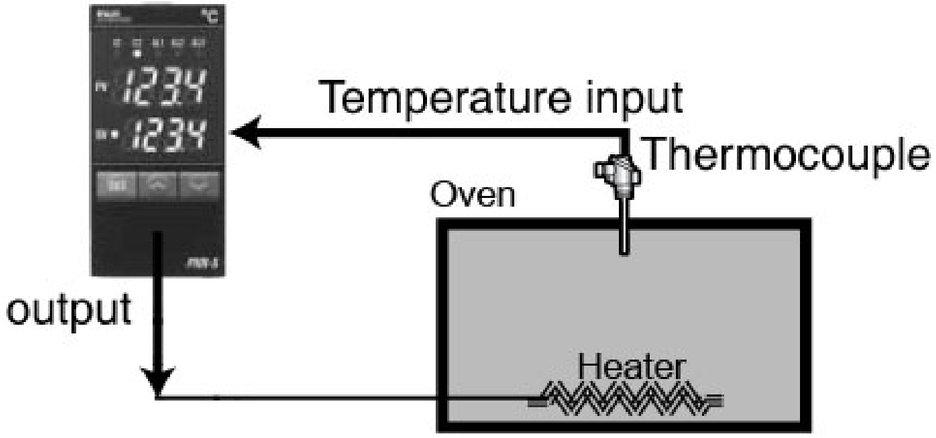
It may make more sense to look at the words in reverse order.



CONTROLLER – This is the key word. A PLC monitors various conditions, and based on these conditions, it determines an outcome.
In other words it has the ability to 'control' the outcome based on the status of different inputs such as sensors, switches, and numeric values from analog signals, etc.



LOGIC – How the PLC determines an outcome is based on the logical rules it has been taught. A simple example: it is night time, AND the door is open, then turn on the light.



PROGRAMMABLE – The PLC is taught the rules to how it should use the input conditions to create an outcome though its programming software. Being programmable makes it versatile, so if needs or conditions change, the PLC can be reprogrammed to meet these changes.

Programmable



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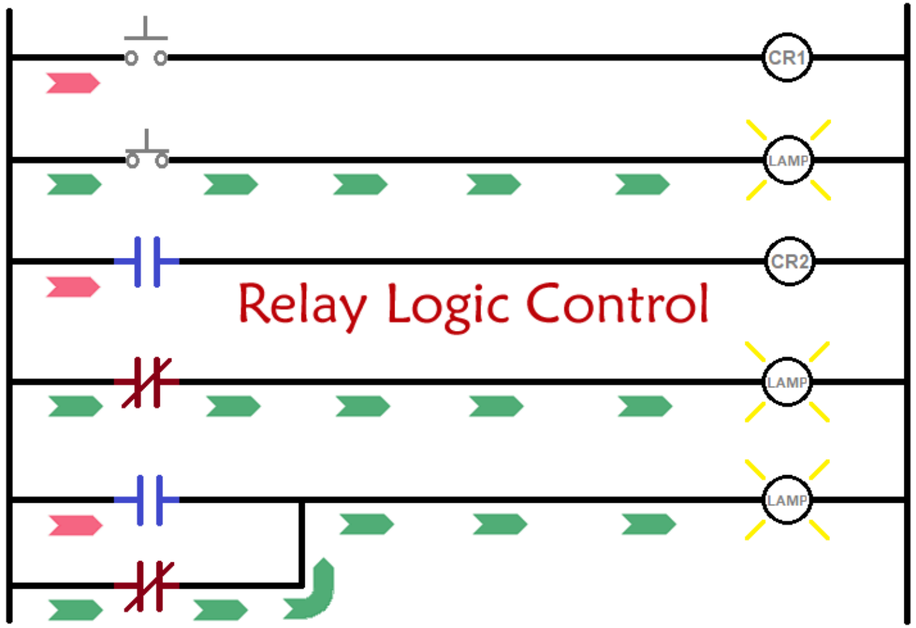
PLC Hardware Components

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CLICK Koyo

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Truth Table

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1



Logical

LOGIC – How the PLC determines an outcome is based on the logical rules it has been taught. A simple example: it is night time, AND the door is open, then turn on the light.

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Programmable



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PLC Hardware Components

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CLICK Koyo

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Controller



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Truth Table

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Programmable



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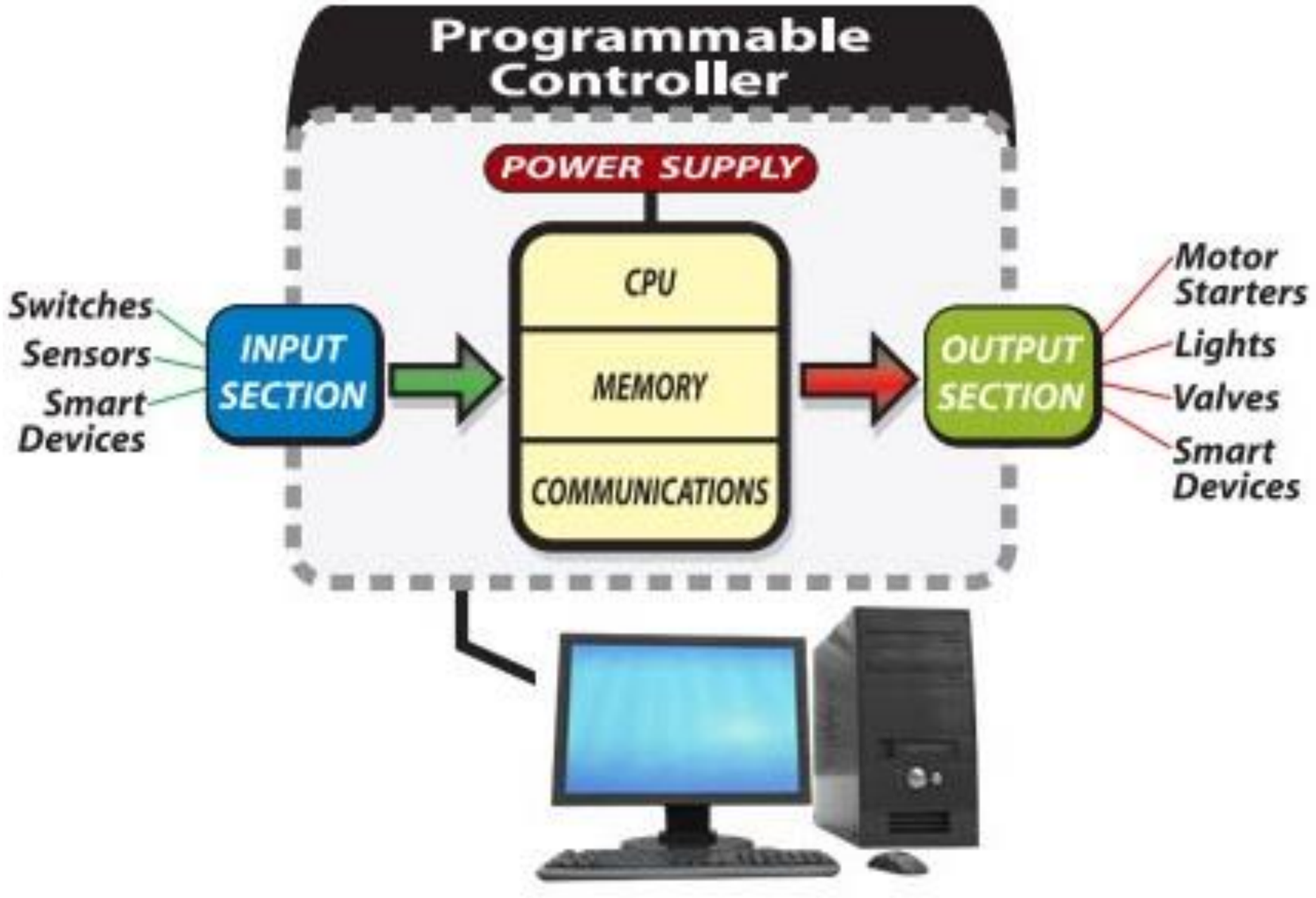
PLC Hardware Components

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PLC Overview:

Four internal areas:

- CPU – Decision making, controls other areas.
- Memory – User's control program stored in nonvolatile section of memory. Also I/O status and data is kept in memory.
- Communications ports – load user's program from PC, also exchange data with external devices, including other PLCs.
- Input/Output – handles interface of signals to real world devices.



PLC Hardware Components



PLC Hardware Components

1. Unit (Micro) Type PLC (iQ-F Series)



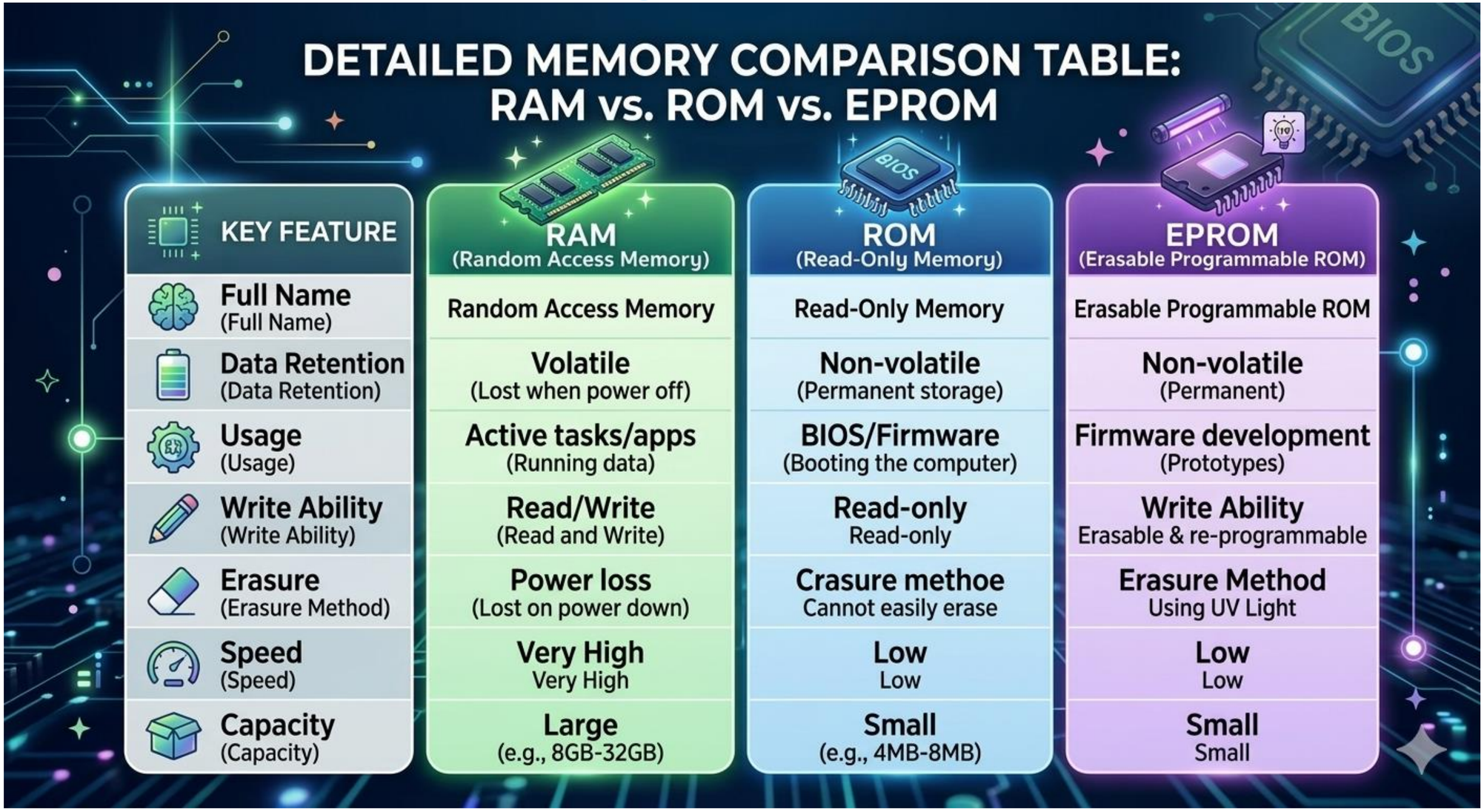
iQ-F Series

2. Modular Type PLC (A, Q , L Series)



iQ-R Series

PLC Hardware Components



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PLC Hardware Components

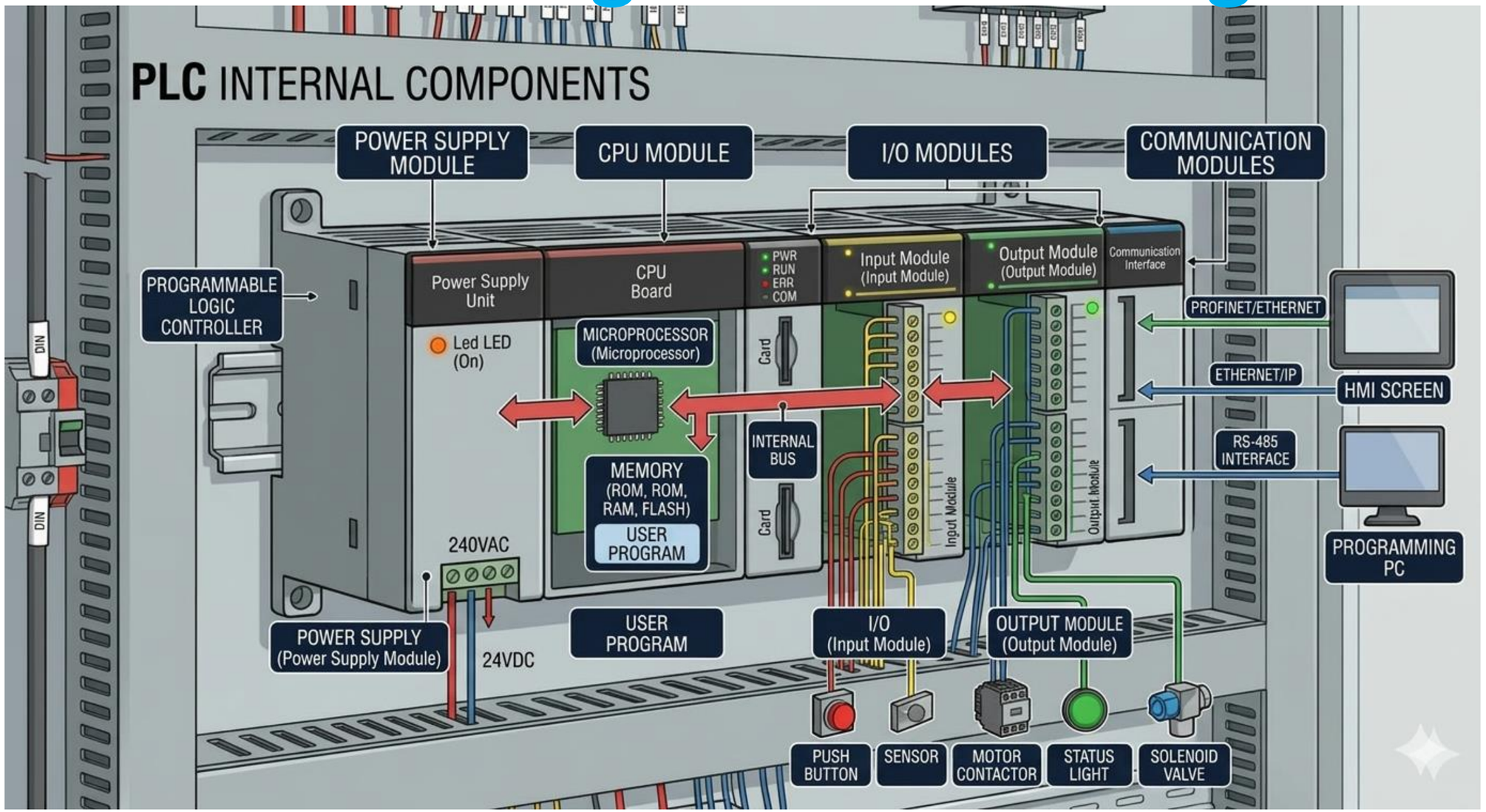
Input Interface



Output Interface

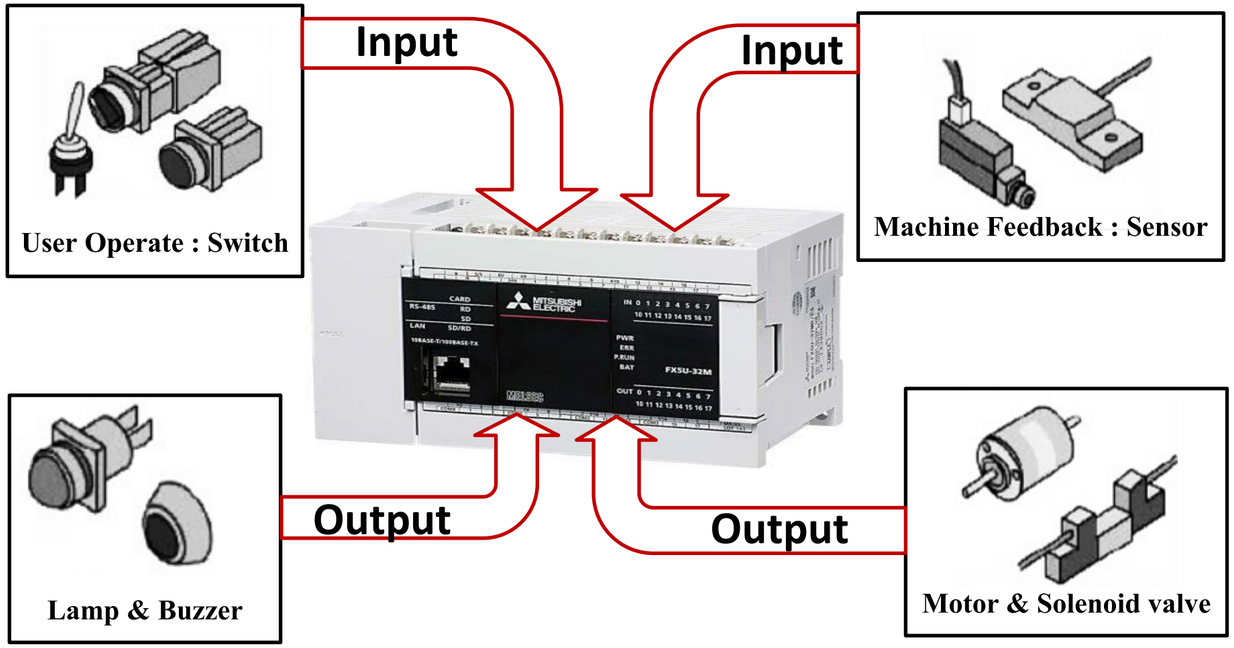


Electrical Wiring and Interfacing

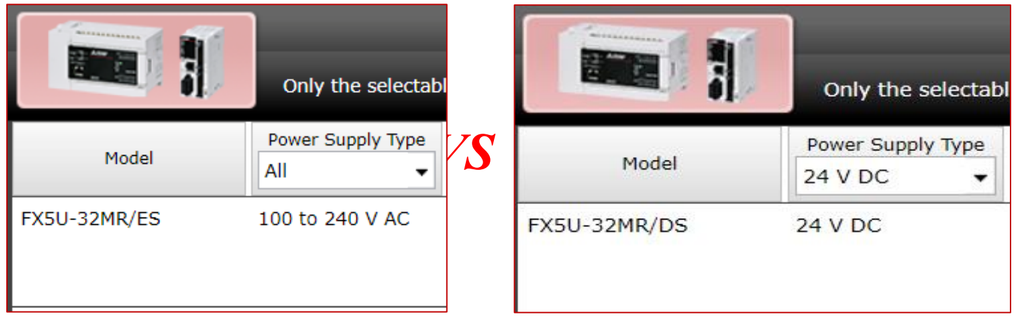


Electrical Wiring and Interfacing

1. Number of Inputs/Outputs



2. Power Supply (24 VDC or 220 VAC)

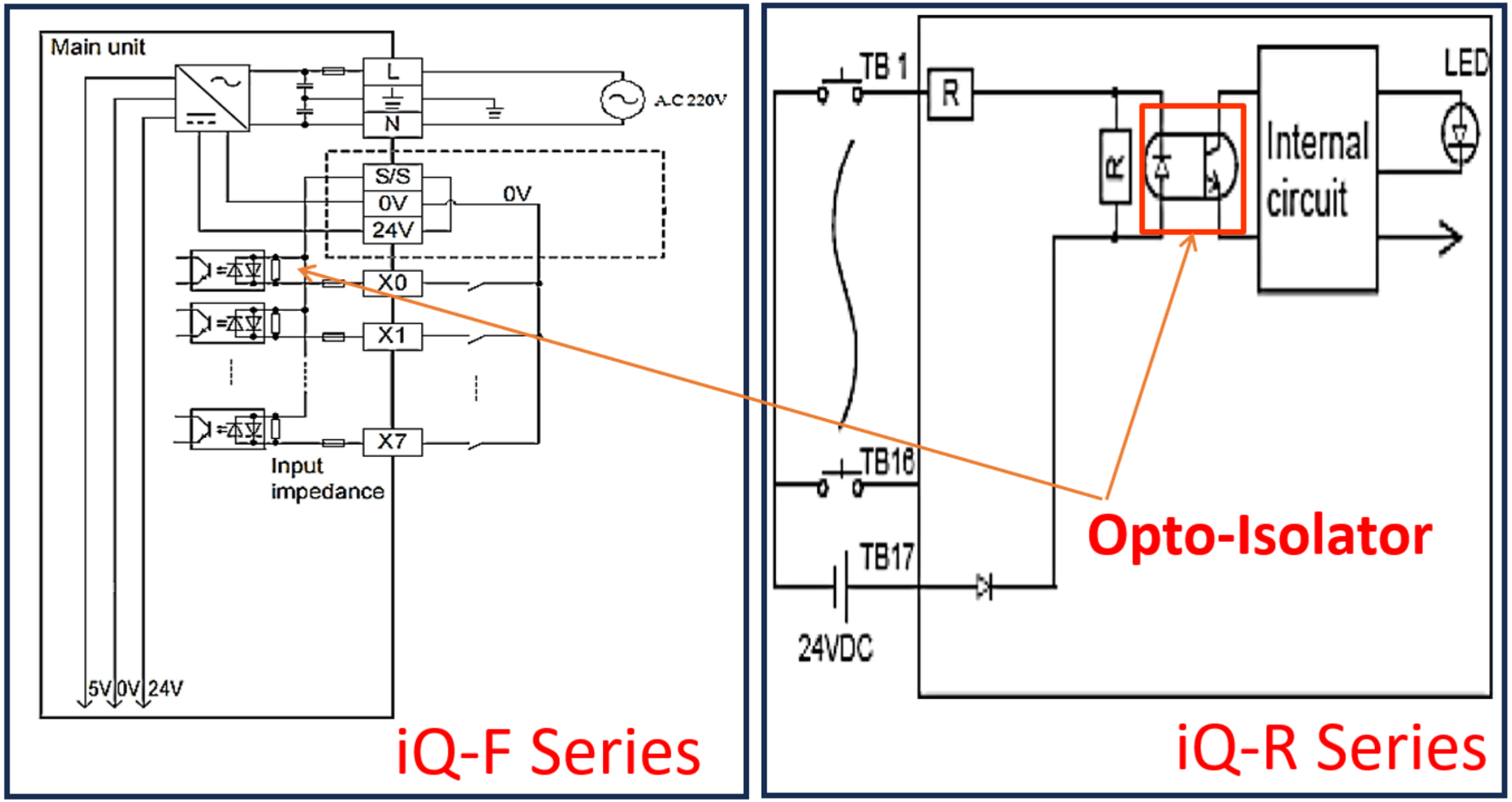






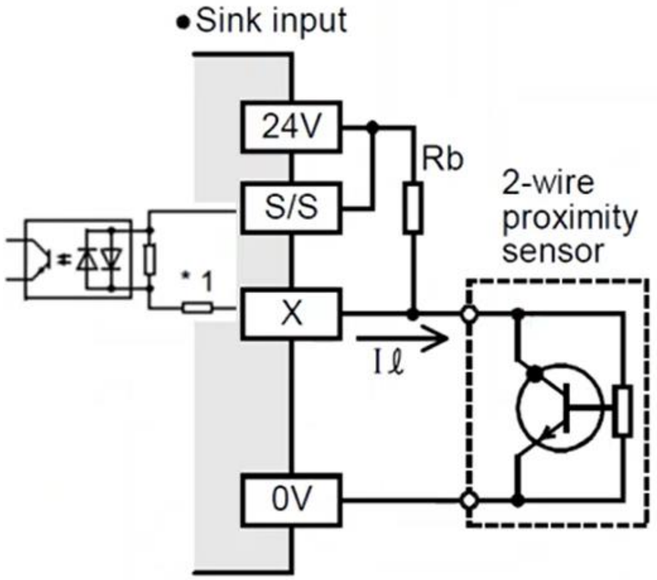
Electrical Wiring and Interfacing

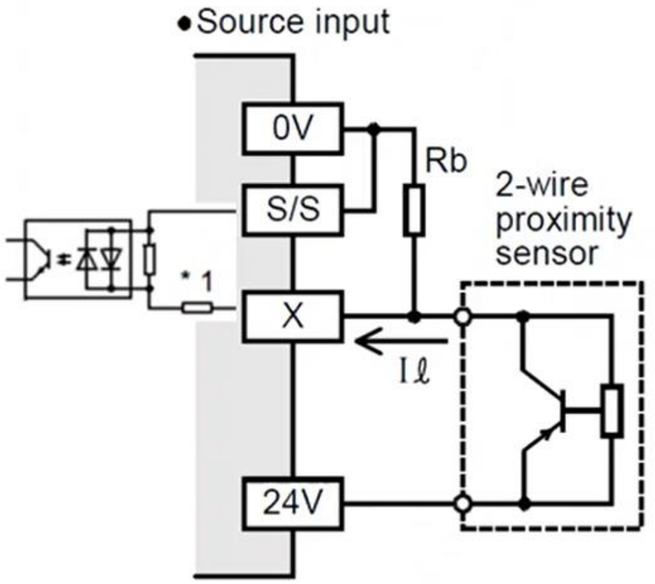
3. Select the type/source of power supply for the input circuit



Electrical Wiring and Interfacing

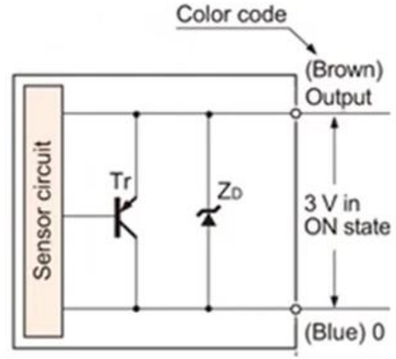
3. Select the type/source of power supply for the input circuit







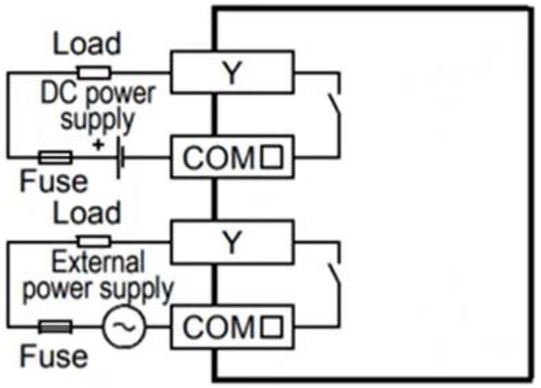




Electrical Wiring and Interfacing

4. Select the type/source of power supply for the output circuit

4.1 Relay output type

Max. load	Resistance load	2 A / point The total load current of resistance loads per common terminal should be the following value. → **For details on the common terminal for each model, refer to the terminal block layout.** • 1 output point/common terminal: 2 A or less • 4 output points/common terminal: 8 A or less • 8 output points/common terminal: 8 A or less
	Inductive load	80 VA → **For the product life, refer to Subsection 4.4.2.** → **For cautions on external wiring, refer to Subsection 12.2.4.**
Output circuit configuration		 A number (1 or more) is entered in □ of [COM□].

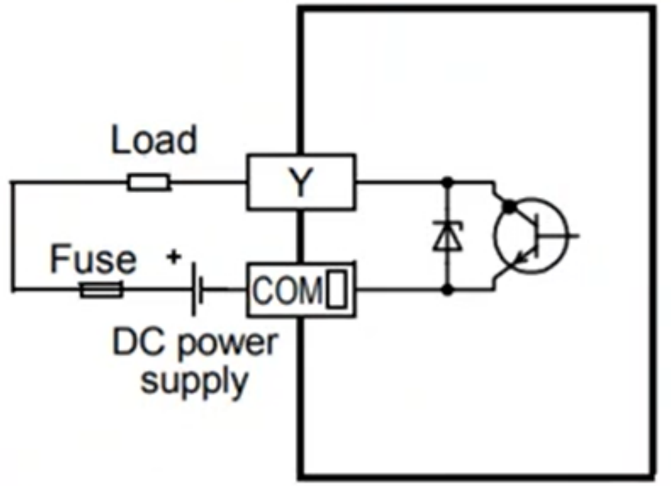
Electrical Wiring and Interfacing

4. Select the type/source of power supply for the output circuit

4.2 Transistor output type

	Resistance load	0.5 A / point The total load current of resistance loads per common terminal should be the following value. → **For details on the common terminal for each model, refer to the terminal block layout.** • 1 output point/common terminal: 0.5 A or less • 4 output points/common terminal: 0.8 A or less • 8 output points/common terminal: 1.6 A or less

4.2.1 sink type

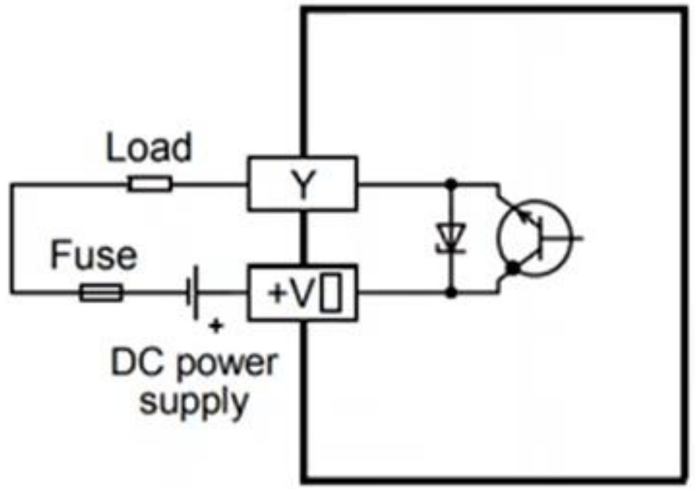
Output circuit configuration	 A common number applies to the □of [COM□].

Electrical Wiring and Interfacing

4. Select the type/source of power supply for the output circuit

4.2 Transistor output type

4.2.2 source type

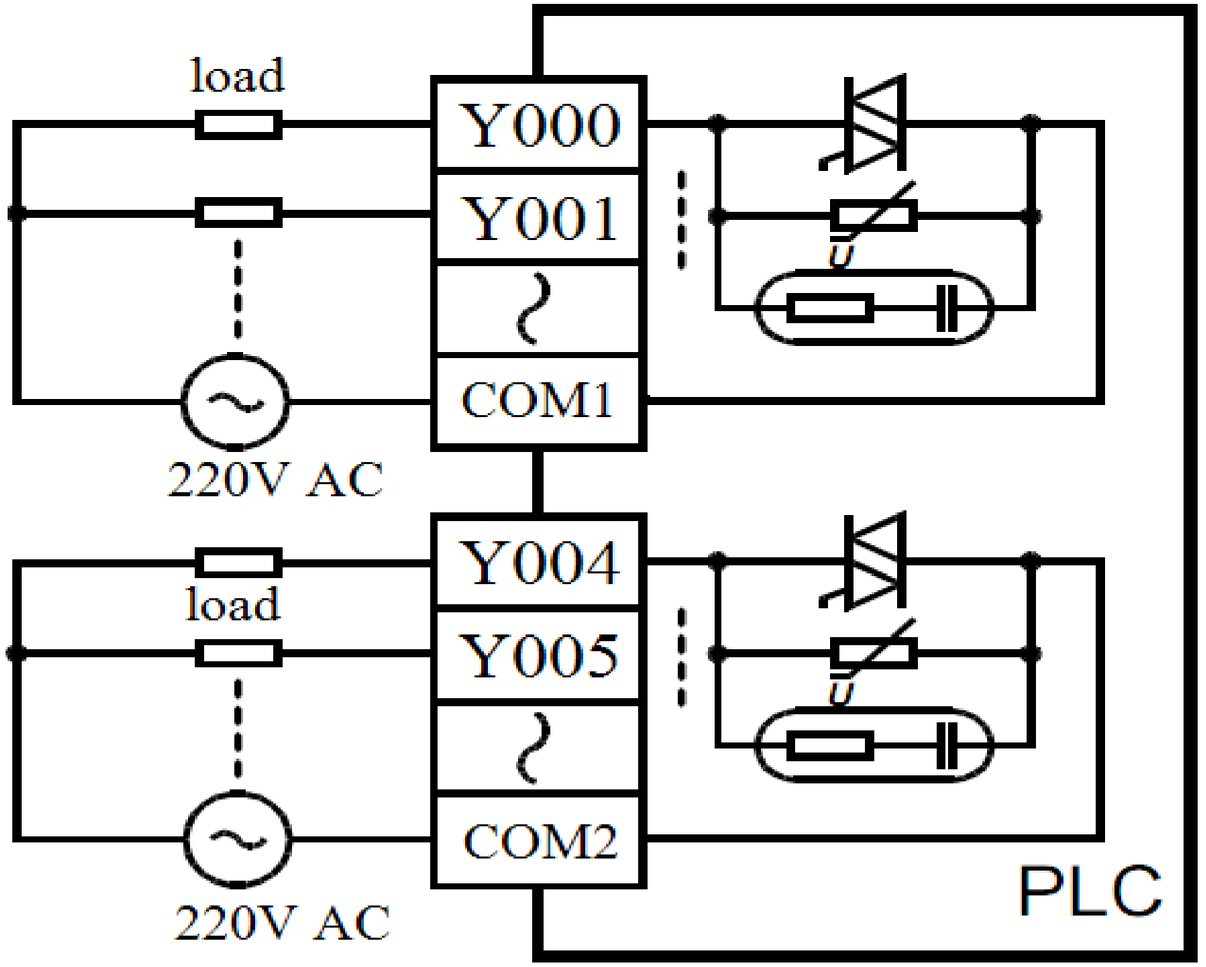
	Resistance load	0.5 A / point The total load current of resistance loads per common terminal should be the following value. → **For details on the common terminal for each model, refer to the terminal block layout.** • 1 output point/common terminal: 0.5 A or less • 4 output points/common terminal: 0.8 A or less • 8 output points/common terminal: 1.6 A or less
Output circuit configuration		 A common number applies to the □ of [+V □].

Electrical Wiring and Interfacing

4. Select the type/source of power supply for the output circuit

4.3 Triac

output type



Electrical Wiring and Interfacing

4. Select the type/source of power supply for the output circuit

Output type	Switching Frequency		Driving a Load	
	High	Low	AC	DC
Relay	-	☑	☑	☑
Transistor	☑	-	-	☑
Triac (SSR)	☑	-	☑	-

Analog I/O Processing and Control

Easy Understanding Table

Type	Full Form	Signal	Used For
DI	Digital Input.	ON/OFF	Sensors, Switches
DO	Digital Output	ON/OFF	Motors, Lamps
AI	Analog Input.	4-20mA	Level, Temp
AO	Analog Output.	4-20mA	Speed, Valves



Practical Power Plant Example



DI: DG Running status

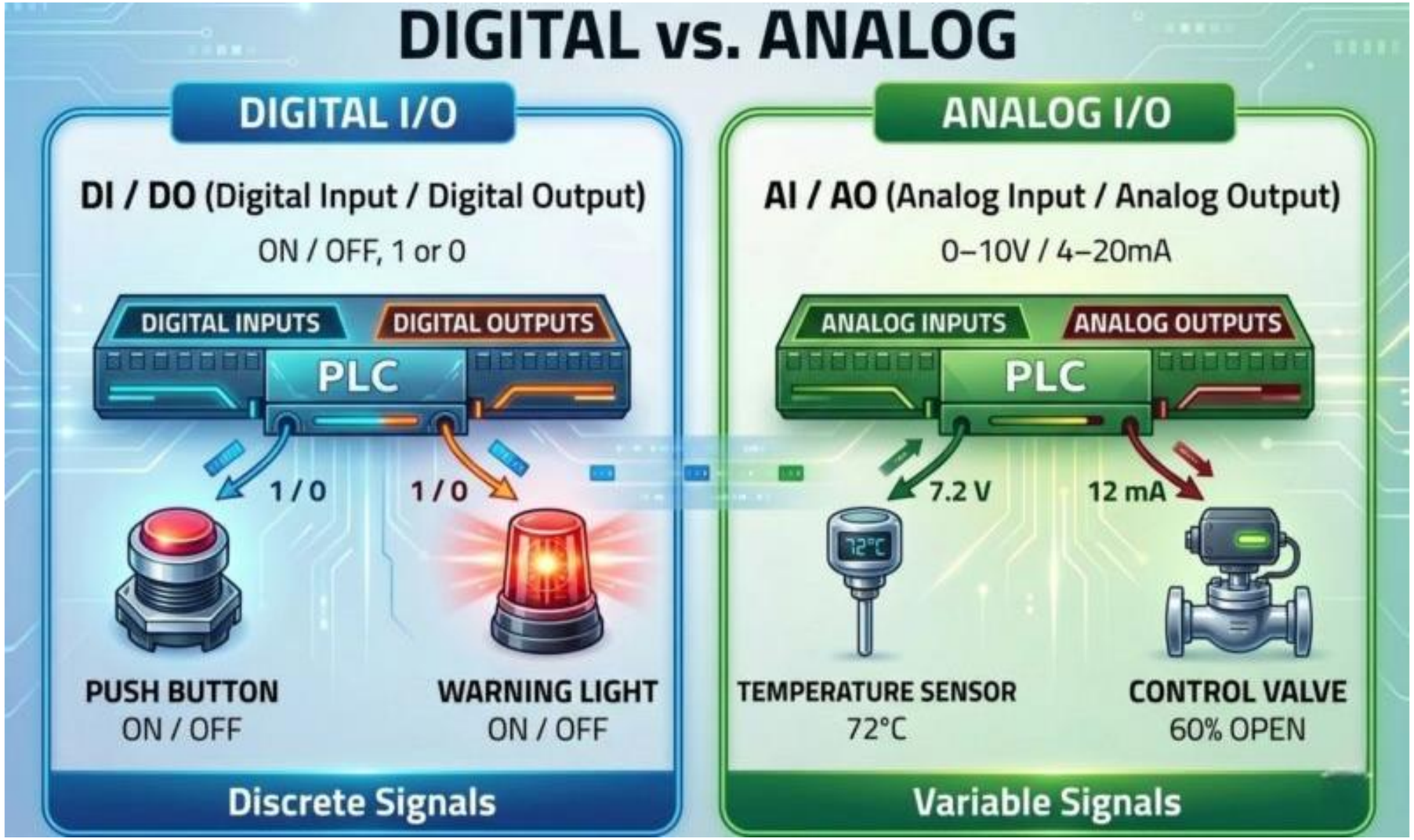
DO: Start DG command

AI: Lube oil pressure

AO: Governor speed control

https://www.facebook.com/photo/?fbid=26495508026716997&set=gm.4240200829579546&idorvanity=1392527111013613

Analog I/O Processing and Control



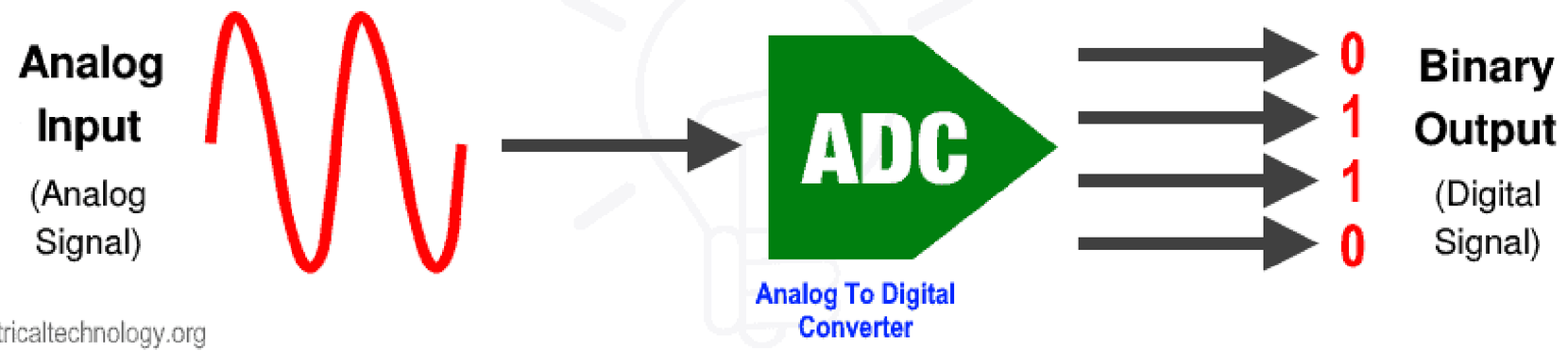
https://www.linkedin.com/posts/ahmed-zayed85_digital-io-vs-analog-io-in-plc-systems-activity-7435593199315767296-BeK9

Analog I/O Processing and Control

ADC

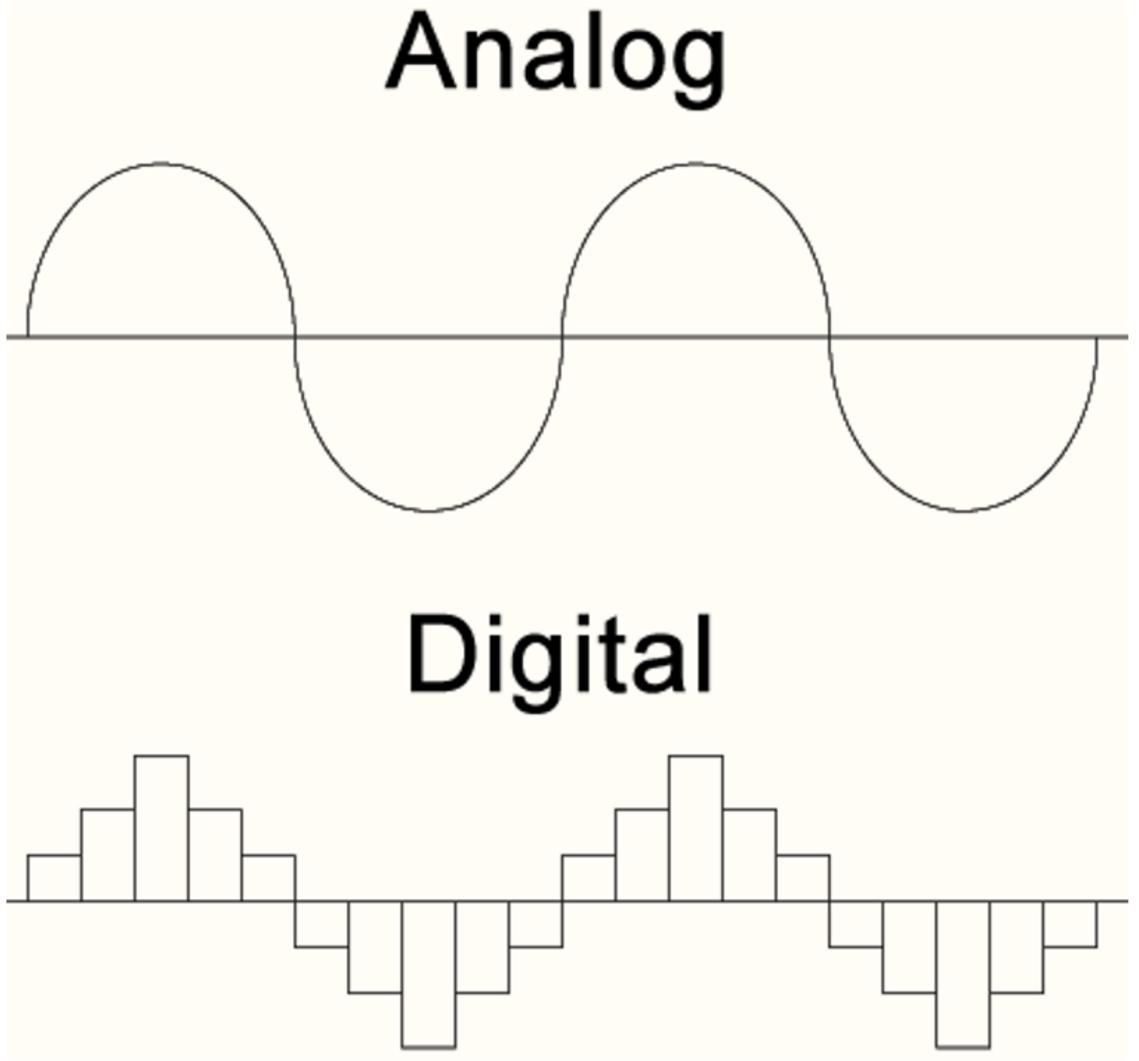
ANALOG TO DIGITAL CONVERTER

Types, Working, Block Diagram & Applications

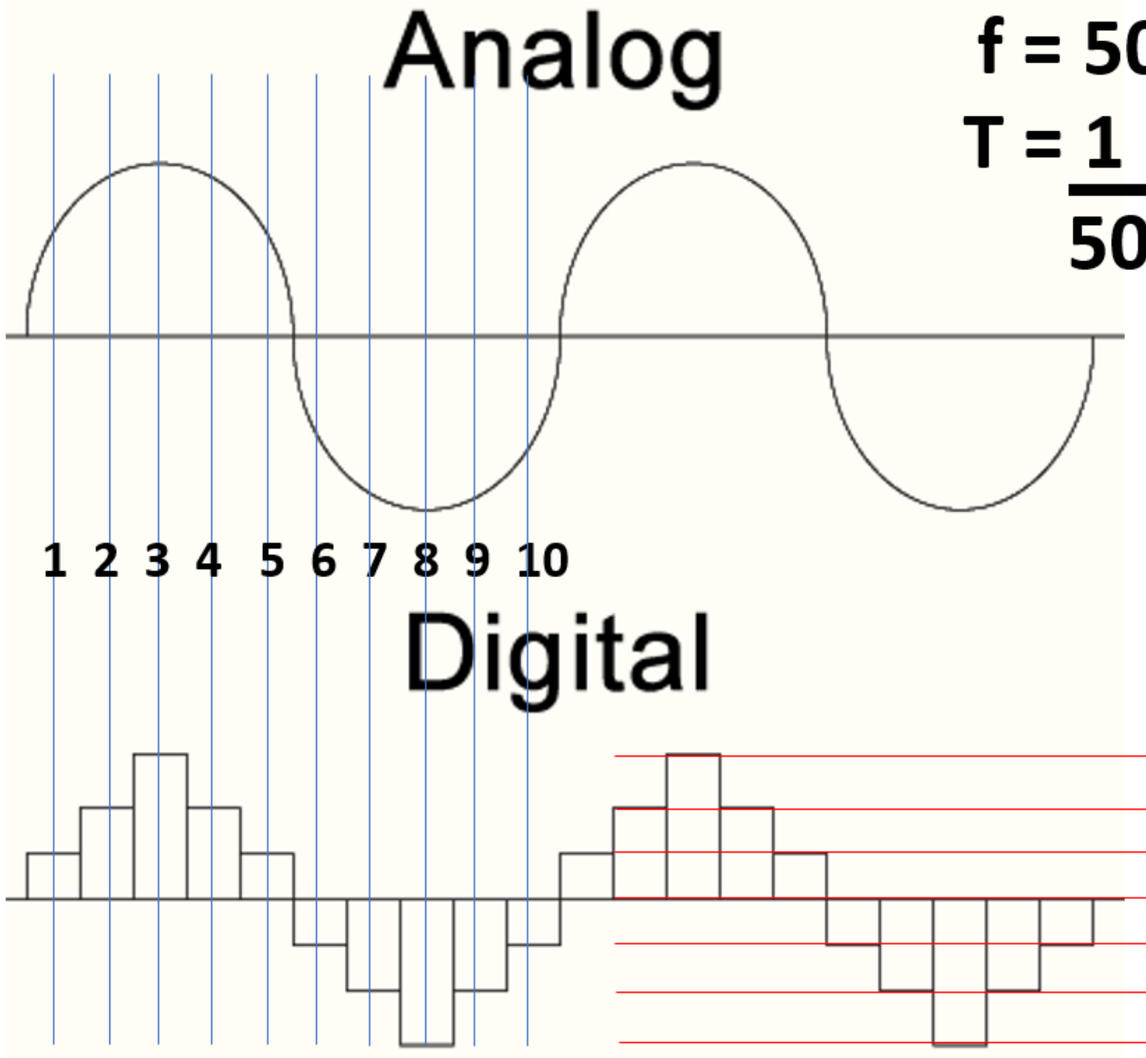


https://www.electricaltechnology.org/2019/02/analog-to-digital-converter-adc.html

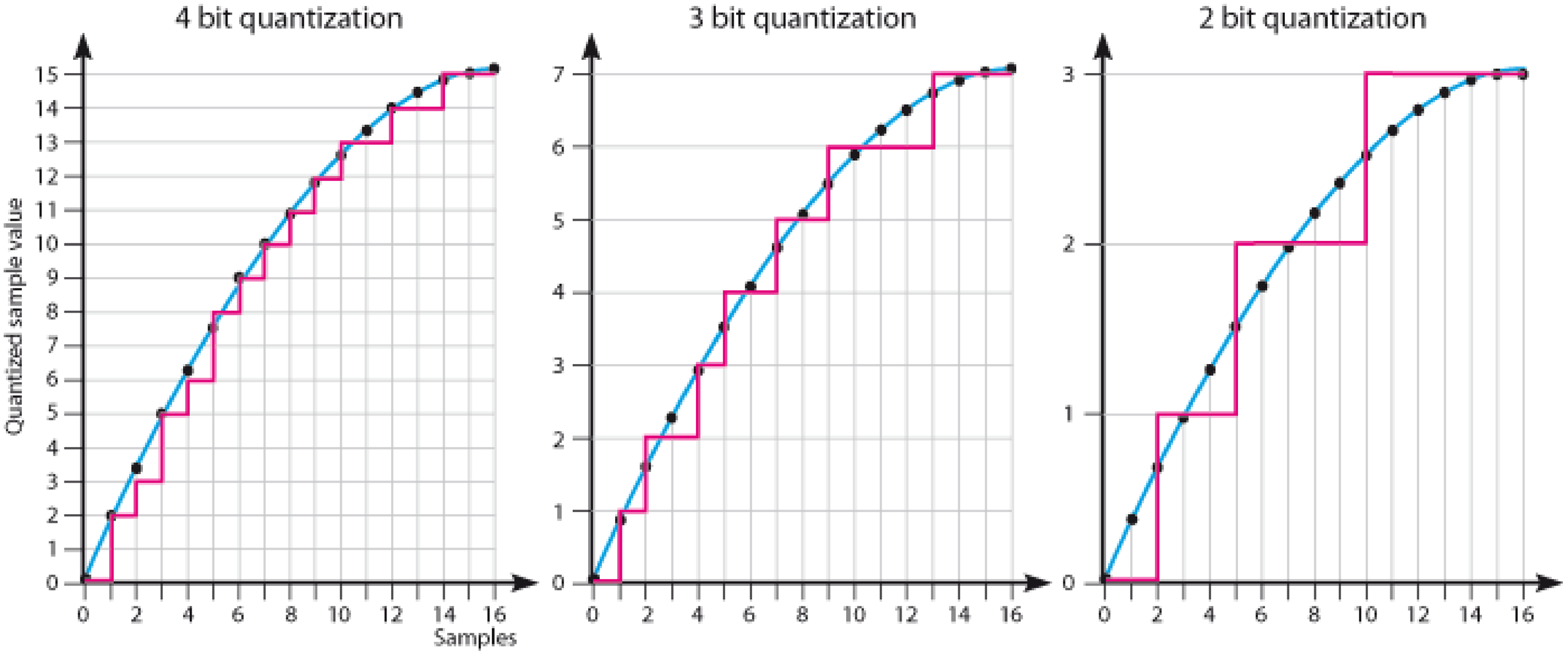
Analog I/O Processing and Control



Analog I/O Processing and Control

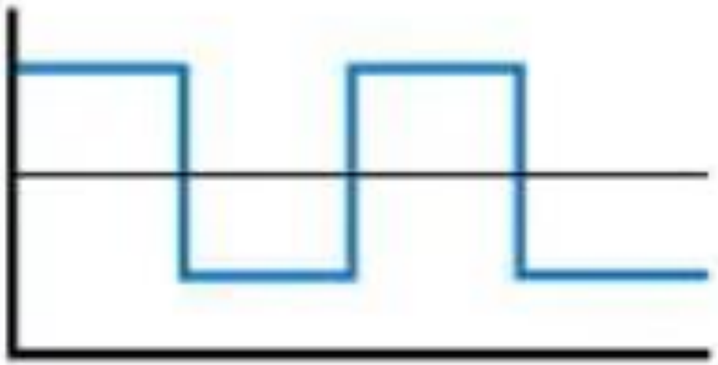


Analog I/O Processing and Control

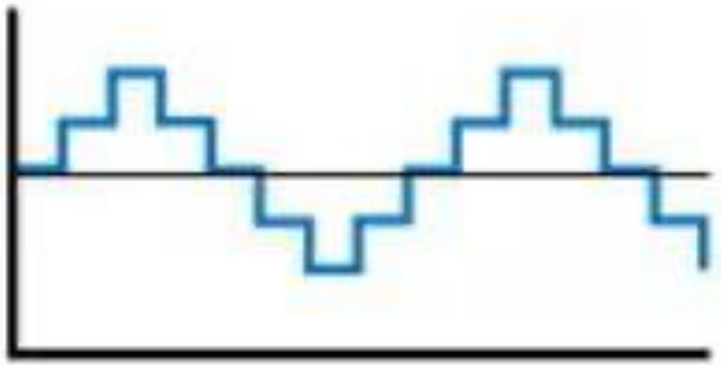


https://electronics.stackexchange.com/questions/190824/sampling-rate-analog-to-digital-converter

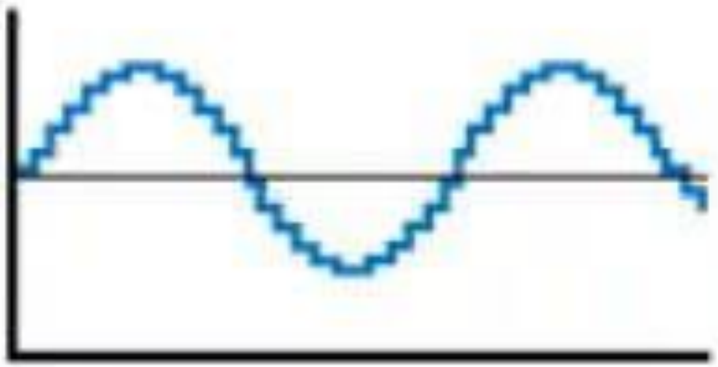
Analog I/O Processing and Control



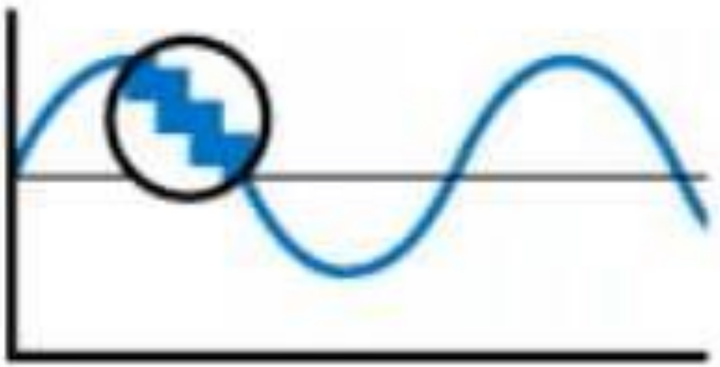
1-bit



2-bit



4-bit



16-bit

https://www.seeedstudio.com/blog/2020/10/20/analog-to-digital-converter-adc-reading-raspberry-pi-analog/?srsltid=AfmBOopjRRLX6P3RZS14OfFA4DwbZQGlYA2IWYi8BPo6A7sJkYbORZN5

Analog I/O Processing and Control

R60AD4

Item	Specifications			
Number of analog input channels	4 channels			
Analog input voltage	-10 to 10VDC (input resistance 1MΩ)			
Analog input current	0 to 20mADC (input resistance 250Ω)			
Digital output value	16-bit signed binary value (-32768 to 32767)			
I/O characteristics, resolution*1	Analog input range		Digital output value	Resolution
	Voltage	0 to 10V	0 to 32000	312.5μV
		0 to 5V		156.3μV
		1 to 5V		125.0μV
		1 to 5V (extended mode)	-8000 to 32000	125.0μV
		-10 to 10V	-32000 to 32000	312.5μV
		User range setting		47.7μV
	Current	0 to 20mA	0 to 32000	625.0nA
		4 to 20mA		500.0nA
		4 to 20mA (extended mode)	-8000 to 32000	500.0nA
		User range setting	-32000 to 32000	190.7nA

16 bit

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Training

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HMI (Human Machine Interface)



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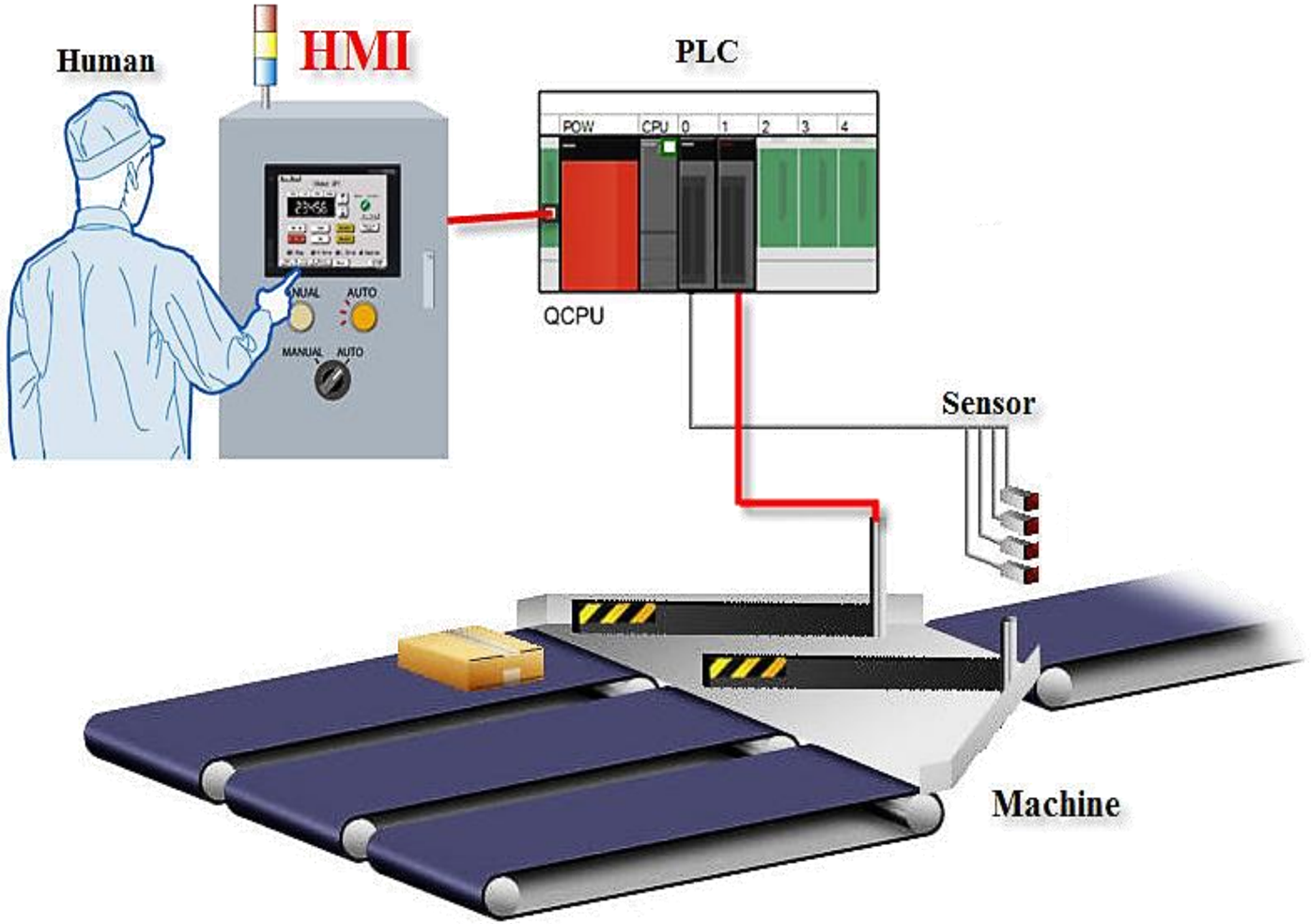


GOT2000

Graphic Operation Terminal



HMI (Human Machine Interface)



It is a user interface or dashboard that connects a person to a machine, system, or device, translating complex data into actionable information.

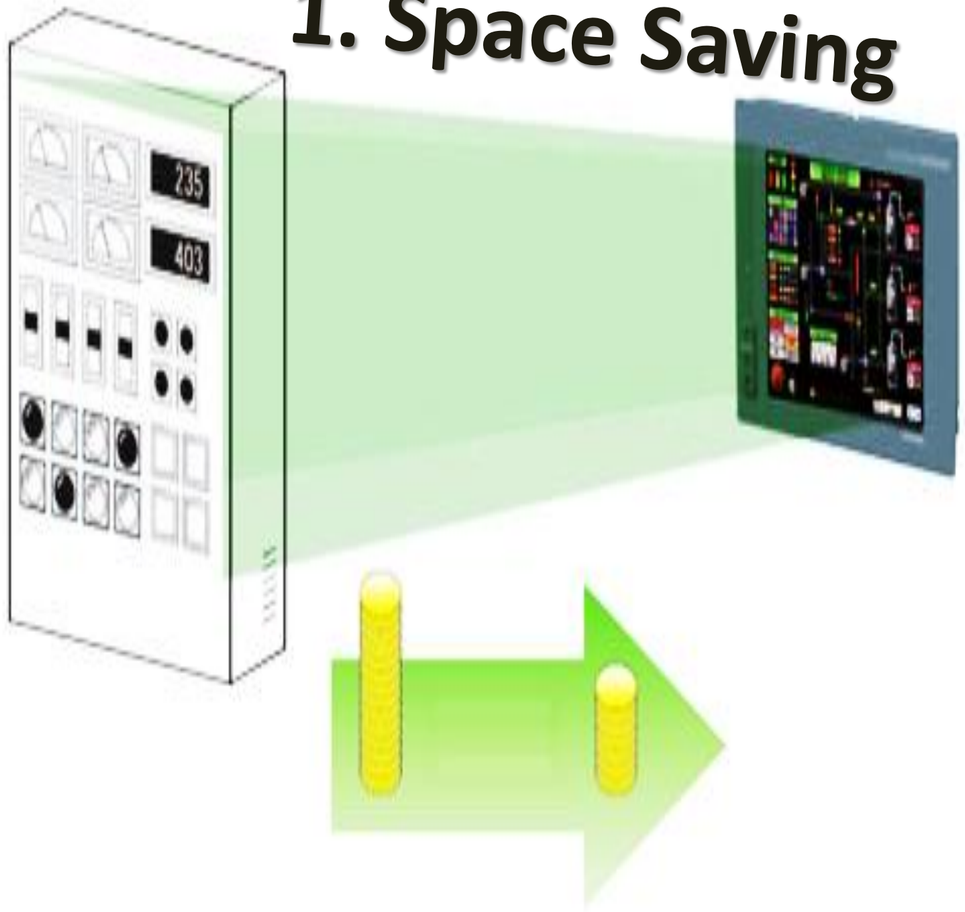
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HMI (Human Machine Interface)



GOT (Graphic Operation Terminal)

1. Space Saving :
   - Reduce touchscreen size to save space
   - Screen sizes from 3.5" to 15"

2. Cost Saving :
   - Reduce design time and costs
   - No hardware costs for the control panel

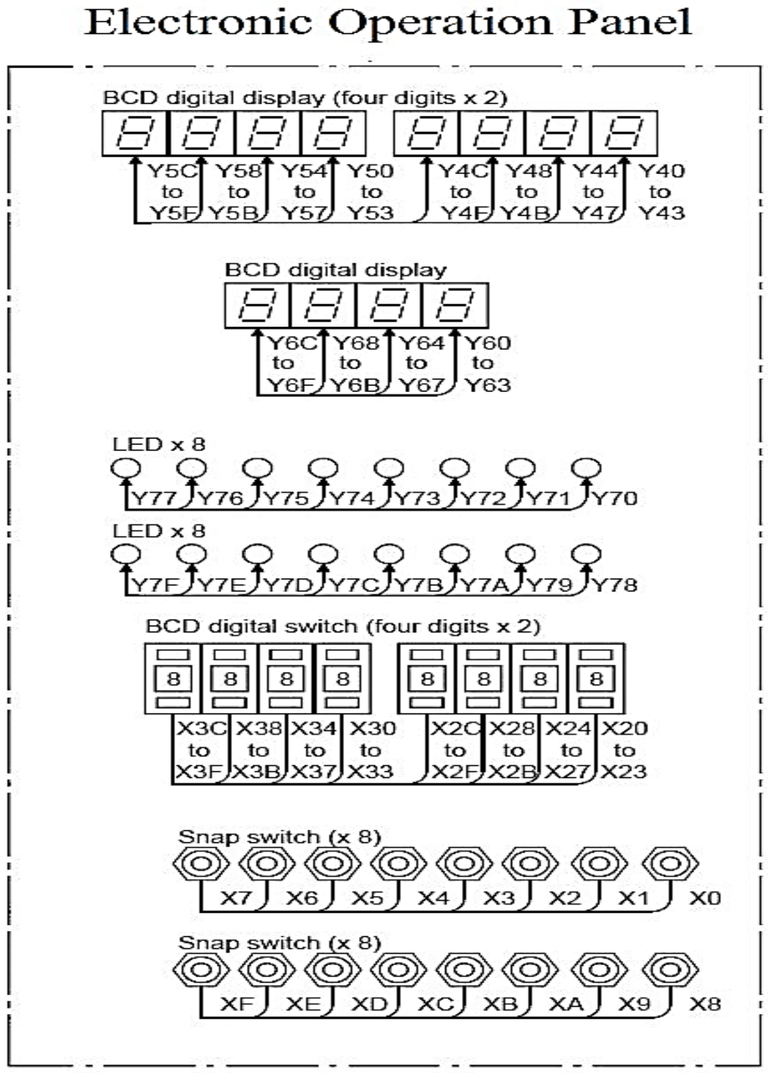
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HMI (Human Machine Interface)



Control Panel



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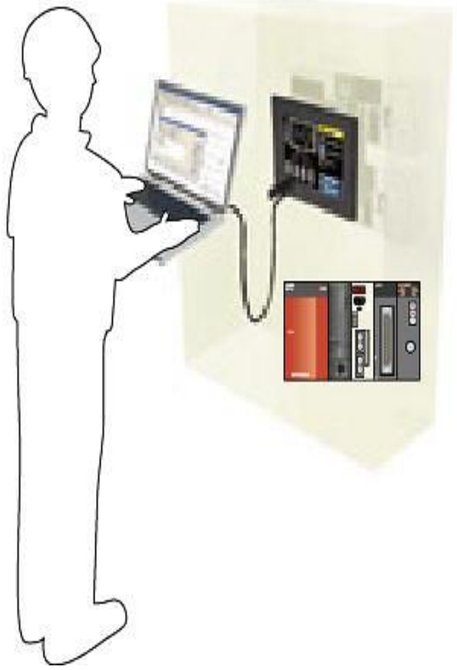
Training

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HMI (Human Machine Interface)







GOT (Graphic Operation Terminal)

3. Design & Modification Saving :

- **The GOT control panel is standard**
- **Screen settings and data can be edited via software**

4. Maintenance Saving:

- The GOT display easily showcases graphics, text, and system notifications

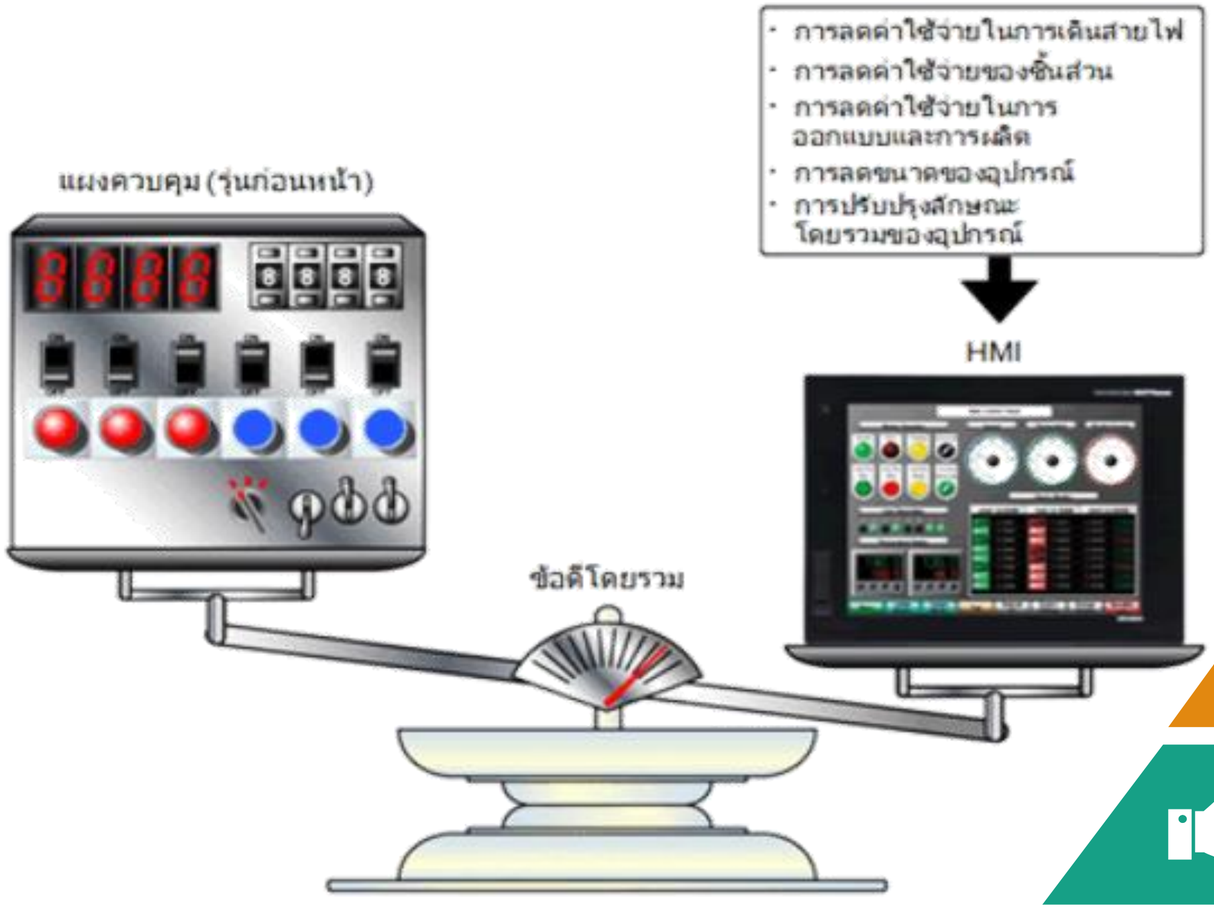
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HMI (Human Machine Interface)



- Space Saving
- Cost Saving
- Design & Modification Saving
- Maintenance Saving

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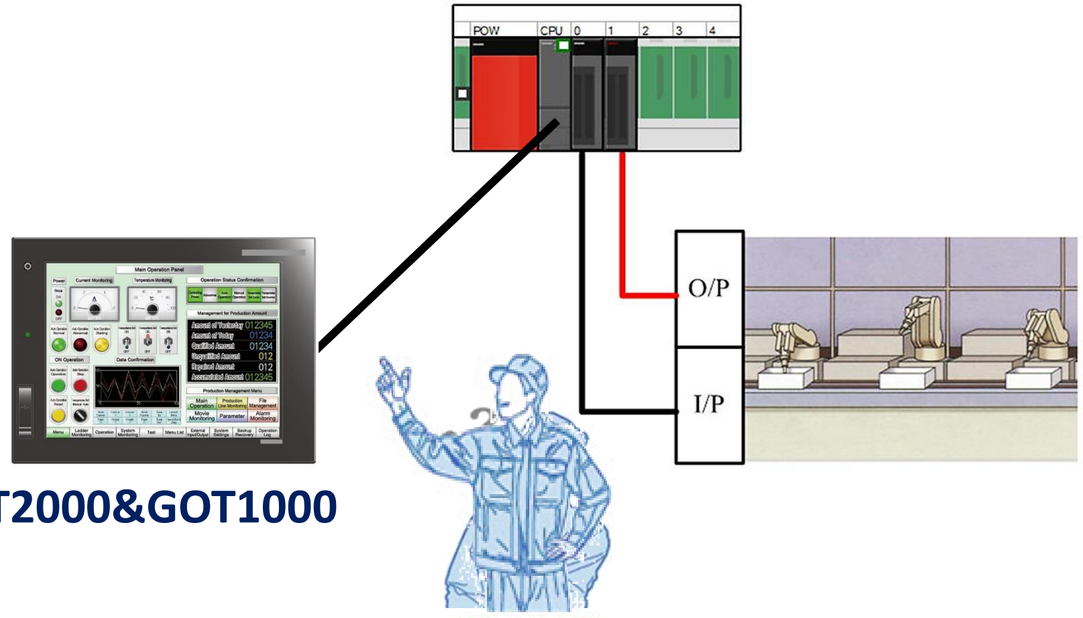
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HMI (Human Machine Interface)

Process Control



GOT2000&GOT1000

MITSUBISHI ELECTRIC

MELSOFT

GOT Screen Design Software

GT Designer3

Version 1

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GT Designer3

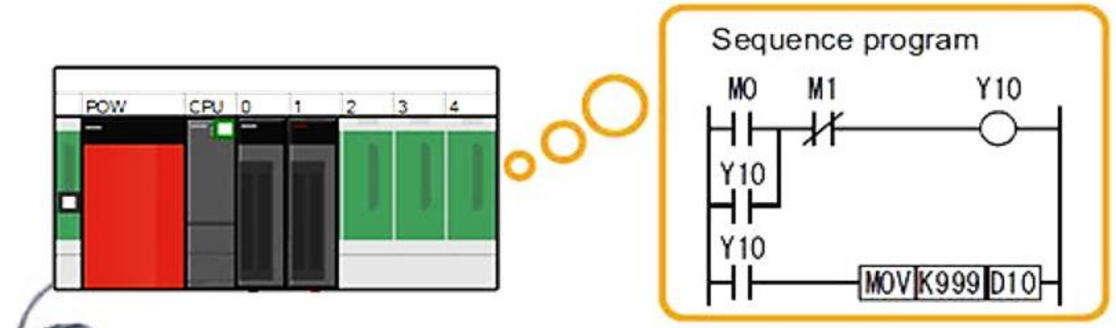
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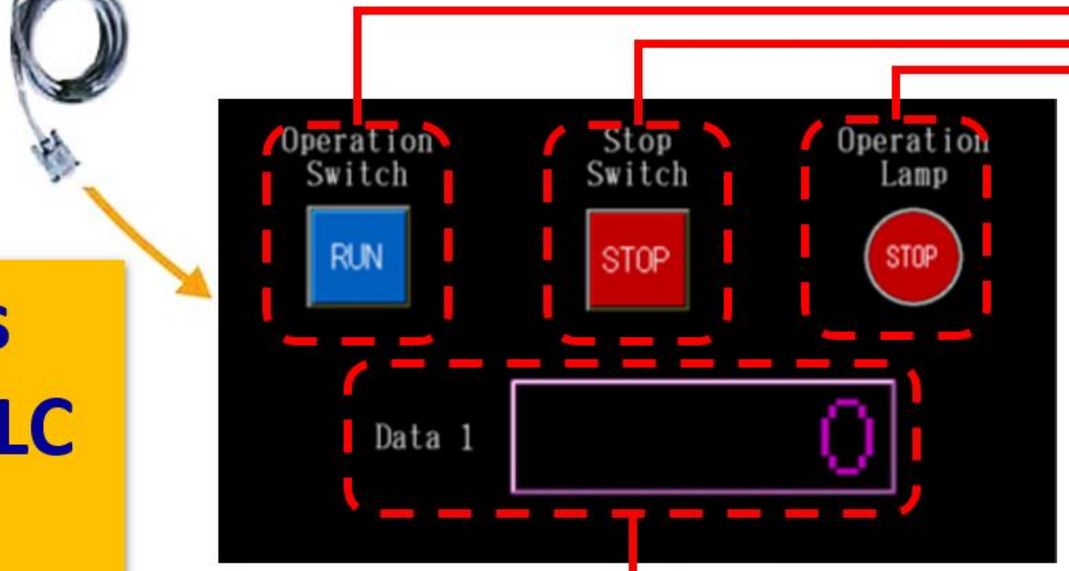
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HMI (Human Machine Interface)







GOT operates based on the PLC program

- **M0 = Operation Switch**
- **M1 = Stop Switch**
- **Y10 = Operation Lamp**
- **D10 = Data in PLC**

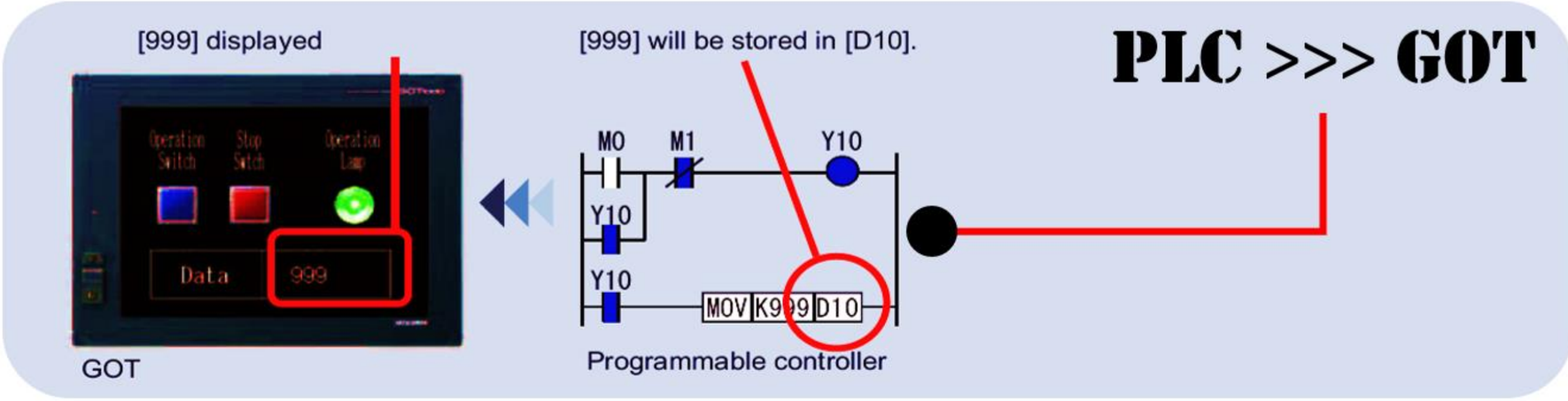
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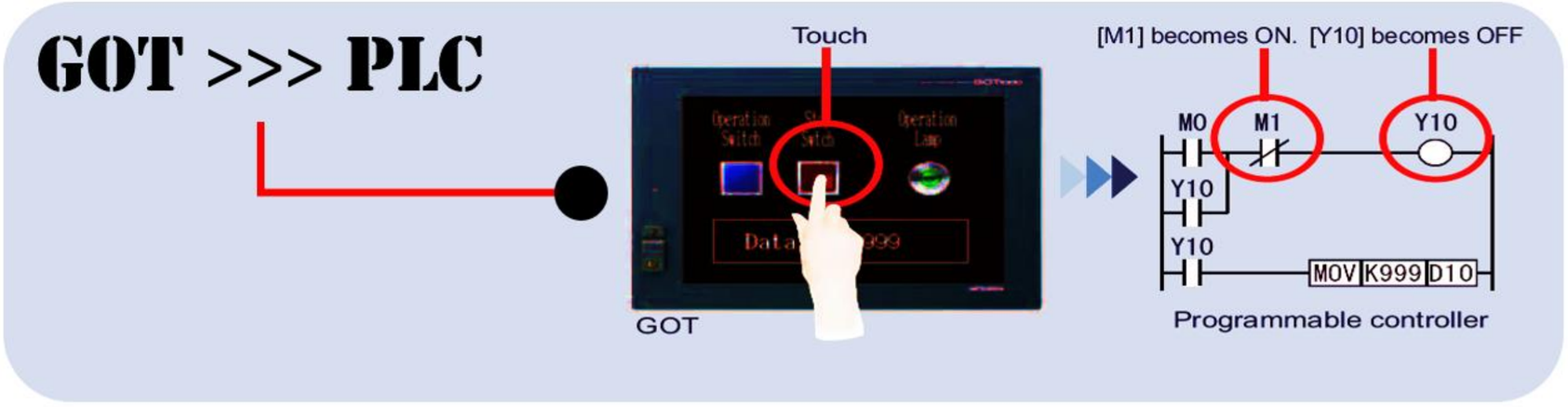
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HMI (Human Machine Interface)





Integration of GOT and PLC

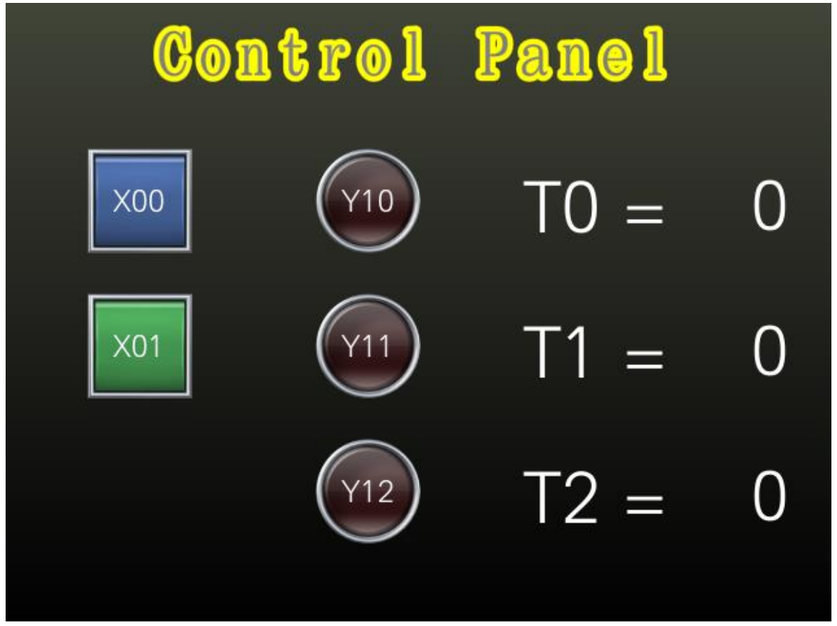
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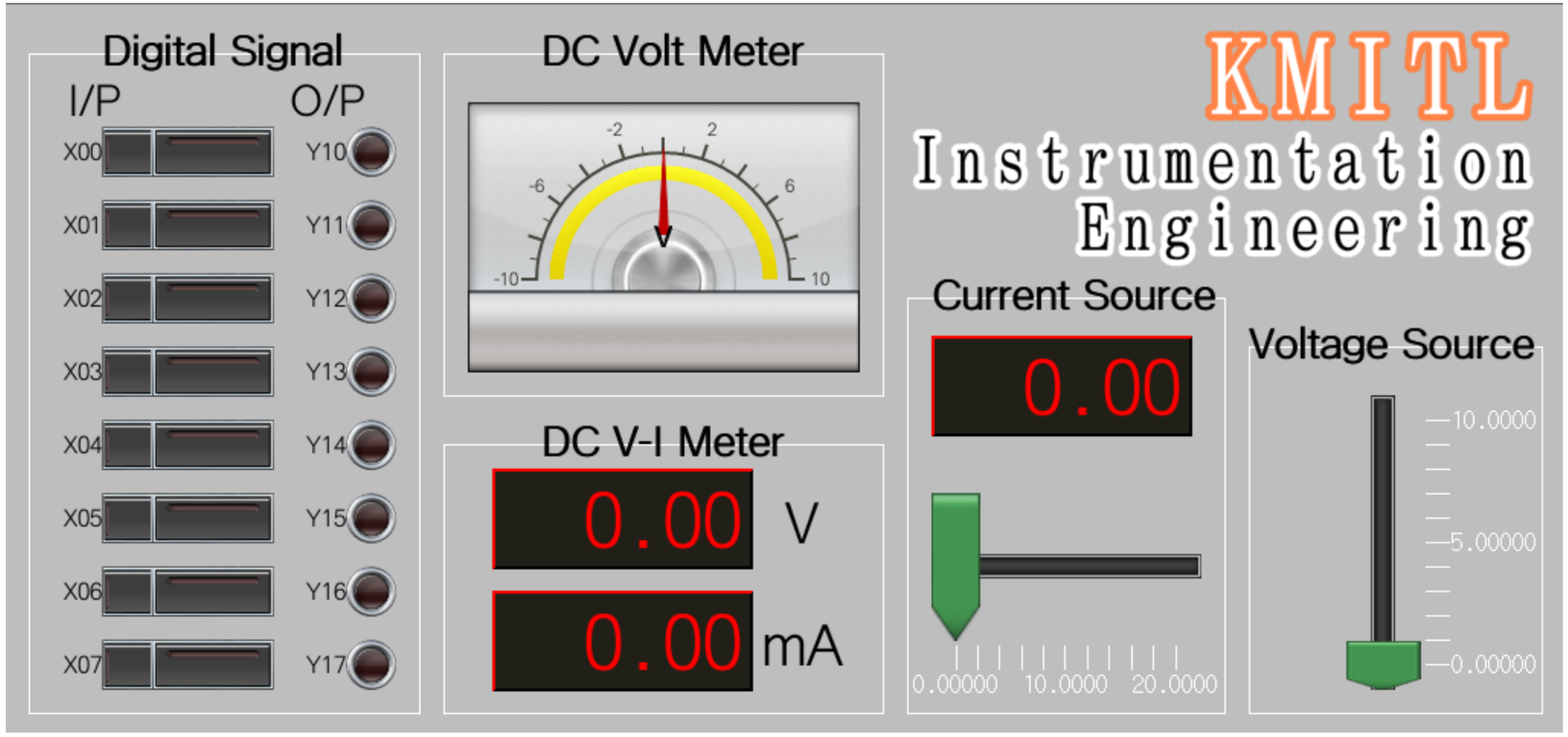
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HMI (Human Machine Interface)





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ICONICS Suite™

GENESIS64™



GENESIS64™ Advanced

- Extensive monitoring of large-scale plant and building systems
- Distributed processing by multiple servers
- Cloud-ready universal connectivity

GENESIS64™



GENESIS64™ Basic SCADA

- All essential SCADA functions
- Server for distributed clients
- Low-cost single-server license

GT SoftGOT2000



GT SoftGOT2000

- Computer-based remote monitoring
- Execute GOT (HMI) functions on a computer

GOT2000



GOT2000 Series

- Compact and highly robust
- Visualize equipment status

Equipment | Production line | Entire factory and building | Multiple production sites

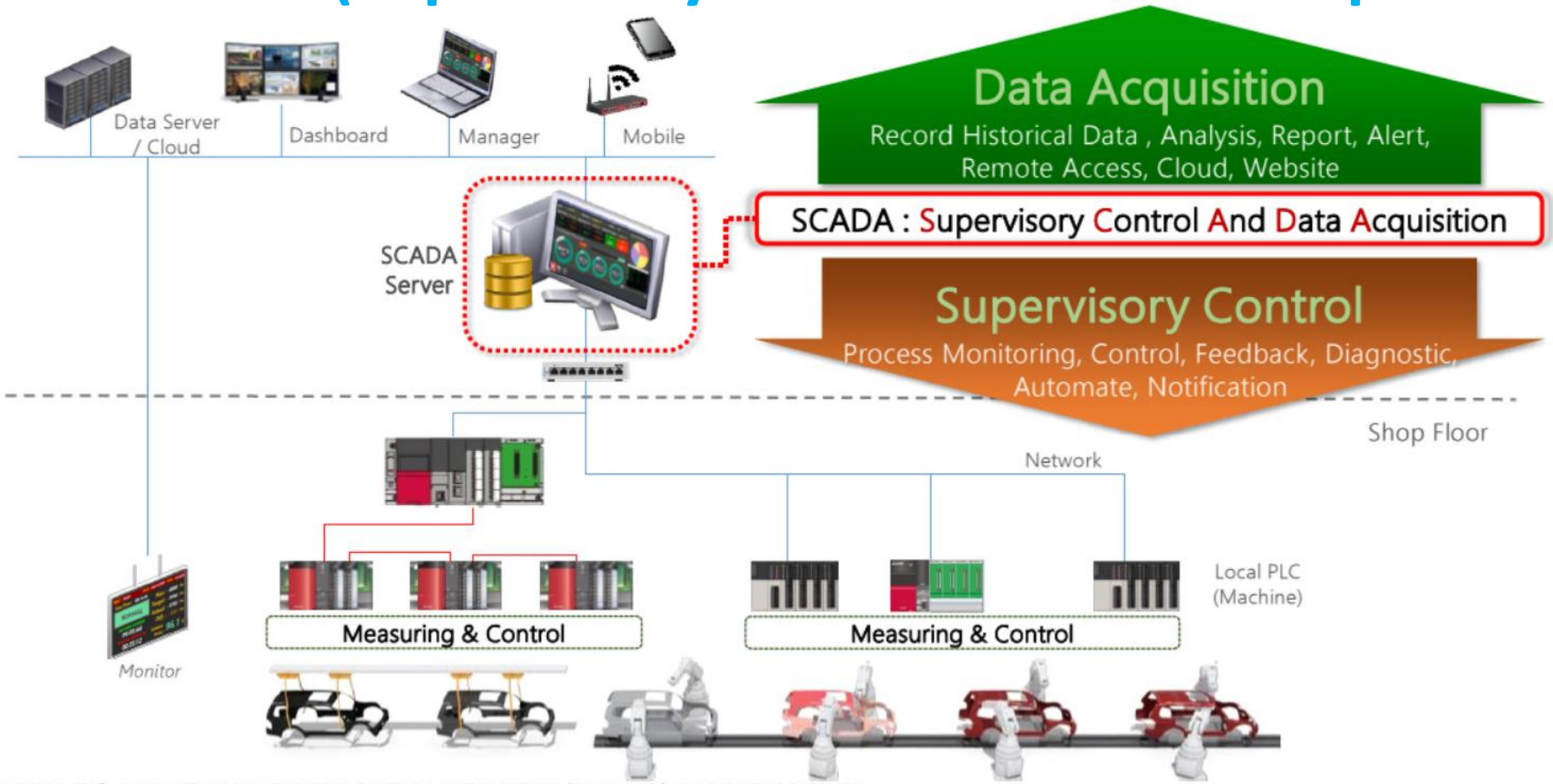
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Training

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SCADA (supervisory control and data acquisition)



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SCADA (supervisory control and data acquisition)





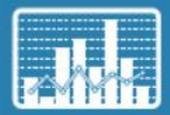
KPIWorX™ MobileHMI™





Alarm Management


Production Management


Data Historian


Energy Management


Fault Detection Artificial Intelligence


Quality and SPC


Scheduling


Reporting

HMI/SCADA	Historian	Analytics	IoT
GENESIS64™	Hyper Historian™	AnalytiX®	IoTWorX™









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SCADA (supervisory control and data acquisition)

GENESIS64™ Basic SCADA

Entry-level SCADA package



- Basic SCADA functions
- State-of-the-art, real-time visualization
- ANSI/ISA-18.2 compliant alarm management
- Robust, reliable historical data collection

GENESIS64™ Advanced

Full-fledged functionality



- Extensive monitoring of enterprise-wide plant and building systems
- Scales to large, distributed systems
- Support for secure cloud communications

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SCADA (supervisory control and data acquisition)

MITSUBISHI ELECTRIC
Changes for the Better

โรงงานผลิตกระดาษ

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SCADA (supervisory control and data acquisition)

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CASE STUDY

กรณีที่กระบวนการผลิตไม่มีความเสถียร

Day1
Energy : 5,000 kWh
Products: 10,000 pcs.

Day2
Energy : 3,000 kWh
Products: 12,000 pcs.

Day3
Energy : 9,000 kWh
Products: 15,000 pcs.

Day4
Energy : 6,000 kWh
Products: 1,500 pcs.



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SCADA (supervisory control and data acquisition)

MITSUBISHI ELECTRIC Changes for the Better

การจัดการพลังงาน (Energy Management)

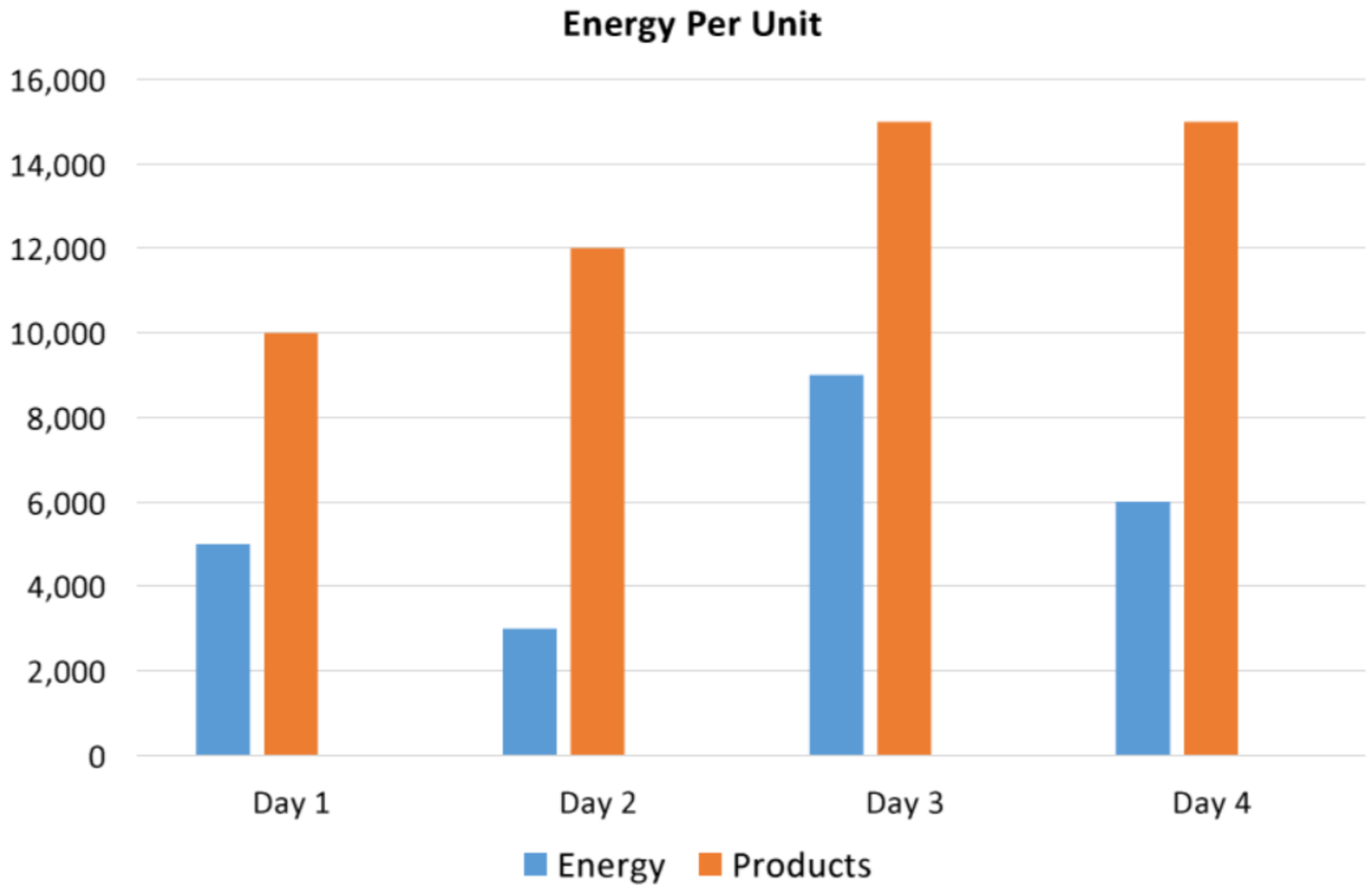
ICONICS

Day 1
Energy : 5,000 kWh
Product : 10,000 PCS.
EPU : 0.5 kWh / Uint

Day 2
Energy : 3,000 kWh
Product : 12,000 PCS.
EPU : 0.25 kWh / Uint

Day 3
Energy : 9,000 kWh
Product : 15,000 PCS.
EPU : 0.6 kWh / Uint

Day 4
Energy : 6,000 kWh
Product : 1,500 PCS.
EPU : 4.0 kWh / Uint



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SCADA (supervisory control and data acquisition)

MITSUBISHI ELECTRIC
Changes for the Better

เกิดอะไรขึ้นกับ DAY 4 ?

ICONICS

Day 4

Energy : 6,000 kWh

Product : 1,500 PCS.

EPU : 4.0 kWh / Uint





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Training

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SCADA (supervisory control and data acquisition)

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Why should we be focusing on Energy Saving?

ICONICS

Key words for saving energy

1. Improve of productivity
2. Visible management
3. Monitoring factory productivity conditions



Effects of Energy Saving







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Measurement and improvement activities are necessary for saving energy



Plan — High specific consumption (waste/worsening productivity) → If improved, ~kwh/day of electricity can be saved

Do — Starting machines too early
Setting appropriate starting time

Check — Before and after data comparison (~kwh/day of electricity saved)

Action — Application of plan to other facilities

Monitoring system can help us

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Training

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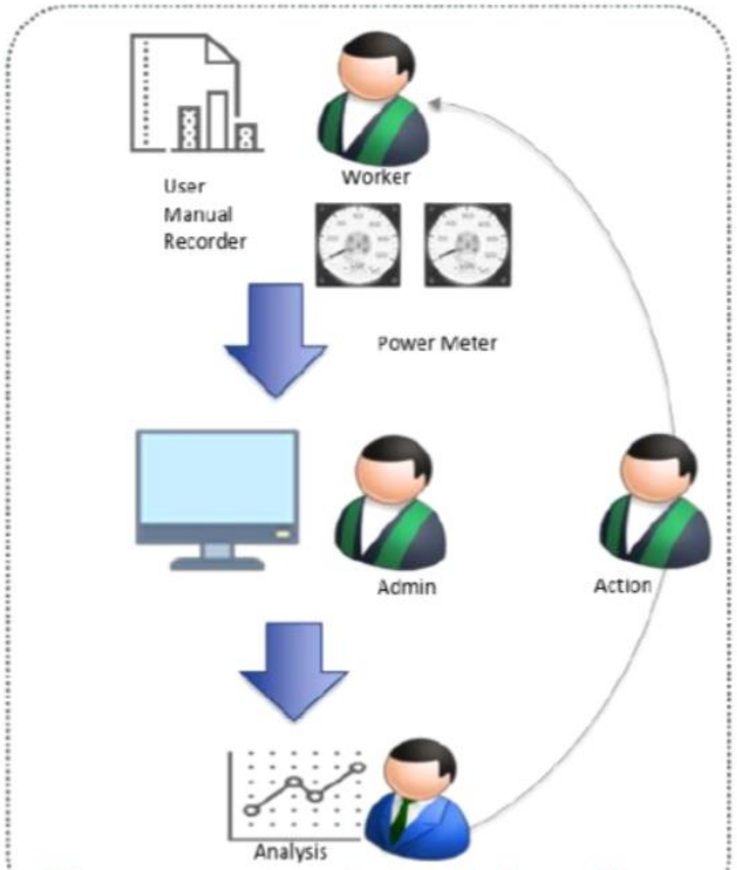
SCADA (supervisory control and data acquisition)

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Changes for the Better

Concept of Energy Saving

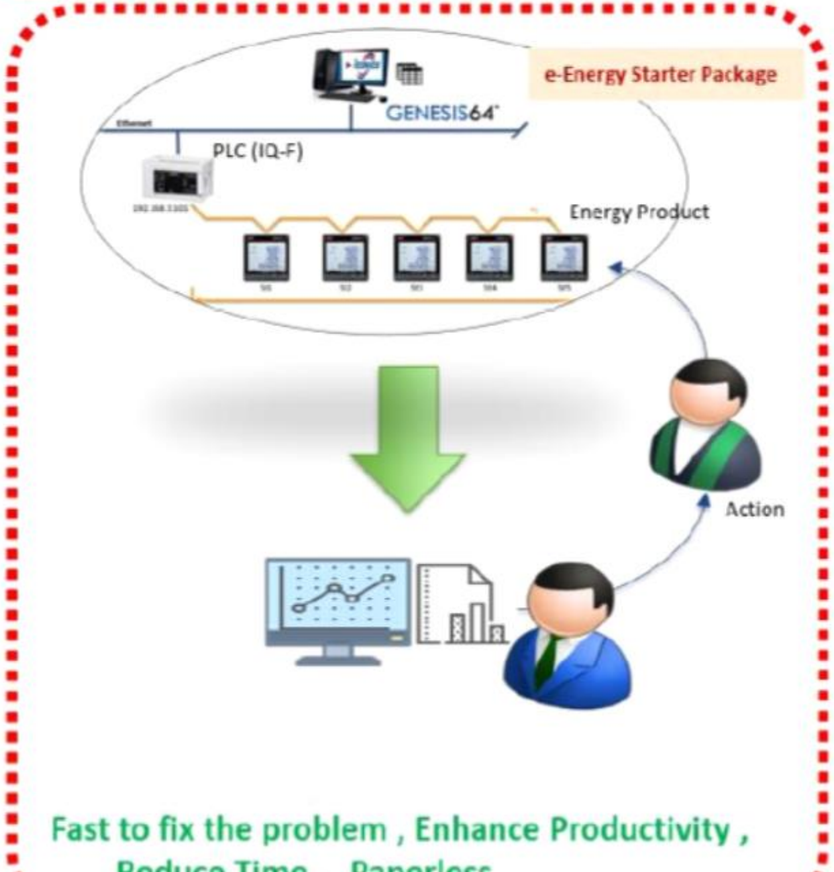
ICONICS

Existing Solution



The process not on time to fix the problem

New Solution



Fast to fix the problem , Enhance Productivity ,
Reduce Time , Paperless

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Training

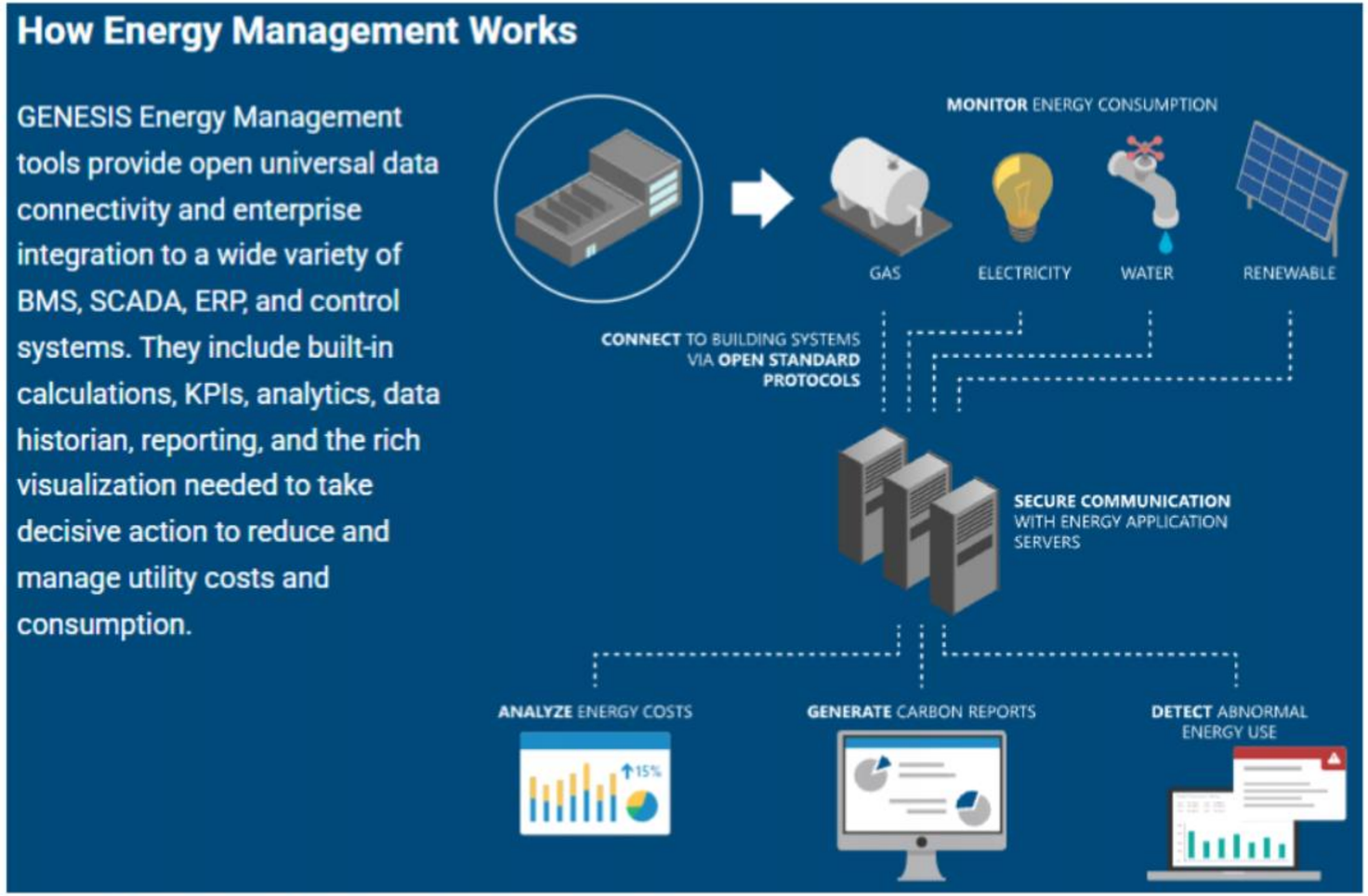
UMES x KMITL x TIRA x ABAC

SCADA (supervisory control and data acquisition)

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Energy Starter Package Detail

ICONICS



https://iconics.com/solutions/energy-management

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PLC Programming and Logic control

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MELSOFT

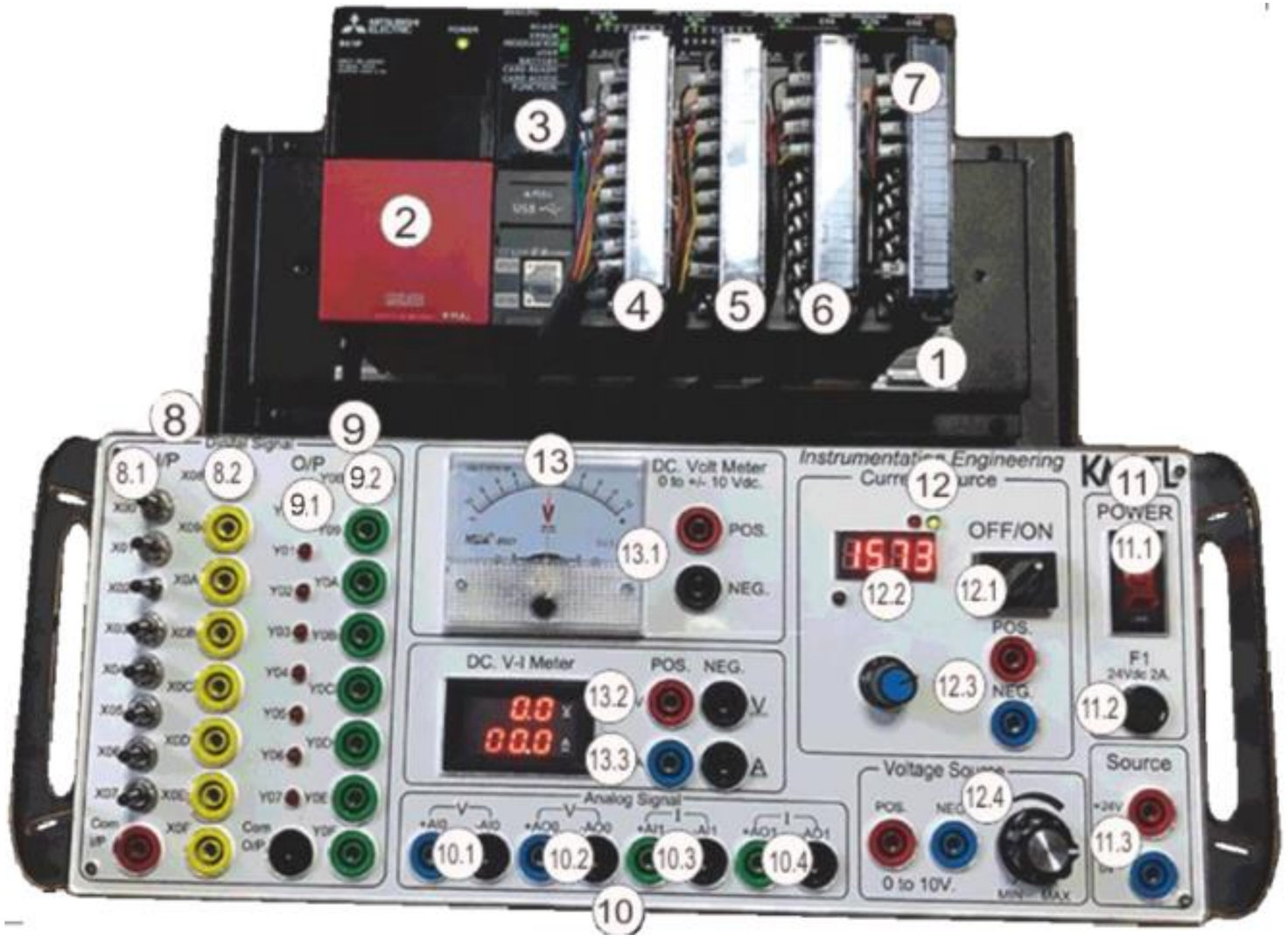
Programmable Controllers Engineering Software

GX Works3

Version 1

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GX Works3

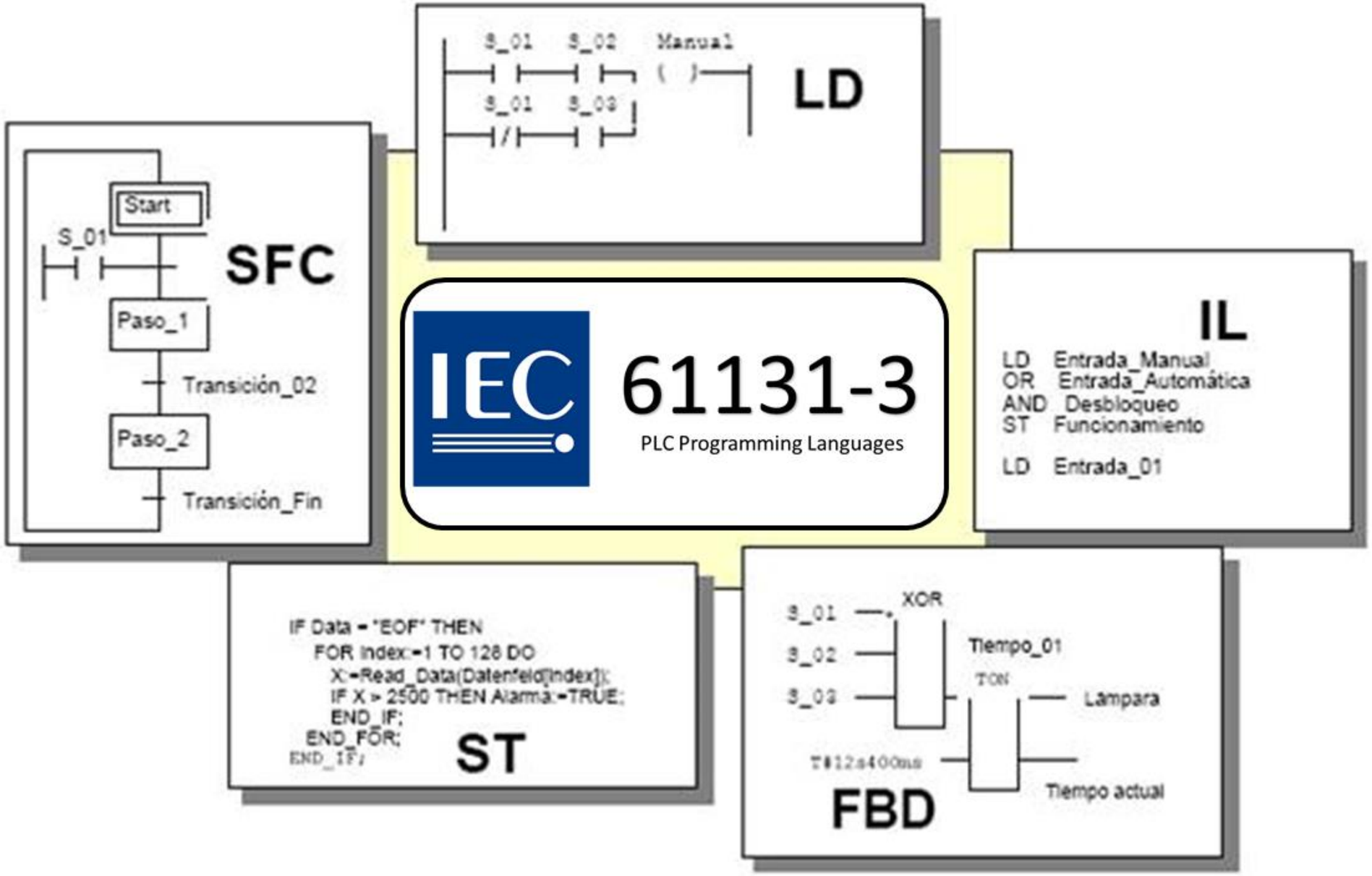


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Training

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Training

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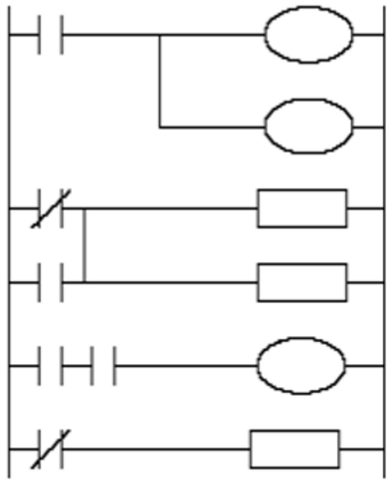
PLC Programming and Logic control

IEC-61131-3

Graphical Language

Textual Language

Ladder Diagram (LD)



Ladder Format

Function Block Diagram (FBD)

Function Block Format

Sequential Function Chart (SFC)

SFC Format

Structured Text (ST)

```
    END_IF;
Setpoint_IN_STAGE_1_FAILED:
    (* During 'STAGE_1_FAILED': '<S1>:119' *)
    IF (stage3_sensor <= 0) OR (stage2_sensor <= 0) THEN
        (* Transition: '(<S1>:150' *)
        (* Transition: '(<S1>:152' *)
        IF stage2_sensor > 0 THEN
            (* Transition: '<S1>:155' *)
            is_c2_Setpoint := Setpoint_IN_STAGES_1_3_FAILED;
            (* Entry 'STAGES_1_3_FAILED': '<S1>:120' *)
            rtb_stage1_setpoint := L0;
            rtb_stage2_setpoint := L0 - overall_target;
            distributed_target := rtb_stage2_setpoint;
        ELSE
            (* Transition: '<S1>:154' *)
            IF stage3_sensor > 0 THEN
                (* Transition: '<S1>:159' *)
                is_c2_Setpoint := Setpoint_IN_STAGES_1_2_FAILED;
                (* Entry 'STAGES_1_2_FAILED': '<S1>:121' *)
                rtb_stage1_setpoint := L0;
                rtb_stage2_setpoint := L0;
                distributed_target := L0 - overall_target;
            ELSE
                guard_0 := TRUE;
            END_IF;
        END_IF;
    ELSE
        guard_0 := TRUE;
    END_IF;
```

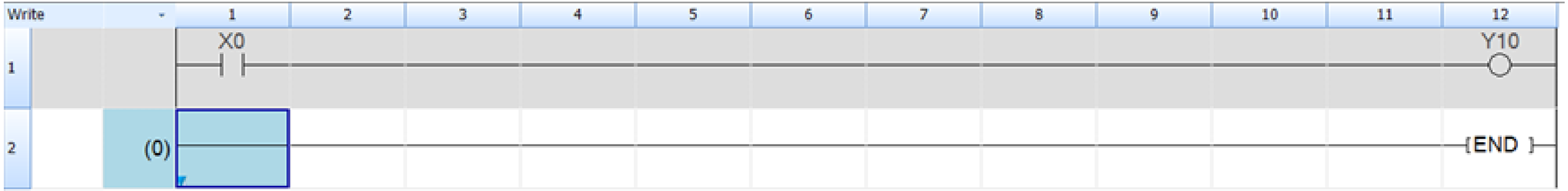
Structure Text Format

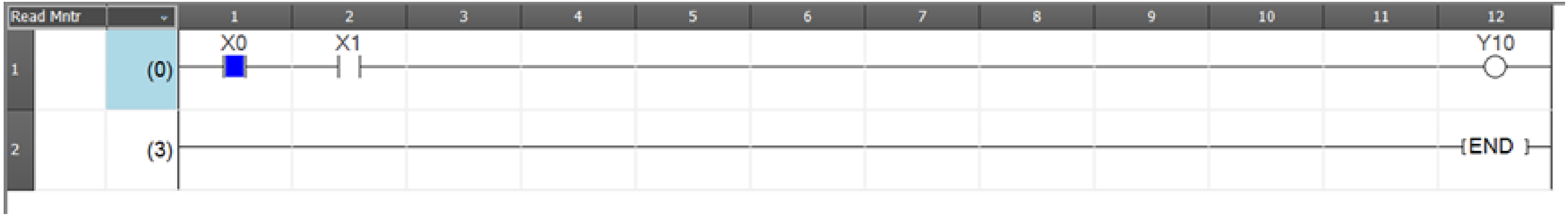
Instruction List (IL)

```
LD    X10
OUT   Y7
AND   M38
SET   S5
LD    X21
OUT   T01
      K40
```

Instruction format

PLC Programming and Logic control





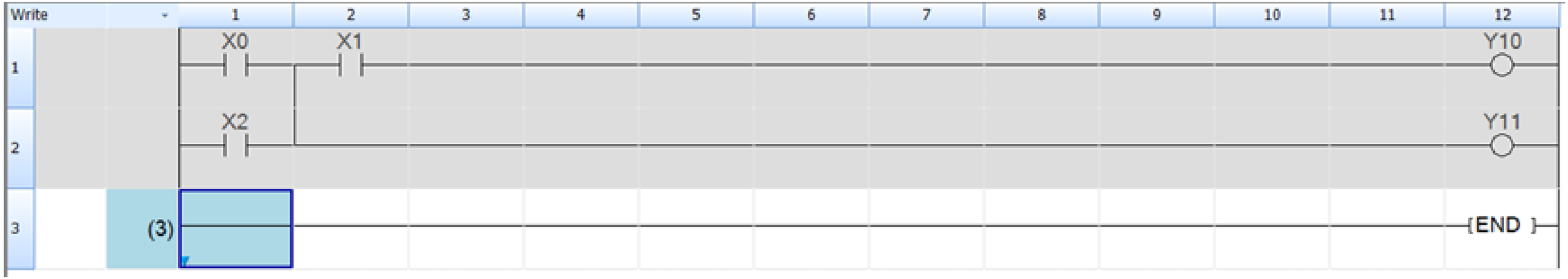
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PLC Programming and Logic control



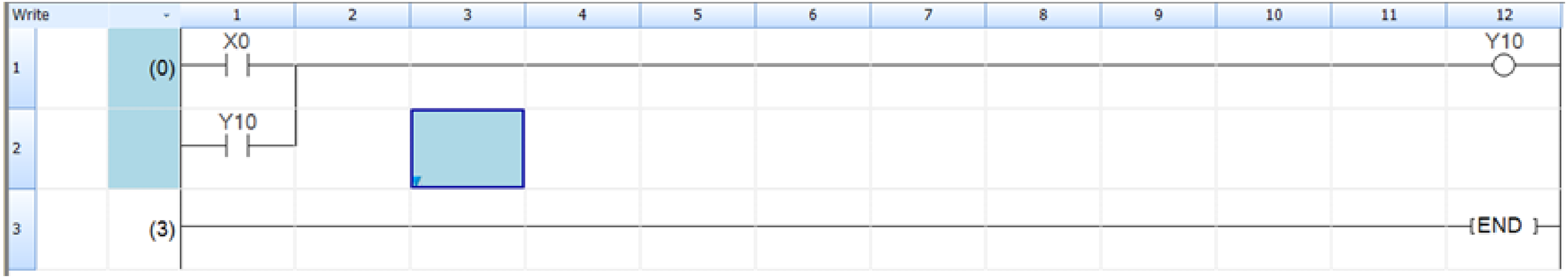
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Training

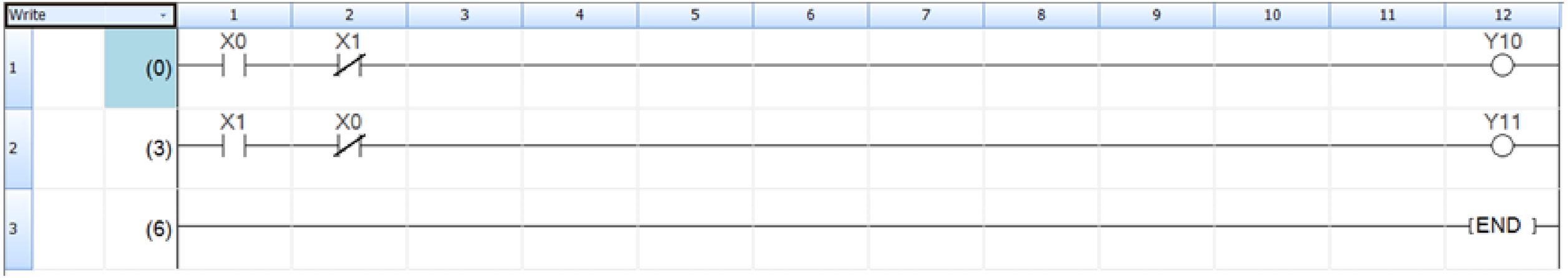
UMES x KMITL x TIRA x ABAC

PLC Programming and Logic control



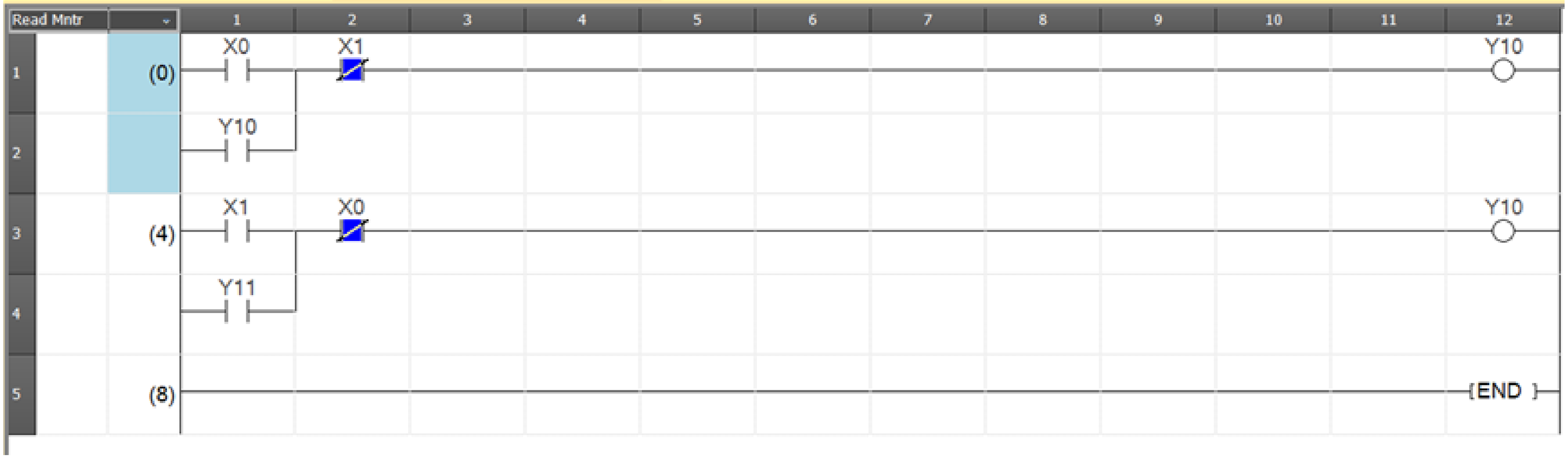
Self holding

PLC Programming and Logic control



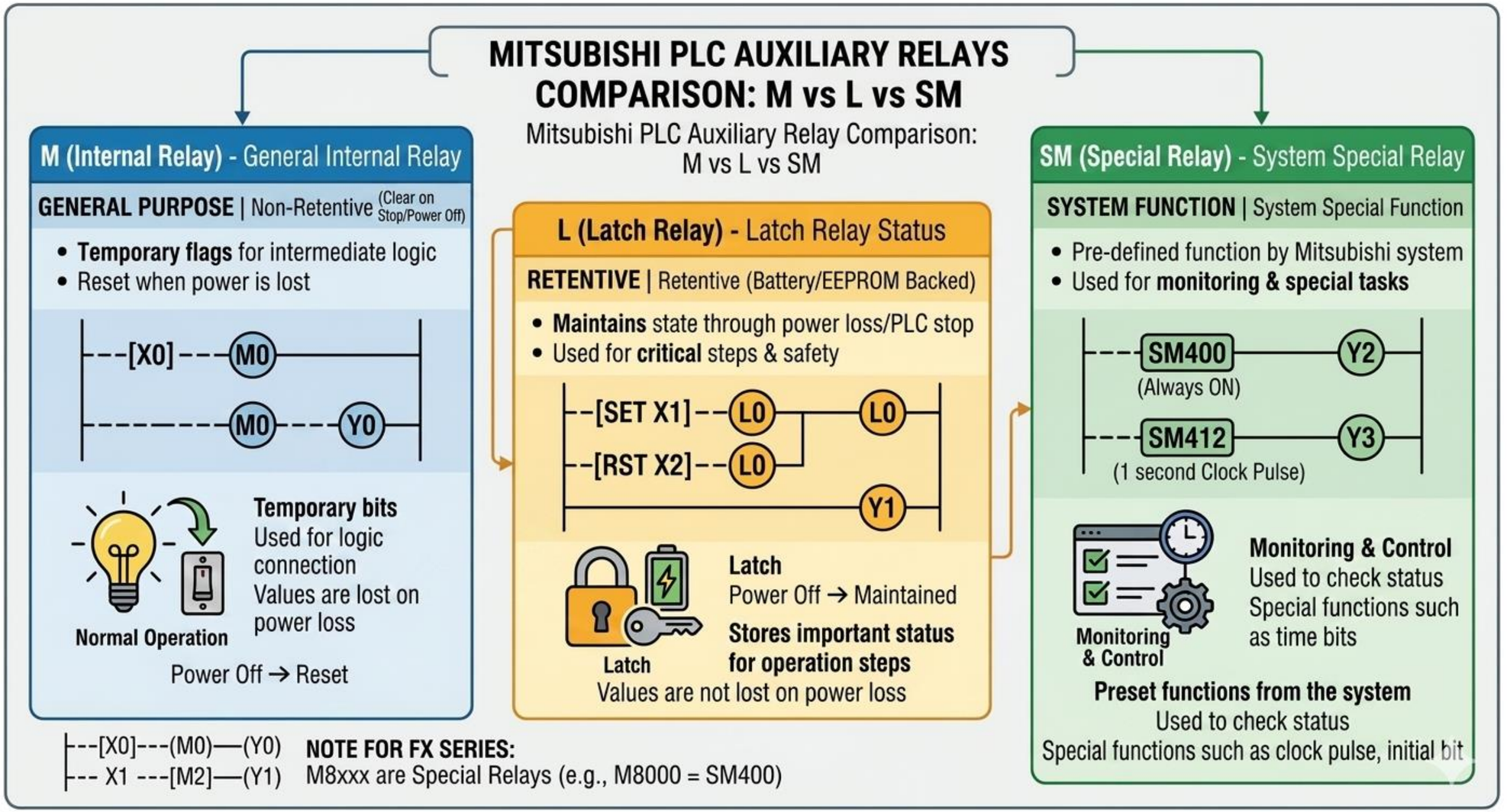
Interlocking

PLC Programming and Logic control



PLC Programming and Logic control

Auxiliary Relay



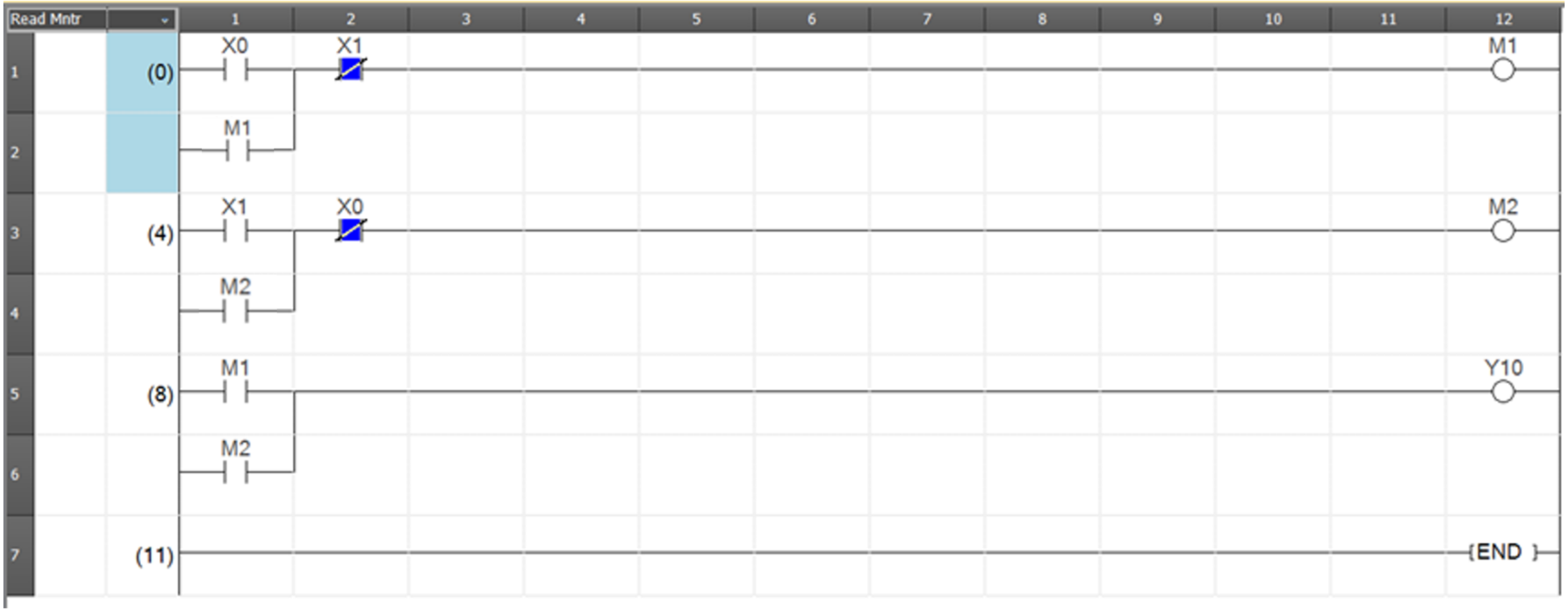
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Training

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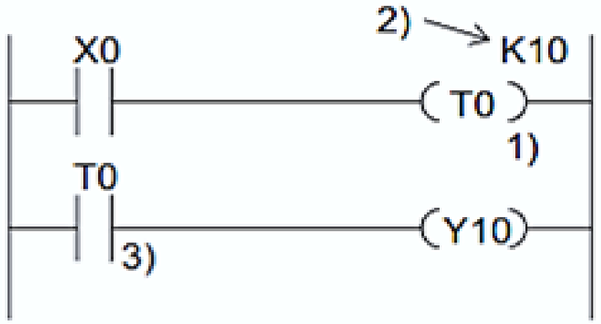
PLC Programming and Logic control

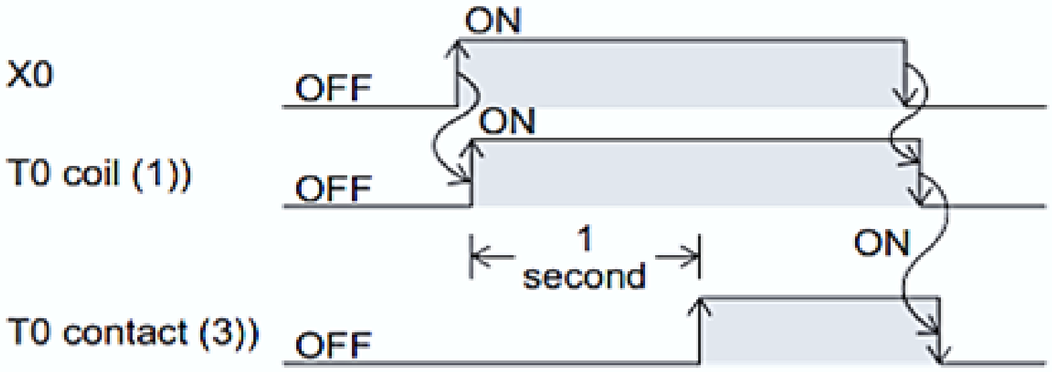


PLC Programming and Logic control

General Timer (T)

Measurement starts when the timer's coil is turned ON. Time up occurs when the current value of the timer matches the setting value and the timer's contact is turned ON. When the timer's coil is turned OFF, the current value becomes "0" and the timer's contact is turned OFF.







PLC Programming and Logic control

1. Push X00 → Y10 = ON

2. and Hold 3 sec

3. Y11 = ON

3 S= 3,000 mS
= 3,000/100
= 30 → K30

Type of timers			
Timer	16 bits	Timer (T)	Low-speed timer
			High-speed timer
		Retentive timer (ST)	Low-speed retentive timer
			High-speed retentive timer

รูปที่ 4.6 ตารางค่าความละเอียดของ Timer ของ PLC iQ-F Series

สำหรับความละเอียด Timer ใน PLC **รุ่น iQ-F Series** นั้นเป็นแบบ 16 bits และมีค่าความละเอียดในการหน่วงเวลาไม่เท่ากันโดยจะแบ่งค่าความละเอียดของการหน่วงเวลาออกได้เป็น 3 ค่าด้วยกันคือ 100 ms, 10 ms และ 1 ms ซึ่งมีการเขียนคำสั่งต่างกัน

Timer 100 ms วิธีการเขียน “OUT T0”

Timer 10 ms วิธีการเขียน “OUTH T0”

Timer 1 ms วิธีการเขียน “OUTHS T0”

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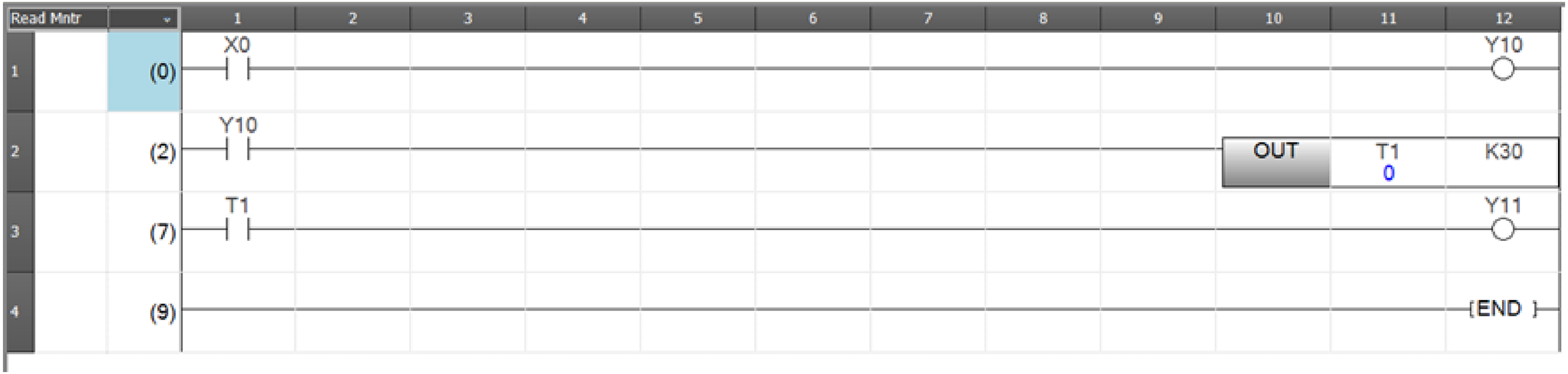
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IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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PLC Programming and Logic control



PLC Programming and Logic control

Counter

Counter type

There is counter (C) that maintains the counter value in 16 bits and the long counter (LC) that maintains the counter value in 32 bits. Counter (C) and long counter (LC) are separate devices. You can set number of device points for each.

■Counter (C)

The counting range is from 0 to 32767.

■Long counter (LC)

The counting range is from 0 to 4294967295.



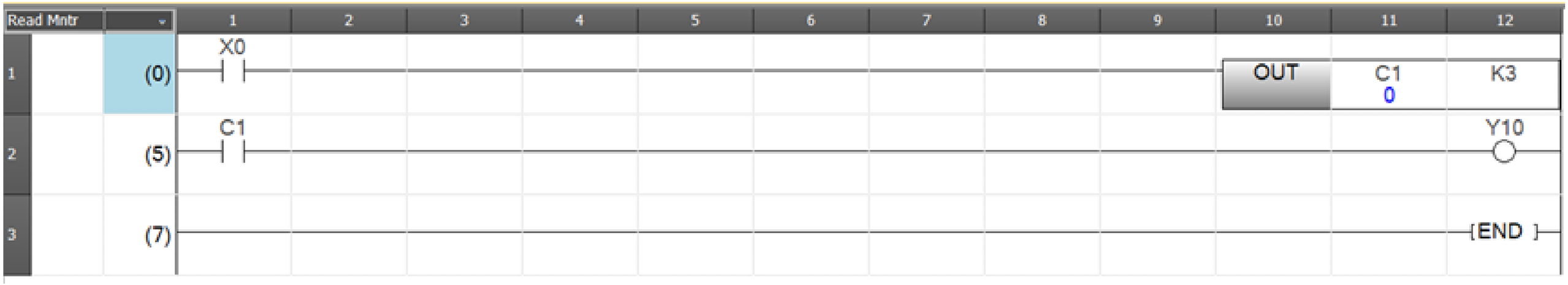
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Training

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PLC Programming and Logic control



Reset CPU

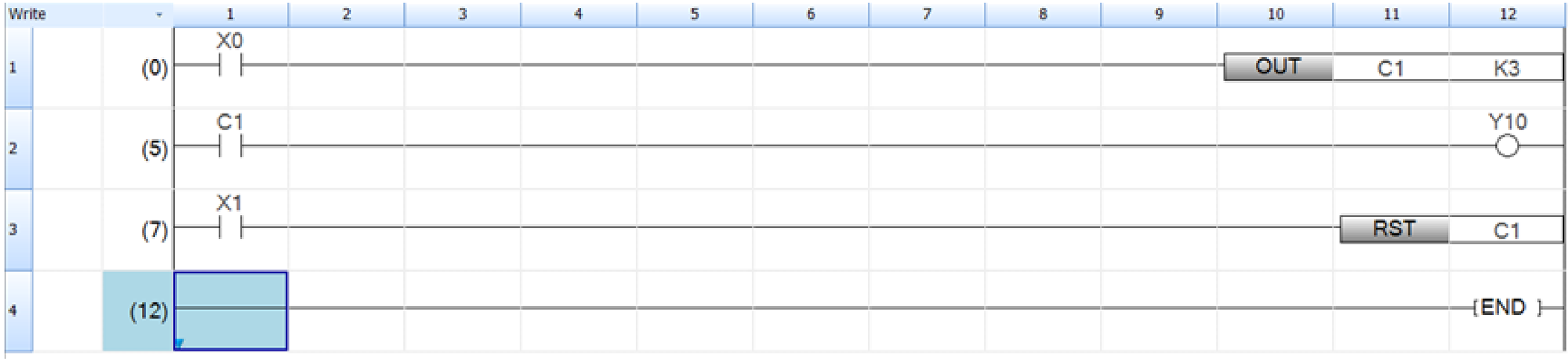
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Training

UMES x KMITL x TIRA x ABAC

PLC Programming and Logic control



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Training

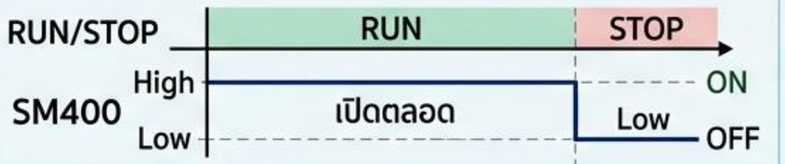
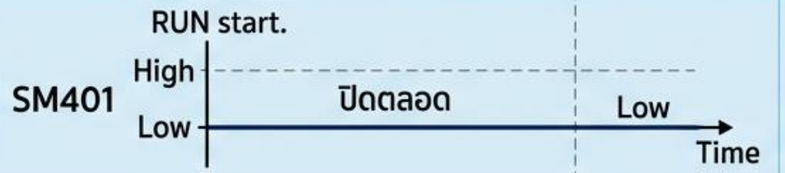
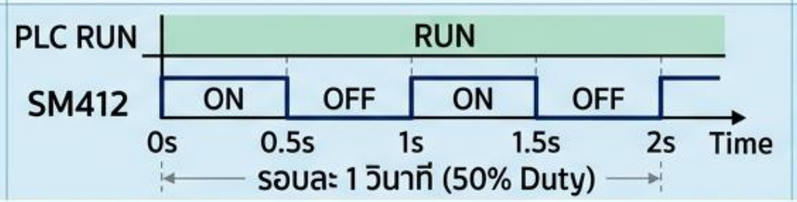
UMES x KMITL x TIRA x ABAC

PLC Programming and Logic control

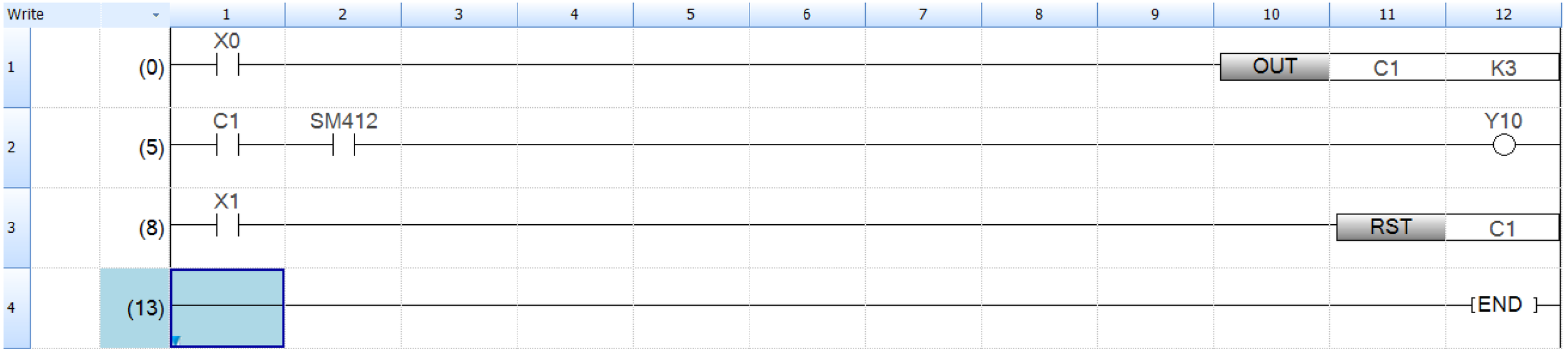
Auxiliary Relay

Special Relay

สรุป SM Special Relay ของ PLC Mitsubishi (iQ Series)

รหัส SM / ชื่อ	ฟังก์ชัน / การใช้งาน	Timing Diagram (ความสัมพันธ์กับสถานะ PLC/เวลา)	การใช้งานหลัก
SM400 (Always ON)	ติดค้างตลอด (Always ON)		• เงื่อนไขเริ่มต้น • บรรทัดคำสั่งที่ต้องทำงานเสมอ (e.g., [MCR], [CALL])
SM401 (Always OFF)	ดับตลอด (Always OFF)		• ตัดการทำงานชั่วคราว • ใช้เพื่อ Debug • ใส่แทน N.C. ในฟังก์ชันบล็อก
SM402 (Initial Pulse)	สัญญาณพัลส์เดียวตอนเริ่ม (Initial Pulse)		• รีเซ็ตค่าเริ่มต้น (Initial Reset) • กำหนดค่าเริ่มต้นให้กับ [D] [D] หรือ [M]
SM412 (1 Second Clock)	สัญญาณนาฬิกา 1 วินาที (1 Second Clock)		• สัญญาณไฟกะพริบแจ้งเตือน • วัดเวลา • ทำให้อุปกรณ์ทำงานเป็นจังหวะ

PLC Programming and Logic control



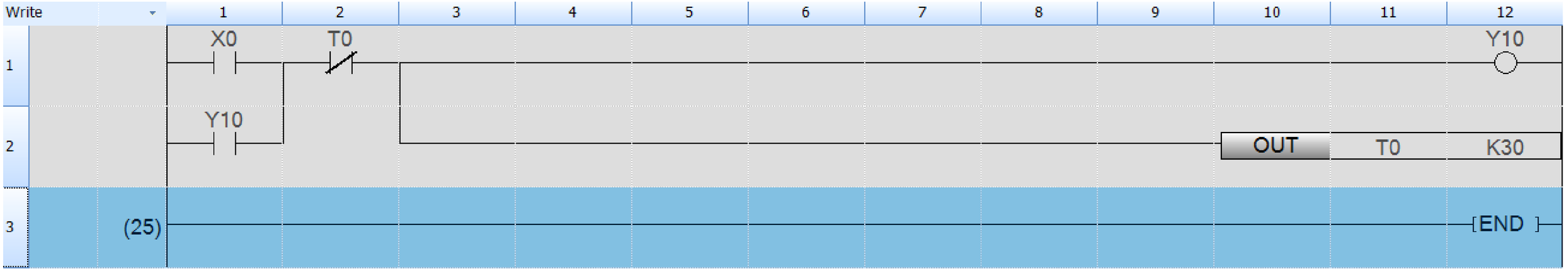
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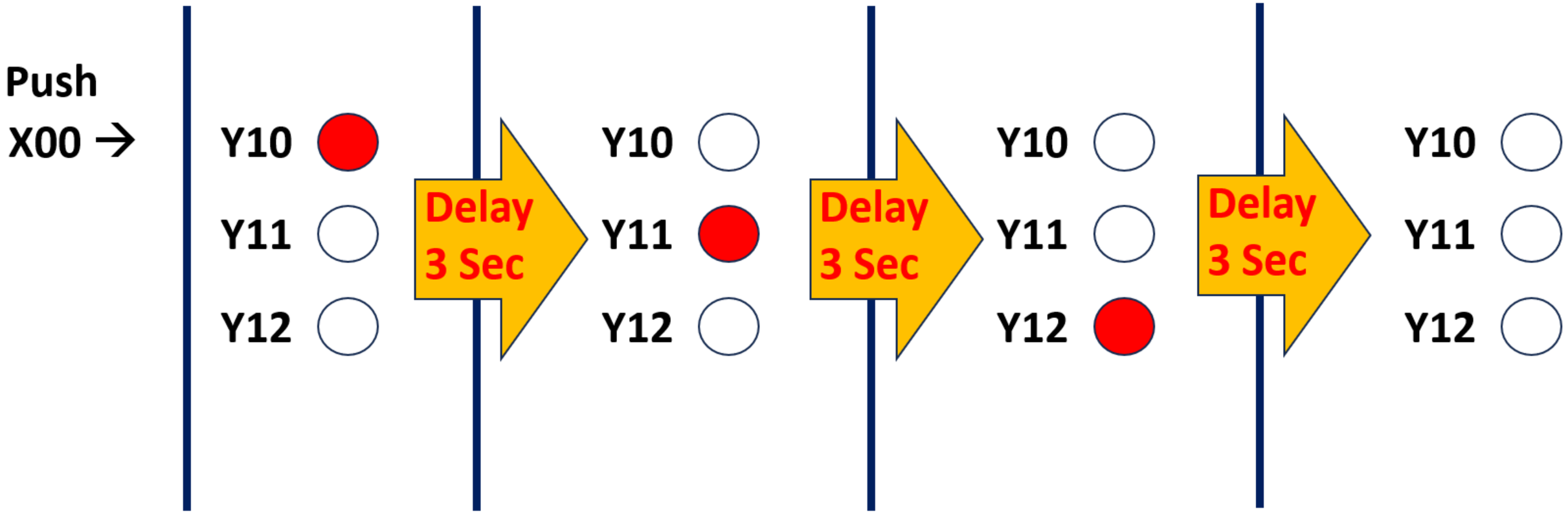
Training

UMES x KMITL x TIRA x ABAC

PLC Programming and Logic control



PLC Programming and Logic control



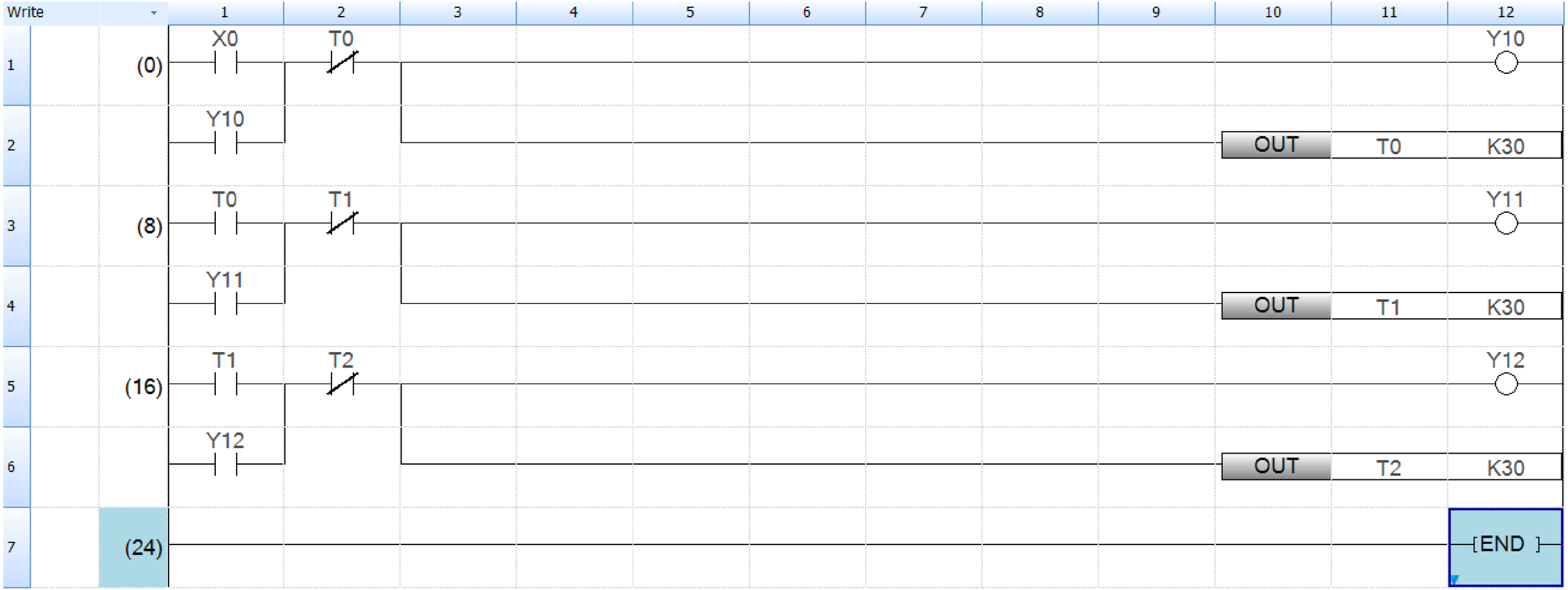
Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

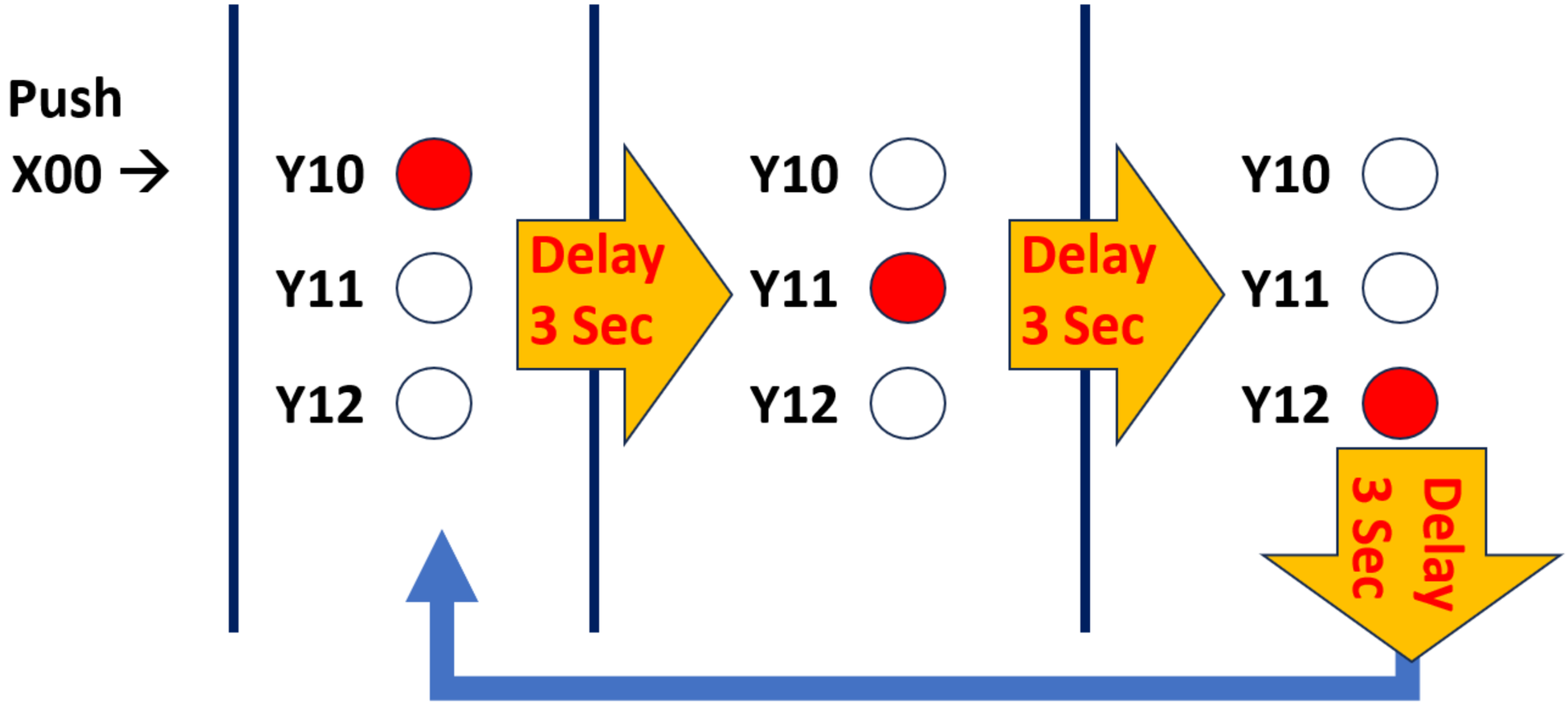
Training

UMES x KMITL x TIRA x ABAC

PLC Programming and Logic control



PLC Programming and Logic control



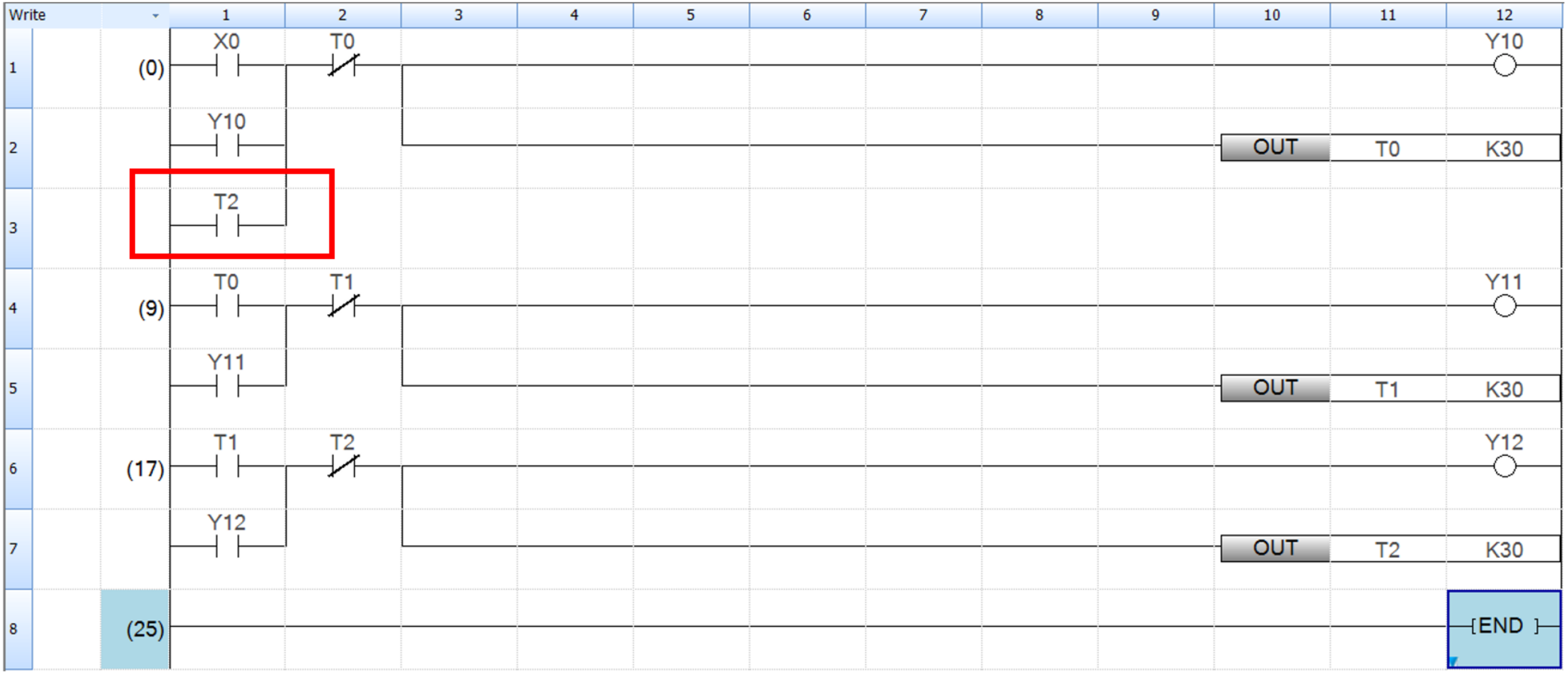
Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

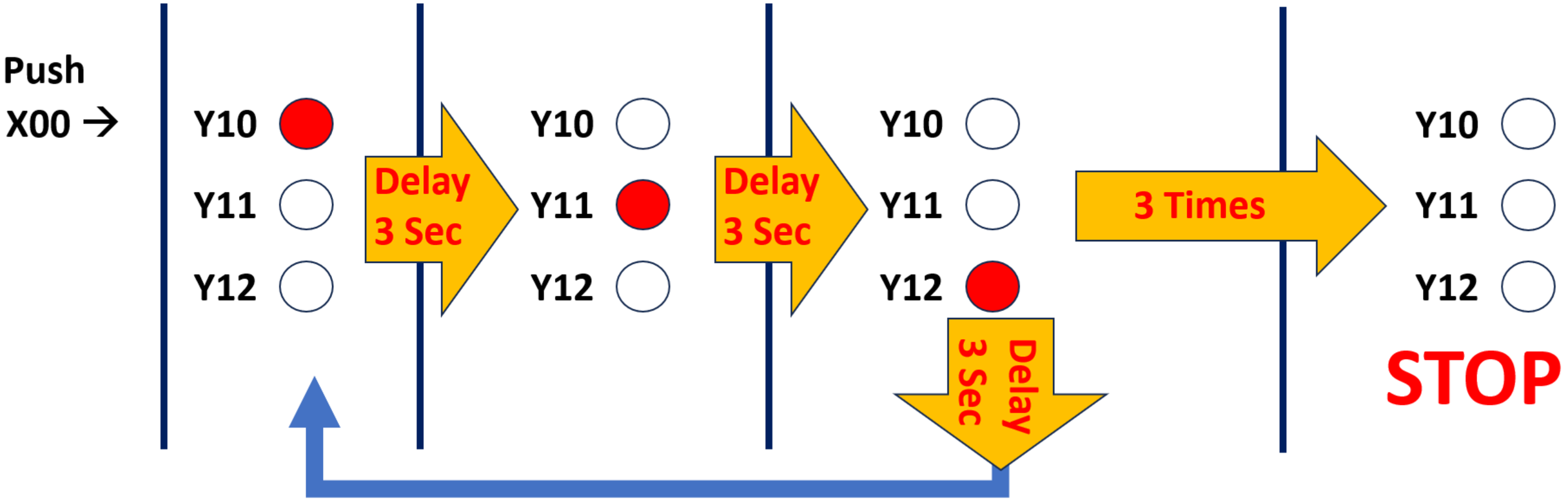
Training

UMES x KMITL x TIRA x ABAC

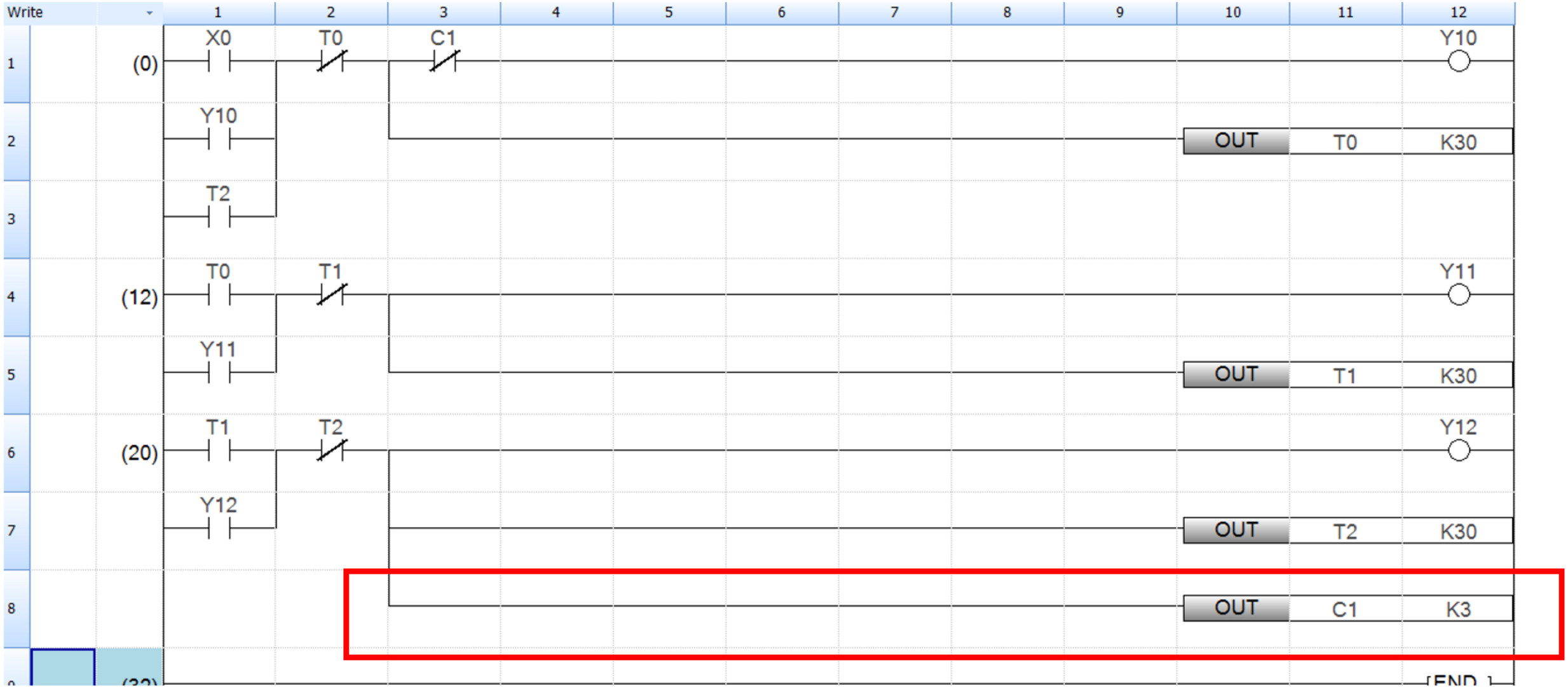
PLC Programming and Logic control



PLC Programming and Logic control



PLC Programming and Logic control



Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

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PLC Programming and Logic control

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Graphical Language

- Ladder Diagram (LD)
- Function Block Diagram (FBD)
- Sequential Function Chart (SFC)

Ladder Format

Function Block Format

SFC Format

Textual Language

- Structured Text (ST)
- Instruction List (IL)

```
    END_IF;
Setpoint_IN_STAGE_1_FAILED:
    (* During 'STAGE_1_FAILED': '<S1>:119' *)
    IF (stage3_sensor <= 0) OR (stage2_sensor <= 0) THEN
        (* Transition: '(<S1>:150' *)
        (* Transition: '(<S1>:152' *)
        IF stage2_sensor > 0 THEN
            (* Transition: '<S1>:155' *)
            is_c2_Setpoint := Setpoint_IN_STAGES_1_3_FAILED;
            (* Entry 'STAGES_1_3_FAILED': '<S1>:120' *)
            rtb_stage1_setpoint := L0;
            rtb_stage2_setpoint := L0 - overall_target;
            distributed_target := rtb_stage2_setpoint;
        ELSE
            (* Transition: '<S1>:154' *)
            IF stage3_sensor > 0 THEN
                (* Transition: '<S1>:159' *)
                is_c2_Setpoint := Setpoint_IN_STAGES_1_2_FAILED;
                (* Entry 'STAGES_1_2_FAILED': '<S1>:121' *)
                rtb_stage1_setpoint := L0;
                rtb_stage2_setpoint := L0;
                distributed_target := L0 - overall_target;
            ELSE
                guard_0 := TRUE;
            END_IF;
        END_IF;
    ELSE
        guard_0 := TRUE;
    END_IF;
```

Structure Text Format

```
LD      X10
OUT     Y7
AND     M38
SET     S5
LD      X21
OUT     T01
        K40
```

Instruction format

PLC Controls with Structured Text (ST)

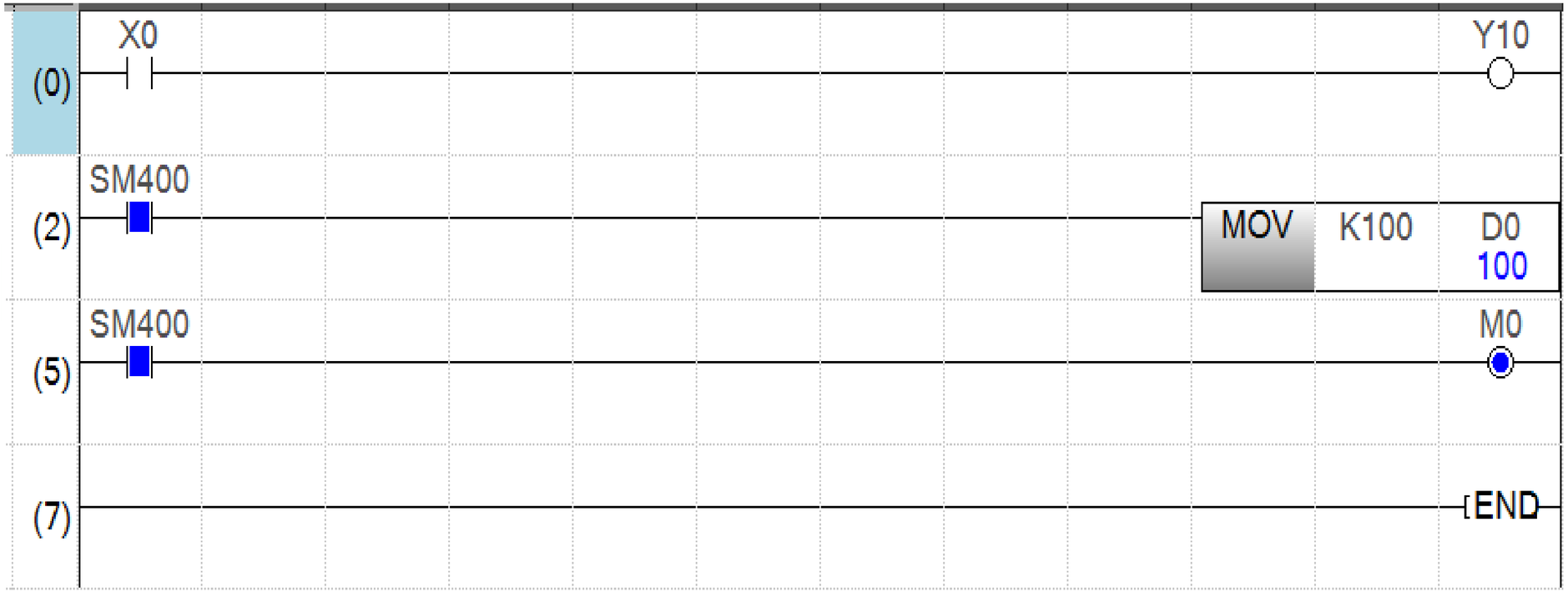
เปรียบเทียบคำสั่ง: Structured Text (ST) vs ภาษา C

ลักษณะการใช้งาน	ภาษา ST (PLC)	ภาษา C	ความแตกต่างสำคัญ
ตัวปิดท้ายประโยค	;	;	Similar
การใส่ Comment	//` หรือ `(* ... *)	//` หรือ `/* ... */	ST uses `(* *)
การกำหนดค่า	:=	=	:=
เปรียบเทียบ "เท่ากับ"	=	==	=
เปรียบเทียบ "ไม่เท่ากับ"	<>	!=	<>
เงื่อนไข	IF...THEN...END_IF	if()...{...}	ST uses keywords not {}
เงื่อนไขหลายทางเลือก	CASE...OF...END_CASE	switch()...case...break	no `break` needed
ลูปแบบกำหนดรอบ	FOR...TO...BY...DO...END_FOR	for(;;){}	`TO` and `DO
ลูปแบบเช็คเงื่อนไข	WHILE...DO...END_WHILE	while({}	Syntax structure is different
ตัวดำเนินการทางตรรกศาสตร์	AND OR NOT	&& || !	ST uses keywords

Structured Text vs C Comparison

PLC Controls with Structured Text (ST)

```
1  Y10 := X00;
2  D0 := 100;
3  M0 := TRUE;
4
```



PLC Controls with Structured Text (ST)

ARITHMETIC EXPRESSIONS IN MITSUBISHI PLC ST PROGRAMMING

Operator	Description (TH)	Description (EN)	Usage Example (ST)	Result/Note
+	การบวก	Addition	Result := A + B;	Adds values.
-	การลบ	Subtraction	Result := Current - Previous;	Subtracts values.
*	การคูณ	Multiplication	Result := Factor * 10;	Multiplies values.
/	การหาร (หารจริง)	Division (Floating Point)	Result_REAL := Value_REAL / Divider_REAL;	Standard division (often for REAL types)
/ (หารตัดเศษ)	การหาร (หารตัดเศษ)	Division (Integer Truncation)	Result_INT := D100 / D102;	For integer types (DINT, INT), truncates decimal part.
MOD	หารเอาเศษ (Modulo)	Modulo (Remainder)	Remainder := Input MOD 7;	Returns the remainder of division.

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STRUCTURED TEXT SUMMARY

PLC Controls with Structured Text (ST)

RELATIONAL EXPRESSIONS IN MITSUBISHI PLC ST PROGRAMMING

Operator	Description (TH)	Description (EN)	Usage Example (ST)	Result (Boolean)
=	เท่ากับ	Equal to	**If** (A = B) **Then** Result_Bool := TRUE;	TRUE if A equals B
<>	ไม่เท่ากับ	Not Equal to	**If** (X <> 100) **Then** Run_Bit := FALSE;	TRUE if X is not 100
<	น้อยกว่า	Less than	**If** (Current_Value < High_Limit) **Then** Status := OK;	TRUE if Current_Value is less than High_Limit
>	มากกว่า	Greater than	**If** (Pressure > Max_Pressure) **Then** Alarm_Active := TRUE;	TRUE if Pressure is greater than Max_Pressure
<=	น้อยกว่าหรือเท่ากับ	Less than or Equal to	**If** (Counter <= Setpoint) **Then** Count_Up := TRUE;	TRUE if Counter is less than or equal to Setpoint
>=	มากกว่าหรือเท่ากับ	Greater than or Equal to	**If** (Level >= Fill_Level) **Then** Valve_Close := TRUE;	TRUE if Level is greater than or equal to Fill_Level

MITSUBISHI ELECTRIC | FA

STRUCTURED TEXT SUMMARY: RELATIONAL & LOGICAL

PLC Controls with Structured Text (ST)

LOGICAL EXPRESSIONS IN MITSUBISHI PLC ST PROGRAMMING

Operator	Description (TH)	Description (EN)	Usage Example (ST)	Result (Boolean)
AND	และ (ตรรกะ)	Logical AND	**If** (Start_Button **AND** Emergency_OK) **Then** Motor_On := TRUE;	TRUE if both inputs are TRUE
OR	หรือ (ตรรกะ)	Logical OR	**If** (Stop_1 **OR** Stop_2) **Then** Stop_System := TRUE;	TRUE if either input is TRUE
NOT	ไม่ (ตรรกะ)	Logical NOT (Inversion)	**If NOT** (Door_Closed) **Then** Warning_Light := TRUE;	Inverts the state (TRUE to FALSE, FALSE to TRUE)
XOR	หรือแบบพิเศษ	Exclusive OR	Result_Bit := BitA **XOR** BitB;	TRUE if inputs are different
&&	(also used in some tools)	(note: Bitwise AND)	**If** (Stop_1 **OR** Stop_2) **Then** Warnings_Light := TRUE;	TRUE if bitwise AND

MITSUBISHI ELECTRIC | FA

STRUCTURED TEXT SUMMARY: RELATIONAL & LOGICAL

PLC Controls with Structured Text (ST)

```
1  Y10 := X00;
2  D0 := 100;
3  M0 := TRUE;
4  D1 := D0 + 100;
5  D5 := (D0 * 5) MOD 3;
```

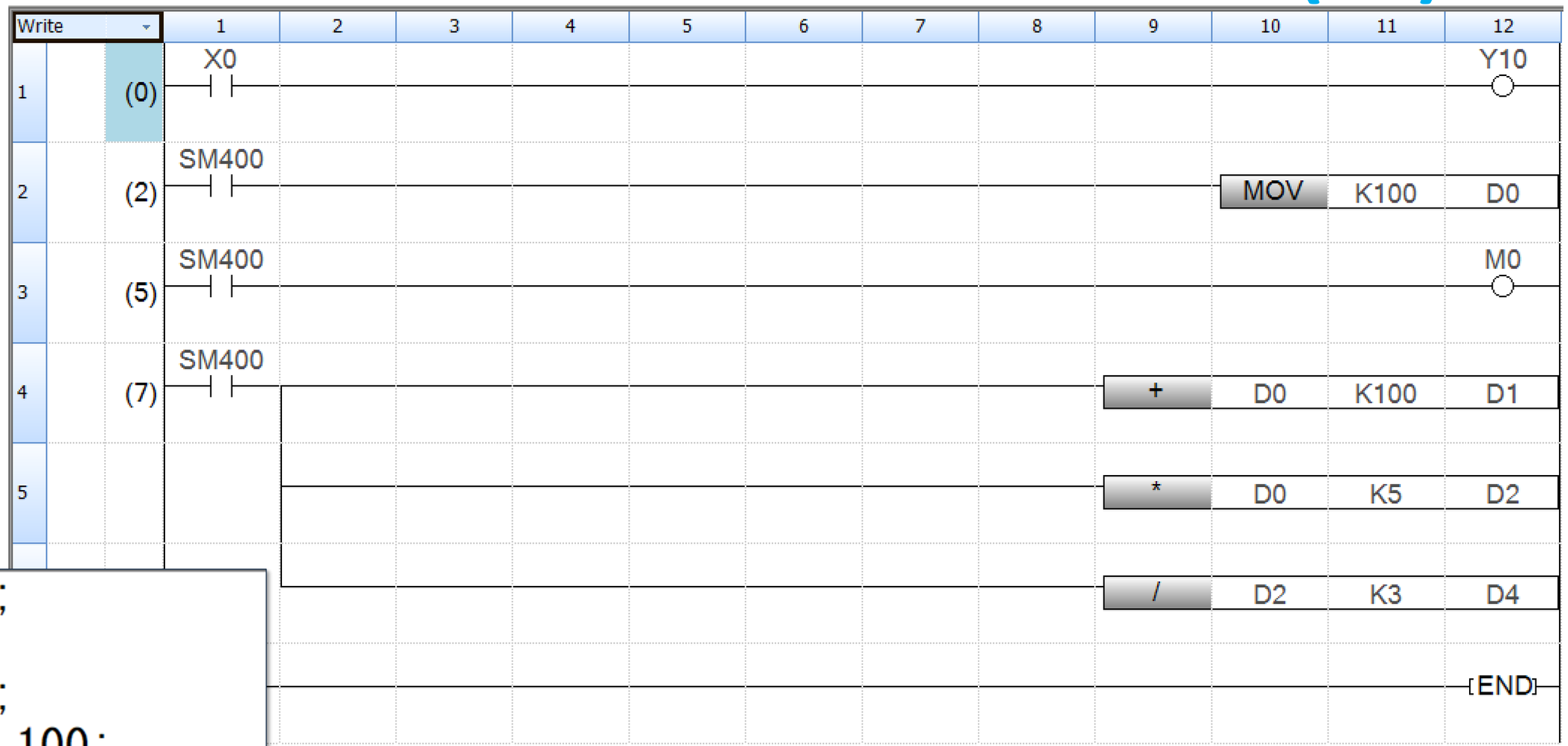
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Training

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PLC Controls with Structured Text (ST)



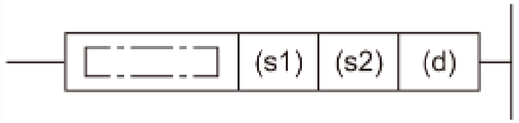

```
1 Y10 := X00;
2 D0 := 100;
3 M0 := TRUE;
4 D1 := D0 + 100;
5 D5 := (D0 * 5) MOD 3;
```

PLC Controls with Structured Text (ST)

/(P)(_U)

RnCPU | RnENCPU | RnPCPU (Process) | RnPCPU (Redundant) | RnPSFCPU (Standard) | RnPSFCPU (Safety) | RnSFCPU (Standard) | RnSFCPU (Safety)

These instructions perform division between the two sets of 16-bit binary data specified.

Ladder	ST*1	
	ENO:=DIVISION(EN,s1,s2,d); ENO:=DIVISIONP(EN,s1,s2,d);	ENO:=DIVISION_U(EN,s1,s2,d); ENO:=DIVISIONP_U(EN,s1,s2,d);

Setting data

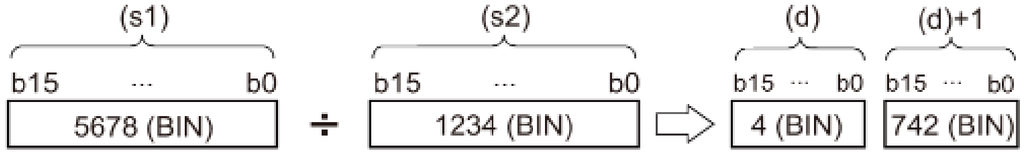
■ Description, range, data type

Operand		Description	Range	Data type	Data type (label)
(s1)	/(P)	Dividend data or the device where dividend data is stored	-32768 to 32767	16-bit signed binary	ANY16_S
	/(P)_U		0 to 65535	16-bit unsigned binary	ANY16_U
(s2)	/(P)	Divisor data or the device where divisor data is stored	-32768 to 32767	16-bit signed binary	ANY16_S
	/(P)_U		0 to 65535	16-bit unsigned binary	ANY16_U
(d)	/(P)	Start device for storing the operation result	—	32-bit signed binary	ANY16_S_ARRAY (Number of elements: 2)
	/(P)_U			32-bit unsigned binary	ANY16_U_ARRAY (Number of elements: 2)
EN		Execution condition	—	Bit	BOOL
ENO		Execution result	—	Bit	BOOL

PLC Controls with Structured Text (ST)

Processing details

- These instructions divide the 16-bit binary data in the device specified by (s1) by the 16-bit binary data in the device specified by (s2), and store the operation result in the device specified by (d).



(d): Quotient
(d)+1: Remainder

- As the operation result, the quotient and remainder are stored in 32 bits. When a bit device is specified, the number of digit-specified bits is used to store the quotient and remainder.

- Quotient · · · Stored in lower 16 bits.
- Remainder · · · Stored in upper 16 bits.

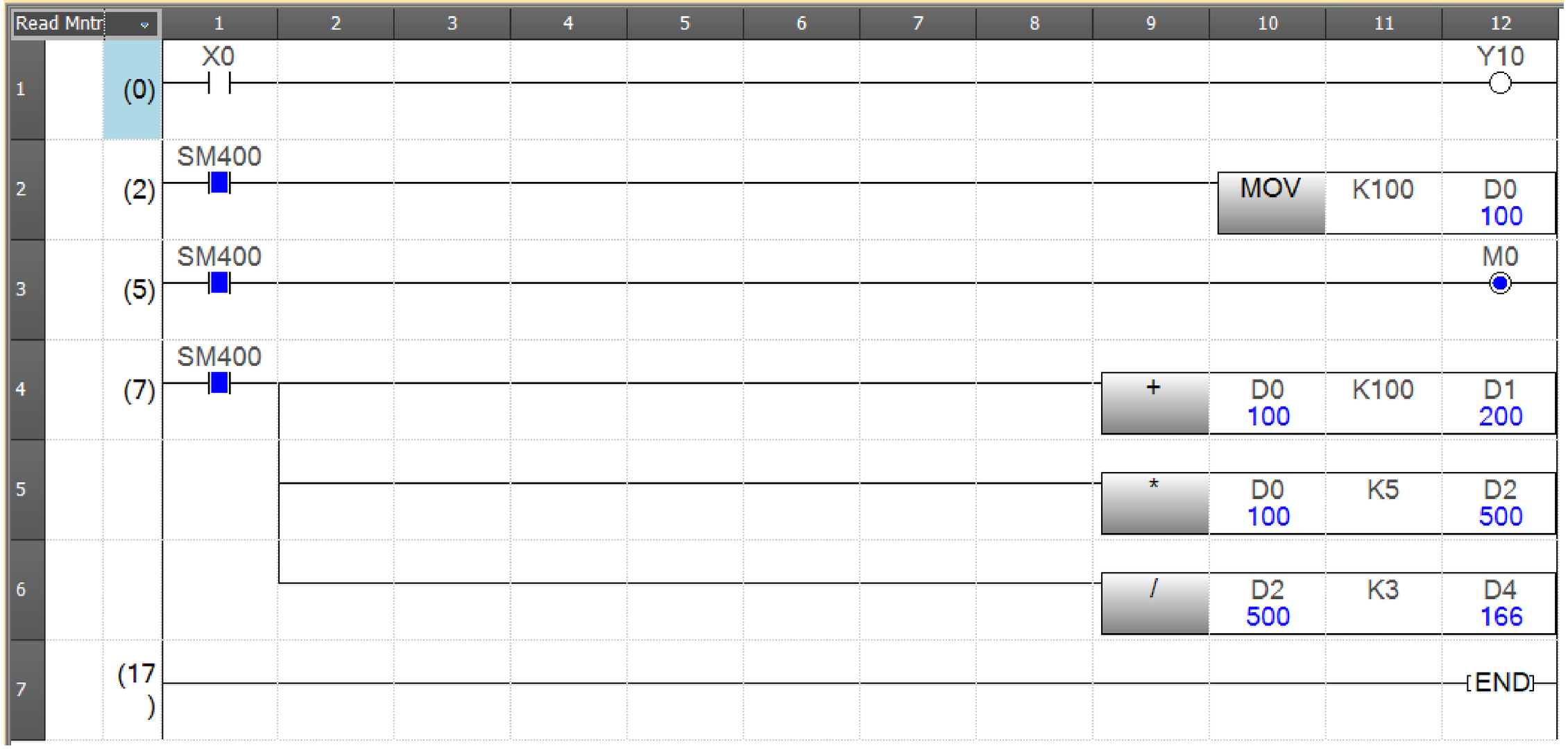
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PLC Controls with Structured Text (ST)



PLC Controls with Structured Text (ST)

Device Name	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	Current Value	
D0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	100	d.
D1	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	0	200	ศ.
D2	0	0	0	0	0	0	0	1	1	1	1	1	0	1	0	0	500	๘.
D3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	..
D4	0	0	0	0	0	0	0	0	1	0	1	0	0	1	1	0	166	ฆ.
D5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	2	..
D6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	..

```
4 D1 := D0 + 100;
5 D5 := (D0 * 5) MOD 3;
```

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PLC Programming and Logic control

IEC-61131-3

Graphical Language

Textual Language

Ladder Diagram (LD)

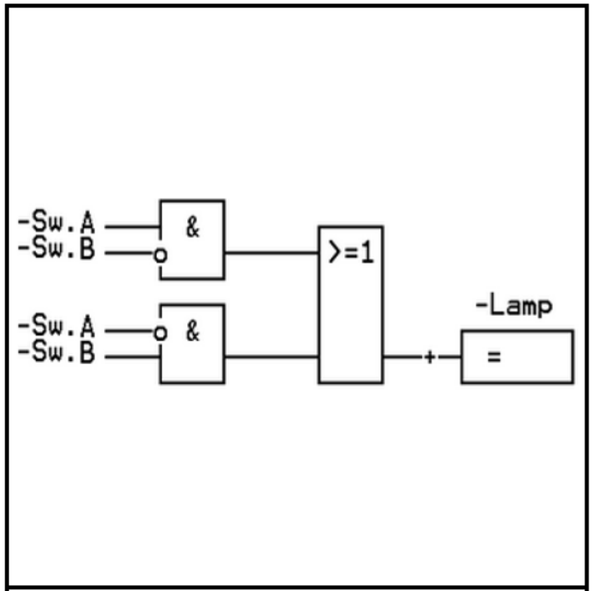
Function Block Diagram (FBD)

Sequential Function Chart (SFC)

Structured Text (ST)

Instruction List (IL)

Ladder Format



Function Block Format

SFC Format

```
  END_IF;
Setpoint_IN_STAGE_1_FAILED:
   (* During 'STAGE_1_FAILED': '<S1>:119' *)
   IF (stage3_sensor <= 0) OR (stage2_sensor <= 0) THEN
      (* Transition: '<S1>:150' *)
      (* Transition: '<S1>:152' *)
      IF stage2_sensor > 0 THEN
         (* Transition: '<S1>:155' *)
         is_c2_Setpoint := Setpoint_IN_STAGES_1_3_FAILED;
         (* Entry 'STAGES_1_3_FAILED': '<S1>:120' *)
         rtb_stage1_setpoint := L0;
         rtb_stage2_setpoint := L0 - overall_target;
         distributed_target := rtb_stage2_setpoint;
      ELSE
         (* Transition: '<S1>:154' *)
         IF stage3_sensor > 0 THEN
            (* Transition: '<S1>:159' *)
            is_c2_Setpoint := Setpoint_IN_STAGES_1_2_FAILED;
            (* Entry 'STAGES_1_2_FAILED': '<S1>:121' *)
            rtb_stage1_setpoint := L0;
            rtb_stage2_setpoint := L0;
            distributed_target := L0 - overall_target;
         ELSE
            guard_0 := TRUE;
         END_IF;
      END_IF;
   ELSE
      guard_0 := TRUE;
   END_IF;
```

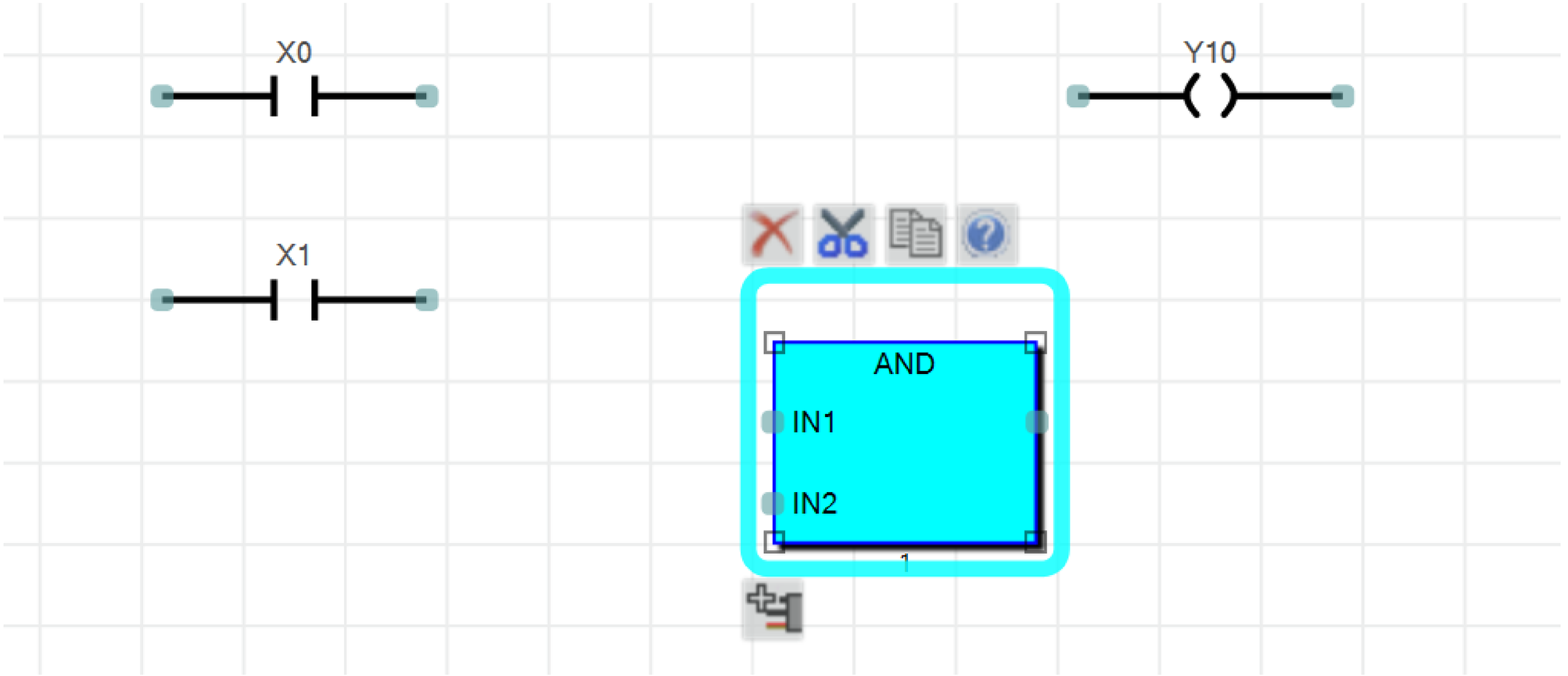
Structure Text Format

```
LD    X10
OUT   Y7
AND   M38
SET   S5
LD    X21
OUT   T01
      K40
```

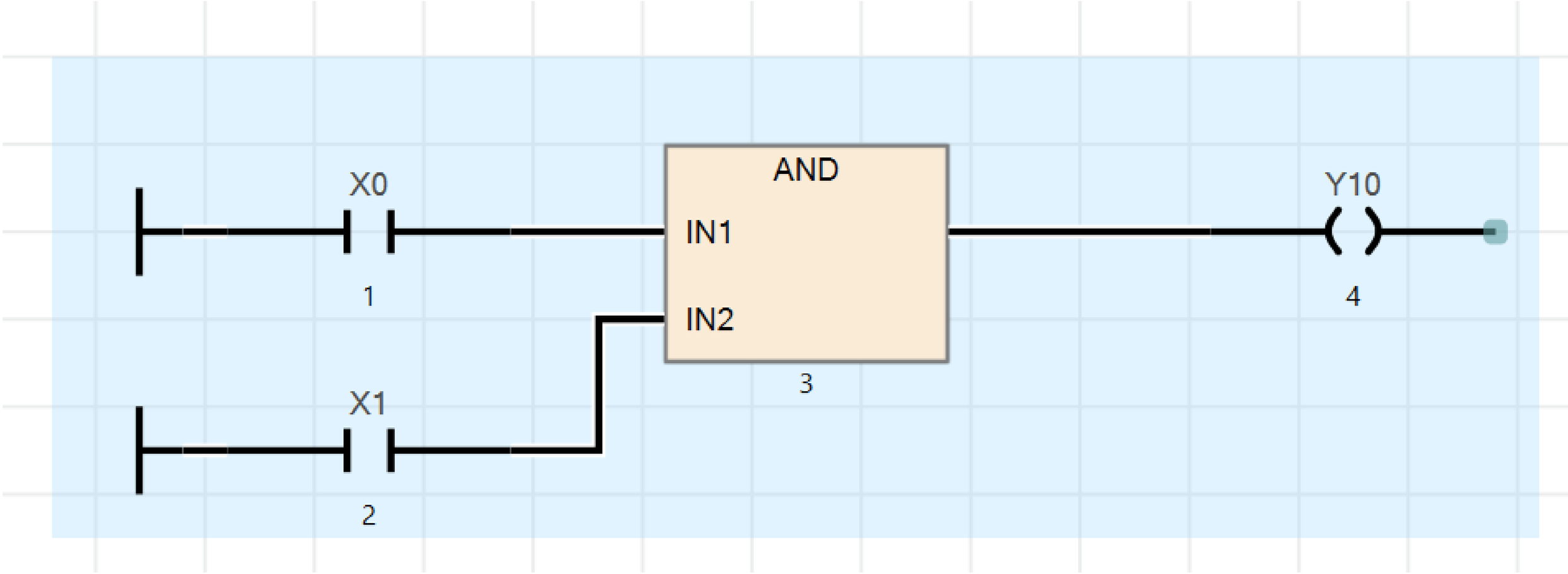
Instruction format

PLC Controls with Function Block Diagram (FBD)

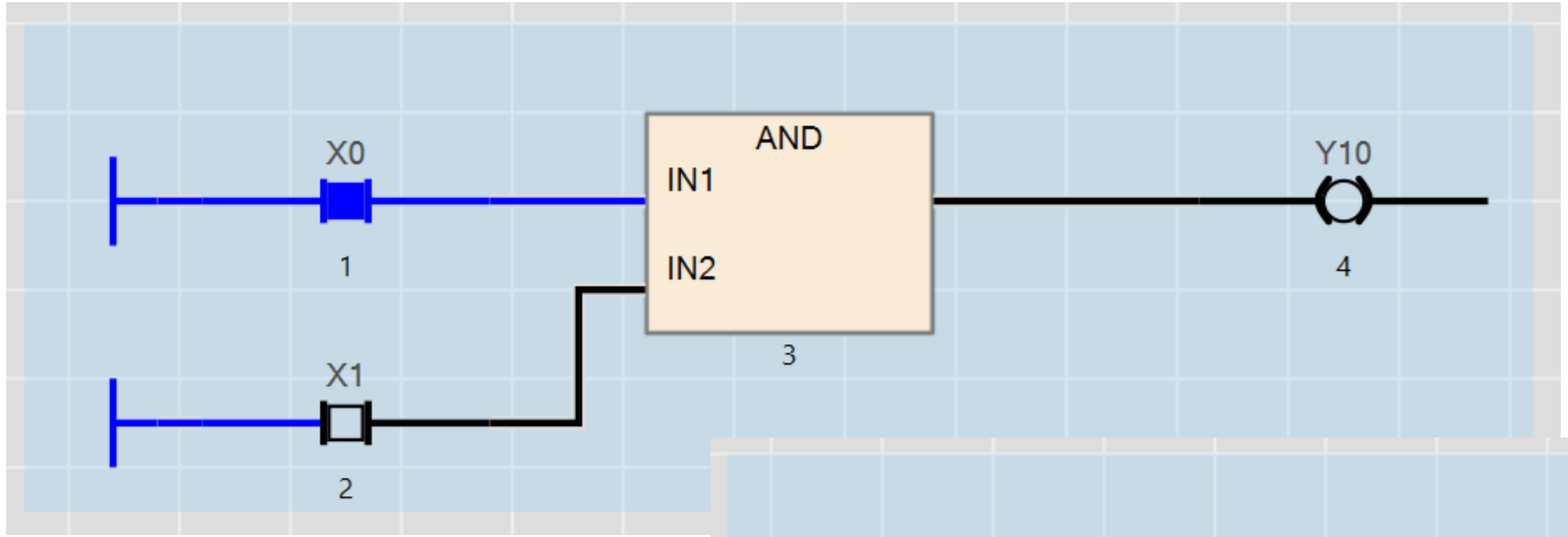
Y10 = X00 and X01

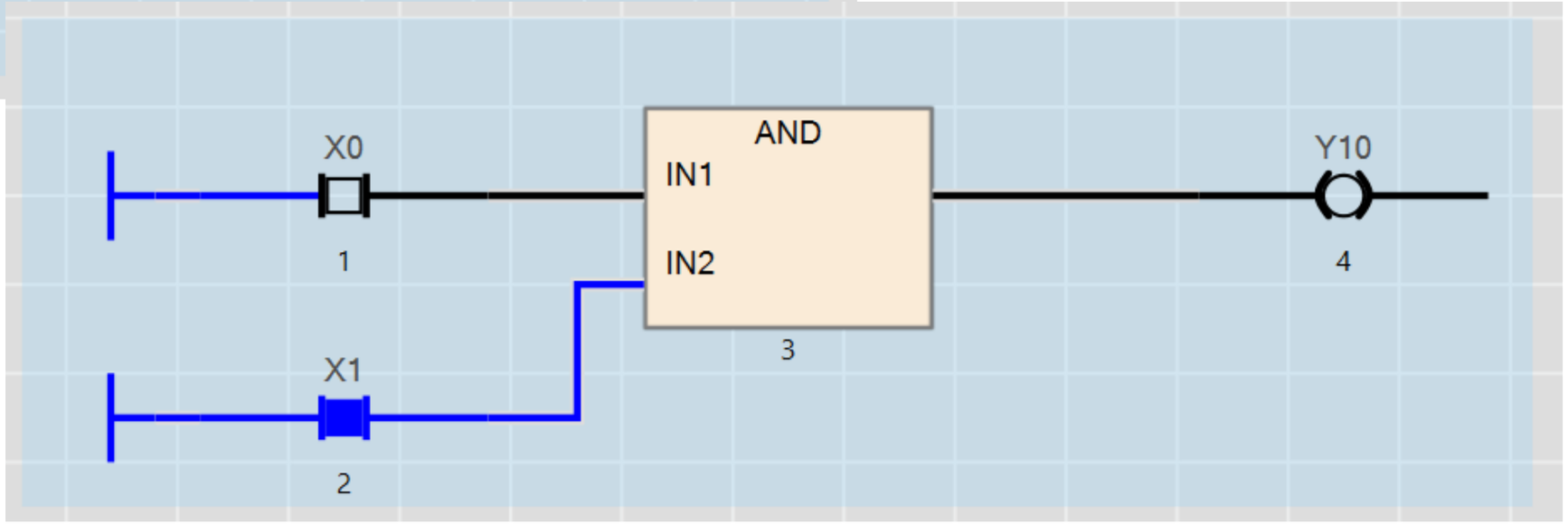


PLC Controls with Function Block Diagram (FBD)



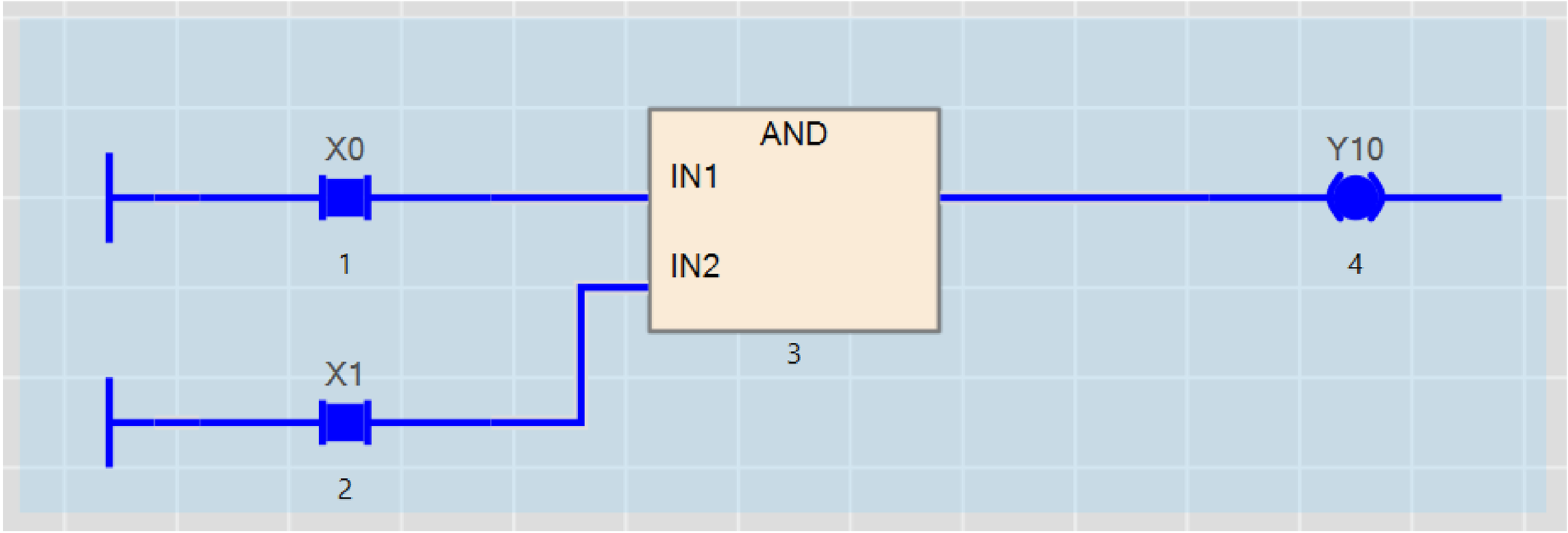
PLC Controls with Function Block Diagram (FBD)



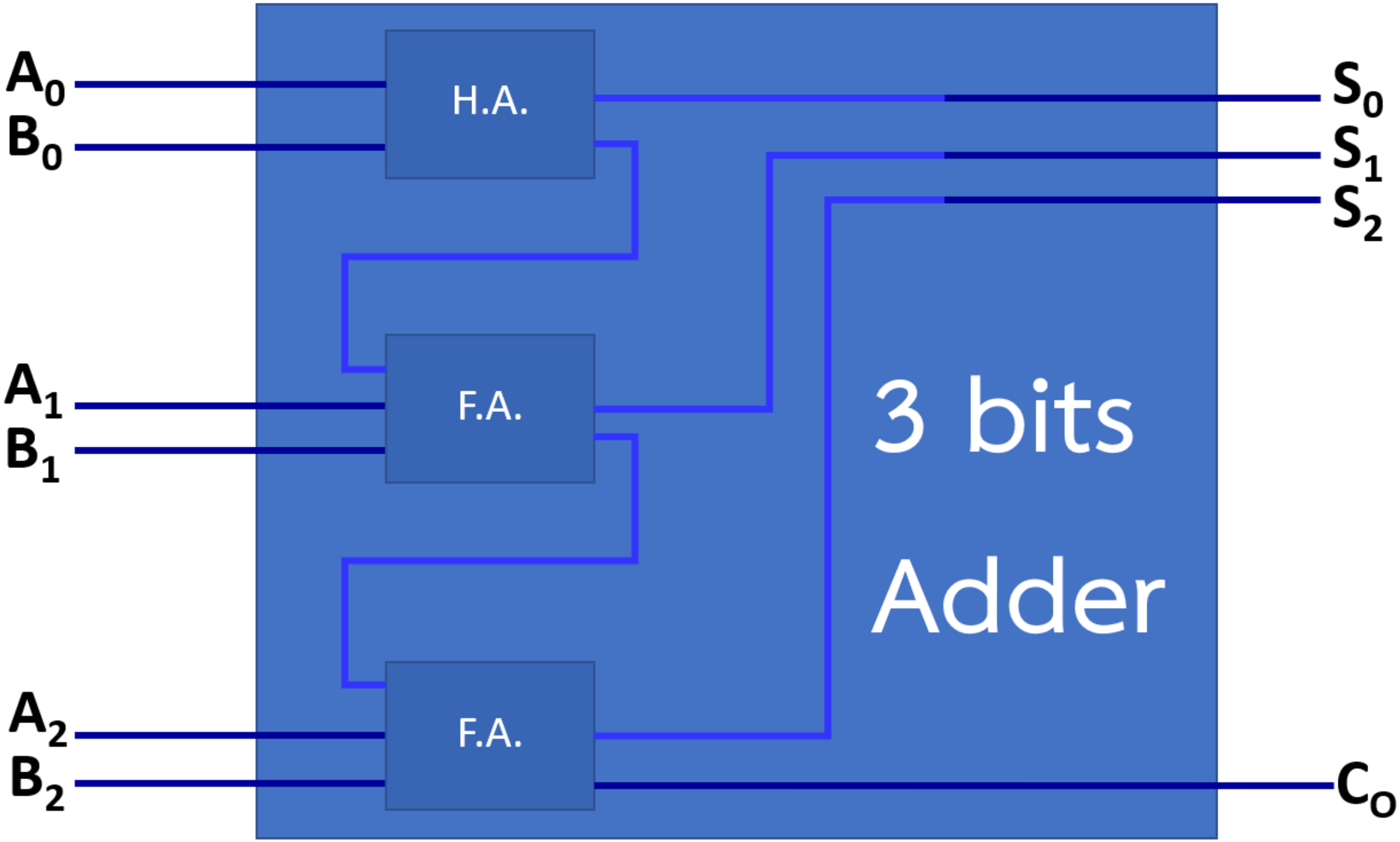


PLC Controls with Function Block Diagram (FBD)

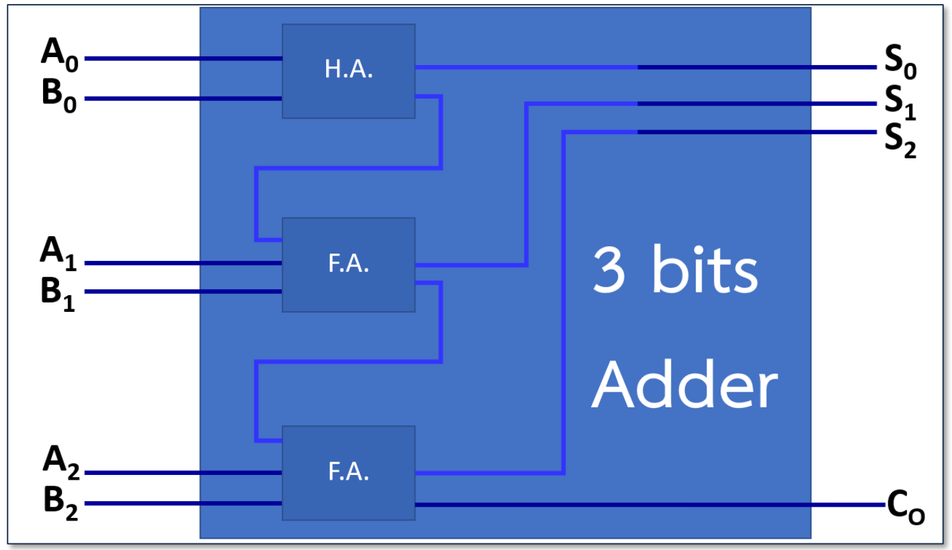
Y10 = X00 and X01

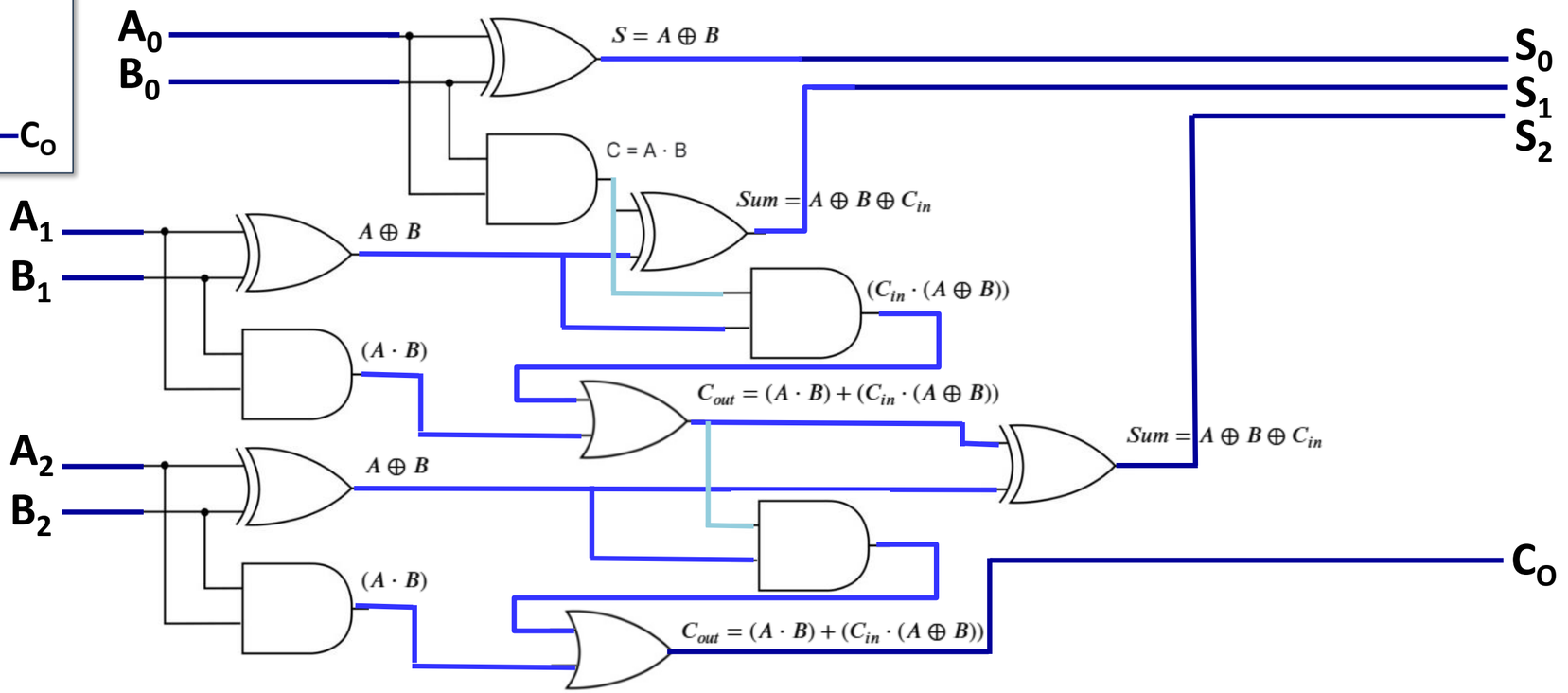


PLC Controls with Function Block Diagram (FBD)

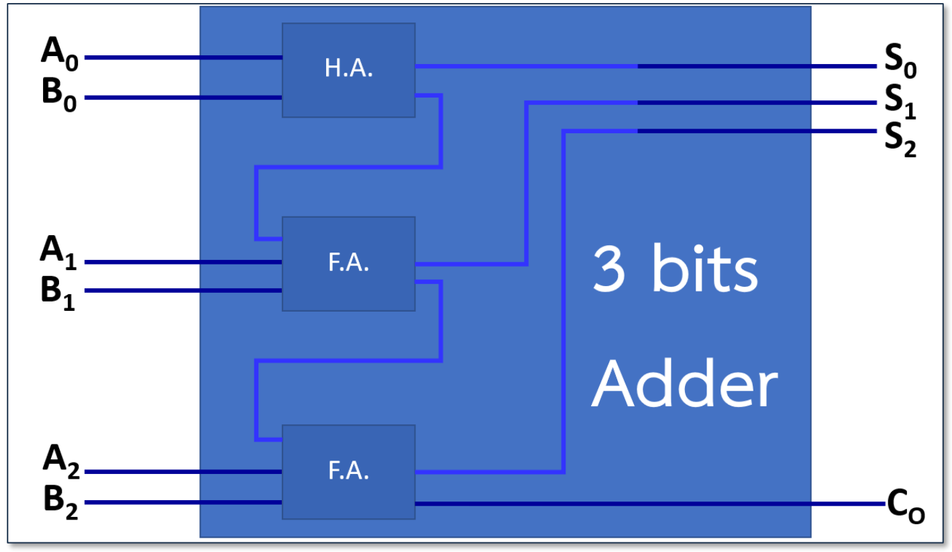


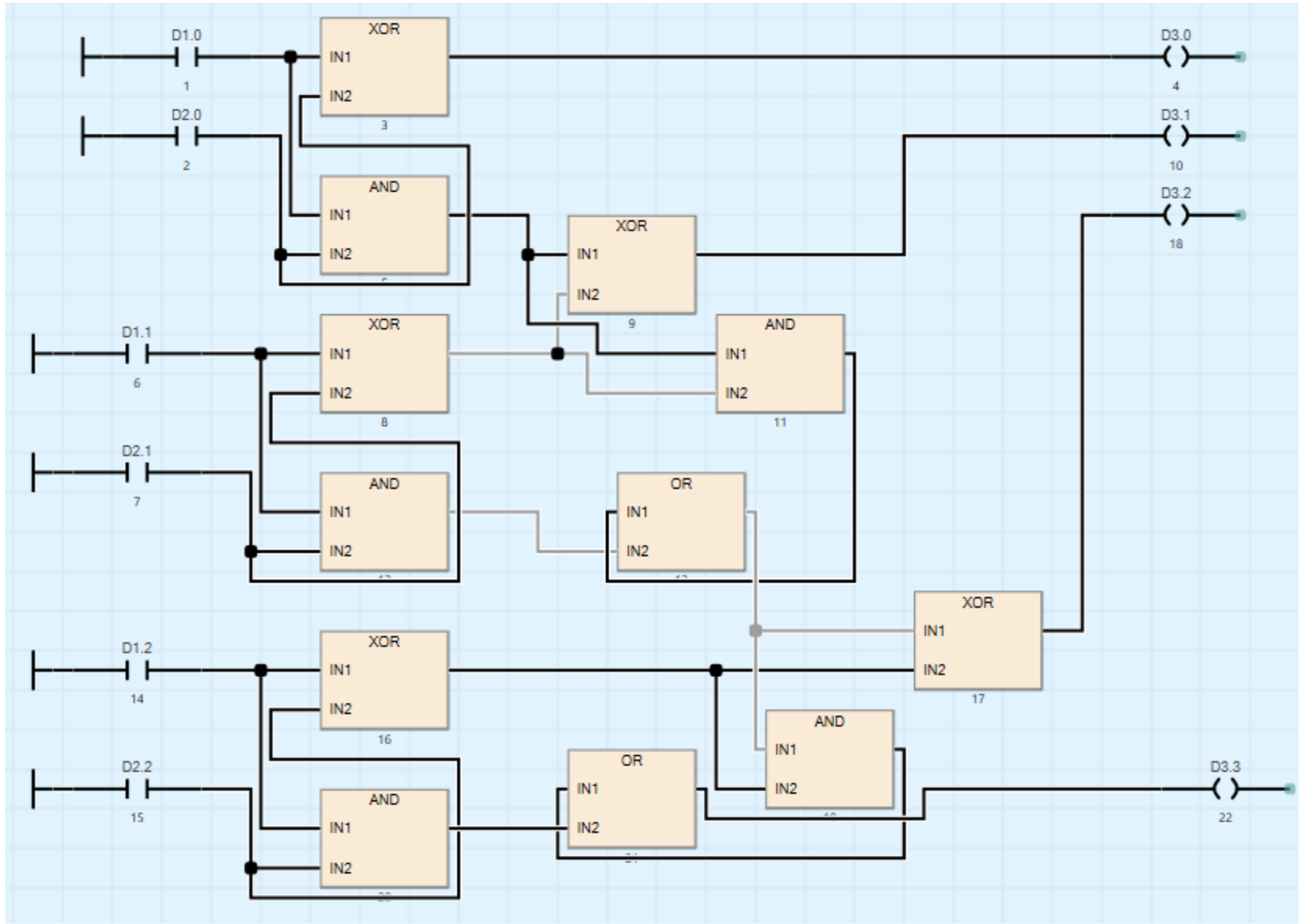
PLC Controls with Function Block Diagram (FBD)



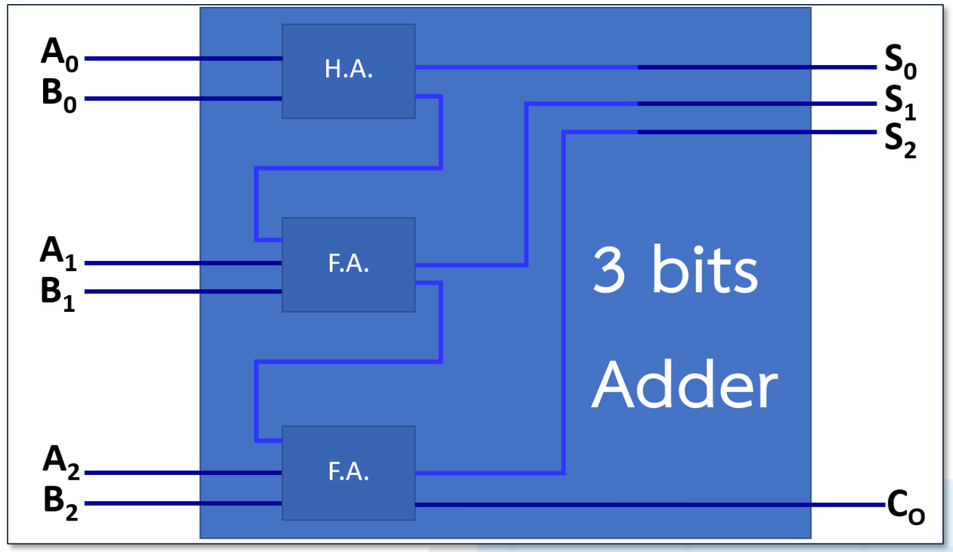


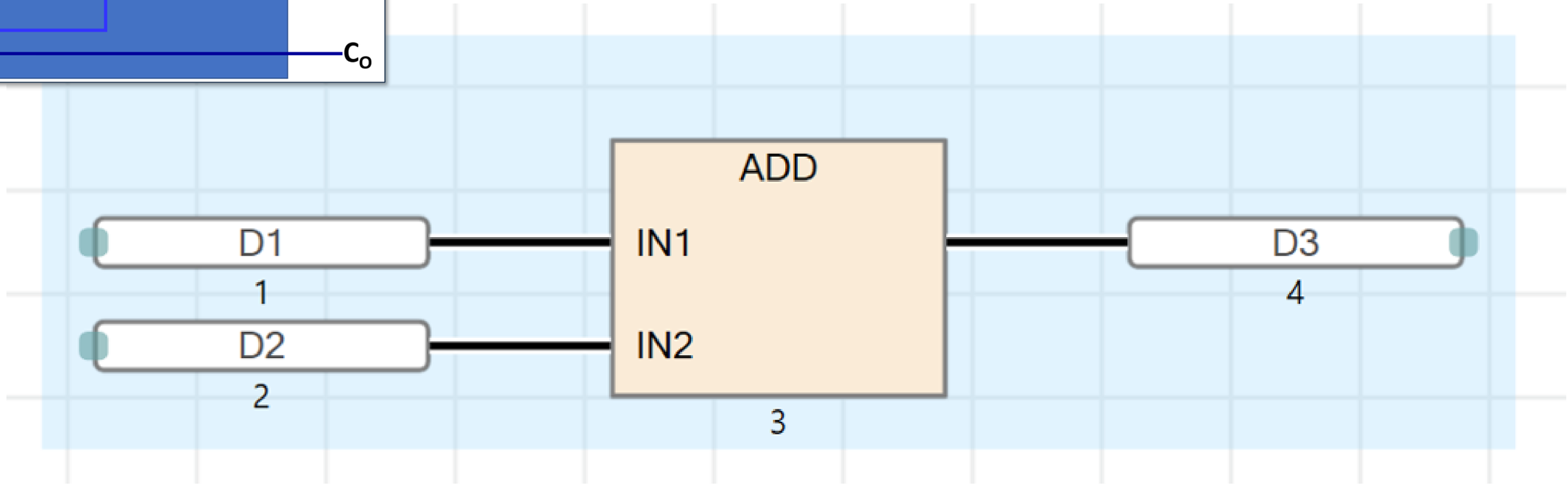
PLC Controls with Function Block Diagram (FBD)





PLC Controls with Function Block Diagram (FBD)





PLC Programming and Logic control

IEC-61131-3

Graphical Language

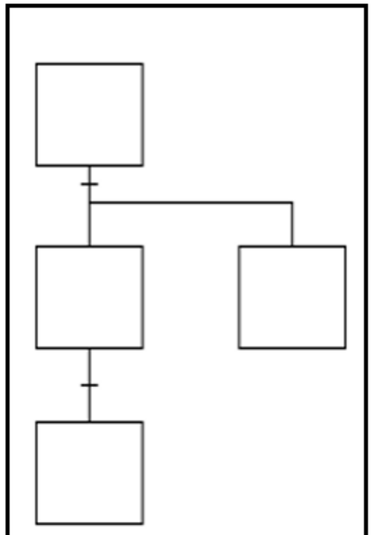
Ladder Diagram (LD)

Ladder Format

Function Block Diagram (FBD)

Function Block Format

Sequential Function Chart (SFC)



SFC Format

Textual Language

Structured Text (ST)

```
    END_IF;
Setpoint_IN_STAGE_1_FAILED:
    (* During 'STAGE_1_FAILED': '<S1>:119' *)
    IF (stage3_sensor <= 0) OR (stage2_sensor <= 0) THEN
        (* Transition: '<S1>:150' *)
        (* Transition: '<S1>:152' *)
        IF stage2_sensor > 0 THEN
            (* Transition: '<S1>:155' *)
            is_c2_Setpoint := Setpoint_IN_STAGES_1_3_FAILED;
            (* Entry 'STAGES_1_3_FAILED': '<S1>:120' *)
            rtb_stage1_setpoint := L0;
            rtb_stage2_setpoint := L0 - overall_target;
            distributed_target := rtb_stage2_setpoint;
        ELSE
            (* Transition: '<S1>:154' *)
            IF stage3_sensor > 0 THEN
                (* Transition: '<S1>:159' *)
                is_c2_Setpoint := Setpoint_IN_STAGES_1_2_FAILED;
                (* Entry 'STAGES_1_2_FAILED': '<S1>:121' *)
                rtb_stage1_setpoint := L0;
                rtb_stage2_setpoint := L0;
                distributed_target := L0 - overall_target;
            ELSE
                guard_0 := TRUE;
            END_IF;
        END_IF;
    ELSE
        guard_0 := TRUE;
    END_IF;
```

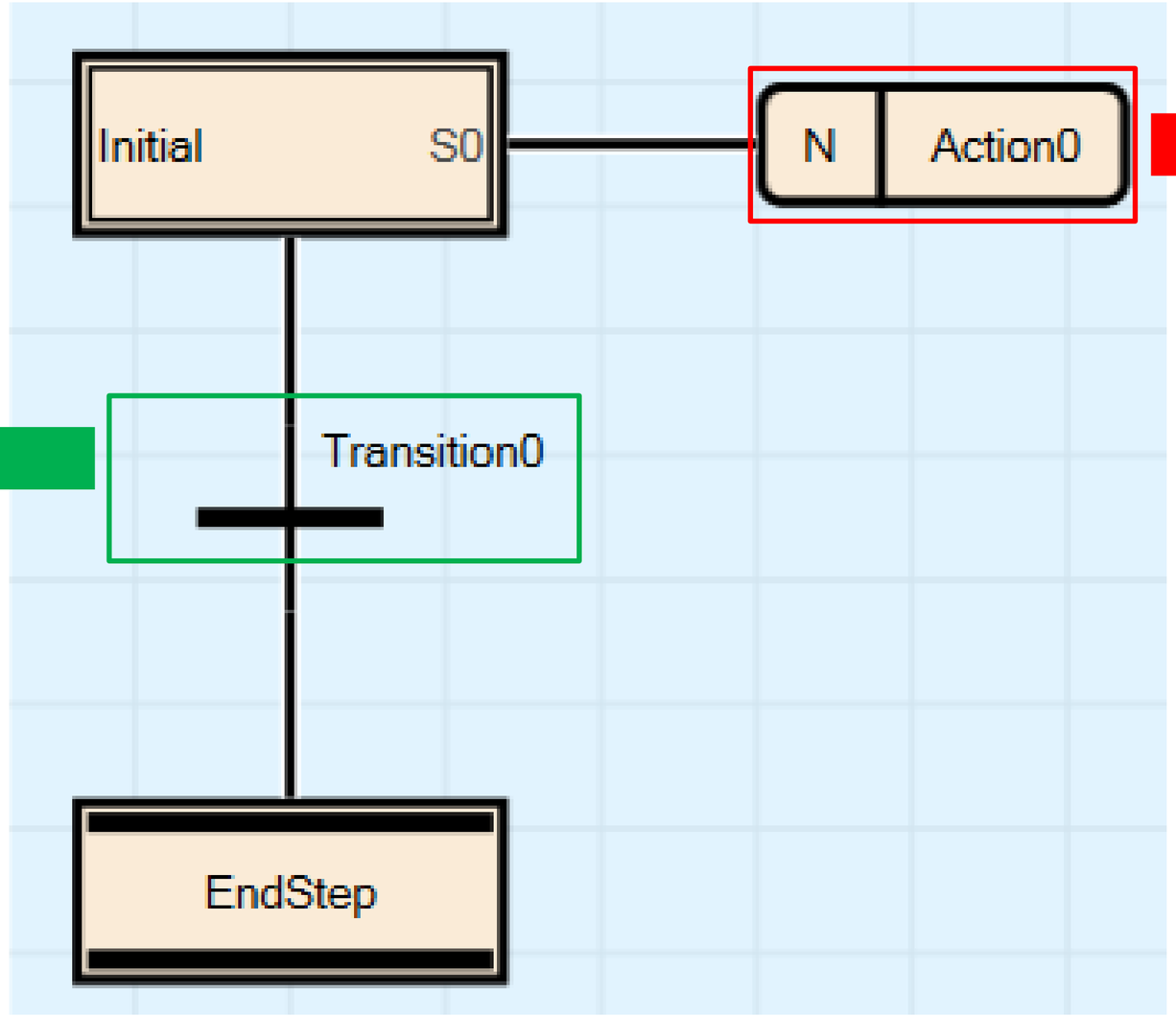
Structure Text Format

Instruction List (IL)

```
LD    X10
OUT   Y7
AND   M38
SET   S5
LD    X21
OUT   T01
      K40
```

Instruction format

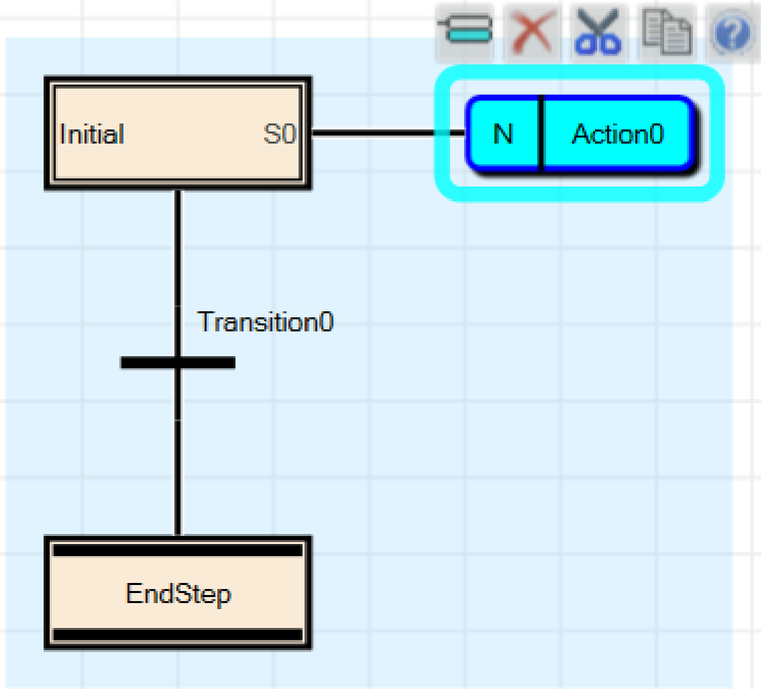
PLC Controls with Sequential Function Chart (SFC)

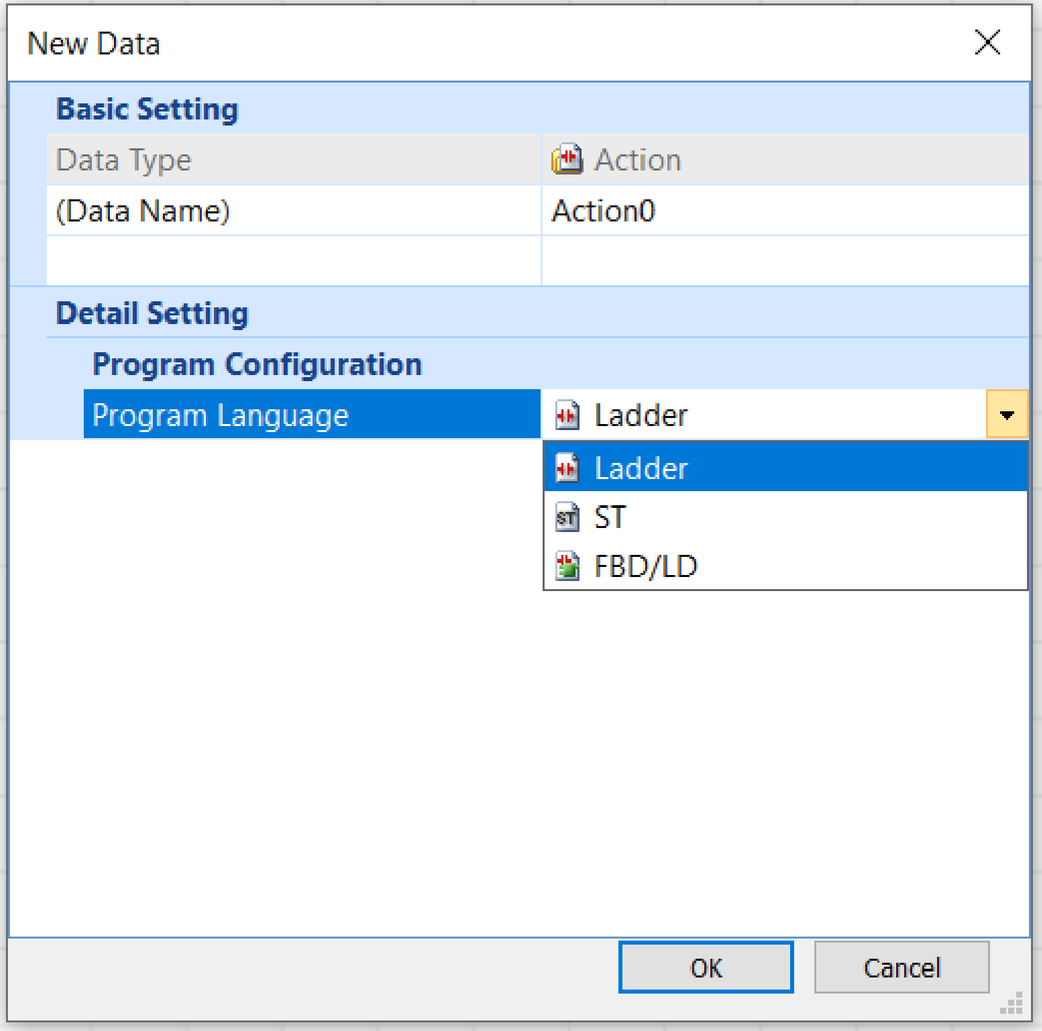


Drives outputs or runs control logic.

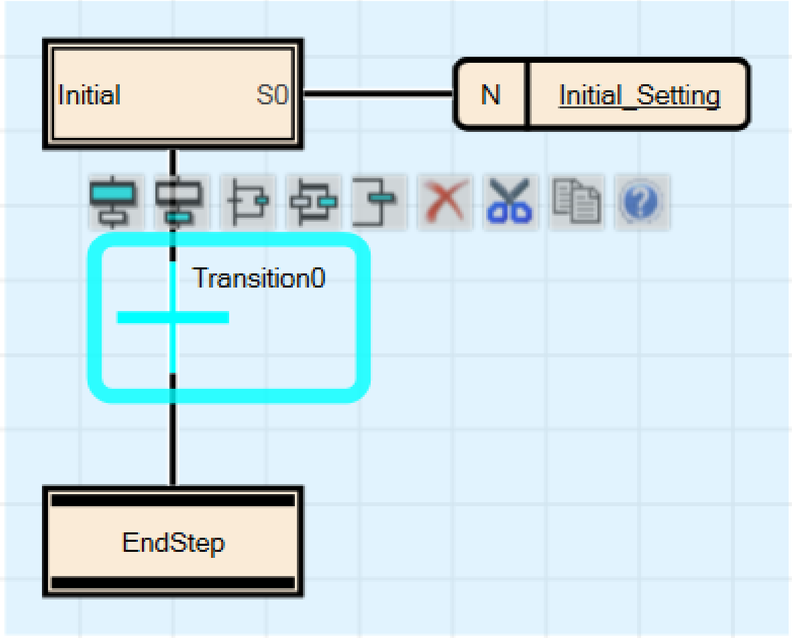
Decides when to move to the next step.

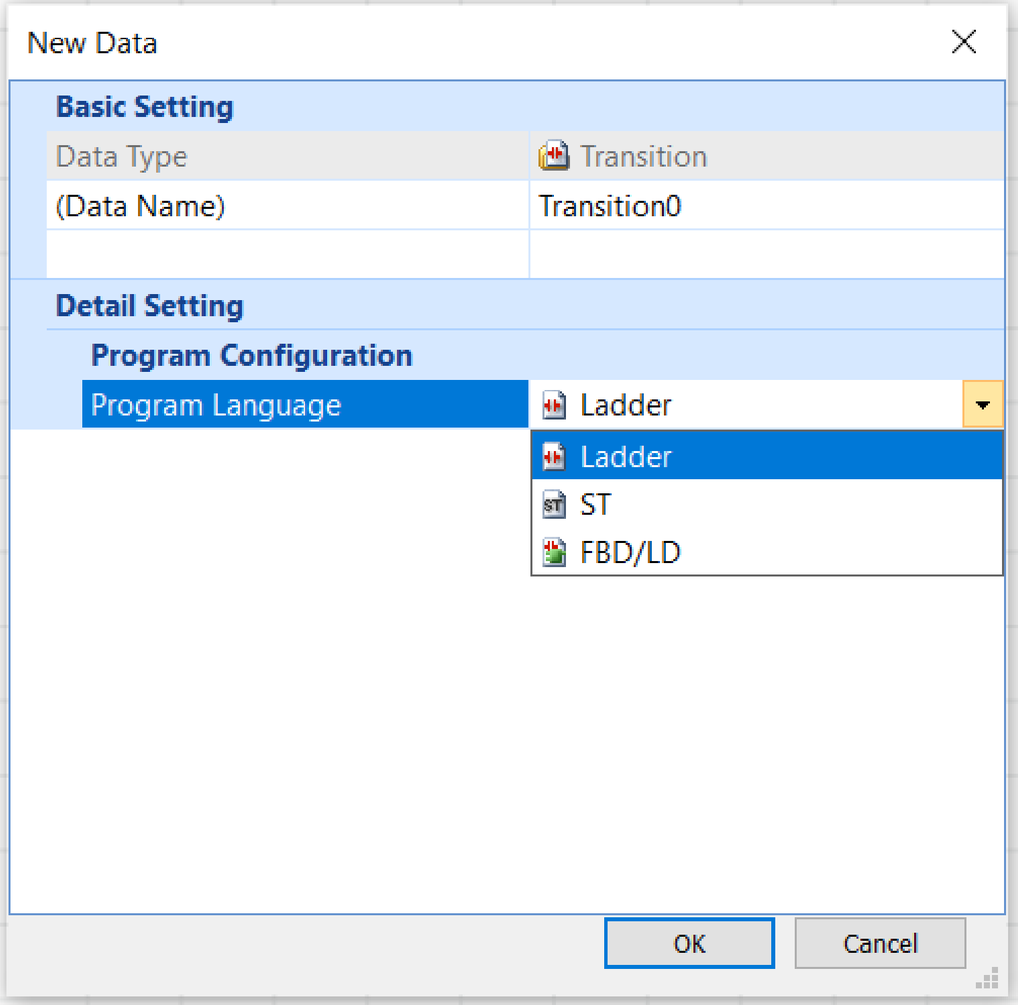
PLC Controls with Sequential Function Chart (SFC)



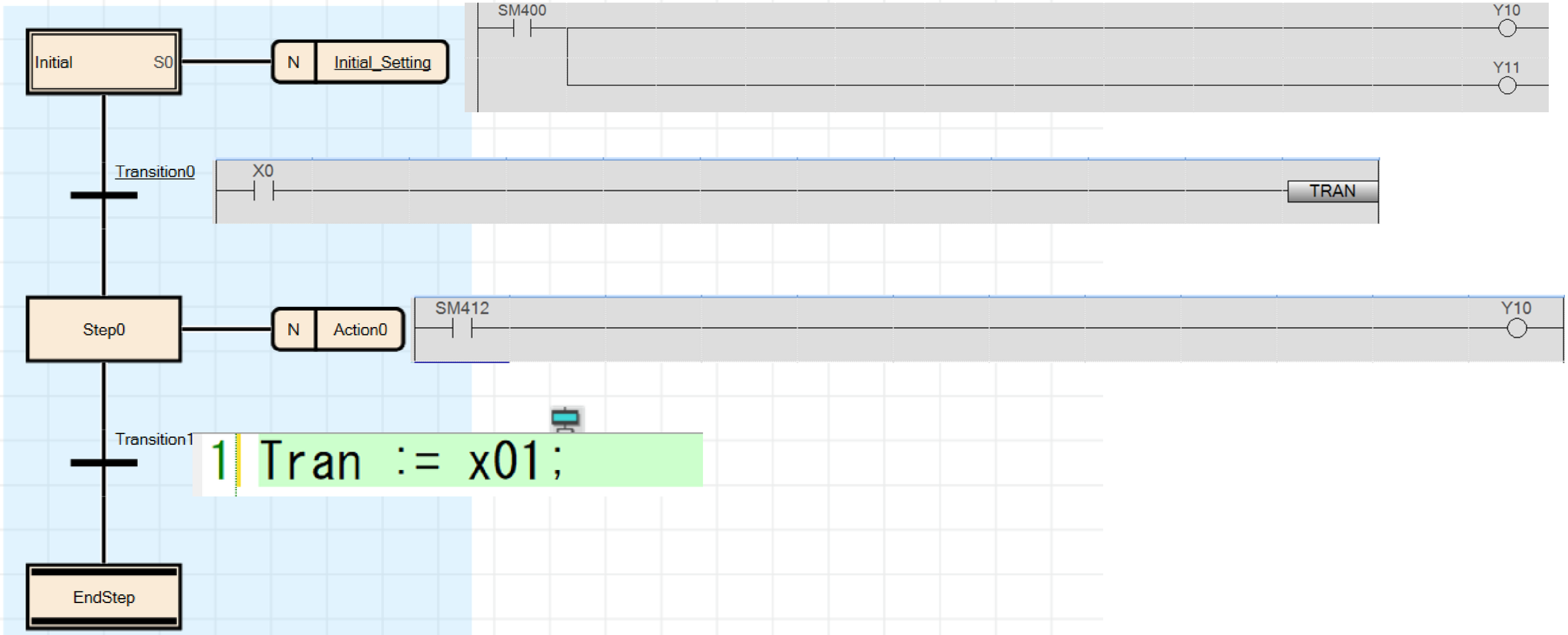


PLC Controls with Sequential Function Chart (SFC)





PLC Controls with Sequential Function Chart (SFC)



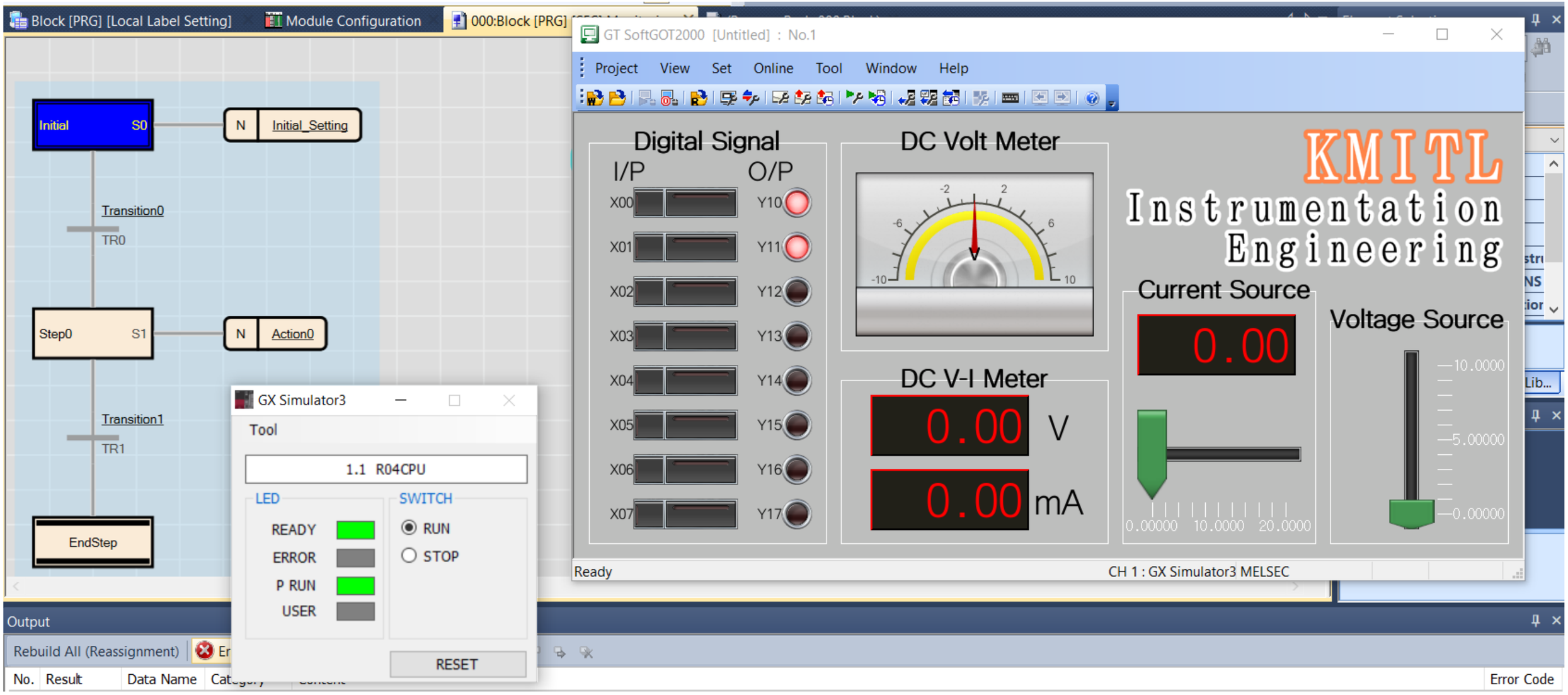
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PLC Controls with Sequential Function Chart (SFC)



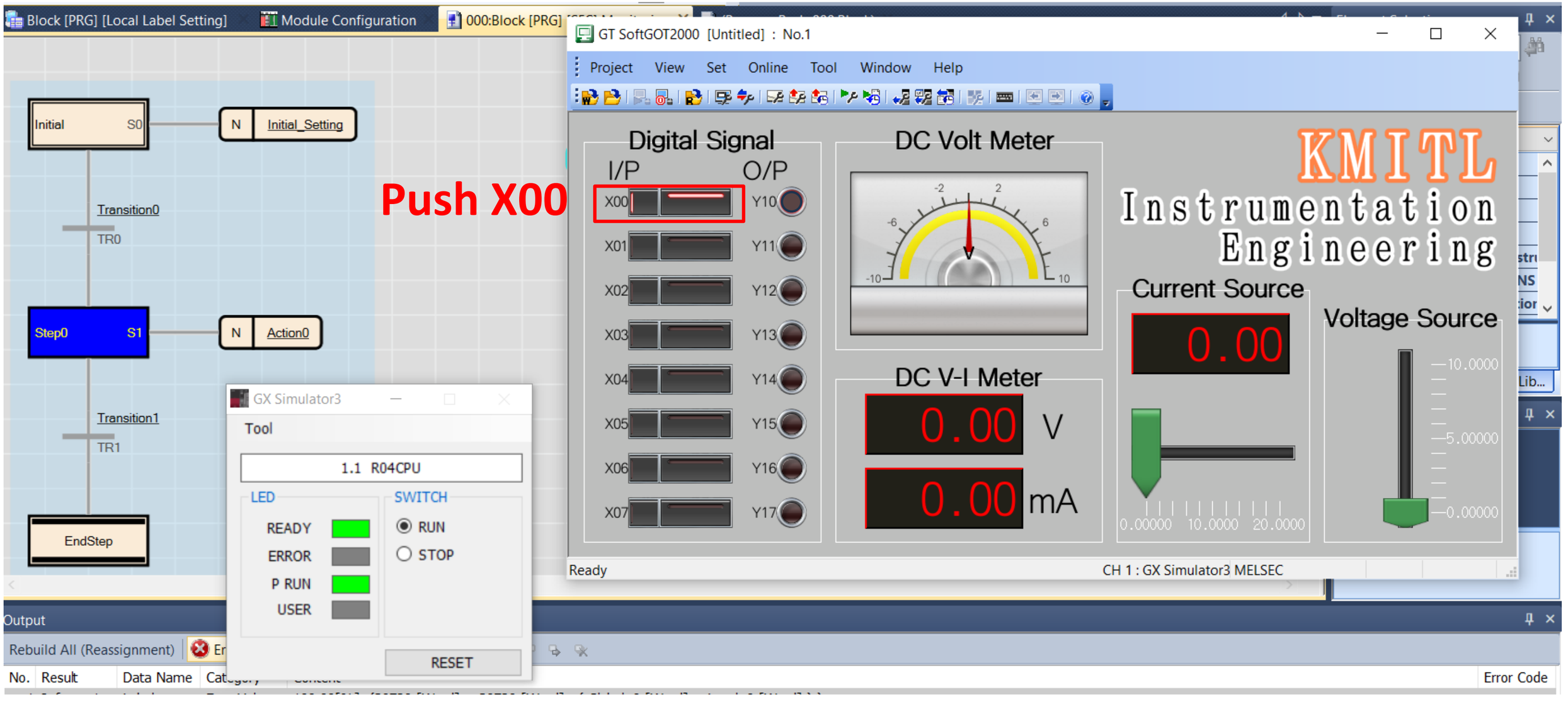
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PLC Controls with Sequential Function Chart (SFC)



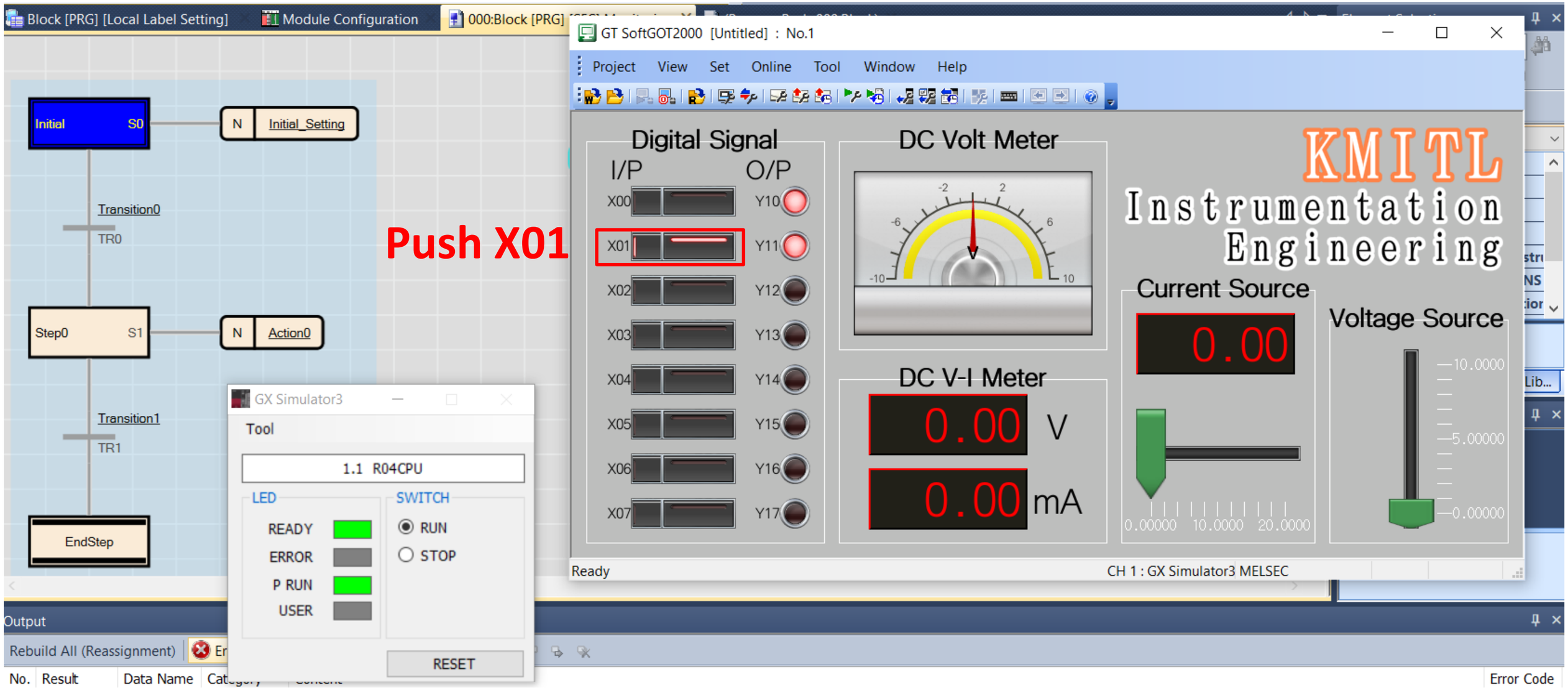
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PLC Controls with Sequential Function Chart (SFC)



Training

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ASSUMPTION UNIVERSITY OF THAILAND
SINCE 1969
มหาวิทยาลัยอัสสัมชัญ

TIRA
THAI INNOVATION AND ROBOTICS ASSOCIATION
สมาคมการสร้างนวัตกรรมและหุ่นยนต์

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กระทรวงการอุดมศึกษา วิทยาศาสตร์ วิจัยและนวัตกรรม
Ministry of Higher Education, Science, Research and Innovation

สอวช
สำนักงานสภานโยบายการอุดมศึกษา
วิทยาศาสตร์ วิจัย
และนวัตกรรมแห่งชาติ

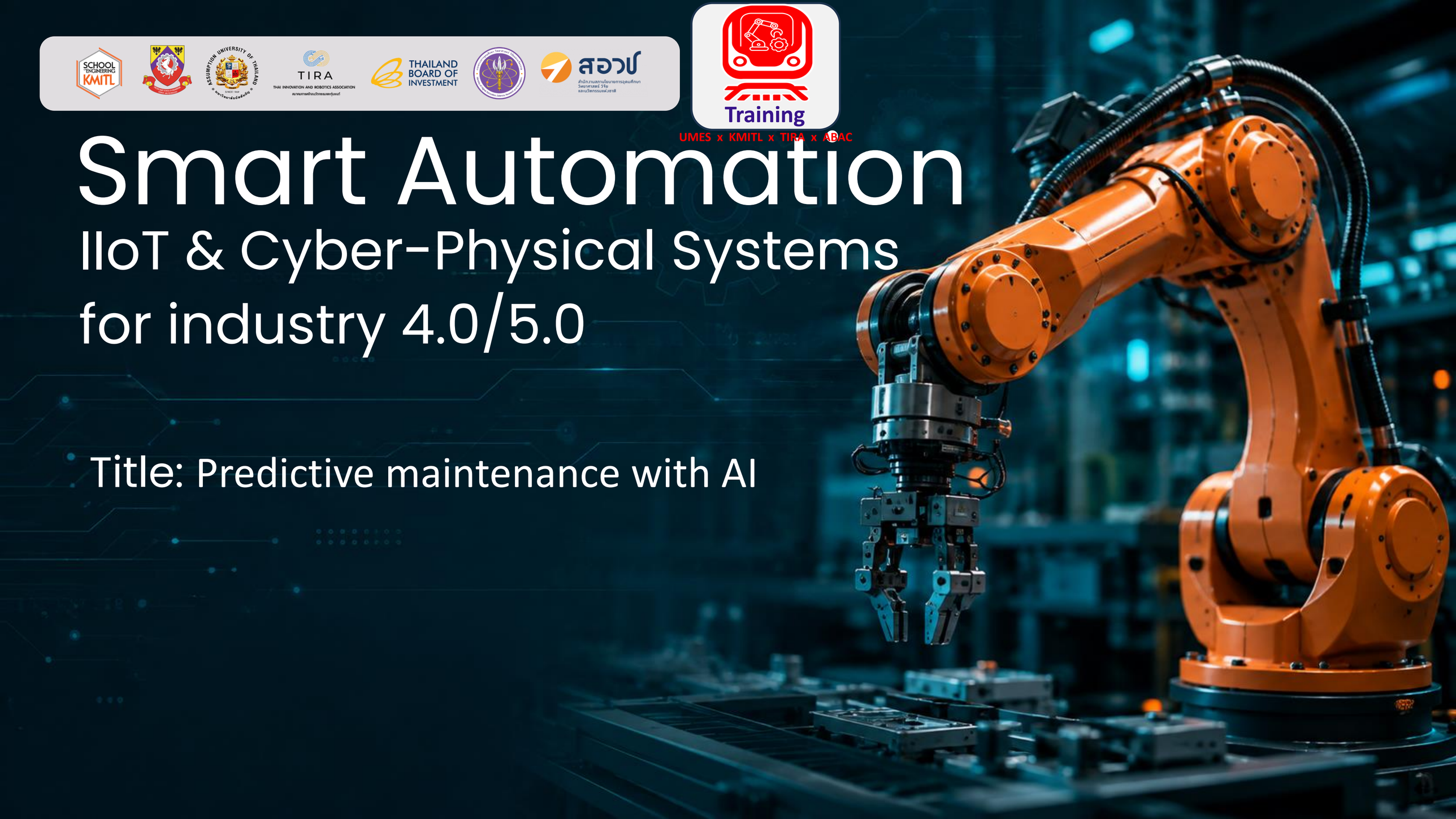
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IIoT & Cyber-Physical Systems for industry 4.0/5.0

Title: Predictive maintenance with AI

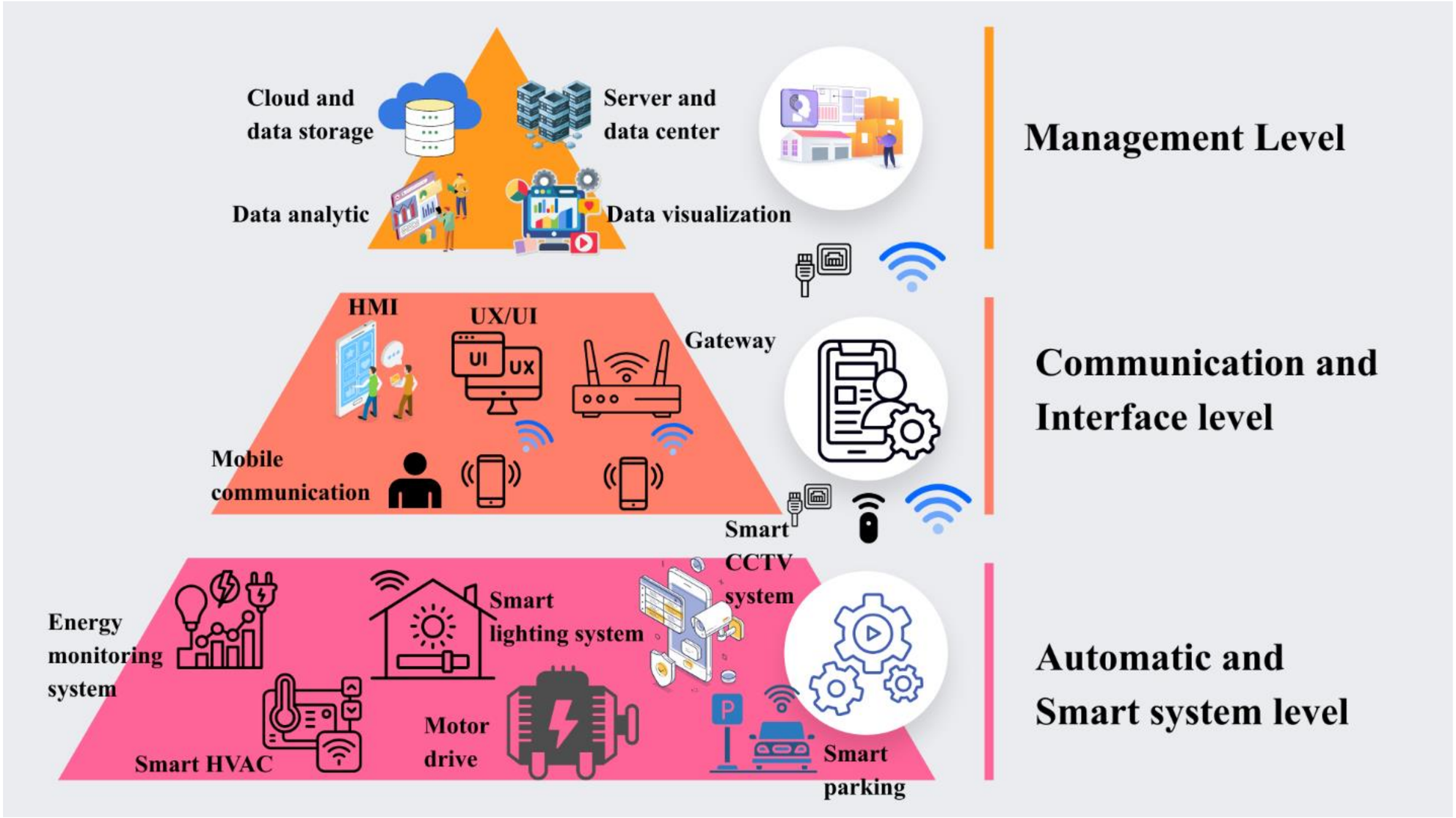


Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

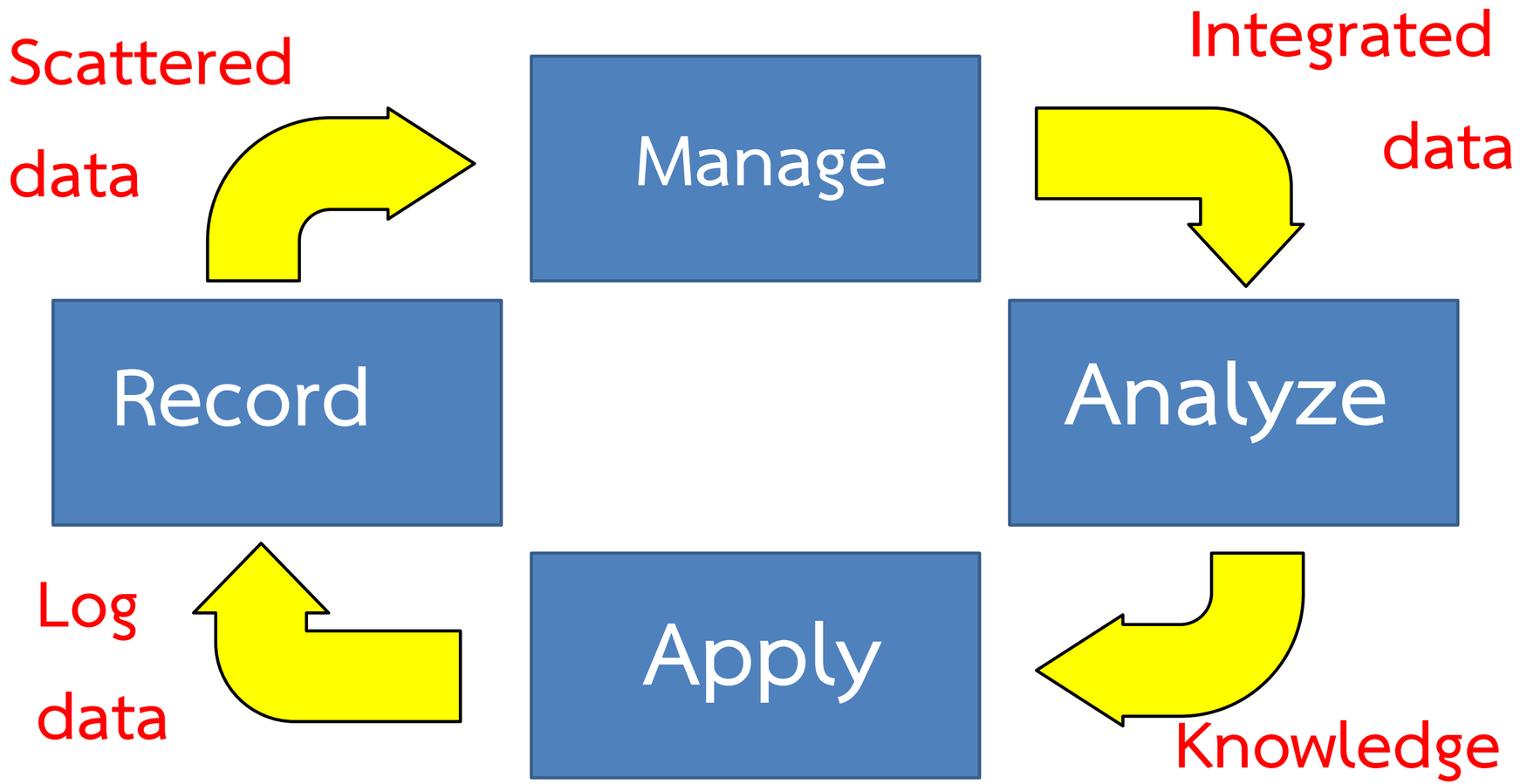
Training

UMES x KMITL x TIRA x ABAC



โครงสร้างของโรงงานอัจฉริยะ

วงจรชีวิตของข้อมูล (Life cycle of data)



นิยามข้อมูลขนาดใหญ่ (Big Data definition)

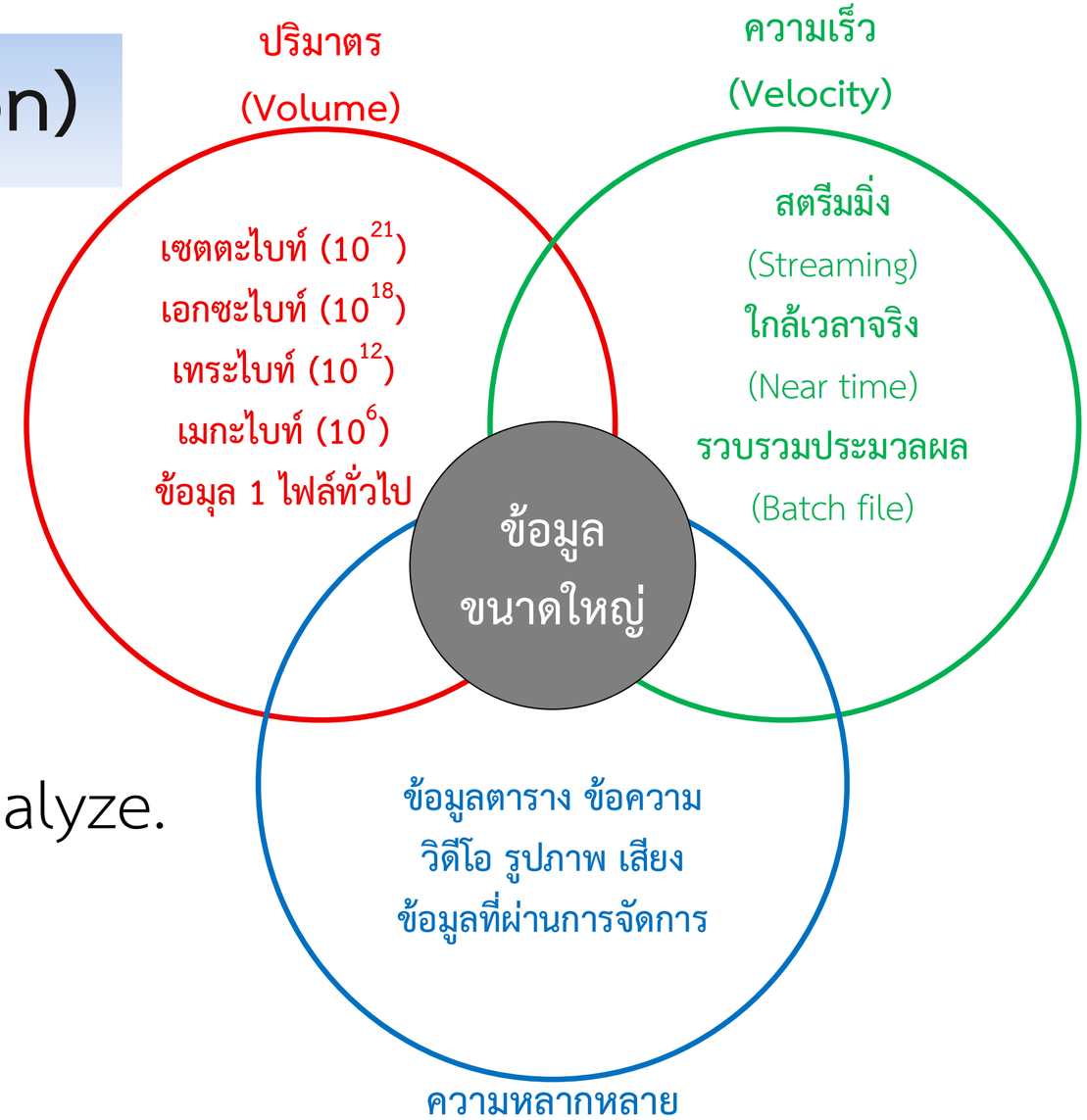
- 3 Vs definition.

- Volume

- Variety

- Velocity

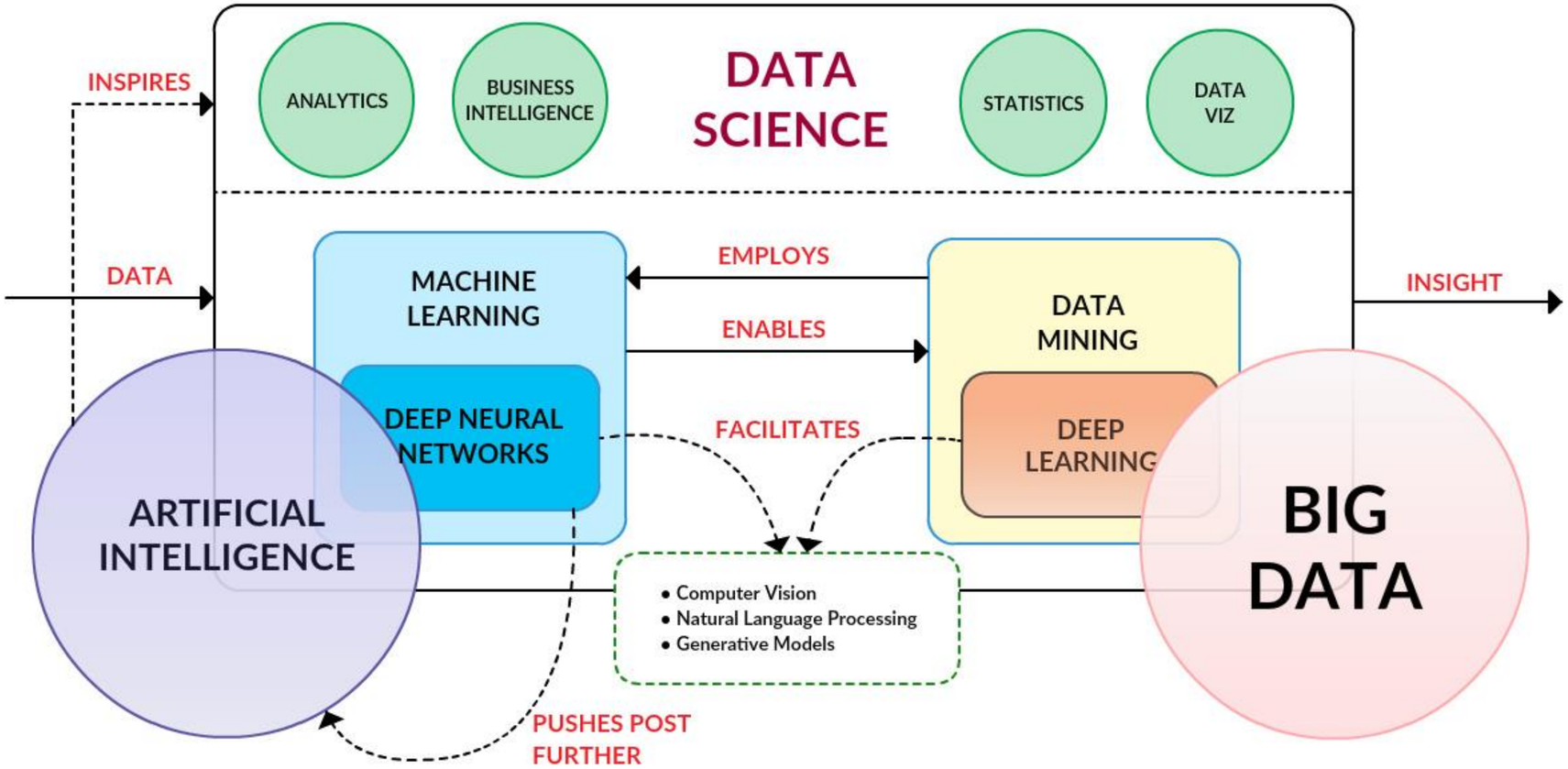
It needs several techniques to manage and analyze.



สาเหตุที่ทำให้เกิดข้อมูลขนาดใหญ่

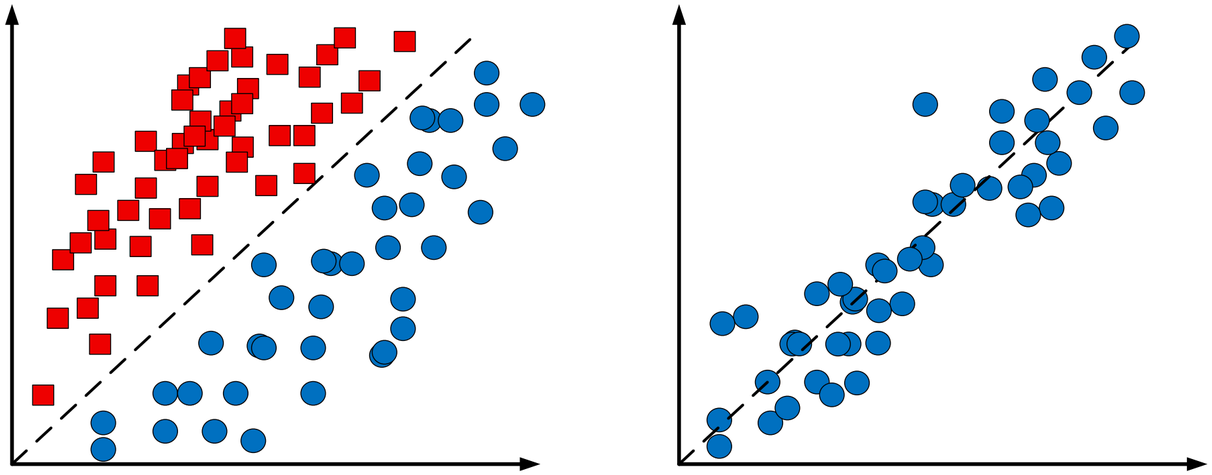
- สมรรถนะของการเก็บข้อมูล
- เสมรรถนะของชุดประมวลผล
- ความหลากหลายของข้อมูล
- การเพิ่มขึ้นของจำนวน Social Network
- การเพิ่มขึ้นของข้อมูลจากระบบ IOT

การบูรณาการของข้อมูลขนาดใหญ่กับปัญญาประดิษฐ์ (Big data & AI)

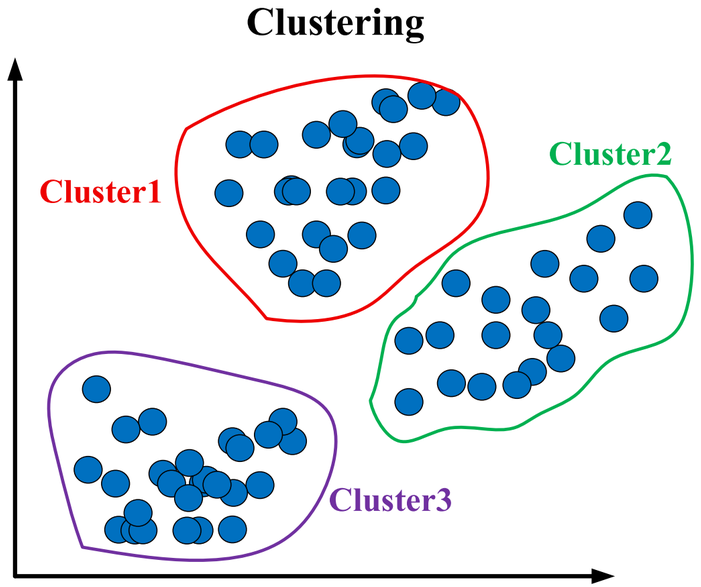


ที่มา: Enthusiasts Mobile App Development and Websites Design Company providing best App Development services (abitstudio.com)

การวิเคราะห์ข้อมูล (Data Analytics)



การเรียนรู้แบบมีผู้สอน (Supervised learning)



การเรียนรู้แบบไม่มีผู้สอน (Unsupervised learning)

ความสัมพันธ์ของข้อมูล (Data Correlation)

Sample Pearson correlation coefficient

$$r(x,y) = \frac{1}{n-1}\sum_{i=1}^{n}\frac{(x_i-\bar{x})}{s_x}\frac{(y_i-\bar{y})}{s_y}$$

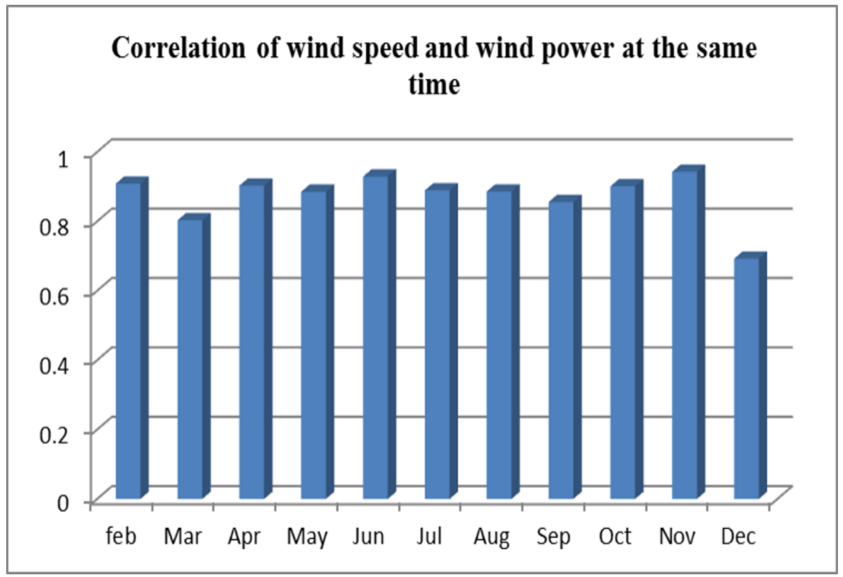
$$s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

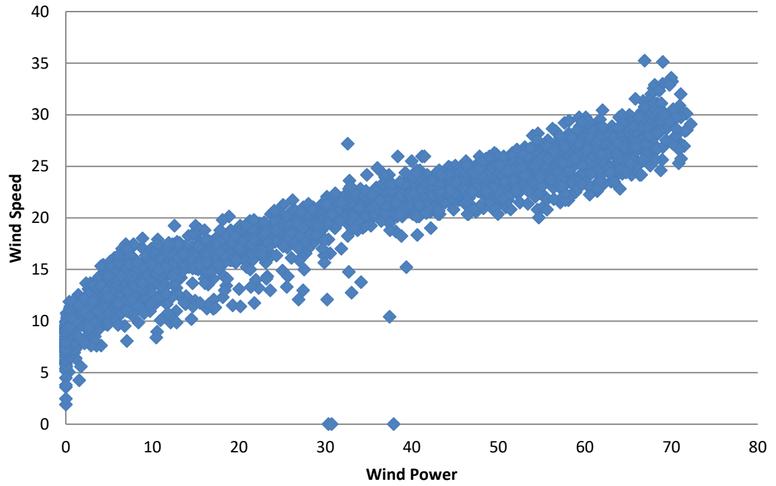
$$s_y = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

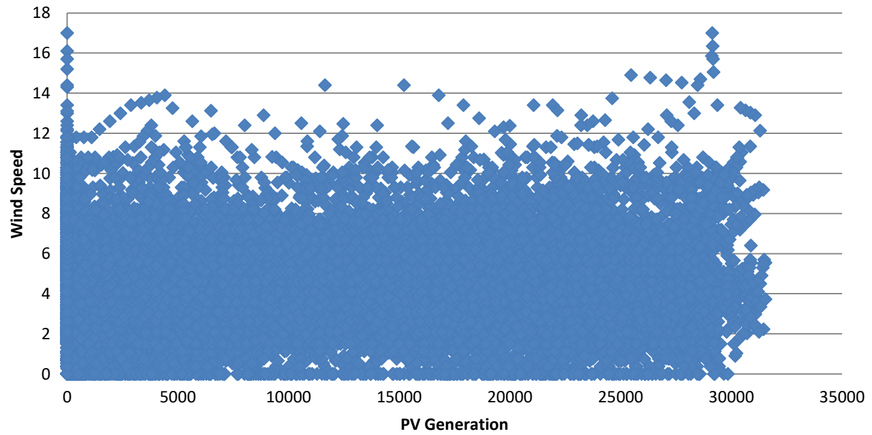
การตีความ

- Exactly –1. มีความสัมพันธ์ในเชิงผกผันสมบูรณ์
- –0.70. มีความสัมพันธ์ในเชิงผกผันสูง
- –0.50. มีความสัมพันธ์ในเชิงผกผันปานกลาง
- –0.30. มีความสัมพันธ์ในเชิงผกผันต่ำ
- 0. **ไม่มีความสัมพันธ์**
- +0.30. มีความสัมพันธ์ต่ำ
- +0.50. มีความสัมพันธ์ปานกลาง
- +0.70. มีความสัมพันธ์สูง
- Exactly +1. มีความสัมพันธ์สมบูรณ์

ตัวอย่างการหาความสัมพันธ์ของข้อมูล







การวิเคราะห์ข้อมูลที่สำคัญต่อการทำ Predictive Maintenance

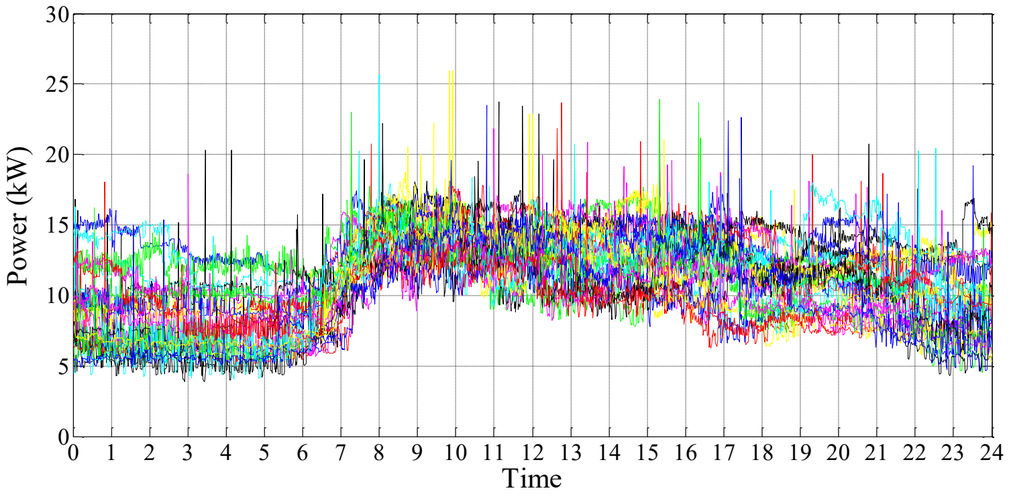
- การทำเหมืองข้อมูล (Data Mining)
- การทำนายข้อมูล (Data Forecasting)

การทำเหมืองข้อมูล (Data Mining)

กระบวนการวิเคราะห์และค้นหารูปแบบ ความสัมพันธ์ หรือแนวโน้มที่ซ่อนอยู่ในชุดข้อมูลขนาดใหญ่ โดยใช้เทคโนโลยีคอมพิวเตอร์ สถิติ และการเรียนรู้ของเครื่อง (Machine Learning) เพื่อเปลี่ยนข้อมูลดิบให้กลายเป็นความรู้

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	DataDateTime	ShiftID	CalcStatus	SendRead	Device	Aggregate	BillingMo	CurrentL1	CurrentL2	CurrentL3	Hz	kVA	kVar	kW	kWh	kWh_Con	PF	THDCurre	THDCurre
2	6/9/2016 0:00	0:00	1		Floor2-Nol		439.62	24.3	7.8	12.6	50.04	3.9	10.2	9.33	1252.62		0.92	0	0
3	6/9/2016 0:01	0:01	1		Floor2-Nol		439.62	24.3	7.8	12.6	50.06	3.87	10.26	9.39	1252.77		0.92	0	0
4	6/9/2016 0:02	0:02	1		Floor2-Nol		439.62	23.7	7.5	12.6	50.07	3.9	9.96	9.06	1252.93		0.92	0	0
5	6/9/2016 0:03	0:03	1		Floor2-Nol		439.62	23.7	7.5	12.6	50.01	3.9	9.96	9.06	1253.06		0.92	0	0
6	6/9/2016 0:04	0:04	1		Floor2-Nol		797.38	23.7	7.5	12	50.03	3.9	9.84	8.94	1253.23		0.92	0	0
7	6/9/2016 0:05	0:05	1		Floor2-Nol		797.38	20.7	7.5	12.3	50	3.84	9.15	8.16	1253.36		0.91	0	0
8	6/9/2016 0:06	0:06	1		Floor2-Nol		797.38	20.7	7.5	12	50.01	3.81	9.15	8.16	1253.5		0.91	0	0
9	6/9/2016 0:07	0:07	1		Floor2-Nol		797.38	20.7	7.5	12	49.99	3.81	9.12	8.16	1253.63		0.91	0	0
10	6/9/2016 0:08	0:08	1		Floor2-Nol		797.38	20.7	7.5	12.6	50	3.81	9.24	8.25	1253.78		0.91	0	0
11	6/9/2016 0:09	0:09	1		Floor2-Nol		797.38	20.7	7.5	15	49.98	4.02	9.72	8.7	1253.9		0.91	0	0
12	6/9/2016 0:10	0:10	1		Floor2-Nol		797.38	23.4	7.5	12.3	50.05	4.08	9.81	8.79	1254.07		0.91	0	0
13	6/9/2016 0:11	0:11	1		Floor2-Nol		797.38	23.4	7.5	12.6	50.09	4.02	9.87	8.88	1254.21		0.91	0	0
14	6/9/2016 0:12	0:12	1		Floor2-Nol		797.38	23.4	7.5	11.1	50.02	3.96	9.57	8.58	1254.36		0.91	0	0
15	6/9/2016 0:13	0:13	1		Floor2-Nol		797.38	23.4	7.5	11.4	50	3.96	9.57	8.58	1254.5		0.91	0	0
16	6/9/2016 0:14	0:14	1		Floor2-Nol		797.38	23.4	7.5	11.1	49.99	3.96	9.6	8.61	1254.65		0.91	0	0
17	6/9/2016 0:15	0:15	1		Floor2-Nol		797.38	23.4	7.5	11.1	50.01	3.93	9.57	8.61	1254.79		0.91	0	0
18	6/9/2016 0:16	0:16	1		Floor2-Nol		797.38	24	7.5	11.1	49.99	3.99	9.75	8.76	1254.95		0.91	0	0
19	6/9/2016 0:17	0:17	1		Floor2-Nol		797.38	24	7.5	11.1	50.07	3.99	9.75	8.79	1255.08		0.91	0	0
20	6/9/2016 0:18	0:18	1		Floor2-Nol		797.38	24.3	7.5	11.4	50.02	4.02	9.78	8.79	1255.23		0.91	0	0
21	6/9/2016 0:19	0:19	1		Floor2-Nol		797.38	21.3	7.5	11.4	49.96	3.75	9.15	8.22	1255.37		0.91	0	0
22	6/9/2016 0:20	0:20	1		Floor2-Nol		797.38	21.3	7.5	11.4	50	3.78	9.15	8.19	1255.51		0.91	0	0
23	6/9/2016 0:21	0:21	1		Floor2-Nol		797.38	21.3	7.5	11.1	50.03	3.78	9.15	8.19	1255.65		0.91	0	0

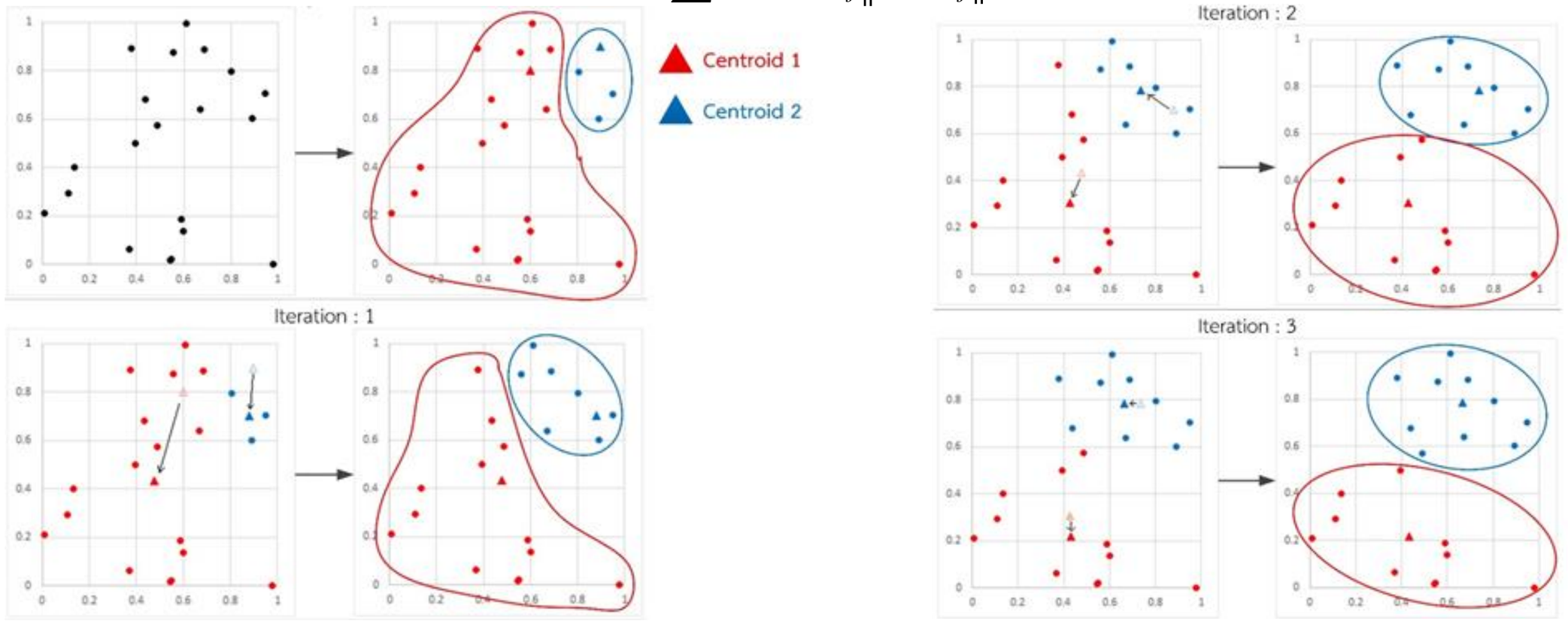
F1_EM F1_N F2_EM F2_N F3_EM F3_N F4_EM F4_N F5 F6 F7_EM F7_N F8 F9 F10 F11 F12



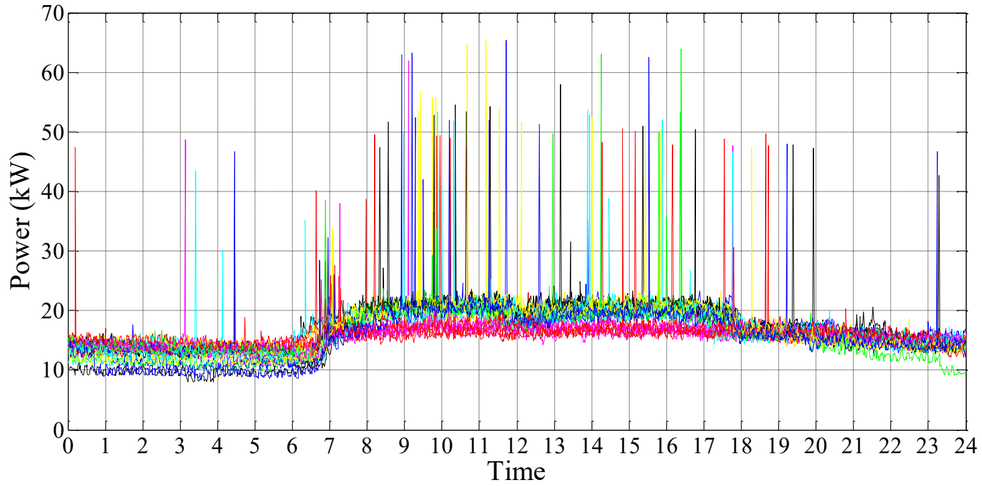
ตัวอย่างของการทำเหมืองข้อมูล (Data Mining)

การจัดกลุ่มโดยค่าเฉลี่ย K (K-mean Clustering)

$$Cost = \sum^{n} (\operatorname{argmin}_j \left\| x_i - c_j \right\|^2)$$



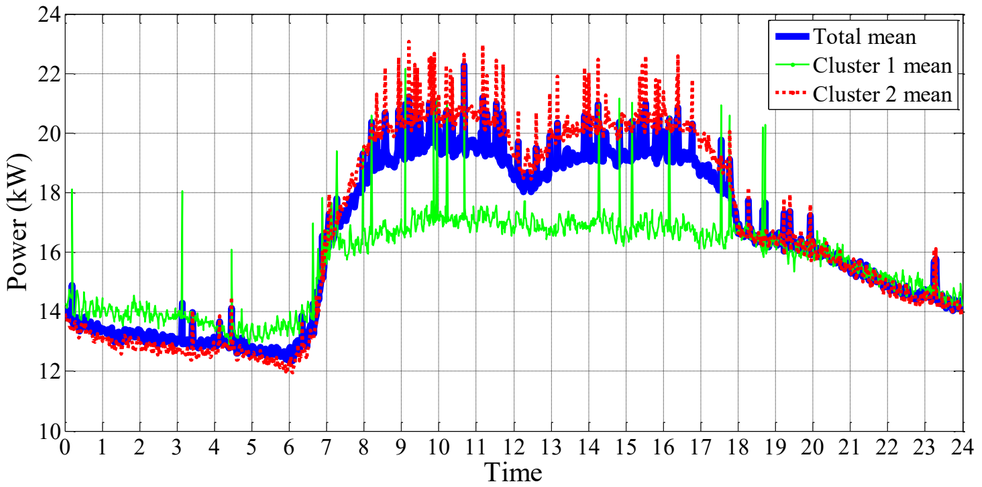
Data



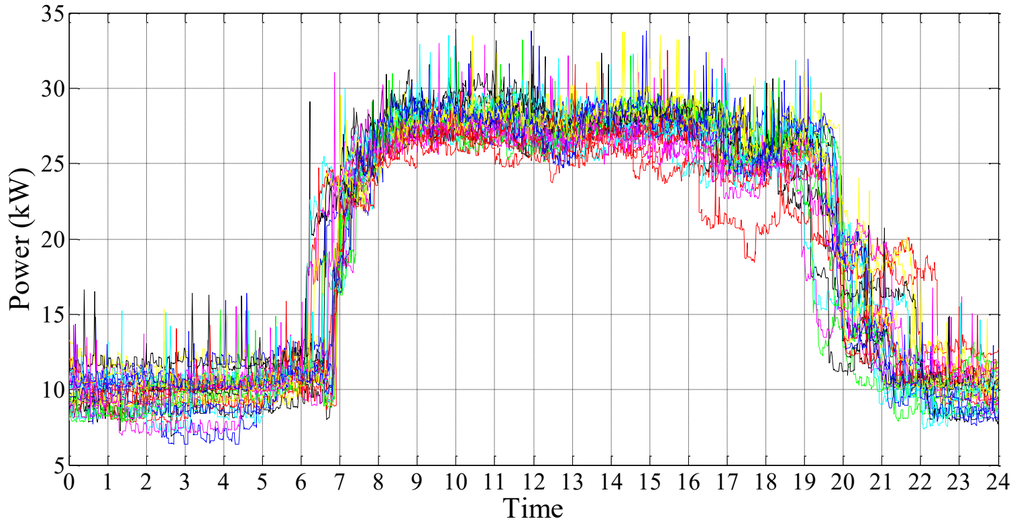
Clustering results

Date / Week	Thu	Fri	Sat	Sun	Mon	Tue	Wed
1	2	2	1	1	2	2	2
2	2	2	1	1	2	2	2
3	2	2	1	1	2	2	2
4	2	2	1	1	2	2	2

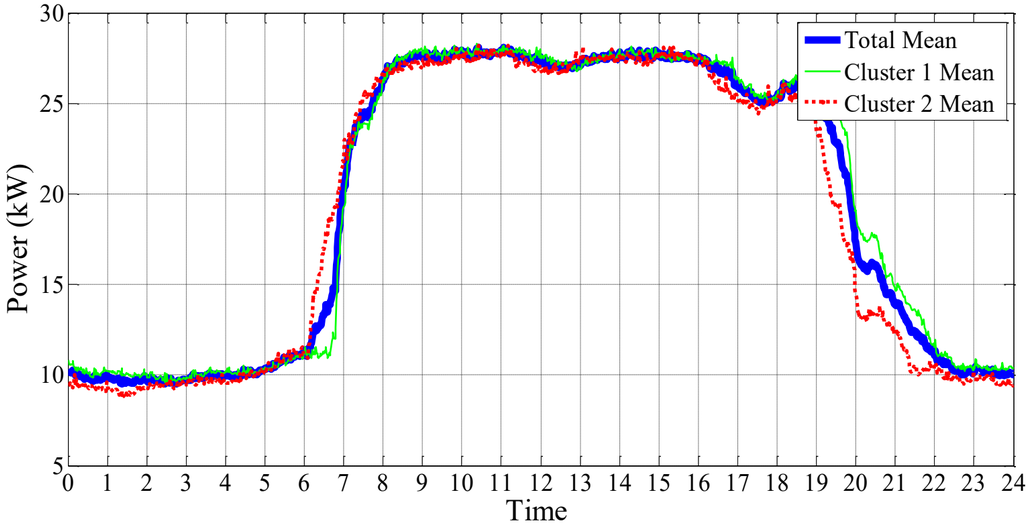
Cluster means



Data



Cluster means

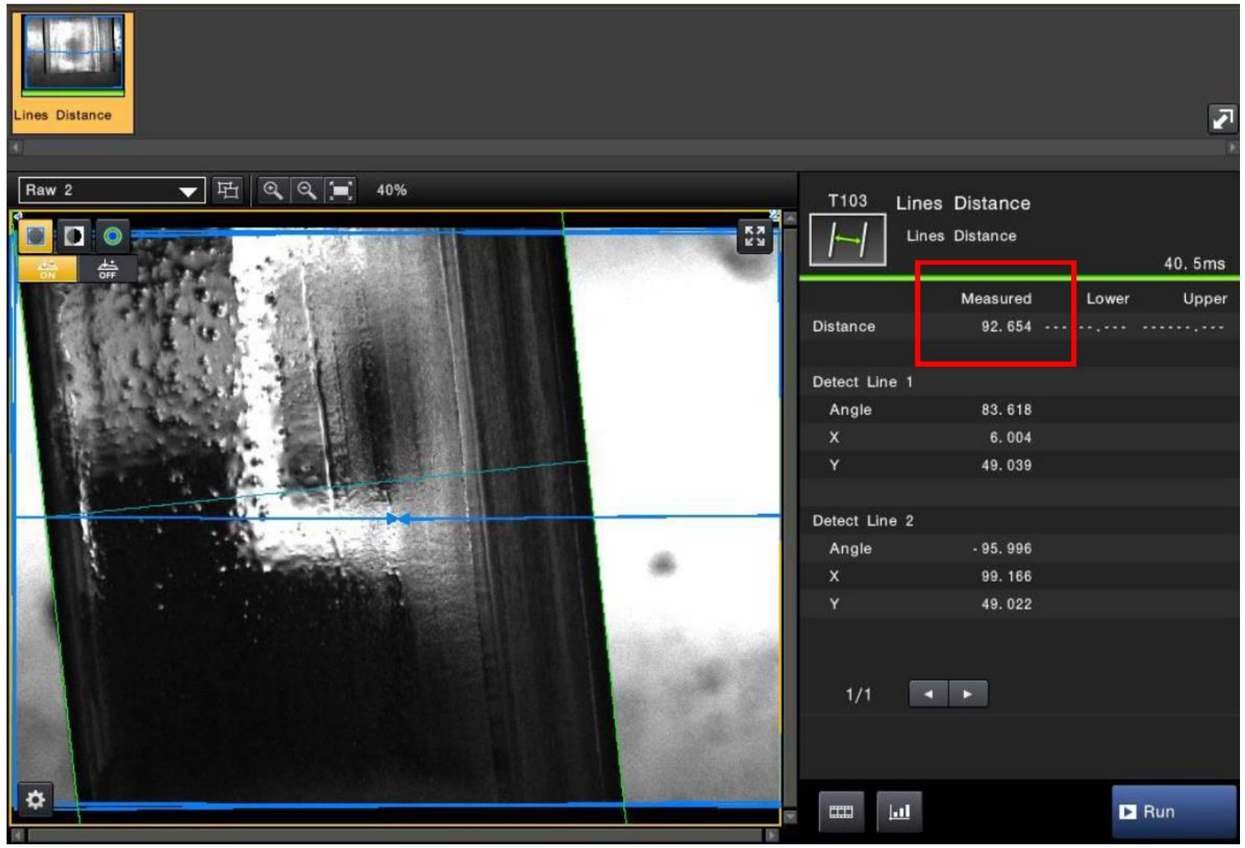


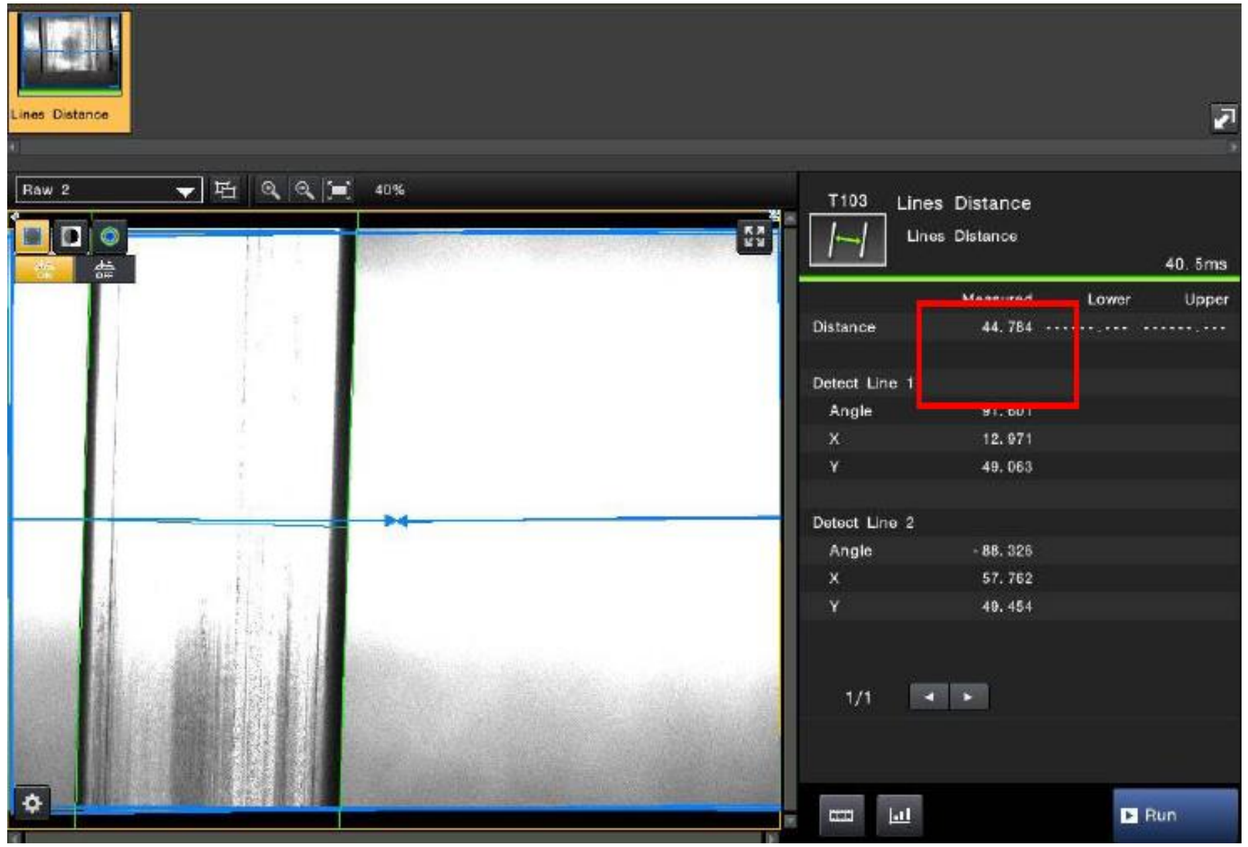
Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

Training

UMES x KMITL x TIRA x ABAC



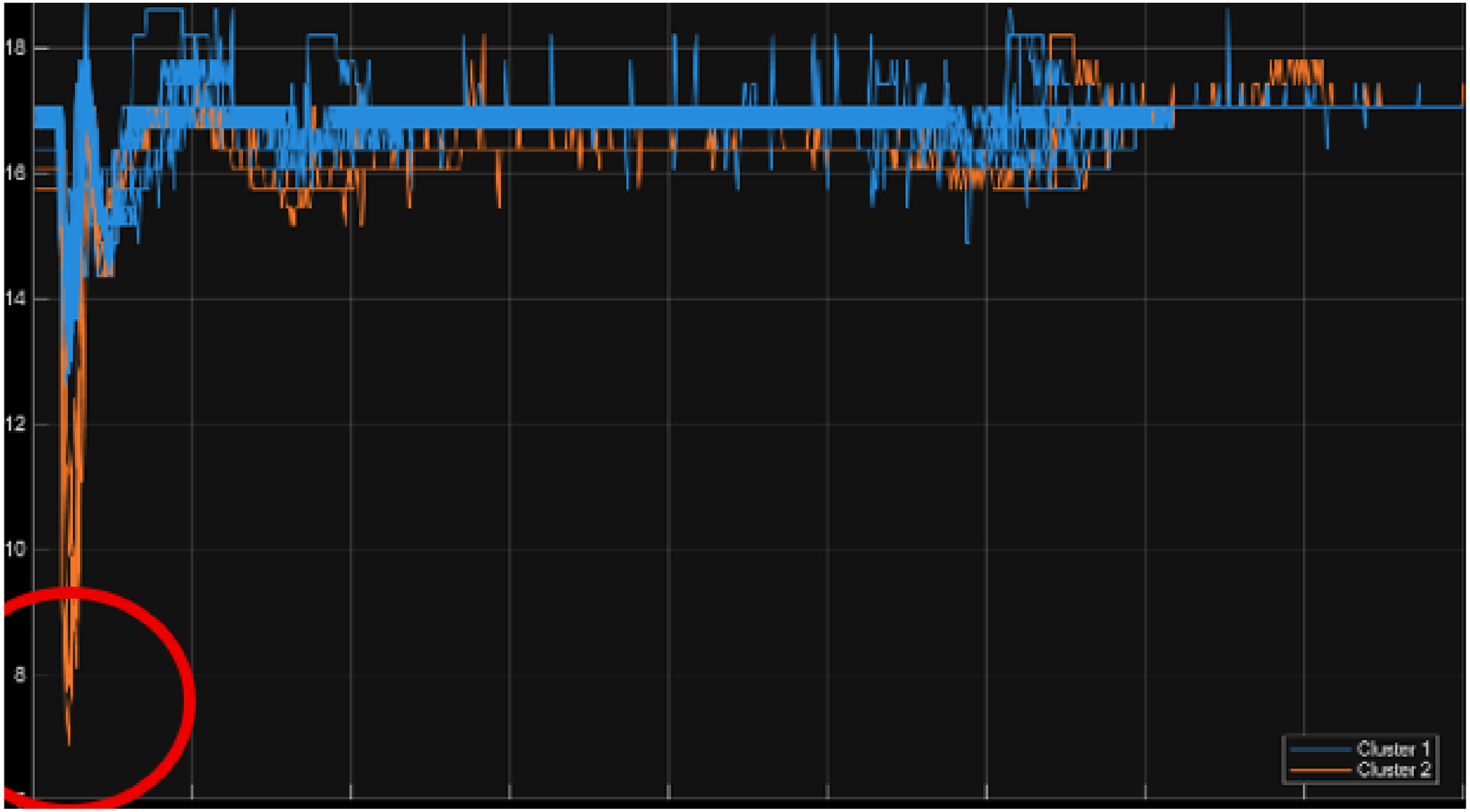


Smart Automation

IIoT & Cyber-Physical Systems for industry 4.0/5.0

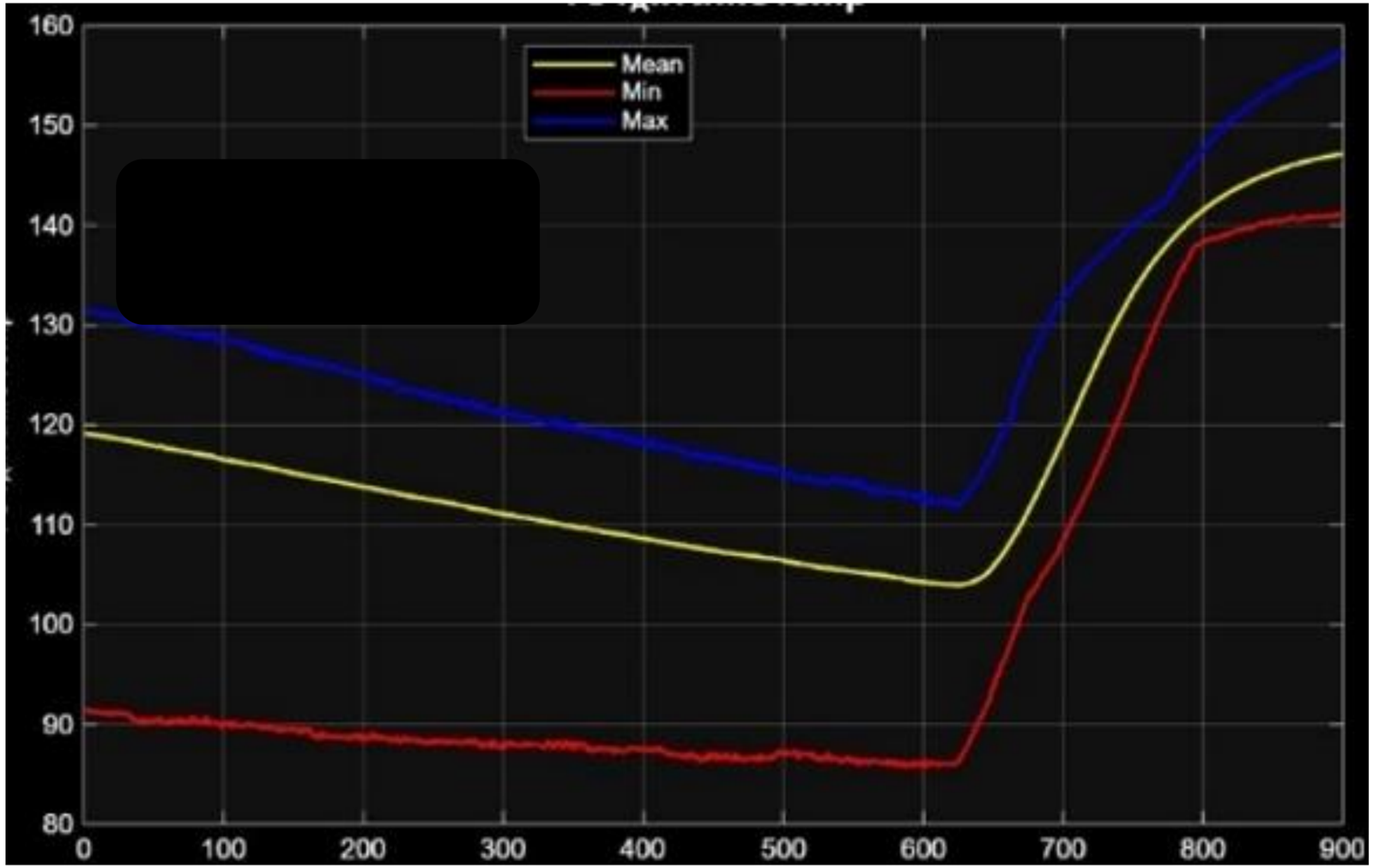
Training

UMES x KMITL x TIRA x ABAC



ข้อมูลที่มีความผิดปกติ

สร้าง Benchmark ให้กับข้อมูล



การทำนายข้อมูล (Data Forecasting)

Very short term forecast : Regression, Time-series, Artificial Intelligent, Machine Learning

Short term forecast : Regression, Time-series, Artificial Intelligent, Machine Learning

Medium term forecast : Time-series, Artificial Intelligent, Machine Learning

Long term forecast : Artificial Intelligent, Machine Learning

หลักการทำหนายข้อมูล (Forecasting principle)

Data

Predictive Model (AI)
- Correlated data
- Historical data
- Training
- Testing

Forecasting Results

ความแม่นยำของการทำนาย (Forecasting Accuracy)

- Mean Relative Error

$$MRE = \frac{1}{N}\sum_{i=1}^{N}\frac{x_{p,i} - x_{a,i}}{x_{\max}} \times 100\ \%$$

วิเคราะห์ทิศทางความผิดปกติ

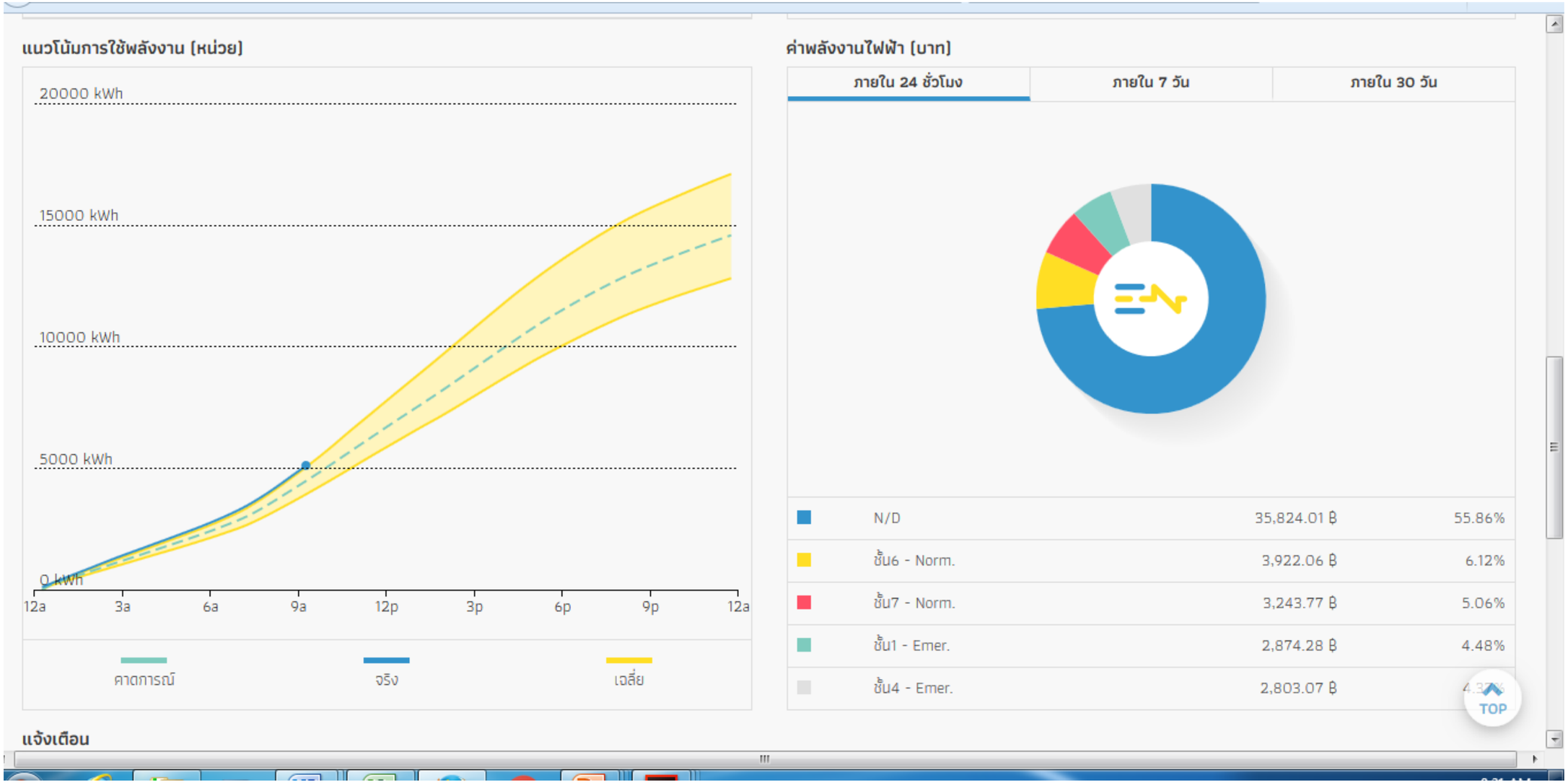
- Mean Absolute Percentage Error

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|x_{p,i} - x_{a,i}\right|}{x_{a,i}} \times 100\ \%$$

วิเคราะห์ความผิดปกติในภาพรวม

ตัวอย่าง AI การทำนายข้อมูล (Data Forecasting)

- Neural Network (NN)
- Support Vector Machines (SVM)
- Long Short-Term Memory (LSTM)
- Gate Recurrent Network (GRU)


แนวโน้มการใช้พลังงาน (หน่วย)
20000 kWh
15000 kWh
10000 kWh
5000 kWh
0 kWh
12a
3a
6a
9a
12p
3p
6p
9p
12a
คาดการณ์
จริง
เฉลี่ย
ค่าพลังงานไฟฟ้า (บาท)
ภายใน 24 ชั่วโมง
ภายใน 7 วัน
ภายใน 30 วัน
N/D 35,824.01 ฿ 55.86%
ชั้น6 - Norm. 3,922.06 ฿ 6.12%
ชั้น7 - Norm. 3,243.77 ฿ 5.06%
ชั้น1 - Emer. 2,874.28 ฿ 4.48%
ชั้น4 - Emer. 2,803.07 ฿
TOP
แจ้งเตือน

ที่มา : Enres.co

การทำนายข้อมูลของข้อมูลที่มีลักษณะผิดปกติ (Anomaly Data Detection)

